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CDR System Operators Course



Bosch Crash Data Retrieval System

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Crash Data Specialists

- Formed in 2011
- Using CDR and Crash Testing for over 12 years
- Experience pre-CDR from 97+

CRASH DATA SPECIALISTS

WWW.CDR-TRAINERS.COM

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Training Overview

- System Operators Course – 2 days
  - CDR System components, vehicle coverage software operation – data collection
  - Kia/Hyundai/Tesla System Operation
- Analysis and Applications Course – 5 days
  - All currently supported OE system nuances
  - Example CDR reports and line-by-line analysis
  - Using Delta-V, PDOF and other parameters in your reconstruction

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Training Overview

- Advanced Reconstruction with CDR Applications

  in partnership with Accident Analysis and Recon – Greg Russell

Additional 5-Day class

  - Working with Delta-v, Restitution
  - Momentum analysis with different “knowns”
  - Laws of Sines and Cosines
- Check NUCPS website for scheduled classes

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Your Background

- Helpful Pre-requisites:
  - Crash Investigator training
  - Awareness of evidence collection procedures
  - Familiarity with basic vehicle systems and components
  - Personal computer and software operation

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Class Outline

- Terms and Conventions
- Software Installation Operation and Activation
- Determining Coverage
- Methods of Collection
- CDR System Components
- Equipment Setup and connection to vehicle / components
- Imaging Process / Saving Imaged Data
- Troubleshooting Steps
- Legal Issues
- Intro to Back-Powering
- Hands-On Practice



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Evaluation and Grading

- Course Evaluation and Grading
  - Classroom participation
  - Hands-on practical experiences
  - Final Examination

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CDR System Operators Course



Overview

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What is CDR / EDR? What are we doing and why?

- **CDR** is the Bosch Tool that allows us to collect **EDR**.
- CDR stands for Crash Data Retrieval
- EDR stands for Event Data Recorder
- We are collecting data stored within vehicle systems to aid in the reconstruction of traffic crashes.



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“CDR”

- **C**rash **D**ata **R**etrieval Tool
  - Manufactured by Bosch
  - Sold in the United States by Crash Data Group only
  - A tool used to image (download) the data from within a vehicle

BOSCH
Invented for life

- **C D R** is the acronym for the tool utilized to image the data from most vehicles
  - Software and Hardware
  - Used for most vehicles that record event data



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How do we Collect the data? (DLC)

- DLC (Data Link Connector) into OBD (On Board Diagnostic Port)





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How do we Collect the data? (D2M)

- ACM Cable (Airbag Control module) aka Umbilical Cable D2M (Direct to Module)





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How does the vehicle communicate with my Computer?

- CanPlus Interface



- CDR900 Interface



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What App or Software do I use?



CDR Legacy Software



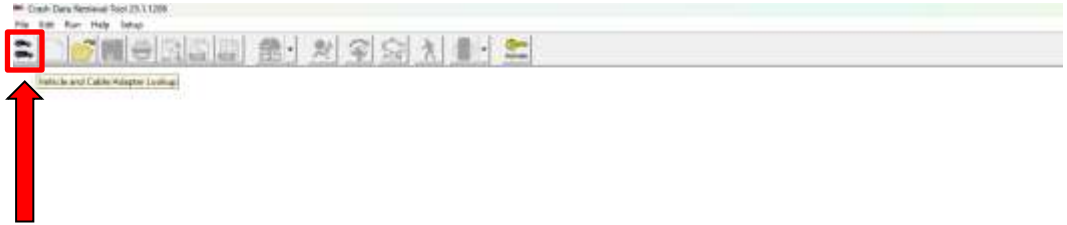
CDR 2

There are currently 2 apps for that!

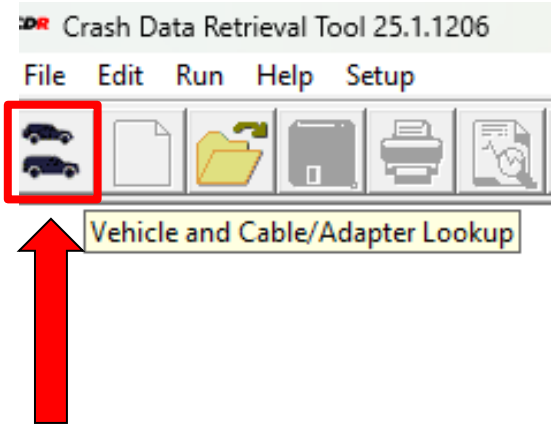
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How do I know which app to use?







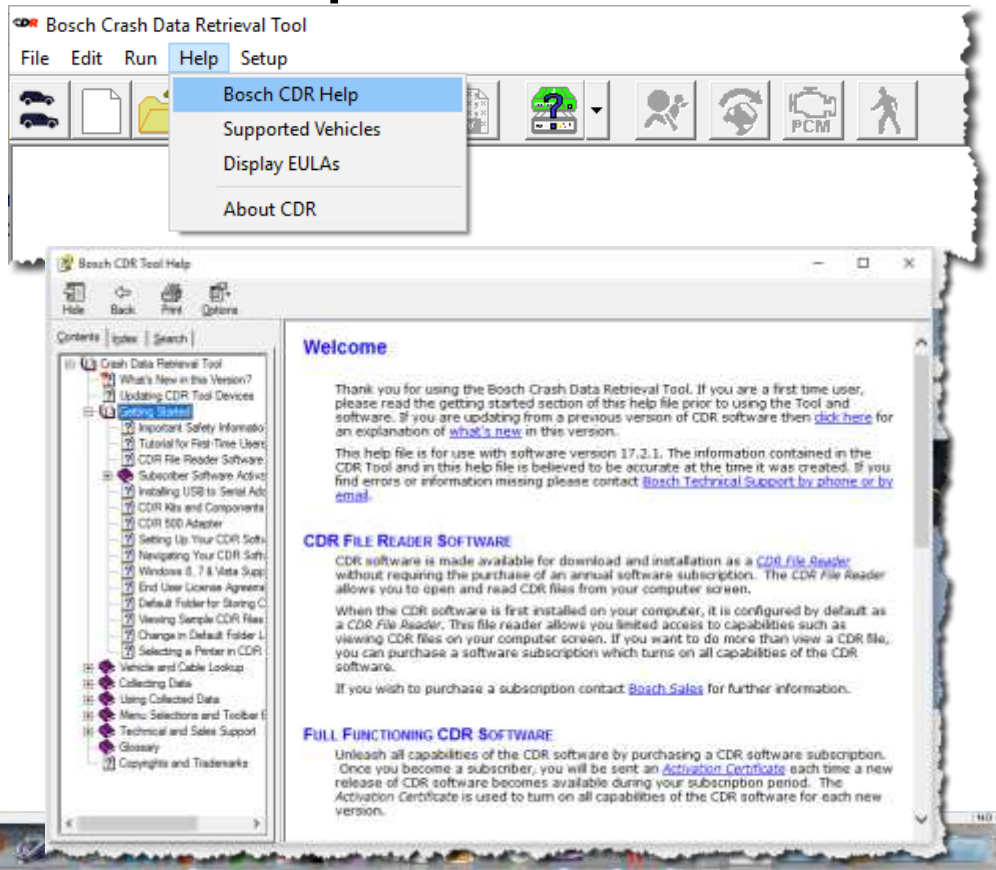
This should always be your first stop. Check the Vehicle Lookup section of the Help File.



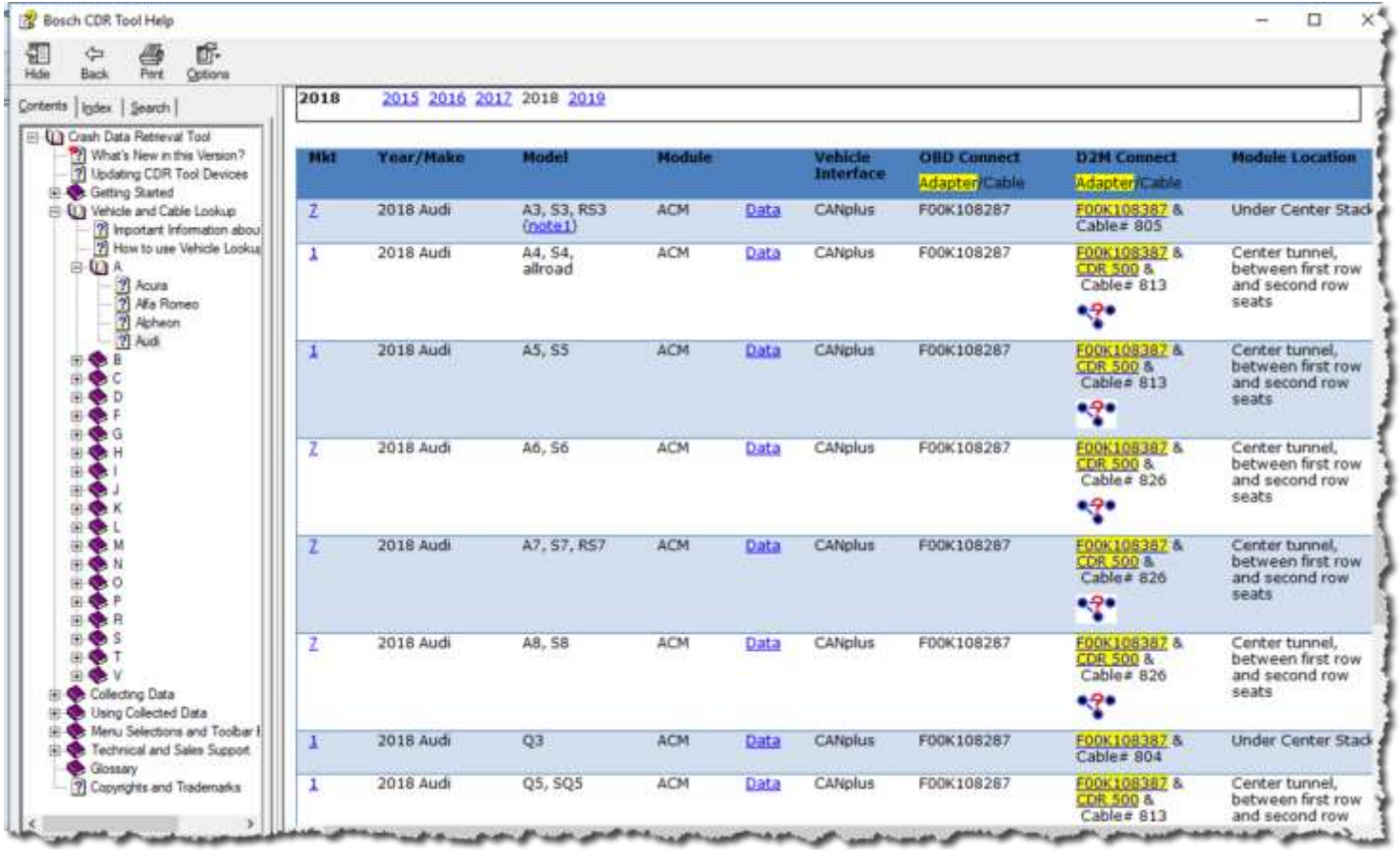
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Bosch CDR Help File

- **Vehicle and Cable Lookup**
- Collecting data
- Using collected data
- Menu and toolbar
- Tech support
- Glossary



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Always Check the Help File

- If you could remember only one thing from this class it would be to always check the supported vehicle and cable lookup section of the Help File.
- You should never do a download without first checking the Help File.

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Introduction to EDR

Courtesy of Crash Data Group



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Event Data Recorders

- Vehicles with Event Data Recorders (EDR's)
- Light Vehicles - passenger cars, SUV's, light trucks, etc.



- Heavy Vehicles – truck-tractors, etc.



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Event Data Recorders

Focus of this course on Light Vehicles, not big trucks

- Cars
- SUV's
- Vans
- Pickup Trucks
- Small Chassis Vehicles
- Ambulances, small busses, small dump trucks, etc.





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Event Data Recorders

- In Light Vehicles: different types of modules that MAY record data, some examples of these are:
  - Airbag Control Modules (ACM)
  - Powertrain Control Modules (PCM)
  - Rollover Sensors (ROS)

- Heavy Vehicles (separate course)
  - Electronic Control Module (ECM)
  - Anti-Lock Brake Systems
  - Radar Systems

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Why EDR's ?

- EDR's Voluntarily installed by many Automakers
- CFR 49, Part 563 – "The Part 563 Rule" specifies data elements to be recorded
  - Time period
  - Resolution
  - Accuracy
  - Effective for vehicles manufactured for sale after 09/2012

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Other Systems

- Hyundai and Kia
  - Read out tool system
  - Separate for each vehicle line





- Tesla



- This course will focus on the Bosch Crash Data Retrieval Tool (CDR)

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CDR System Operators Course



Terms and Conventions

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Types of Download Data and Modules

- Airbag Control Modules (ACM)
- Powertrain Control Modules (PCM)
- Rollover Sensors (ROS)
- Pedestrian Protection Module (PPM)
- Active Safety System Data

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Airbag Control Module

- ACM: Airbag Control Module
  - The generic name for the control module for airbags and related restraint systems
  - Each Manufacturer uses a different name for their ACM
  - This includes both primary and supplemental restraint devices





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Airbag Control Module



- The primary function of the ACM is to:
  - Sense a developing collision
  - Decide if any devices need to be deployed
  - Deploy the devices as appropriate
- Crash data recording is a secondary function
- Often compared to Flight Data Recorder
  - Flight Data Recorder performs NO safety function
  - There to record crash data



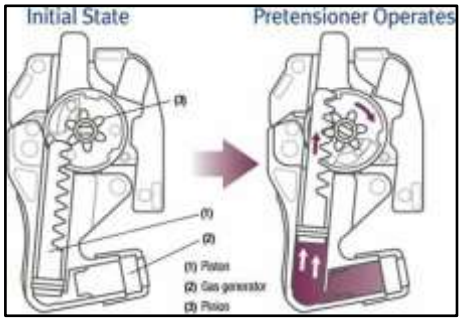
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Airbag Control Module

- Supplemental Restraint Systems
  - Airbags – Frontal, side curtain, torso, knee, etc.



  - Seatbelt Pretensioner



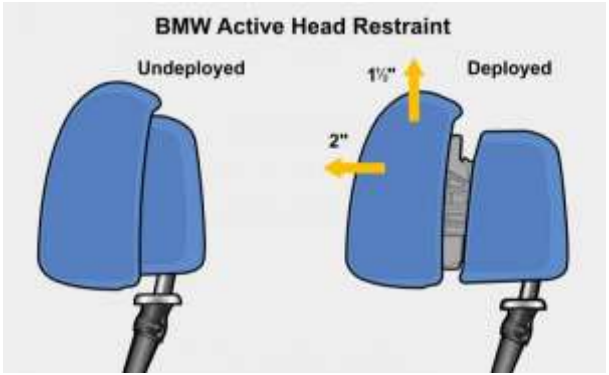




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Airbag Control Module

- Supplemental Restraint Systems
  - Active Head Restraint System







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Airbag Control Module aka...

- SRS – Supplemental Restraint System
- ORC – Occupant Restraint Controller
- RCM – Restraint Control Module
- SDM – Sensing Diagnostic Module

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Powertrain Control Module

- PCM: Powertrain Control Module
  - All vehicles have some type of Powertrain Control Module
  - Ford is the only one that recorded data within the PCM
    - Controls the powertrain vehicle components
    - Event data recorder may record information
    - 2003 through 2011 – Some vehicles
      - Part 563 – 2012



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Rollover Sensor

- ROS: Rollover Sensor
  - Some vehicles are equipped with rollover sensors
    - Side curtain airbags, etc.
  - GM is the only one that recorded data within the ROS
    - 2007 through 2015 – Some SUV's
    - Can be imaged DLC or Direct to Module



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Pedestrian Protection Module

- ECU: Electric Control Unit
  - Controls electronic subsystems within a vehicle
- PPM: Pedestrian Protection Module
  - A dedicated ECU which integrates the pedestrian protection system algorithm to detect a pedestrian impact and deploy devices which increase the gap between the vehicle's hood and engine to help cushion pedestrian impact



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Active Safety System Data

- FCM: Forward Facing Camera Module
  - Images captured leading to event
- ACSM: Advanced Crash Safety Module
  - Automated System Control Data
  - Tracked Target and Alert Data
  - Driver Selection and Action Data
  - Vehicle Action Data



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Front Camera Modules



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Active Safety System Data

- ACSM: Advanced Crash Safety Module
  - Potential Data Includes
    - Driver input data
    - Automatic system data
      - Diagnostics, automatic braking
      - Situational data – Construction zones, motorcycles, etc.
      - Lane position, etc.
      - Suppression reasoning



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Active Safety Control Module

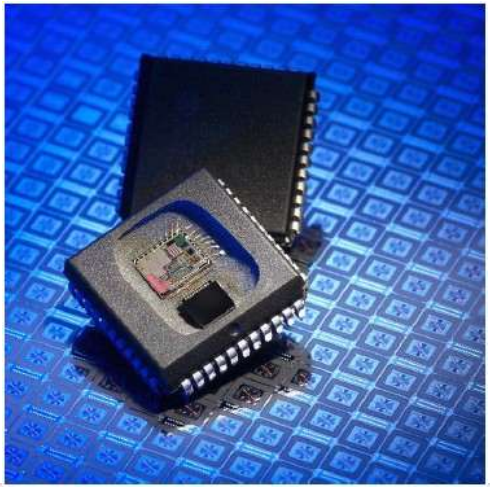


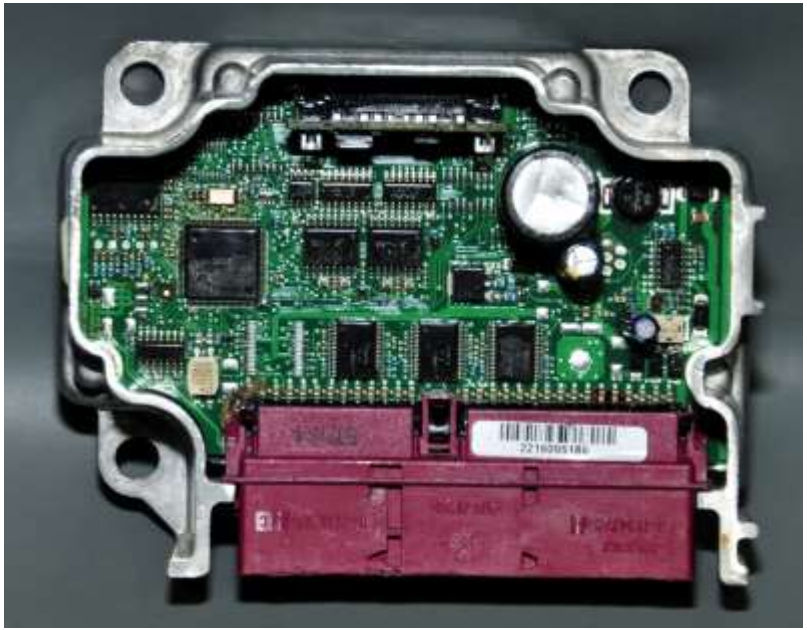


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Where is data stored?

- The data is stored to Flash Memory / EEPROM inside the ACM / PCM / ROS





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How is data stored?

- The data is stored in hexadecimal format and is translated by the CDR Software and presented in the CDR report

```
DPID $32
00 FF 00 04 00 00 00

DPID $35
78 00 00 00 00 00 00

DID $01
41 55 32 33 34 31 54 4D 44 49 52 44 35 39 38 4E

DID $03
41 54 32 33 34 31 54 4D 42 51 44 32 54 39 38 4E

DID $05
41 48 32 33 34 30 54 38 43 34 52 33 4C 44 38 4C

DID $07
41 4A 32 33 34 30 54 38 39 56 45 30 35 44 38 4C

DID $09
30 30 30 30 30 30 54 36 44 49 46 42 4C 39 38 4E

DID $0B
30 30 30 30 30 30 54 4D 44 49 4B 42 54 39 38 4E

DID $0D
30 30 30 30 30 30 54 30 30 30 30 30 30 30 30 30

DID $0F
30 30 30 30 30 30 54 30 30 30 30 30 30 30 30 30

DID $30
01 00 01 01

DID $90
32 47 31 46 43 31 45 56 33 41 39 2A 2A 2A 2A 2A 2A

DID $9A
03 01

DID $B4
41 53 33 36 34 35 54 30 38 32 36 31 30 30 38 38
```

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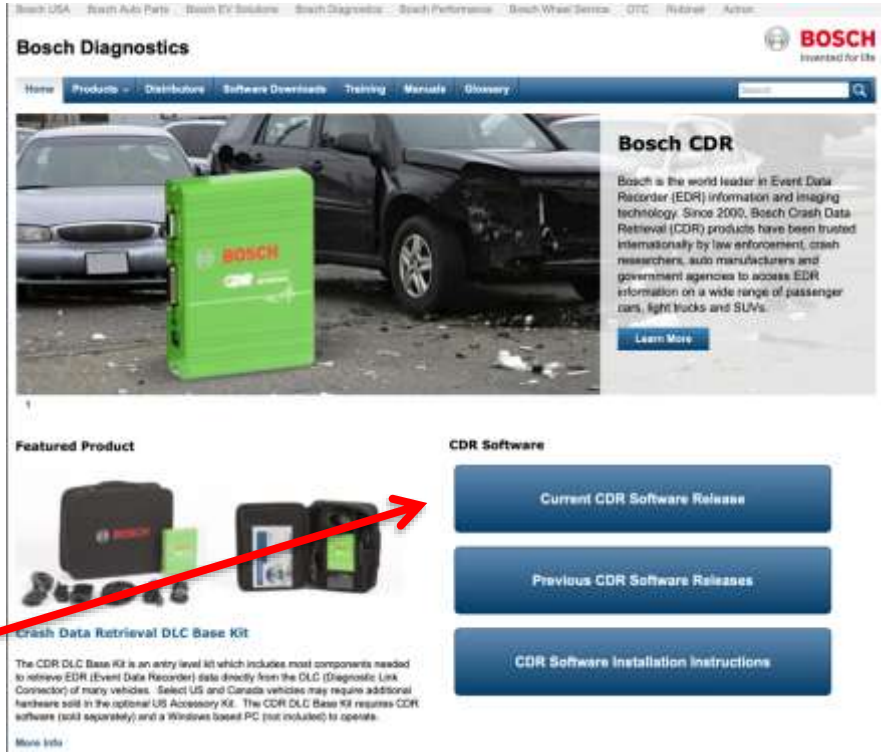


Software Installation and Operation

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Software Installation

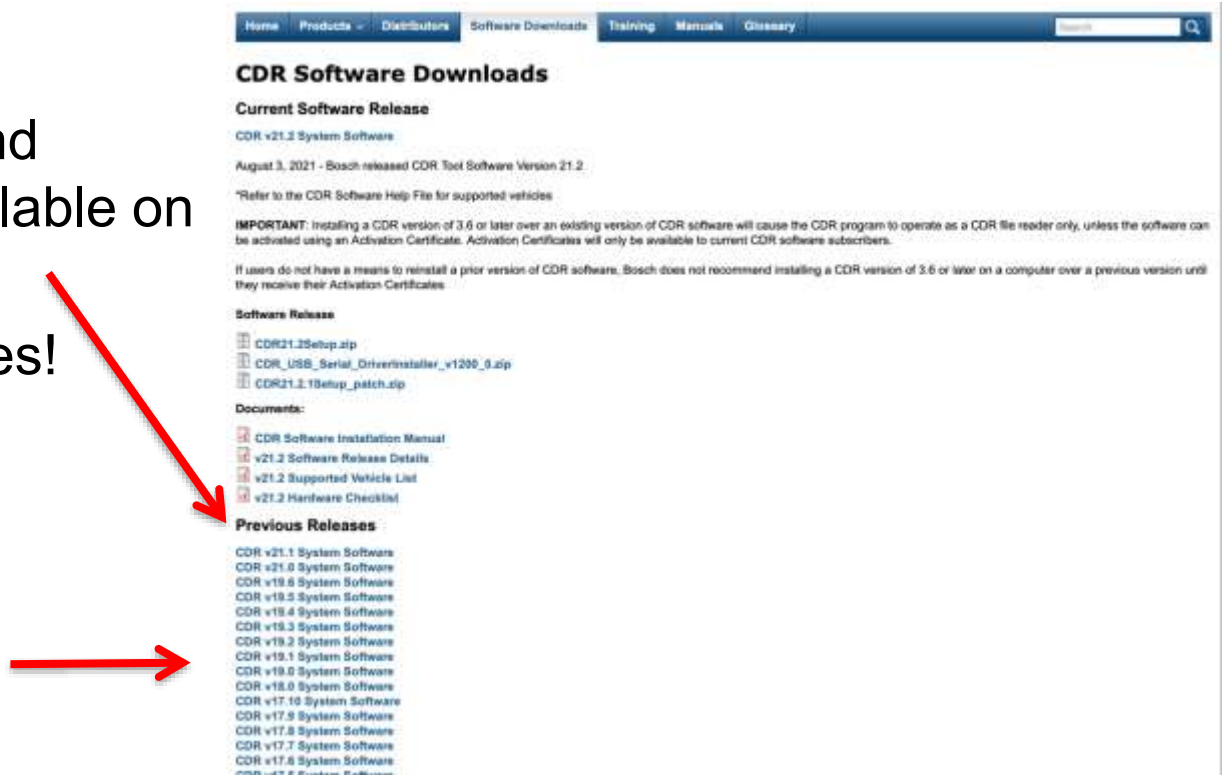
- Install the most recent version of CDR software on your PC. You can find the latest software on the Bosch Diagnostics CDR website at the following location: https://cdr.boschdiagnostics.com/cdr/
- Select “Current CDR Software Release”



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Previous Versions

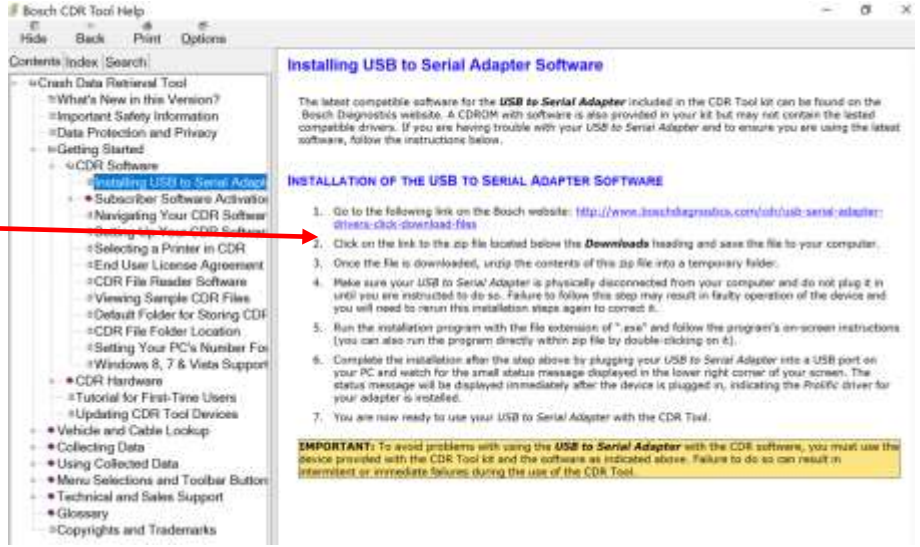
- Previous Release and Patches remain available on Bosch Website
- Keep your Certificates!



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Installing the USB to Serial Adapter Drivers

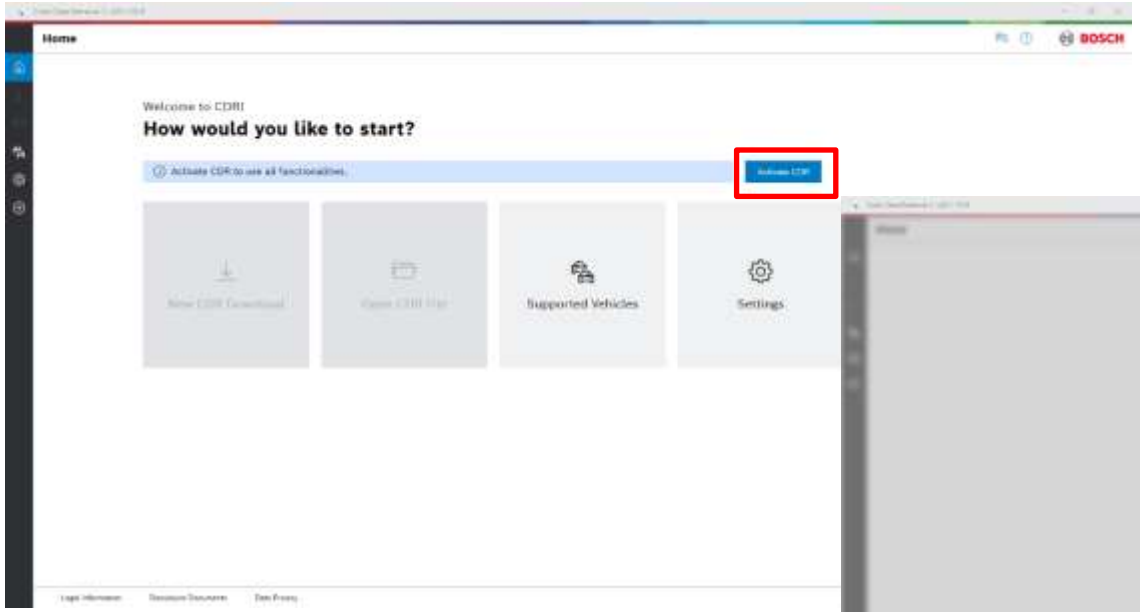
- USB to Serial Adapter Drivers ***should only*** be installed from the link in the CDR Software help file.
- Updates found on the internet may cause the USB Serial convertor to not work.

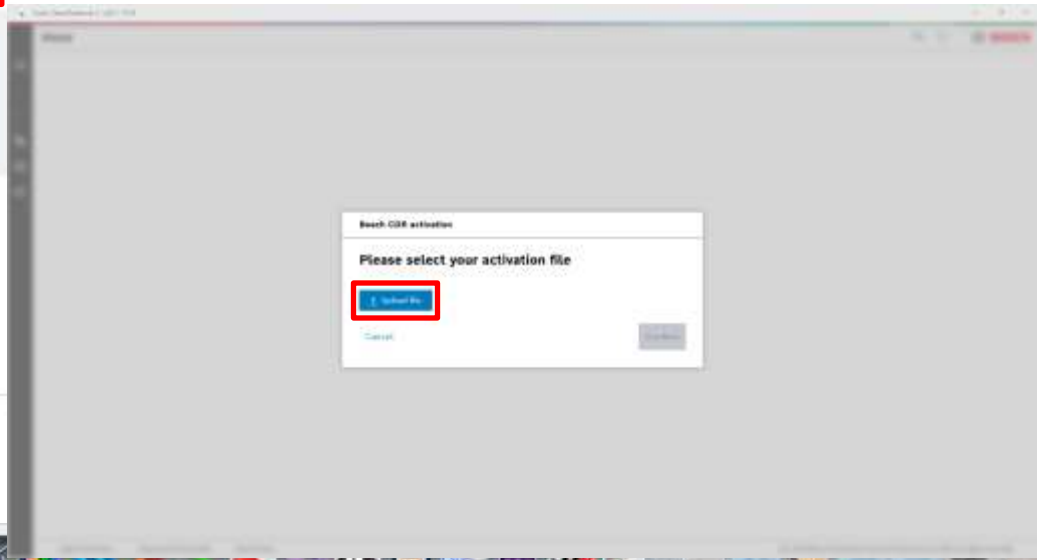


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Activating the Software

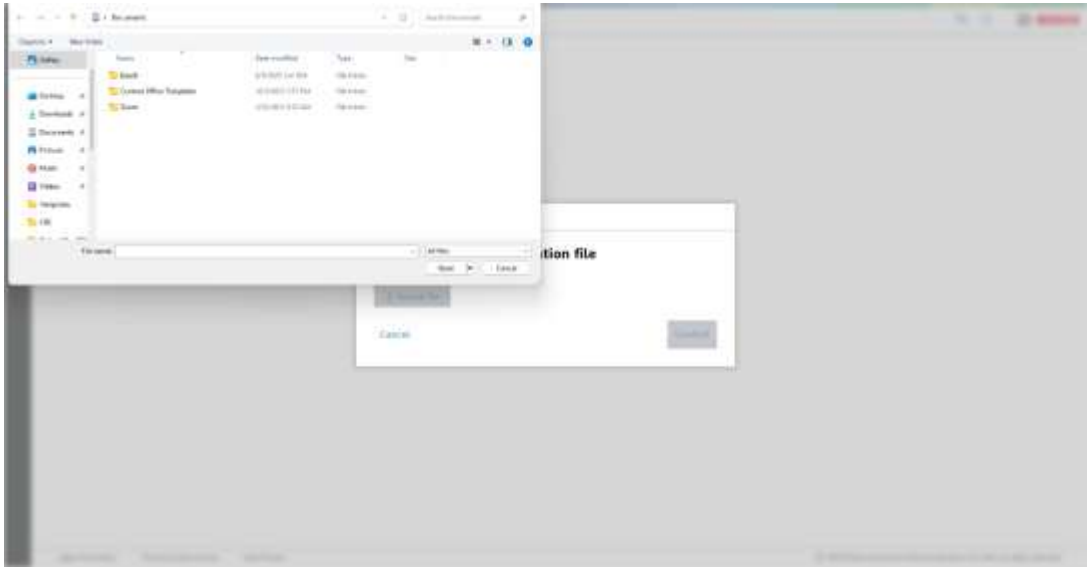




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Activating the Software

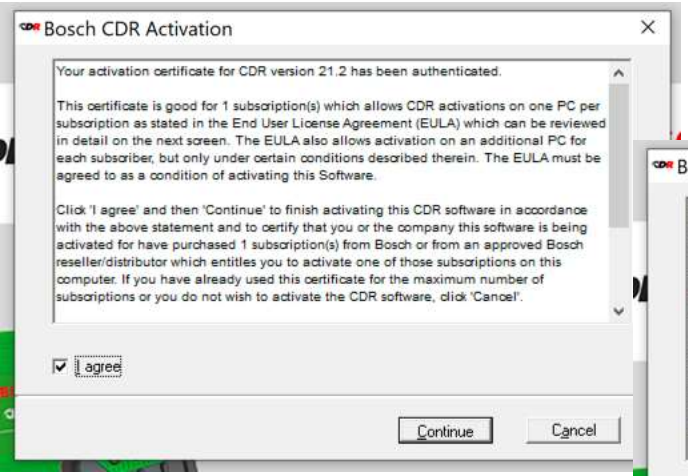


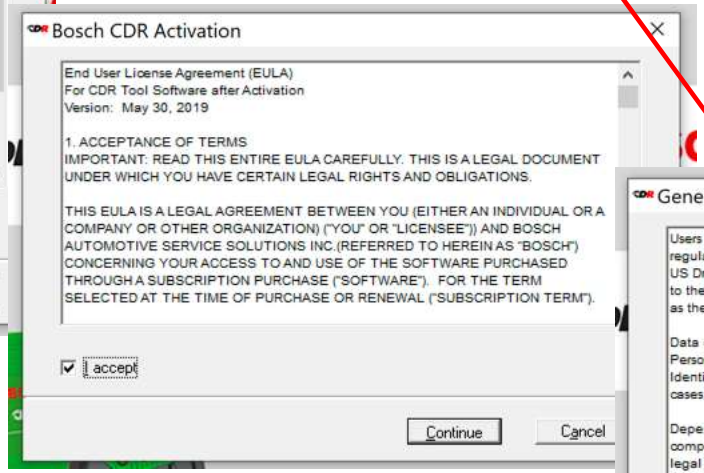
- Navigate to .CTF file and activate software.
- This will activate both CDR2 and the Legacy Software

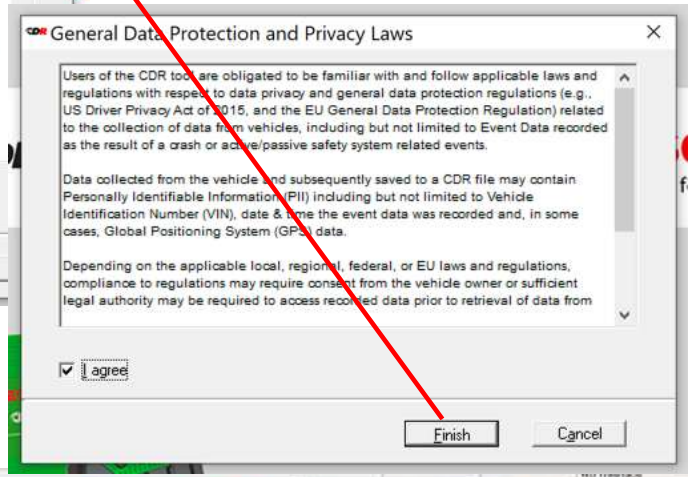
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After Activating the Software







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Updating CDR Software

- Updates do not overwrite CDR files
- Updates just add to the software already installed
- Most recent new software versions only update previous versions and can be installed on top of current version
- Updates don’t overwrite CDR files stored in the Windows Bosch Program Directory

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CDR System Operators Course



CDR System Components

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Minimum System Requirements

- PC based* computer running windows XP or later
  - *Will also run on a Mac using Parallels or Fusion with a Windows installation
- Serial port or a USB port for interface connection to computer

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Interface Module



- The Interface Module
  - A part of the CDR System which is comparable to a modem, a part that facilitates communication between a computer and an accessible vehicle's module using the CDR System software
- The Interface Module simply allows your laptop computer to communicate with the module from a vehicle
  - It sends and receives information at a specific data transfer rate
  - It does not translate anything, nor does it change or alter anything



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CanPlus Interface Versions

- “Vetronix” Blue Metal
- “CANPlus:
- “ETAS” on back



DISCONTINUED!

- ‘Bosch’ Green Metal
- “CANPlus”
- ***Both the same inside***



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CanPlus Interface Module

- The Interface Module
  - Three connections
    - Computer (PC)
      - Serial to USB
      - 9 pin connector
    - Vehicle (Cable for Vehicle Data)
      - 15 pin to Vehicle or Module
    - Power



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CanPlus Interface Module

- The Interface Module
  - Interface Power
    - Directly from the DLC
    - From Wall Power (AC)
    - From Vehicle or Jump Box (DC)
  - When there is power at the Interface Module, the green light will illuminate
    - This means there is power here (at the Interface Module) ONLY!



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Canplus Interface Module

- The Canplus Interface Module
  - Connects to a computer using a Serial to USB adapter cable
  - Or serial cable on an older computer



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CDR 900 - Interface

- Next Generation CDR Interface
- USB / Wireless
- Supports all legacy protocols
- Second channel of CAN added
- Ethernet
- CAN FD
- **Integrated ACM Adapter**

- FlexRay (CDR 500) still required
- CANplus interface will still be required for some applications (i.e. Ford PCM and some legacy coverage)



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CDR 900 Interface Kit

- Includes:
  - VCI
  - I/F Powering Cable
  - OBD II Cable
  - Legacy Cable Adapter
  - 10ft USB Cable
  - *Wifi Modules
  - Extension Cable

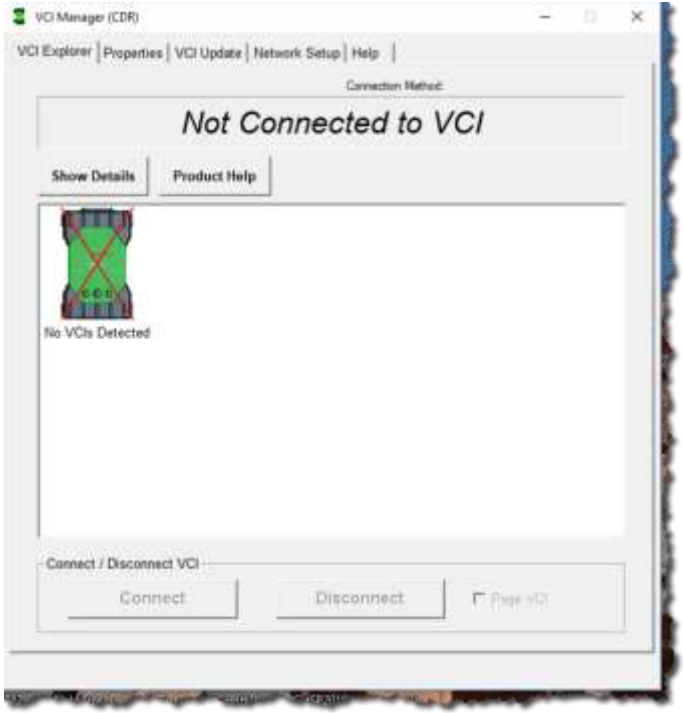


*MUST use modules provided – does ***NOT*** currently support peer-to-peer from laptop built-in wifi

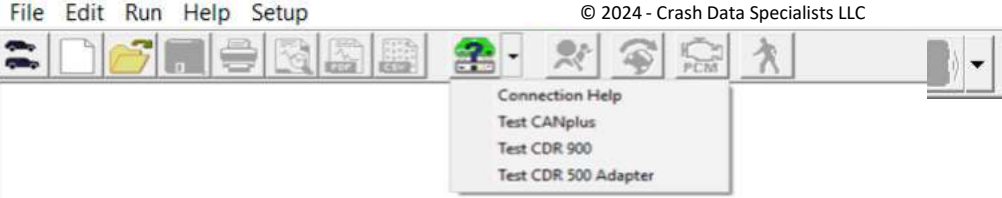
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CDR 900 Interface

- Requires specific setup steps
- Must “register” interface to your PC
- Only 1 x CDR900 can be registered to any PC at a time
- Can switch back-and-forth any time you need to
- Use VCI Manager



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Connection Help – CDR I/F/500/900

- Available from Help File and Icon Menu Bar
- Will assist you with connections for Both Interfaces and the CDR 500
- If in doubt, Test Communications once setup complete, before retrieving data



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Adapters

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Adapters

- There are multiple adapters that have been engineered for the CDR tool to accommodate new and changing technology in vehicles.
- These adapters are ONLY for the CanPlus (with one exception) because the CDR900 has technology that allows internal switching to adapt to different vehicles.

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CanPlus DLC Adapters

- Diagnostic Link Connector (DLC) Adapters
  - Certain adapters utilized when collecting data via the DLC method (explained in detail later)
  - DLC Adapter, DLC 3 Adapter or DLC 4 Adapter



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CanPlus Adapters

- ACM Adapter
  - Gray in color
  - Must be used when collecting data from some ACM's. For Direct to Module collection only



- PCM Adapter
  - Black in color
  - ALWAYS used when collecting PCM data whether the method is DLC or D2M.



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CDR 500

- CDR 500 Adapter
  - Adapter utilized to retrieve data from vehicles with a FlexRay communication network
  - Can be used with the CanPlus or the CDR900
  - Different than the CAN or K-Line Network
  - Enables communication between CDR Software and Module



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Adapters

- ACK Adapter
  - Yellow Direct to Module cable utilized for D2M imaging on some Stellantis vehicles.



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ACM/Umbilical/D2M Cables

- Direct to Module – MANY Individual Cables
- Some are still available; some have been discontinued
- Full list of equipment is available for inventory purposes in the CDR Help File



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Kit Organization



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Cables

- Diagnostic Link Connector (DLC)
  - Preferred Method and Most Used





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Additional Items for Your CDR kit

- The kit you develop will have to fit your situation/needs but would likely include:
  - Thin work gloves/extrication gloves
  - Wrench/socket set
    - 8/10/12/13mm ratchet-wrenches
    - 8/10/12/13mm deep and shallow sockets (1/4" drive)
    - 1/4 u-joint
    - 1/4 stubby ratchet
    - 1/4" short and long extensions
    - 7mm socket and a star screwdriver for trim
    - deep TORX sockets



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Additional Items for Your CDR kit

- The kit you develop will have to fit your situation/needs but would likely include:
  - Pliers
  - Slotted screwdriver
  - Adjustable wrench
  - 12v “booster box” and/or jumper cables
  - Long “penny” nails
  - Paint or “Sharpie” marker
  - Cutting tools
    - Wire cutters, carpet cutter, blade knife, both a "hooked" and normal blades



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CDR System Operators Course



Vehicles with EDR Capabilities

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CDR Covered Vehicles

- General Motors Family
  - 1994 to Present



Oldsmobile
GMC
PONTIAC
HOLDEN
HSV
CORVETTE
Geo
BUICK
CHEVROLET
SATURN
Cadillac
HUMMER

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CDR Covered Vehicles

- Ford Motor Company Family
  - 2001 to Present





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CDR Covered Vehicles

- Fiat Chrysler Automobile Family
  - 2005 to Present

FCA
FIAT CHRYSLER AUTOMOBILES

DODGE
FIAT
Jeep
LANCIA
RAM
SRT
CHRYSLER

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CDR Covered Vehicles

- Toyota Family
  - 2001 to Present







  - Then came Part 563….

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CDR Covered Vehicles

- Volkswagen Group
  - 2009 to Present



- Mazda Motors
  - 2011 to Present



- Volvo
  - 2011 to Present



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CDR Covered Vehicles

- Honda Motor Company
  - 2012 to Present

HONDA

ACURA

HONDA

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CDR Covered Vehicles

- Nissan Motors
  - 2012 to Present

NISSAN
MOTOR COMPANY

NISSAN

INFINITI.

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CDR Covered Vehicles

- Bayerische Motoren Werke Auto Group
  - BMW
  - 2013 to Present



- Porsche
  - 2019 to Present



PORSCHE

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CDR Covered Vehicles

- Rolls-Royce
  - 2013 to Present
- Maserati
  - 2014 to Present
- Mercedes-Benz
  - 2014 to Present

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CDR Covered Vehicles

- Mini
  - 2014 to Present



- Audi Auto Group
  - 2015 to Present



- Alfa Romeo
  - 2015 to Present



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CDR Covered Vehicles

- Lamborghini
  - 2015 to Present



- Bentley Motors
  - 2016 to Present



- Subaru
  - 2012 to Present
  - Other tool…



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CDR Covered Vehicles

- Isuzu
  - 1998 to 2009
- Suzuki
  - 2007 to 2014
- Saab
  - 2005 to 2012
- Mitsubishi
  - 2007-2010
- Sterling Trucks
  - 2008-2009

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Other (Non-CDR) Covered Vehicles

- Hyundai Motor Group
  - 2013 to Present

HYUNDAI MOTOR GROUP

HYUNDAI

KIA

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Other (Non-CDR) Covered Vehicles

- Tesla
  - 2013 to Present

TESLA MOTORS

- Others with assistance (on occasion)

LAND ROVER

JAGUAR

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CDR System Operators Course

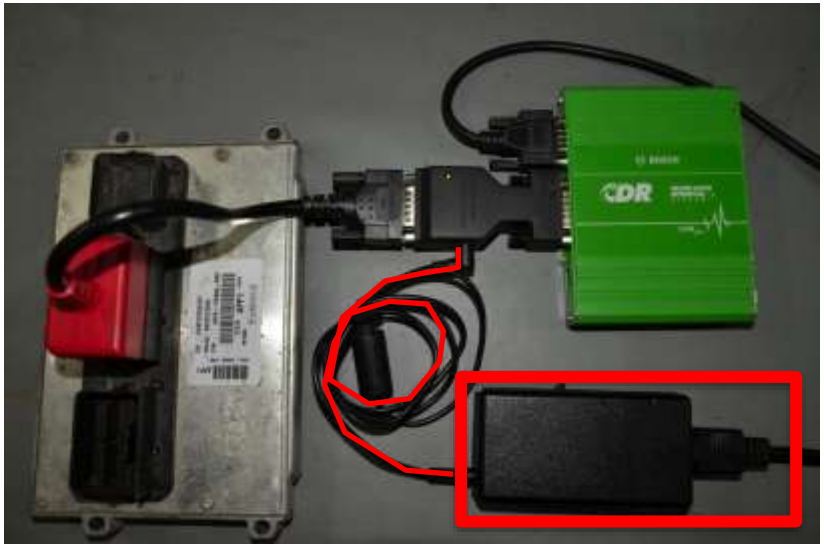


Powering the CDR Interface

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AC Power Supply

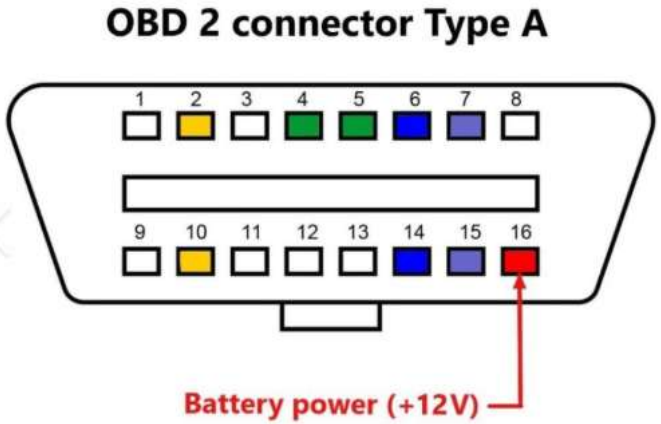
- The “Power Brick” style is the most current and is required for PCM imaging
- 12 volts DC either by conversion from AC or from an automotive DC access point
- For DLC downloads vehicle battery can be boosted or the system can be back powered



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OBD Port

- When using the DLC cable connected to the vehicle OBD Port (on a live vehicle) the interface is powered by pin 16 of the OBD Port. No external power supply is needed.



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Booster Boxes

Schumacher DSR168



NOCO GB150



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NOCO GB150

- NOCO Booster packs allow power two ways:
  - Acts as traditional booster battery
  - Has 12volt out via cigarette lighter adapter
  - Some have built-in Voltmeter
- Special “Full Power” Mode



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Vehicle Boosting Considerations

- If the vehicle battery is dead:
  - Disconnect the vehicle battery
    - At least one terminal
- Dead vehicle battery boosting causes “equalization”
  - Booster battery charge life is shortened considerably



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CDR System Operators Course

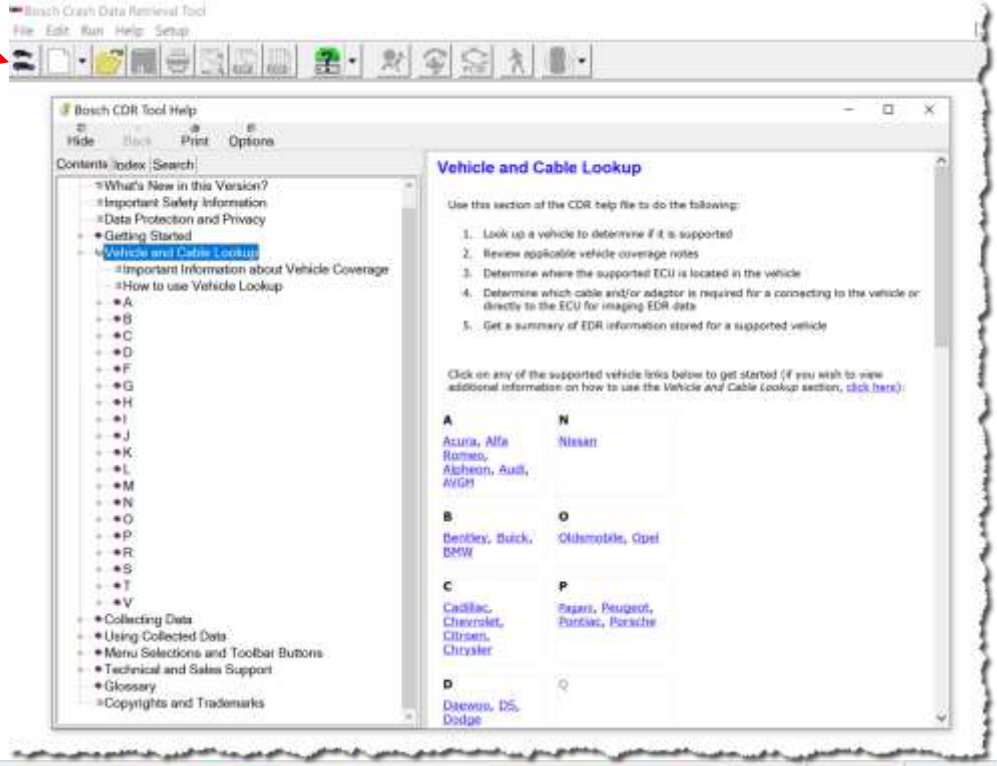


Determining Vehicle Coverage

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Determining Coverage

- CDR Help File / Vehicle and Cable Lookup should be referred to as the authoritative source for CDR Support
- Updated with every new release



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Help File Hyperlinks

Mkt	Year/Make	Model	Module		Vehicle Interface	OBD/DLC Connect Adapter/Cable	D2M Connect Adapter/Cable	Module Location
4,13	2020 Chevrolet	Equinox	ACM (note 6)	Data	CANplus	F00K108287	F00K108454	Center tunnel
			FCM (Note1) how to image	Data	CDR 900	CDR 900 - click here	CDR 900 - Cable# 856	Center - top interior of windshield
4	2020 Chevrolet	Express (note 6)	ACM	Data	CANplus	F00K108287	F00K108454	Center tunnel
4	2020 Chevrolet	Impala (note 6)	ACM	Data	CANplus	F00K108287	F00K108454	Center tunnel
4,13	2020 Chevrolet	Malibu	ACM (note 6)	Data	CDR 900	CDR 900 - click here	Cable# F00K108454 CDR 900 - click here (use listed cable above)	Center tunnel
			ASCM (Note1)	Data	CDR 900	CDR 900 - click here	CDR 900 - Cable# 833	In the luggage compartment, under the left side trim panel
4,5	2020 Chevrolet	Silverado (including Medium Duty)	ACM (note 6)	Data	CDR 900	CDR 900 - click here	Cable# F00K108454 CDR 900 - click here (use listed cable above)	Center tunnel
			FCM (Note1) how to image	Data	CDR 900	CDR 900 - click here	CDR 900 - Cable# 856	Center - top interior of windshield

- Click **Data** for what data is available
- Which Interface is required – most are “CANplus”
- Cable for DLC Download – most are ‘287’
- D2M Cable - varies

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File Edit Run Help Setup

Market Code – Pay Attention!

Bosch CDR Tool Help

Hide Back Print Options

Contents Index Search

- What's New in this Version?
- Important Safety Information
- Data Protection and Privacy
- Getting Started
- Vehicle and Cable Lookup
  - Important Information about Vehicle Coverage
  - How to use Vehicle Lookup
  - A
  - B
  - C
  - D
  - F
    - Ferrari
    - Fiat
    - Ford
    - Freightliner
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  - M
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Supported Ferrari Vehicles

Use this section of the help file to determine if a vehicle is supported by the CDR tool and look up the correct cable (and i
some cases, adapter) to use for retrieving EDR data directly from the ACM.

Click here for tips on how to use this Supported Vehicles help topic

General Ferrari Application Notes:

When imaging data through the OBD connector, the vehicles listed below require the vehicle's engine start button ON wi
engine running during EDR imaging.

2021

Mkt	Year/Make	Model	Module	Vehicle Interface	OBD/DLC Connect Adapter/Cable	D2M Connect Adapter/Cable	Module Location
6	2021 Ferrari	812 GTS / Competizione	ACM	Data CDR 900	1699200615 click here for connection diagram	Cable ID# 819 click here for connection diagram	Under Instrument Panel, Center
6	2021 Ferrari	F8 Spider	ACM	Data CDR 900	1699200615 click here for connection diagram	Cable ID# 819 click here for connection	Under Instrument Panel, Center

Market Code for Supported Vehicles

Market Code	Market Description
1	USA and Canada
2	USA
3	Mexico
4	North American Countries (NAFTA)
5	Australia/New Zealand
6	China
7	All markets the vehicle is sold
8	Brazil
9	Japan
10	Europe
11	Canada
12	Puerto Rico
13	S. Korea

There is one or more *Market Code* designations specified for each vehicle brand and model listed in
the *Vehicle and Cable Lookup* section in this help file. The CDR Tool is designed to retrieve data fro
the vehicles listed which are built for the markets indicated.

If user's attempt to retrieve EDR data from vehicles built for markets not specified, the CDR Tool
may display errors including but not limited to the following:

- This vehicle presently not supported
- This ECU or Module is currently not supported
- ECU communications errors between the CDR interface module and the vehicle

Attempts made to retrieve data from vehicles built for markets the CDR Tool does not support by
entering a VIN from a market it is built for (VIN spoofing) may result in inaccurate CDR report data
to violations and possibly incorrect or erroneous retrieval of data from the ECU.

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Help File Hyperlinks

			ACM (note 6)	Data	CANplus	F00K108287	F00K108454	Center tunnel
4,13	2020 Chevrolet	Equinox	FCM (Note1) how to image	Data	CDR 900	CDR 900 - click here	CDR 900 - Cable# 856	Center - top interior of windshield
4	2020 Chevrolet	Express (note 6)	ACM	Data	CANplus	F00K108287	F00K108454	Center tunnel
4	2020 Chevrolet	Impala (note 6)	ACM	Data	CANplus	F00K108287	F00K108454	Center tunnel
4,13	2020 Chevrolet	Malibu	ACM (note 6)	Data	CDR 900	CDR 900 - click here	Cable# F00K108454 CDR 900 - click here (use listed cable above)	Center tunnel
			ASCM (Note1)	Data	CDR 900	CDR 900 - click here	CDR 900 - Cable# 833	In the luggage compartment, under the left side trim panel
4,5	2020 Chevrolet	Silverado (including Medium Duty)	ACM (note 6)	Data	CDR 900	CDR 900 - click here	Cable# F00K108454 CDR 900 - click here (use listed cable above)	Center tunnel
			FCM (Note1) how to image	Data	CDR 900	CDR 900 - click here	CDR 900 - Cable# 856	Center - top interior of windshield

GM Vehicle Coverage Notes

The numbers below coincide with the note reference numbers called out in the list of supported vehicles.

1. Connect directly to the SDM for EV1 electric cars.
2. Does not record pre-crash data.
3. 1999 Saturn vehicle's DLC port has a non-standard wiring configuration and does not interface with the Bosch OBD II interface cable (02002837 or F00K108287). Please connect directly to the SDM, or contact Bosch for more information.
4. These trucks have different SDMs depending upon body style. If the eighth digit of the VIN is either F, J, or R use the 02002888 cable, all others use 02002829 cable.
5. For DLC data collection, use the 12 pin DLC cable (P/N: 02003090).
6. A key is not required to image these vehicles through the DLC, so you can leave ignition off or on during imaging.
7. If the first 3 digits of the VIN are 1GT, 2GT, 3GT, 1GD, 2GD, 3GD, 1GC, 2GC, 3GC, 1GB, 2GB, 3GB and the production sequence number (last 6 digits of the VIN) is 500000 or lower, use **02003003** cable, otherwise, use the **02003320**.

GM Regular Production Option (RPO) Code Designation for FCM

To determine if this vehicle is equipped with an FCM supported by the CDR Tool, you can see the camera located at the top interior of windshield and/or verify the RPO code includes **&UEU-UGN** in the vehicle option list.

Note: for model year 2019 pickup trucks, FCM is supported only on 1500 Silverado and Sierra trucks.

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Cables and Adapters Required

- Adapters are in Yellow Highlight
  - Two Types:
    - D2M* i.e. “387” Adapter
    - DLC* i.e. “709” “784” and other DLC Adapters
- D2M* Cable and DLC* Cable required are listed

2016 | 2011 2012 2013 2014 2015 2016 2017 2018

Mkt	Year/Make	Model	Module		Vehicle Interface	OBD Connect Adapter/Cable	D2M Connect Adapter/Cable	Module Location
1, 8	2016 Fiat	500 (includes electric) (note 14)	ACM	Data	CANplus	F00K108709 DLC Adapter & F00K108287	F00K108387 & F00K108710	Center Stack
4	2016 Fiat	500L (note 14)	ACM	Data	CANplus	F00K108709 DLC Adapter & F00K108287	F00K108387 & F00K108785	Center console, forward of shifter
7	2016 Fiat	500X (note 14)	ACM	Data	CANplus	DLC 3 Adapter & F00K108287	F00K108387 & Cable #819	Under center stack
7	2016 Fiat	Freemont (note 13)	ACM	Data	CANplus	F00K108287	F00K108387 & F00K108385	Center Stack
			PPM	Data	CANplus	F00K108287	F00K108387 & F00K108516	Under Left Front Seat

D2M* = Direct To Module
DLC* = via Diagnostic Link Connnector

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CDR System – Methods of Connection



DLC Downloads

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Why DLC ?

- The best method of data retrieval is via the DLC
  - ACM remains connected to sensors and other devices
  - Reduces the possibility of overwriting pre-existing trouble codes (DTC's)

Removing an ECU with Stored EDR Data from a Vehicle

The preferred practice as recommended by the automobile manufacturers whose vehicles are accessible by the CDR system and Bosch is to avoid removing the ECU for data collection whenever reasonably possible. Below are some guidelines that may help in facilitating this practice:

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DLC Downloads

Courtesy of Crash Data Group



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Before we get started, Safety First!

- Crashed vehicles are full of hazards!
  - Safety Equipment for Your CDR Toolbox
    - PPE
    - Gloves
    - Goggles
    - Proper Tools









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The DLC Download Ecosystem

- Diagnostic Link Connector (DLC)
  - Preferred Method and Most Used – Sample Connection



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DLC Connection Method



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3 types DLC



GM 12-Pin DLC Cable
Discontinued 1/2011



CDR OBDII Cable
Replaced by one W/VPS

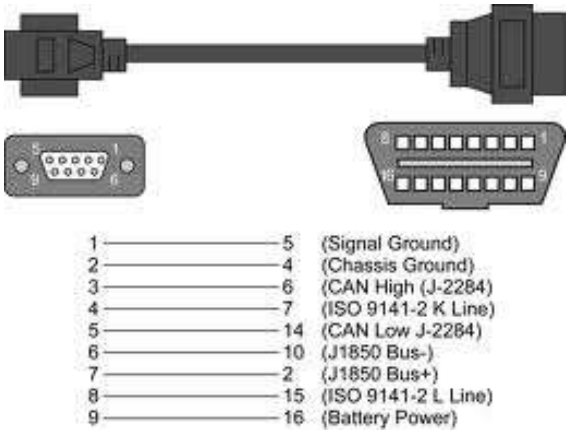


CDR OBDII Cable W/VPS
Current OBD Cable

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The OBD Port

- Access port to the vehicles communication system
- May also be referred to as OBD II port / J1962 Connector or DLC





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DLC Downloads - CDR 900 Interface

- NO Adapters are required
- CDR900 handles switching CAN pinouts internally
  - (that's was the adapters did)
- Can NOT use for PCM data retrieval – MUST use CANPlus
- Remember to wait for the "Beep" that the CDR900 interface has completed "boot up"



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Adapters used for DLC Access

- Fiat DLC Adapter – (Yellow)
  - F-00K-108-709



- DLC2 (Black) / 3 &4Adapter (Grey)
  - F-00K-108-784 / DLC3 &4 in help file



- Ford PCM Adapter (Black)
  - F-00K-108-221



- NO DLC Adapters Required with CDR900!!

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F-00K-108-709 DLC Adapter

Use only when called for in Help File

Supported Fiat Vehicles

Use this section of the help file to determine if a vehicle is supported by the CDR tool and look up the correct cable (and in some cases, adapter) to use for retrieving EDR data directly from the ACM.

Click here for tips on how to use this *Supported Vehicles* help topic

General Fiat Application Notes:

All supported vehicles listed below require the vehicle's ignition key to be switched on prior to and during EDR record imaging through the vehicle's DLC connector.

2012

Make	Model	Module	Adapter/Cable	Module Location
Fiat	500	ACM CSVI	F00K108387 & F00K108710 For DLC imaging, use the F00K108709 DLC Adapter & F00K108287 DLC Cable	Center Stack
Fiat	Freemont	ACM CSVI	F00K108387 & F00K108385	Center tunnel



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DLC 2 / 3 / 4 Adapter

- Required by some vehicles for DLC download
- Check Help File listings

1	Ford	Transit Connect (includes postal and BEV vehicles)	ACM	Data	F00K108387 & F00K108778 For DLC imaging, use the F00K108784 DLC Adapter & F00K108287 DLC Cable	
7	Fiat	500X (note 14)	ACM	Data	F00K108387 & Cable #819 For DLC imaging, use the DLC 3 Adapter & F00K108287 DLC Cable	Under center stack





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CDR System – Methods of Connection



D2M Downloads

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Direct to Module Method



- Direct to module cable is connected between the ACM and the CDR interface
- Power is applied to CDR interface
- Power light on CDR interface should then be illuminated

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Direct to Module Downloads

Courtesy of Crash Data Group



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D2M Adapters

- There are several adapters that may be required:
  - Chrysler ACK Adapter
    - F-00K-108-790



CHRYSLER

  - ACM Adapter (Gray)
    - F-00K-108-387
    - *NOT REQUIRED WITH CDR 900*



Various OE's – see Help File Coverage Listing

  - Ford PCM Adapter (Black)
    - F-00K-108-221 (CDR900 Does NOT support PCM Data Retreival!!)



Ford

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D2M Adapters

- When performing direct-to-module imaging and an adapter is required:
  - Power is applied to the ***Adapter,*** not the CDR Interface





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D2M - CDR 900 Interface

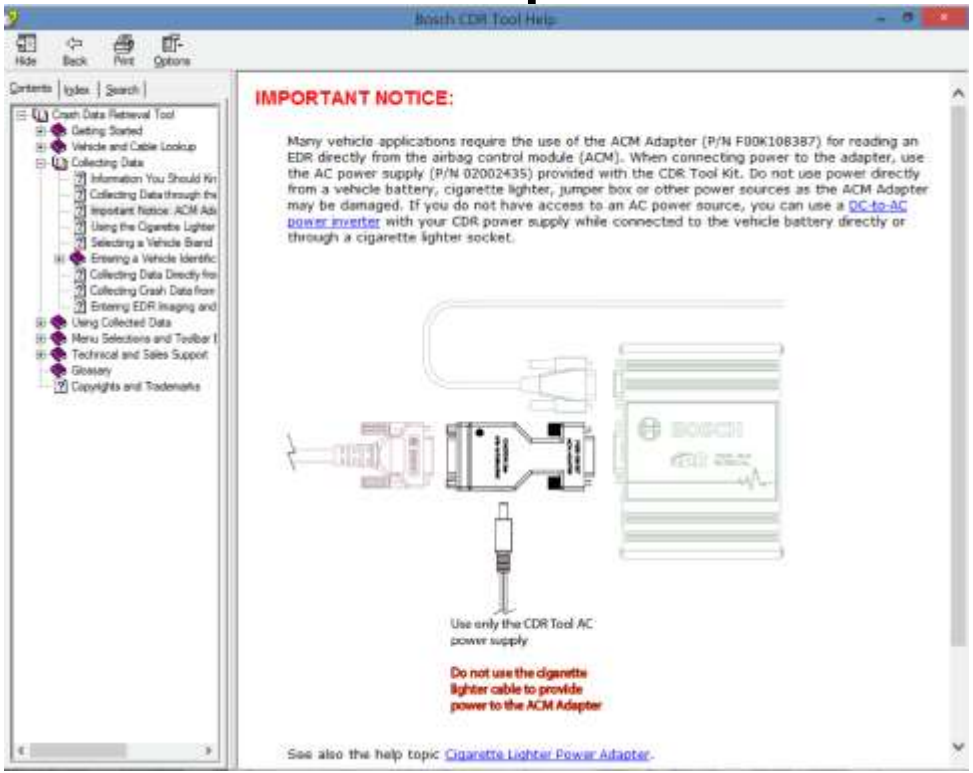
- When using the CDR900 for D2M power is applied to the CDR900 power port
- Remember to wait for the “beep” upon plugging in power to confirm “boot up” is complete
- Status light will change from Amber to Green when “Collect Data” is in progress



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IMPORTANT - ACM Adapter

- ***CAUTION!!***
- Do NOT use a battery to power the ACM adapter!
- High current draw on ACM start up may ***DAMAGE*** the ACM Adapter!!



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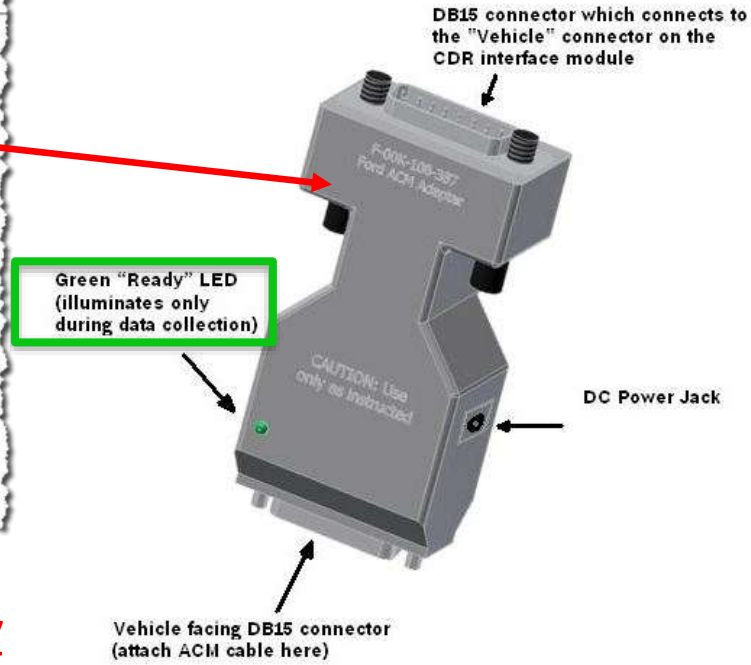
ACM Adapter – F-00K-108-387

2018 2006 2007 2008 2009 2010 2011 2012 2013 2014 2015 2016 2017 2018 2019

Mkt	Year/Make	Model	Module		Vehicle Interface	OBD/DLC Connect Adapter/Cable	D2M Connect Adapter/Cable	Module Location
?	2018 Chrysler	300	ACM	Data	CANplus	F00K108287	F00K108387 & F00K108598	Center Stack
?	2018 Chrysler	Pacifica	ACM	Data	CANplus or CDR 900	CANplus - F00K108287 CDR 900 - click here	CANplus - F00K108387 & F00K108785 CDR 900 - click here (use listed cable above and no ACM adapter)	Center Console

2019 2006 2007 2008 2009 2010 2011 2012 2013 2014 2015 2016 2017 2018 2019

Mkt	Year/Make	Model	Module		Vehicle Interface	OBD/DLC Connect Adapter/Cable	D2M Connect Adapter/Cable	Module Location
?	2019 Chrysler	Pacifica	ACM	Data	CANplus or CDR 900	CANplus - F00K108287 CDR 900 - click here	CANplus - F00K108387 & F00K108785 CDR 900 - click here (use listed cable above and no ACM adapter)	Center Console



Used on direct-to-module imaging only

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ACM Adapter?

2018 2006 2007 2008 2009 2010 2011 2012 2013 2014 2015 2016 2017 2018 2019

Mkt	Year/Make	Model	Module		Vehicle Interface	OBD/DLC Connect Adapter/Cable	D2M Connect Adapter/Cable	Module Location
7	2018 Chrysler	300	ACM	Data	CANplus	F00K108287	F00K108387 & F00K108598	Center Stack
7	2018 Chrysler	Pacifica	ACM	Data	CANplus or CDR 900	CANplus - F00K108287 CDR 900 - click here	CANplus - F00K108387 & F00K108785 CDR 900 - click here (use listed cable above and no ACM adapter)	Center Console

2019 2006 2007 2008 2009 2010 2011 2012 2013 2014 2015 2016 2017 2018 2019

Mkt	Year/Make	Model	Module		Vehicle Interface	OBD/DLC Connect Adapter/Cable	D2M Connect Adapter/Cable	Module Location
7	2019 Chrysler	Pacifica	ACM	Data	CANplus or CDR 900	CANplus - F00K108287 CDR 900 - click here	CANplus - F00K108387 & F00K108785 CDR 900 - click here (use listed cable above and no ACM adapter)	Center Console

- NOT Required if using CDR900!



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Chrysler ACK. Adapter

- ACK Adapter in-line with ACM Umbilical Cable
- Power to CDR Interface NOT ACK
- Power LED’s illuminate on both ACK and CDR Interface



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CDR500 – FlexRay Adapter



The CDR 500 Adapter enables the CDR Tool to retrieve EDR data directly from ECUs with a FlexRay communications network. This advanced technology is much different than traditional CAN or K-Line networks currently supported by the CDR Tool.

APPLICATION

The first application of the CDR 500 will be for select BMW and MINI vehicles. In the near future, the CDR 500 will be required for direct-to-module EDR downloads on Audi and other vehicles.

The CDR 500 Adapter will be called out for each vehicle that requires it for direct to module imaging just as the other cables and adapters are called out. The CDR software will prompt users to connect the adapter if it is not connected already and provides a link to the hook-up diagram.

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CDR500 – FlexRay Adapter

INPUTS/OUTPUTS

This section defines each of the connectors on the CDR 500. Some of the items are not used as defined below but may be in the future.



1. **LAN** - *This connector is not used at this time*
2. **SD Card** - *This connector is not used at this time*
3. **USB-B** - This port is used only for reprogramming the adapter. The CDR Tool software will provide instructions on when to connect the USB cable to this connector
4. **USB-A** - *This connector is not used at this time*
5. **LEDs 1-8** - Activity indicator lights
6. **R-CAN1 switch** - The adapter operates in any switch position
7. **R-CAN2 switch** - The adapter operates in any switch position
8. **6-36V DC** - The DC power jack is compatible with the CDR Tool power supply. The CDR 500 device currently does not need to be powered through this jack during reprogramming or operation. It may be used in the future capabilities.

ATTENTION: When operating the CDR 500, use only the AC power supply provided with the CDR Tool Kit when powering the device directly or through the ACM Power Adapter. Do not use another power source including a vehicle battery. Failure to do so may cause damage to the adapter and void the warranty.

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CDR500 – FlexRay Adapter

ATTENTION: When operating the CDR 500, use only the AC power supply provided with the CDR Tool Kit when powering the device directly or through the ACM Power Adapter. Do not use another power source including a vehicle battery. Failure to do so may cause damage to the adapter and void the warranty.



1. **X2** - This connector is where the ACM connector plugs directly into the CDR interface and the
2. **X1** - When directed to do so, connect the CDR 500 interface cable in to this connection and connected the other end to the CDR ACM Power Adapter

Click here for information on how to connect the CDR 500 Adapter.

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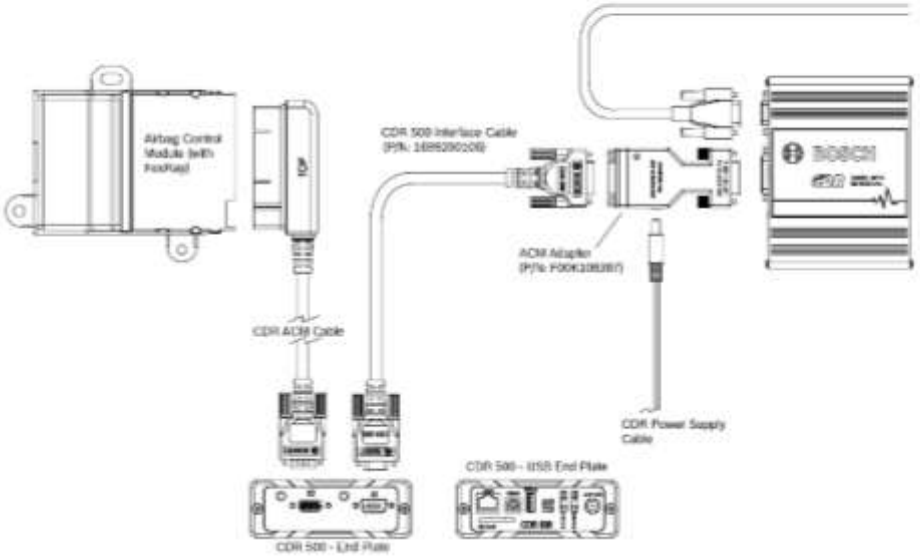
File Edit Run Help Setup

CDR500 – FlexRay Adapter

CDR 500 Connection Diagram

The CDR 500 Adapter connection diagram is shown below. Power to the CDR 500 is provided through the ACM Adapter (F00K108387) and controlled by the CDR application.

IMPORTANT: Always use the *AC Power Supply* that came in your CDR Tool Kit. The use of any other power source may cause permanent damage to your CDR 500.



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Downloads requiring CDR500

Courtesy of Crash Data Group



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Direct to Module Method – GM Booster



- Umbilical cable is connected between the ACM and the CDR interface thru the identified power adapter
- Power is applied to adapter power port
- Power light on CDR interface should then be illuminated
- Discontinued – may not be in your CDR System kit

File Edit Run Help Setup

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CDR System Operators Course



Determining ACM Location

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ACM Locations

- Under Seat
  - Older GM vehicles
  - May be Driver or Passenger



- Center Stack
  - Various Supported OE's
  - Typically under radio / heater stack



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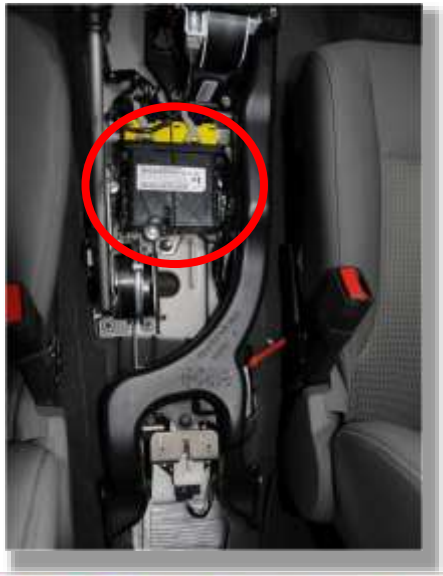
ACM Locations

- Center Tunnel
  - Anywhere from Front to Back





- Under Center Console
  - May before or aft of shifter





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Check the Supported vehicle file

General Nissan Application Notes:

The CDR Tool software will accept most Nissan VINs entered by the user and the *Collect Airbag Control Mo* no longer be grayed out. **This is not an indicator that the vehicle is supported.** Only the vehicles listed below are supported as indicated by the list and any associated coverage notes.

All supported vehicles listed can be imaged through the vehicle's DLC connector using the CDR OBDII DL (F00K108287) unless otherwise stated.

All supported vehicles require the vehicle's ignition key to be switched on prior to and during EDR record the vehicle's DLC connector.

2012 2012 2013 2014 2015 2016 2017

Mkt	Make	Model	Module		Adapter/Cable	Module Location
1	Nissan	Frontier (note 1)	ACM CSV	Data	F00K108387 & F00K108780	Center Tunnel, Between Front Seats
1	Nissan	Versa Hatchback (note 1)	ACM CSV	Data	F00K108387 & F00K108780	Center Tunnel, Between Front Seats
1	Nissan	Xterra (note 1)	ACM CSV	Data	F00K108387 & F00K108780	Center Tunnel, Between Front Seats

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CDR System Operators Course



ACM Location - Exceptions

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BMW ACM = ACSM

- BMW ACM = ACSM
  - Advanced Crash Safety Module
  - *White / Black Plastic ACSM*:
    - no internal sensors - relies on network of external sensors



- DO NOT CONFUSE with ICM
  - Integrated Chassis Module located under console
  - “Airbag” written on it
  - Sensor only – NO DATA





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BMW ACSM Location

- From the CDR Help File:

Bosch CDR Tool Help

Show Back Print Options

General BMW Application Notes:

All supported vehicles listed can be imaged through the vehicle's DLC connector using the CDR OBDII DLC Cable (F00K108287) unless otherwise stated

All supported vehicles require the vehicle's ignition key to be switched on prior to and during EDR record imaging through the vehicle's DLC connector.

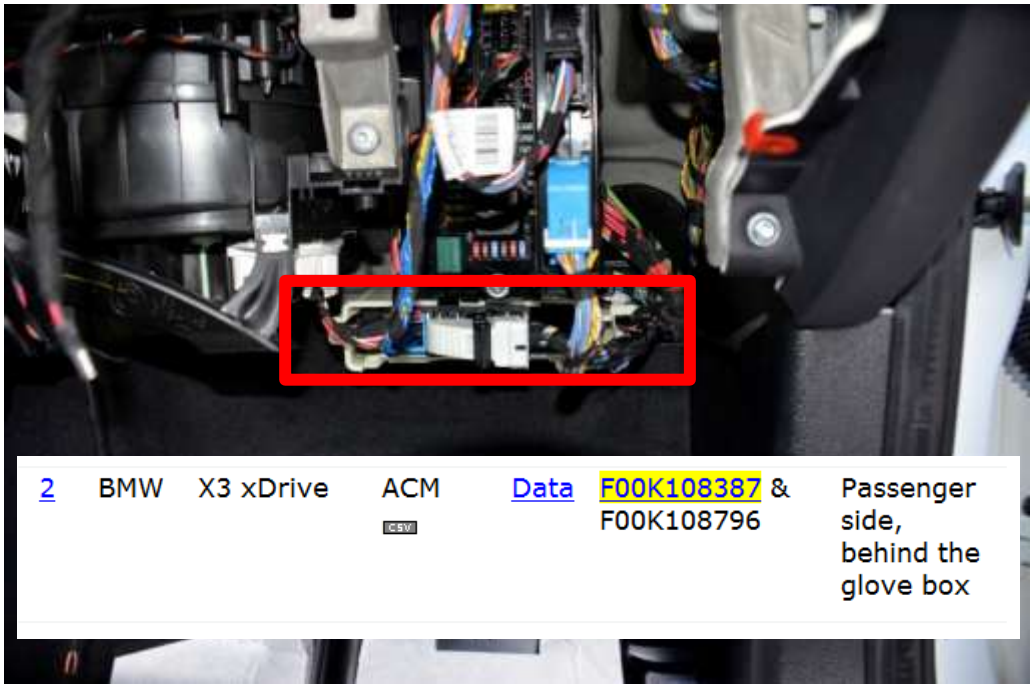
2013 2013

Mkt	Make	Model	Module		Adapter/Cable	Module Location
2	BMW	Active Hybrid 3, 5 & 7 (sedan)	ACM	Data	F00K108387 & F00K108796	Passenger side, behind the glove box
2	BMW	M5 (sedan)	ACM	Data	F00K108387 & F00K108796	Passenger side, behind the glove box
2	BMW	M6 (coupe, convertible)	ACM	Data	F00K108387 & F00K108796	Passenger side, behind the glove box



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2014 BMW X3 – XDrive 28i



2	BMW	X3 xDrive	ACM	Data	F00K108387 & F00K108796	Passenger side, behind the glove box

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BMW ACSM Location

- ACSM Location: on supporting structure of the cockpit module on the driver side.



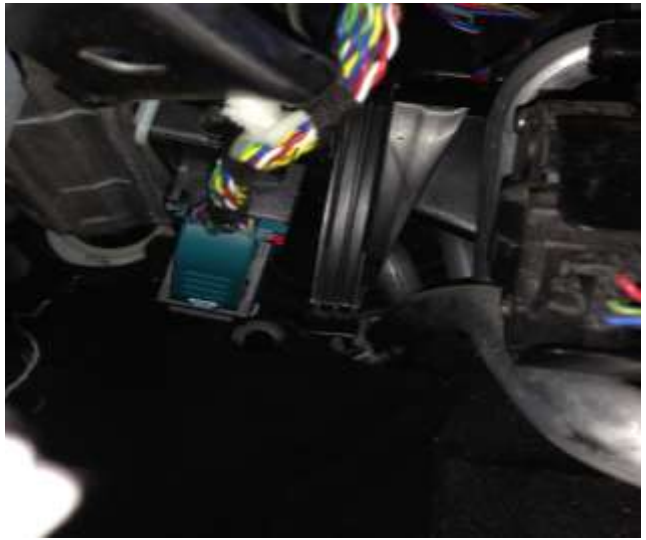
- ACSM Location: Center Console
  - Note proximity to ICM!



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BMW ACSM Location

- ACSM Location: on supporting structure of the cockpit module on the driver side.



- ACSM Location: Center Console
  - Note proximity to ICM!



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Newer BMW & Mini - Metal ACM

- Center Tunnel Location
- Ahead of or behind (under) shifter
- Does not have the “1//2 round” shield over connector
- May have “ACSM” on label



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Manufacturer Specific Considerations

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Jeep Driver Monitoring System Module / Driver Camera Module Support







Requires CDR900 & CDR version 23.4 or newer

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Driver Monitoring System (DMSM) / Driver Camera Module (DCM) – Some Jeeps

- Support for readout of images captured by the Driver Monitoring System stored in the Driver Monitoring System
- Module (DMSM) or Driver Camera Module (DCM)
  - 2022~2024 Jeep Grand Cherokee, Grand Wagoneer and Wagoneer



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CHRYSLER

Driver Monitoring System

- Part of the *Hands-Free Active Driving Assist*
- *R*ecords driver camera images within the ECU (upon certain triggers such as airbag deployment)
- DMSM module records 3 events and can store up to 5 images for each event
- CDR tool retrieves the recorded image data and stores the data in the CDR
- Images can be exported from the CDR file.
- CDR file does not include a CDR report with data elements, just image data.
- Images are exported in a .YUV format and be viewed using a third-party image viewing software

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DCM

CHRYSLER

Driver Monitoring System (DMSM) / Driver Camera Module (DCM) – Some Jeeps

Grand Cherokee DCM Imaging:

The Grand Cherokee driver camera images can be retrieved from the DCM module directly or by connecting to a vehicle connector located behind the knee airbag below the steering column.

On Vehicle Connection:

1. Remove the cover below the steering column and the knee airbag (see figure 1)
2. Unplug the 8 pin connector below the steering column (see figure 2)
3. See table below figure for for connecting cable 851 to the vehicle
4. Plug in 12VDC power into the CDR 900 (see figure 4)

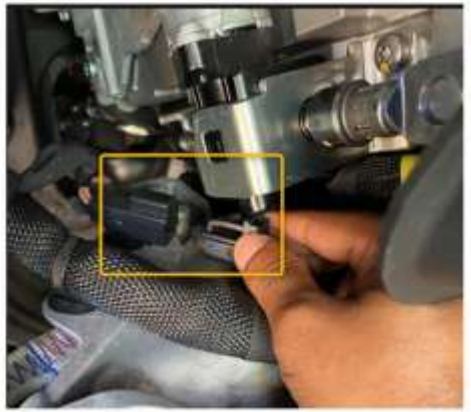
Direct to Module Connection:

1. Remove the DMSM module which is located behind the center of the dash and mounted to the HVAC housing. Refer to vehicle's service manual for the removal of the DMSM module
2. Connect cable 851 as per the figure 4 and table to make the connections to the DMSM module to the CDR 900
3. Plug in 12VDC power into the CDR 900 (see figure 4)

Figure 1: Grand Cherokee: Knee Airbag below the steering column



Figure 2: Grand Cherokee: connector for imaging DCM



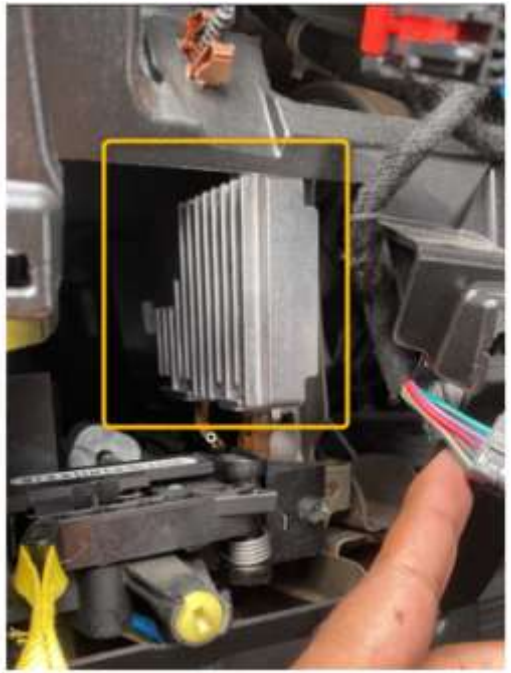
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Driver Monitoring System (DMSM) / Driver Camera Module (DCM) – Some Jeeps

Grand Wagoneer/Wagoneer DCM Imaging:

Direct to Module Connection:

1. Remove the DMSM module which is located below the steering column behind the access panel (see Figure 3). Refer to vehicle's service manual for the removal of the DMSM module
2. Connect cable 851 as per the figure 4 and table to make the connections to the DMSM module to the CDR 900
3. Plug in 12VDC power into the CDR 900 (see figure 4)



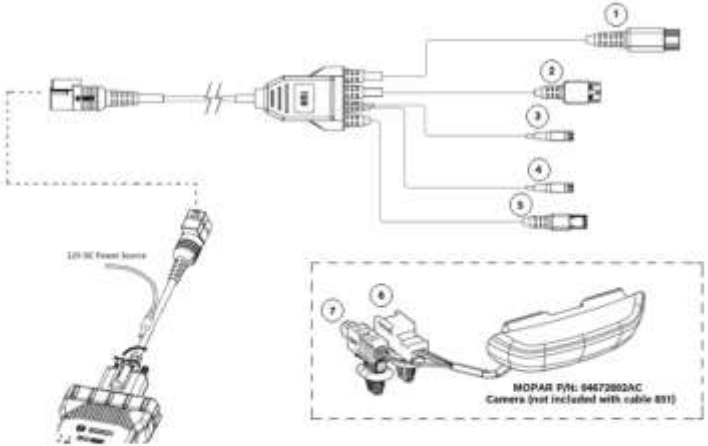
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CHRYSLER

Driver Monitoring System (DMSM) / Driver Camera Module (DCM) – Some Jeeps

- CDR Cable ID 851 For direct to module downloads.
- Vehicle camera module must be connected
- Spare Camera:
  - MOPAR P/N: 04672802AC

- *Note: CDR Cable 851 availability expected in Q1 of 2024; sold only in US market (Checking on Canada)*

Figure 4: Cable ID 851, CDR Driver Cam (P/N 1686200984) and camera



Connector (figure 4)	Direct to Module	On Vehicle	Description
1		x	connect to the 6 pin connector on vehicle (see figure 2)
2	x		12 pin connector connects to the DMSM module
3	x		Coax cable, connects to the DMSM module
4	x		Coax cable, connects to the camera (connects to 7)
5	x		4 pin connector connects to camera (connects to 6)
6	x		Camera 4 pin connector (connects to 5)
7	x		Camera coax cable (connects to 4

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Manufacturer Specific Considerations

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Powertrain Control Module

- PCM = PowerTrain Control Module
  - Controls Engine and Transmission
- 12.5 / 25.4 seconds or 6.5 minutes of crash data on some 2003-2010 Ford / Lincoln & Mercury vehicles

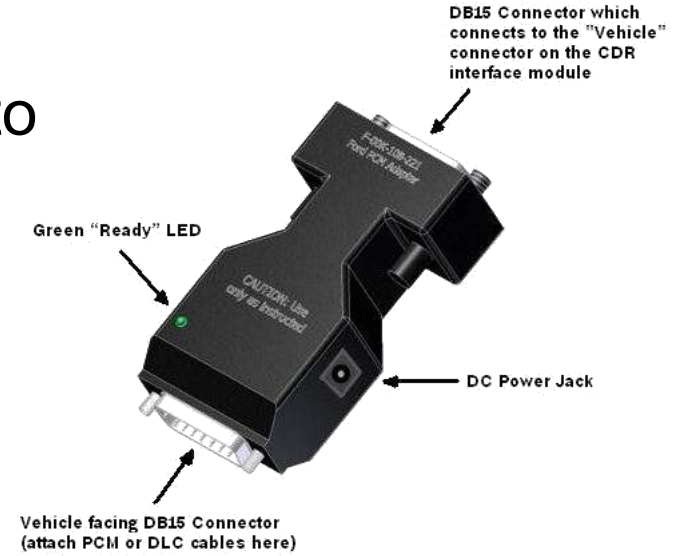




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PCM Adapter F-00K-108-221

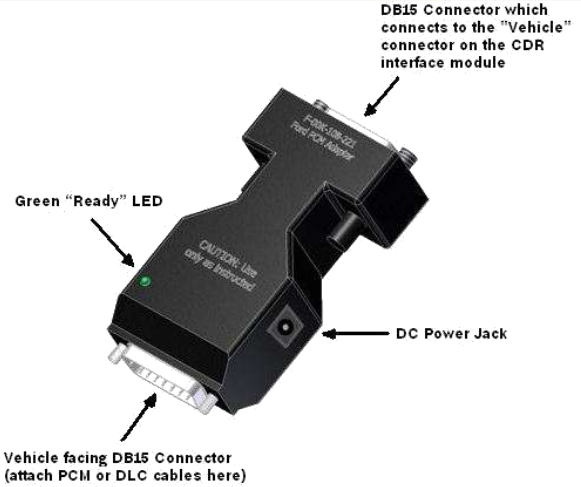
- Must be used on both DLC and Direct To Module imaging
- The adapter sends a signal (VPS) to the PCM to prevent it from overwriting data if the restraint deployment signal (RDS) was not received by the PCM.
- CDR900 Does NOT support PCM data retrieval!



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PCM Adapter F-00K-108-221

Mkt	Make	Model	Module	Adapter/Cable	Module Location
1	Ford	Crown Victoria	ACM	F00K108225	Center Tunnel
			PCM	F00K108221 & F00K108222	Driver Shock Tower





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PCM Connectivity - DLC

- DLC downloads
  - Ensure there's power available and communication with CDR Interface before turning the key "on"
    - May have to disconnect the PCM to check power status
  - Have the adapter in place and check for the "green power light" on both



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DLC PCM Connection Method



- With Key off, OBD II cable is connected to CDR interface thru the PCM Adapter
- Both Lite on CDR interface and PCM adapter should illuminate
- Serial cable is connected to laptop

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PCM Adapter F-00K-108-221

- The OBDII with VPS cable is required for DLC downloads
- When downloading the PCM via the, no external power to the PCM Adapter is required
- The DLC will supply power to the PCM Adapter and the CDR interface.
- You MUST use a CANPlus Interface with the PCM Adapter!!!
- PCM is NOT supported on CDR900

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PCM Considerations

- For Ford vehicles with accessible PCMs, powering up the car without the PCM Adapter present and properly connected to the PCM either over the vehicle bus (DLC) or by direct cable can result in the PCM data being overwritten
- Remember, not all PCM data is “locked” after a crash
- If the PCM Adapter is not available or during vehicle power up testing, unplug the PCM

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Disconnect wiring to “safe” PCM

- Locate the PCM and remove ALL connectors.
- PCM will not have power when key is turn ON.
- Check available power
- Check wiring
- Attempt DLC Download of PCM as applicable



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PCM Connectivity Direct to Module

- Direct to PCM access
  - Connect cables, have the adapter in place and then check for the “green power light” on the PCM Adapter often
  - May be the “safer method” of collecting data when in-car communication or power is questionable



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Manufacturer Specific Considerations

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GM - ROS / Roll Over Sensor

- Satellite module that senses forces that may result in roll over curtain and related restraints may deploy
- Usually results in data in ACM from same event



“A



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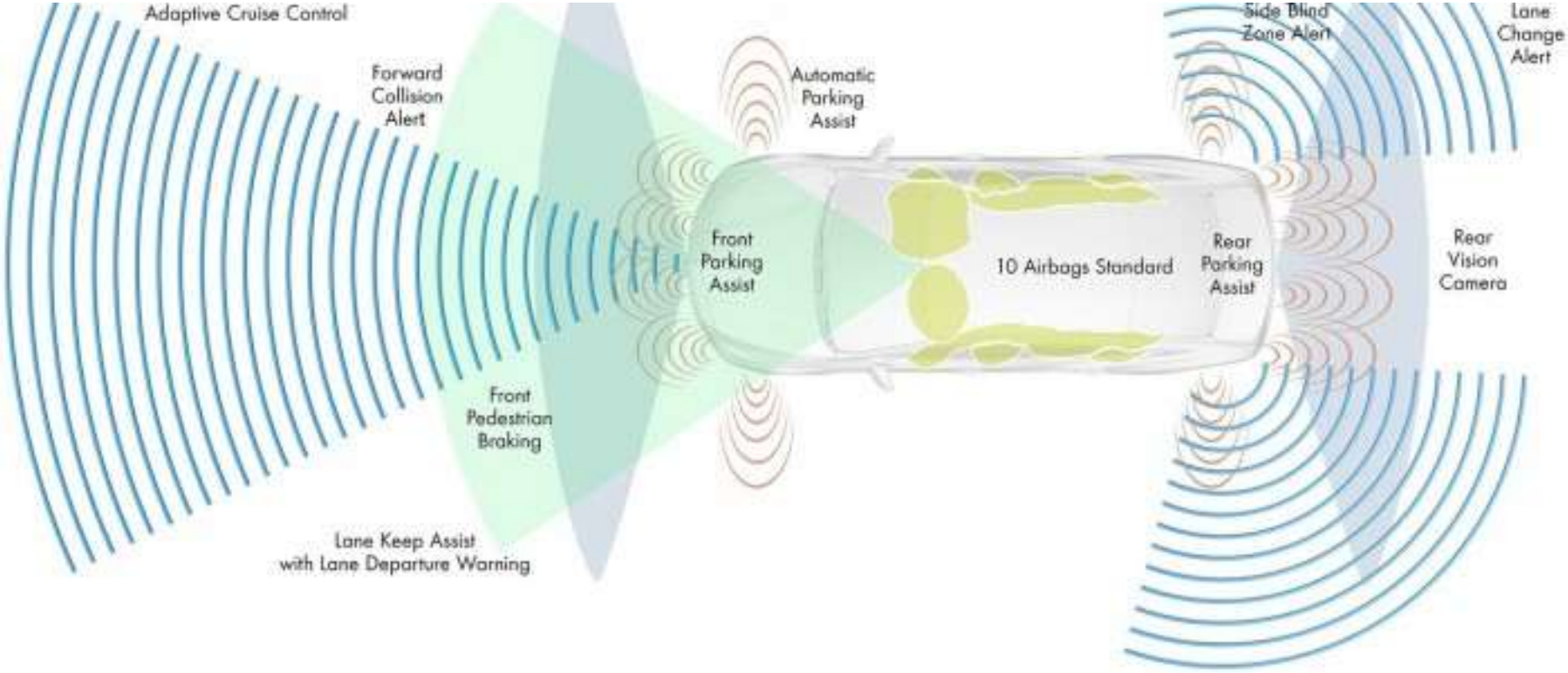
GM ROS Access

- The ROS icon becoming “live” is not always an absolute indication the vehicle is equipped with a supported ROS
  - Look for the “airbag” indication on the “B” pillar but even that doesn’t necessarily mean it’s supported



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GM Active Safety Systems



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ASCM Functions

- Enhanced Automatic Emergency Braking uses the Active Safety Control Module, Frontview Camera and Long Range Radar Module.
- Some vehicle also use Short Range Radar Modules
- The ASCM collections information from the various sensors and fuses it together to better understand objects around the vehicles
- Provides varies features including: Forward Collision Alert, Automatic Emergency Braking, Forward Pedestrian Braking, and sometimes Super Cruise

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GM Active Safety Control Module

- Currently On “Super Cruise” equipped and select other vehicles
- Other GM Brands / Models *may* be equipped

4	2021 Cadillac	XT4 (note 6)	ACM	Data	CANplus or CDR 900	CANplus - F00K108287 CDR 900 - click here	CANplus - F00K108454 CDR 900 - click here (use listed cable above)	Center tunnel
			ASCM (Note1)	Data	CDR 900	CDR 900 - click here	CDR 900 - Cable# 833	In the rear cargo storage area, under the left rear trim panel, at the front of the wheel well.

- ASCM Icon becomes “LIVE” if a supported VIN is entered / read
- MUST check for presence of ASCM as not all models are equipped / supported!!

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GM Active Safety Control Module

- READ THE NOTES
- 2 Note areas
  - Vehicle Model
  - Specific ECU Notes
    - (ECU = Electronic Control Unit)

4	2021 Cadillac	XT4 (note 6)	ACM	Data	CANplus or CDR 900	CANplus - F00K108287 CDR 900 - click here	CANplus - F00K108454 CDR 900 - click here (use listed cable above)	Center tunnel
			ASCM (Note1)	Data	CDR 900	CDR 900 - click here	CDR 900 - Cable# 833	In the rear cargo storage area, under the left rear trim panel, at the front of the wheel well.

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GM Active Safety Control Module

4	2021 Cadillac	XT4 (note 6)	ACM	Data	CANplus or CDR 900	CANplus - F00K108287 CDR 900 - click here	CANplus - F00K108454 CDR 900 - click here (use listed cable above)	Center tunnel
			ASCM (Note1)	Data	CDR 900	CDR 900 - click here	CDR 900 - Cable# 833	In the rear cargo storage area, under the left rear trim panel, at the front of the wheel well.

GM Regular Production Option (RPO) Code Designation for ASCM

To determine if the vehicle is equipped with an ACSM supported by the CDR Tool, refer to the RPO code information listed below:

Note 1: is equipped if RPO code **UGN** is present in the vehicle option list

Note 2: is equipped if both RPO codes **UGN** and **UKL** are present in the vehicle option list

Note 3: is equipped if RPO code **UGN** is is present and **UKL** is *NOT* present in the vehicle option list

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RPO Codes / QR Code Lookup



repairlinkshop.com

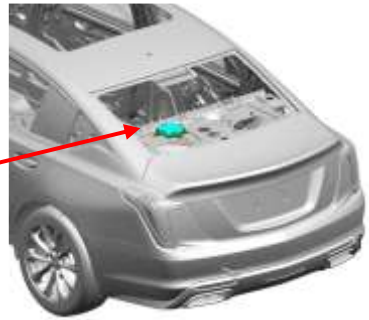
2GNAXL... 226,07,20,X ,XR
...Q C4P DLF
...NA GND H72 IOR
...NB5 NGT NTB N52 RJA
...UZ6 U2J V8E WML ZP5 Z49
...M 5CX 6X1 7X1 8X2 9X2,8624,,,,



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ASCM Notes & RPO Codes

Mkt	Year/Make	Model	Module		Vehicle Interface	OBD/DLC Connect Adapter/Cable	D2M Connect Adapter/Cable	Module Location
4	2020 Cadillac	CT4	ACM	Data	CDR 900	CDR 900 - click here	CDR 900 - Cable# 845	Center tunnel
			ASCM (Note1)	Data	CDR 900	CDR 900 - click here	CDR 900 - Cable# 854	In the luggage compartment, rear upper shelf
4	2020 Cadillac	CT5	ACM	Data	CDR 900	CDR 900 - click here	CDR 900 - Cable# 845	Center tunnel
			ASCM (Note1)	Data	CDR 900	CDR 900 - click here	CDR 900 - Cable# 854	In the luggage compartment, rear upper shelf
4	2020 Cadillac	CT6	ACM (note 6)	Data	CDR 900	CDR 900 - click here	Cable# F00K108454 CDR 900 - click here (use listed cable above)	Center tunnel
			ASCM (Note2) how to image	Data	CDR 900	CDR 900 - click here	CDR 900 - Cable# 834	In the luggage compartment, right center, mounted to the right side in the battery well. The secondary ASCM is on the left.
			ASCM (Note3)	Data	CDR 900	CDR 900 - click here	CDR 900 - Cable# 833	In the luggage compartment, right center, mounted to the right side in the battery well



GM Regular Production Option (RPO) Code Designation for ASCM

To determine if the vehicle is equipped with an ACSM supported by the CDR Tool, refer to the RPO code information listed below:

Note 1: is equipped if RPO code **UGN** is present in the vehicle option list

Note 2: is equipped if both RPO codes **UGN** and **UKL** are present in the vehicle option list

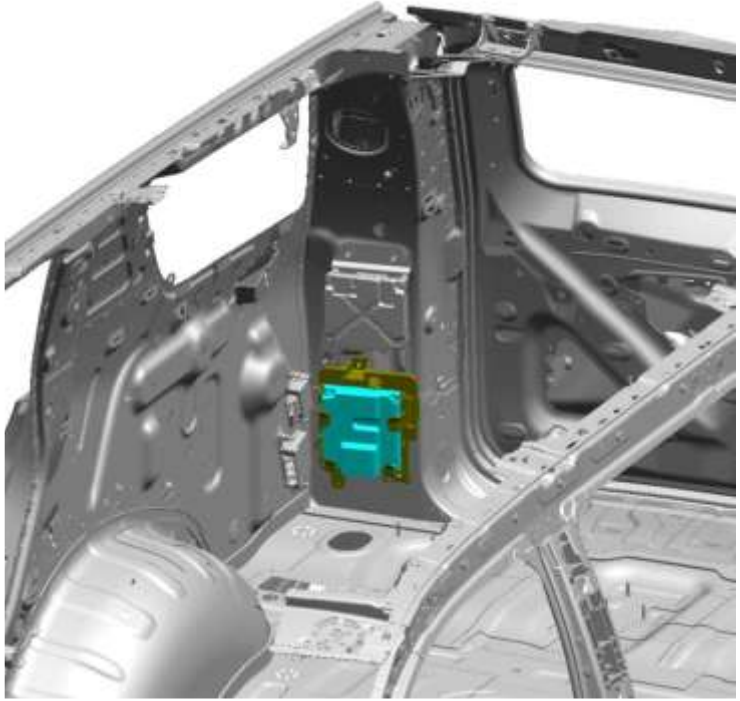
Note 3: is equipped if RPO code **UGN** is is present and **UKL** is *NOT* present in the vehicle option list

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ASCM Locations

4	2023 Chevrolet	Tahoe	ACM	Data	CDR 900	CDR 900 - click here	CDR 900 - Cable# 852	Center Tunnel
			ASCM (Note1)	Data	CDR 900	CDR 900 - click here	CDR 900 - Cable# 854	In the rear compartment, left side, underneath rear quarter glass window, behind trim
			ASCM (Note1)	Data	CDR 900	CDR 900 - click here	CDR 900 - Cable# 855	In the rear compartment, left side, underneath rear quarter glass window, behind trim

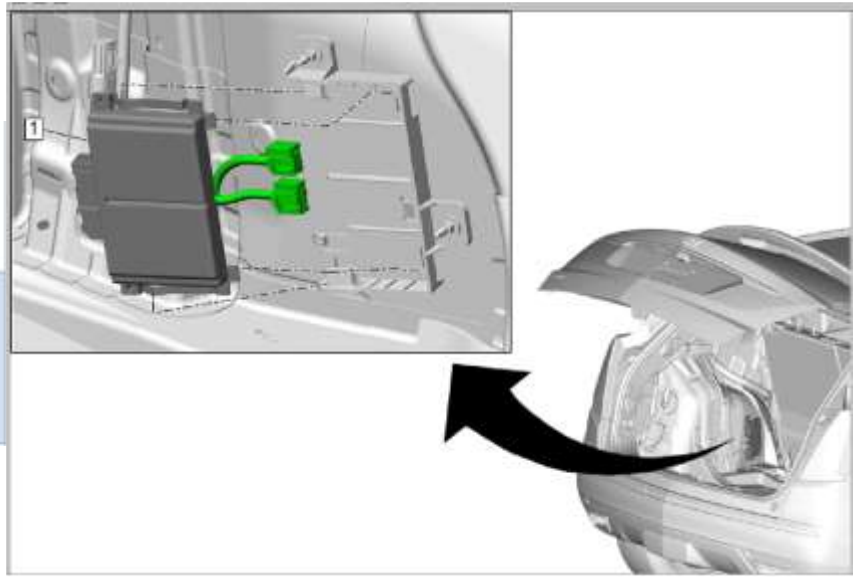


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ASCM Locations

				ACM (note 6)	Data	CDR 900	CDR 900 - click here	Cable# F00K108454 CDR 900 - click here (use listed cable above)	Center tunnel
4	2022 Cadillac	XT4		ASCM (Note1)	Data	CDR 900	CDR 900 - click here	CDR 900 - Cable# 833	In the rear cargo storage area, under the left rear trim panel, at the front of the wheel well.



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ASCM Locations

			ACM (note 6)	Data	CDR 900	CDR 900 - click here	Cable# F00K108454 CDR 900 - click here (use listed cable above)	Center tunnel
4	2022 Cadillac	XT4	ASCM (Note1)	Data	CDR 900	CDR 900 - click here	CDR 900 - Cable# 833	In the rear cargo storage area, under the left rear trim panel, at the front of the wheel well.



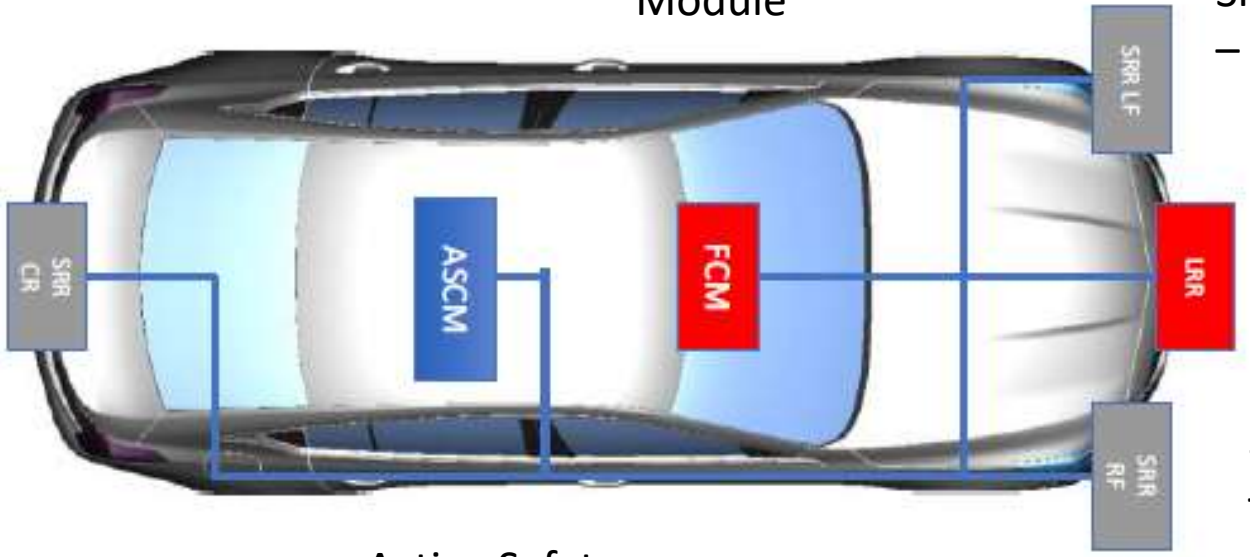
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ASCM Components

Front Camera Module

Short Range Radar – Left Front

Short Range Radar - Rear



Long Range Radar

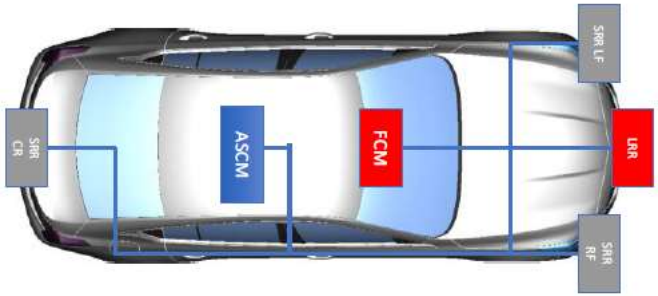
Short Range Radar – Right Front

Active Safety Control Module

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ADAS EDR and RPOs (Vehicles without Super Cruise)

- Vehicle equipped with ASCM:
  - Includes RPOs UEU (Forward Collision Alert) and UGN (Automatic Emergency Braking)
- **<u>ADAS EDR is contained within the ASCM</u>**
- No EDR or images are stored in the FCM
  - Designation = Is equipped, if RPO Code UGN is present



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ADAS EDR and RPOs (Vehicles without Super Cruise)

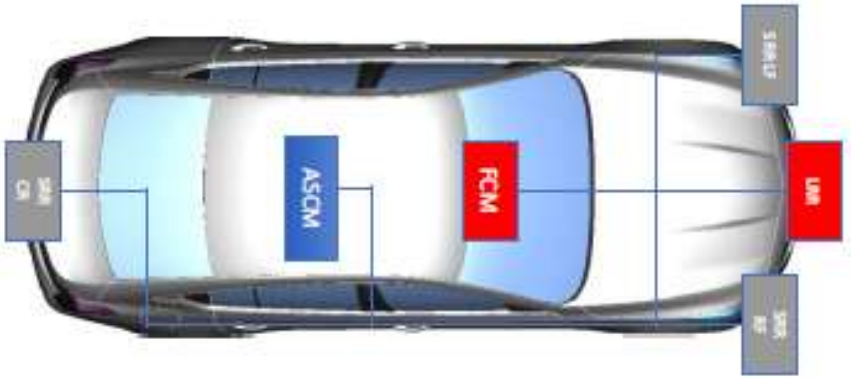
- Vehicle equipped with an FCM only
  - Includes RPO UEU (Forward Collision Alert)
- **<u>ADAS EDR is contained within the FCM</u>**
  - (Designation = Is equipped, if RPO Code UEU is present <u>without UGN</u>)



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ADAS EDR and RPOs
(Vehicles with Super Cruise Available)

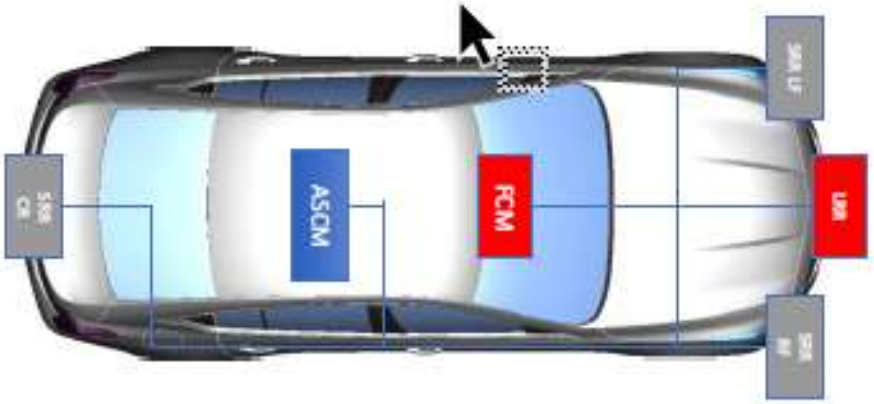
- Vehicle equipped with ASCM
- With Super Cruise
  - Includes RPOs UEU (Forward Collisio Alert), UGN (Automatic Emergency Braking), and UKL (Super Cruise)
  - **<u>ADAS EDR is contained within the ASCM</u>**
  - No EDR stored in the FCM
  - (Designation = Is equipped if RPO Code <u>UGN is present with UKL</u>)



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ADAS EDR and RPOs (Vehicles with Super Cruise Available)

- Vehicle equipped with ASCM – with out Super Cruise
  - Includes RPOs UEU (Forward Collision Alert) and UGN (Automatic Emergency Braking)
  - **<u>ADAS EDR is contained within the ASCM</u>**
  - No EDR stored in the FCM
  - (Designation = Is equipped if RPO Code <u>UGN is present without UKL</u>)



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ADAS EDR and RPOs (Vehicles with Super Cruise Available)

- Vehicle equipped with an FCM only
  - Includes RPO UEU (Forward Collision Alert)
  - **ADAS EDR is contained within the FCM**
  - (Designation = Is equipped if RPO Code UEU is present without UGN)



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Front Camera Module

- Front Camera Modules are located behind the rear-view mirror
- Remove the plastic covering to expose the camera if D2M is required





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Front Camera Module

- Not ALL FCM are supported at this time
- More coverage is expected to be added for some older and newer vehicles
- Some cameras can NOT be supported





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GM Help File information

			ACM	Data	CANplus	F00K108287	F00K108454	Center tunnel
4,13	2020 Chevrolet	Equinox (note 6)	FCM (Note1) how to image	Data	CDR 900	CDR 900 - click here	CDR 900 - Cable# 856	Center - top interior of windshield

- Pay Attention to “notes” – 2 different areas
  - Note 6 under “Equinox” tells us no key is required
  - Note 1 under FCM tells about RPO codes to determine if a vehicle is equipped with a support camera

GM Regular Production Option (RPO) Code Designation for FCM

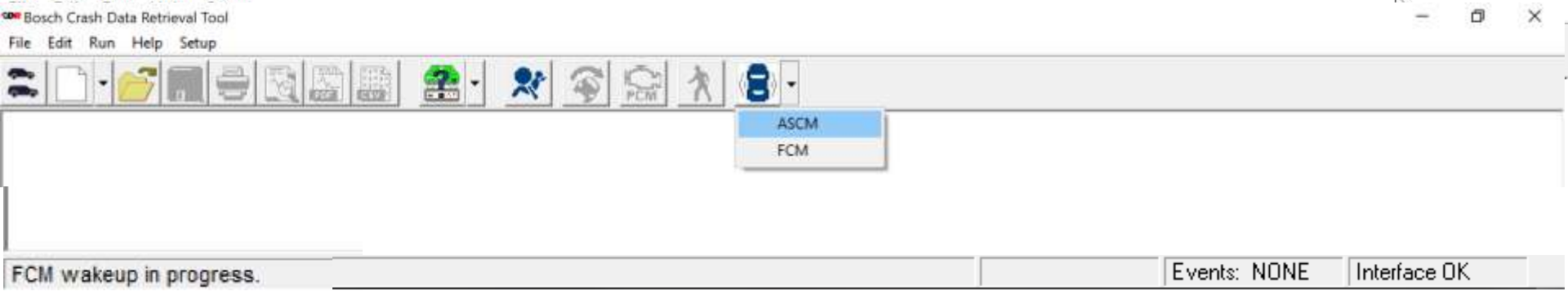
To determine if this vehicle is equipped with an FCM supported by the CDR Tool, you can see the camera located at the top interior of windshield and/or verify the RPO code includes **&UEU-UGN** in the vehicle option list.

Note: for model year 2019 pickup trucks, FCM is supported only on 1500 Silverado and Sierra trucks.

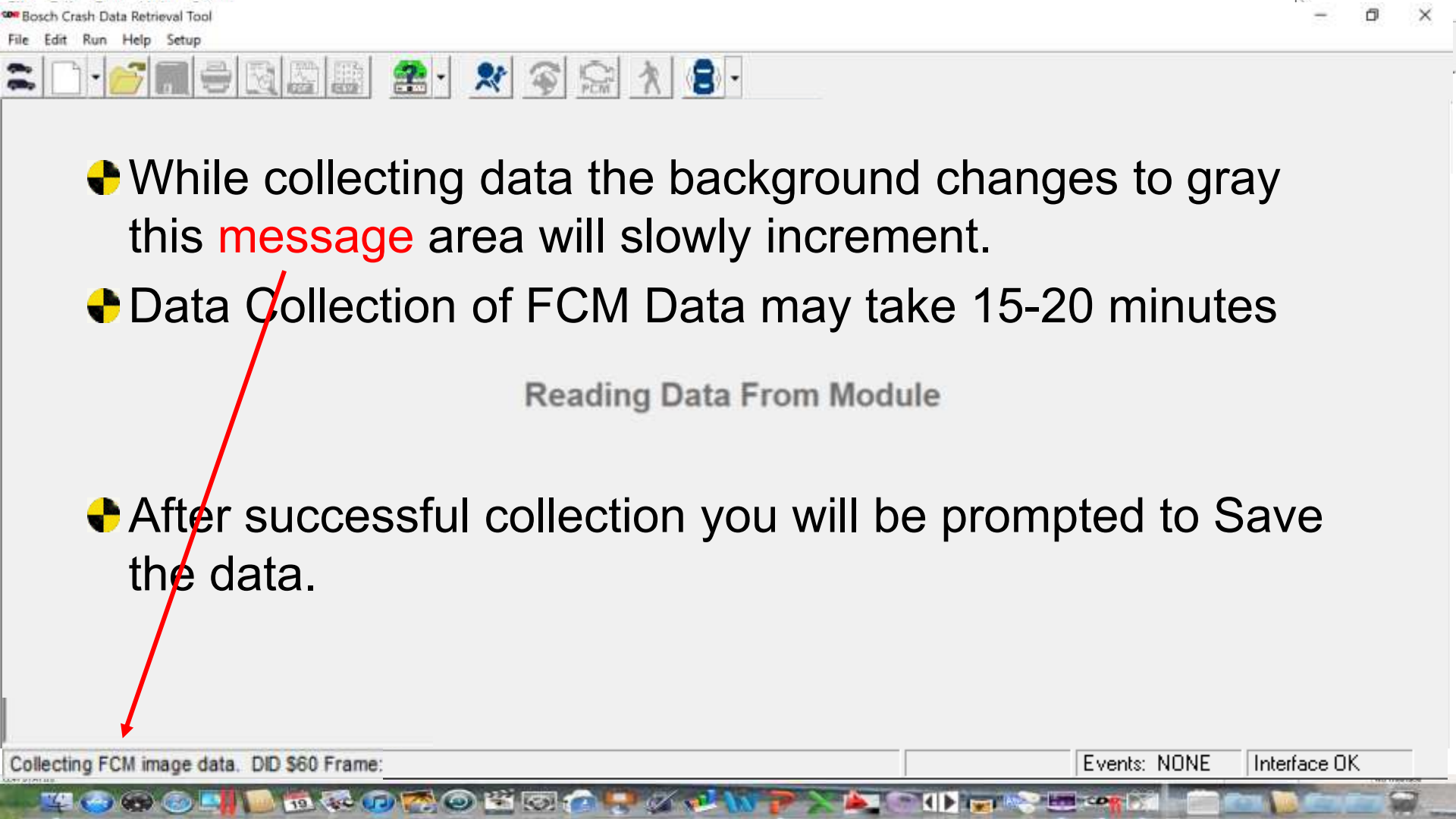
MUST have UEU but NOT UGN



- After inputting or reading the VIN – check for Active Safety Components that ***MAY*** be present for additional download!
- Just because the option is available does NOT mean the vehicle is equipped with these systems or a supported system!
- CHECK the Vehicle Coverage Listing in Help File!



- Once collect data from FCM is clicked, watch for this message.
- The FCM will be woken up via CAN messaging to facilitate data collection


Bosch Crash Data Retrieval Tool
File Edit Run Help Setup
While collecting data the background changes to gray this message area will slowly increment.
Data Collection of FCM Data may take 15-20 minutes
Reading Data From Module
After successful collection you will be prompted to Save the data.
Collecting FCM image data. DID $60 Frame:
Events: NONE
Interface OK

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FCM Image Export

- Once Images are collected they within the .FCM file created by CDR when saved.
- Click Here to Export the images.
- Images are Exported as ".PGM" files
  - **P**ortable **G**ray **M**ap file format
- Images must be "Flipped" for proper viewing orientation

File Edit Run Help Setup
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FCM Image Viewing

- Images “As the Camera Saw Them”

2020 Chevrolet Equinox - 2GNAXVEX6L6112470_FCM_AtEvent1.PGM
2020 Chevrolet Equinox - 2GNAXVEX6L6112470_FCM_AtEvent2.PGM
2020 Chevrolet Equinox - 2GNAXVEX6L6112470_FCM_PostEvent1.PGM
2020 Chevrolet Equinox - 2GNAXVEX6L6112470_FCM_PostEvent2.PGM
2020 Chevrolet Equinox - 2GNAXVEX6L6112470_FCM_PreEvent1.PGM
2020 Chevrolet Equinox - 2GNAXVEX6L6112470_FCM_PreEvent2.PGM





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FCM Image Viewing

- Images corrected for viewing
- Irfanview (Free!) or similar image handler for .PGM files







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CDR System Operators Course



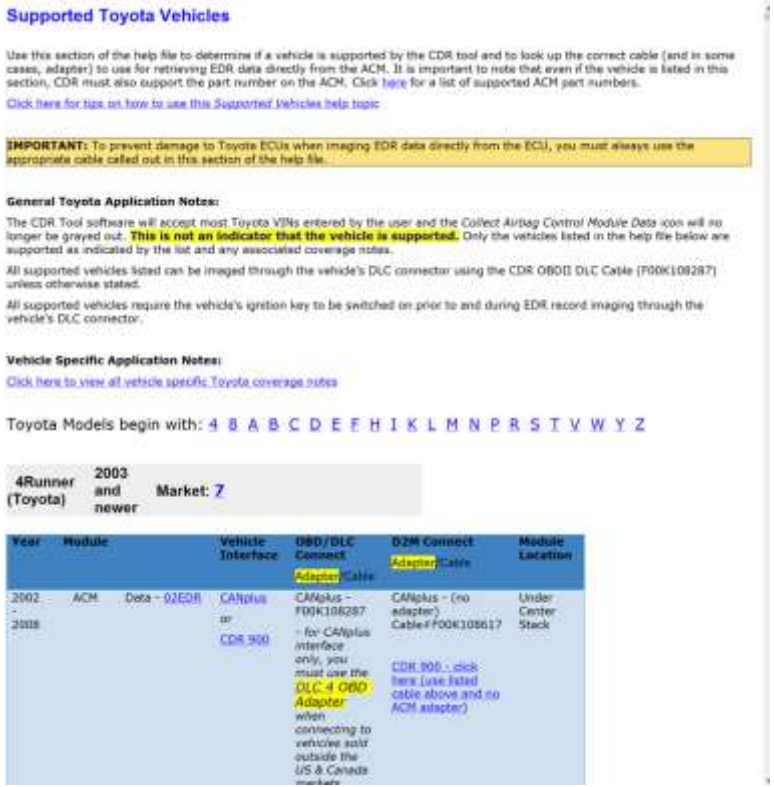
Toyota – Lexus – Scion
Vehicle Coverage Considerations

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File Edit Run Help Setup

Toyota Coverage Lookup Considerations

- Toyota has near world-wide coverage
- Difference in USA/Canada Model Years and “Other Markets”
- Different lookup procedure



Supported Toyota Vehicles

Use this section of the help file to determine if a vehicle is supported by the CDR tool and to look up the correct cable (and in some cases, adapter) to use for retrieving EDR data directly from the ACM. It is important to note that even if the vehicle is listed in this section, CDR must also support the part number on the ACM. Click here for a list of supported ACM part numbers.

Click here for tips on how to use this Supported Vehicles help topic

IMPORTANT: To prevent damage to Toyota ECUs when imaging EDR data directly from the ECU, you must always use the appropriate cable called out in this section of the help file.

General Toyota Application Notes:

The CDR Tool software will accept most Toyota VINs entered by the user and the *Collect Airbag Control Module Data* icon will no longer be grayed out. **This is not an indicator that the vehicle is supported.** Only the vehicles listed in the help file below are supported as indicated by the list and any associated coverage notes.

All supported vehicles listed can be imaged through the vehicle's DLC connector using the CDR OBDII DLC Cable (F00K108287) unless otherwise stated.

All supported vehicles require the vehicle's ignition key to be switched on prior to and during EDR record imaging through the vehicle's DLC connector.

Vehicle Specific Application Notes:

Click here to view all vehicle specific Toyota coverage notes

Toyota Models begin with: 4 8 A B C D E F H I K L M N P R S T V W Y Z

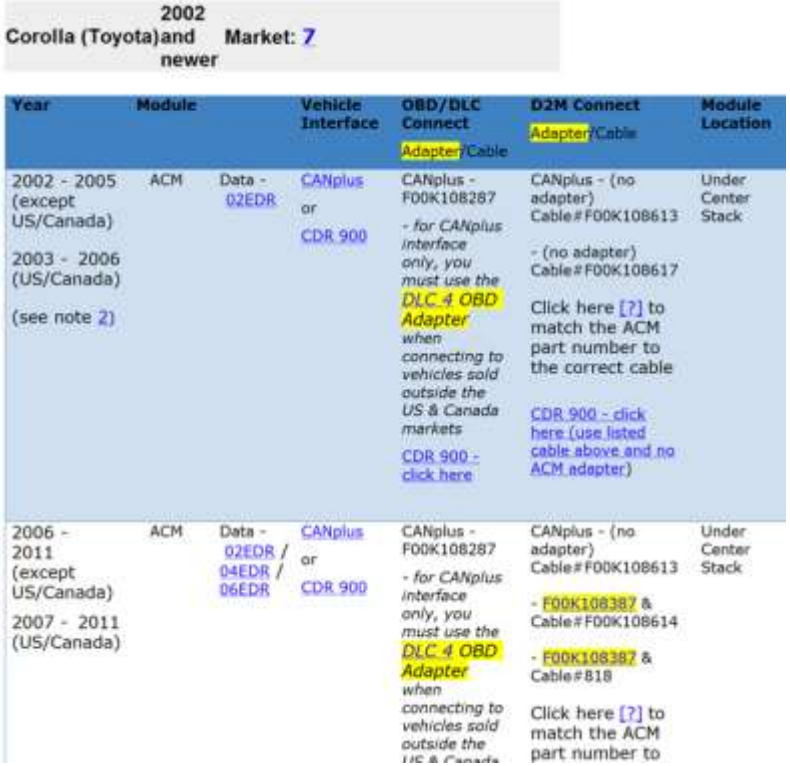
4Runner (Toyota)	2003 and newer	Market: Z

Year	Module		Vehicle Interface	OBD/DLC Connect Adapter/Cable	D2M Connect Adapter/Cable	Module Location
2002 - 2008	ACM	Data - 02EDR	CANplus or CDR 900	CANplus - F00K108287 - for CANplus interface only, you must use the DLC 4 OBD Adapter when connecting to vehicles sold outside the US & Canada	CANplus - (no adapter) Cable # F00K108617 CDR 900 - click here (use listed cable above and no ACM adapter)	Under Center Stack

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Toyota – Model Year Considerations

- Close Attention to the coverage years and markets indicated
- Market Code 7 = All Markets the Vehicle is Sold
- US/Canada Listed differently



2002 Corolla (Toyota) and newer **Market: 7**

Year	Module		Vehicle Interface	OBD/DLC Connect Adapter/Cable	D2M Connect Adapter/Cable	Module Location
2002 - 2005 (except US/Canada) 2003 - 2006 (US/Canada) (see note 2)	ACM	Data - 02EDR	CANplus or CDR 900	CANplus - F00K108287 *- for CANplus interface only, you must use the DLC 4 OBD Adapter when connecting to vehicles sold outside the US & Canada markets* CDR 900 - click here	CANplus - (no adapter) Cable# F00K108613 - (no adapter) Cable# F00K108617 Click here [?] to match the ACM part number to the correct cable CDR 900 - click here (use listed cable above and no ACM adapter)	Under Center Stack
2006 - 2011 (except US/Canada) 2007 - 2011 (US/Canada)	ACM	Data - 02EDR / 04EDR / 06EDR	CANplus or CDR 900	CANplus - F00K108287 *- for CANplus interface only, you must use the DLC 4 OBD Adapter when connecting to vehicles sold outside the*	CANplus - (no adapter) Cable# F00K108613 - F00K108387 & Cable# F00K108614 - F00K108387 & Cable# 818 Click here [?] to match the ACM part number to	Under Center Stack

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Toyota Coverage Multiple Cables Indicated

- Some may indicate multiple cables
- Some require ACM (“387”) adapter
- Lookup now requires ACM Part Number for Direct to Module (D2M) access
- Lookup access via “[?]” Hyperlink

Corolla (Toyota) 2002 and newer Market: 7

Year	Module		Vehicle Interface	OBD/DLC Connect Adapter/Cable	D2M Connect Adapter/Cable	Module Location
2002 - 2005 (except US/Canada) 2003 - 2006 (US/Canada) (see note 2)	ACM	Data - 02EDR	CANplus or CDR 900	CANplus - F00K108287 *- for CANplus interface only, you must use the DLC 4 OBD Adapter when connecting to vehicles sold outside the US & Canada markets* CDR 900 - click here	CANplus - (no adapter) Cable# F00K108613 - (no adapter) Cable# F00K108617 Click here [?] to match the ACM part number to the correct cable CDR 900 - click here (use listed cable above and no ACM adapter)	Under Center Stack
2006 -	ACM	Data -	CANplus	CANplus -	CANplus - (no	Under

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ACM Part Number

- Only Necessary for D2M Access!
- DLC is via VIN and is not ACM specific



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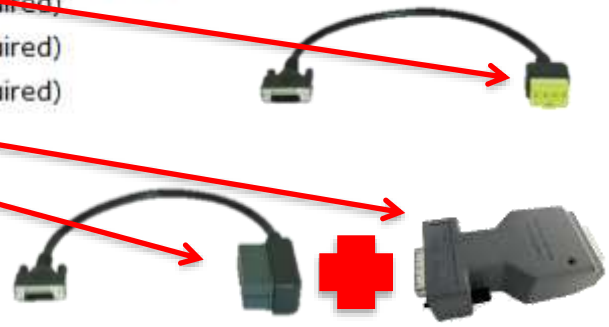
File Edit Run Help Setup

Toyota ACM Notes

Toyota/Lexus/Scion CDR Cable Lookup by ACM Part Number:

If the vehicle listed in the *Vehicle and Cable Lookup* section of the help file lists multiple CDR cable references, refer to this chart to determine the correct cable to use for a given part number.

89170-08050	F00K108617	(adapter not required)
89170-08060	F00K108617	(adapter not required)
89170-08070	F00K108617	(adapter not required)
89170-08080	F00K108617	(adapter not required)
89170-08090	F00K108614	F00K108387
89170-08091	F00K108614	F00K108387
89170-08100	F00K108614	F00K108387
89170-08120	F00K108614	F00K108387
89170-08130	F00K108616	F00K108387
89170-08140	F00K108616	F00K108387



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CDR System Operators Course



CDR Legacy Software Operation

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Bosch Crash Data Retrieval Tool

File Edit Run Help Setup

- New
- Open Ctrl+O
- Close
- Properties
- Save Ctrl+S
- Save without VIN Sequence Number
- Save CDR Report as PDF
- Save Engineering Translation Report As PDF
- Save CDR Report as CSV Text File
- Preview
- Print Setup
- Set Printer for Optimized PDF Output
- Print
- Exit

Bosch Crash Data Retrieval Tool

File Edit Run Help Setup

- Case Information
- Comments
- Vehicle Identification Number
- Copy to Clipboard

Bosch Crash Data Retrieval Tool

File Edit Run Help Setup

- Collect ACM Data
- Collect PCM Data
- Collect ROS Data
- Collect PPM Data
- Collect Active Safety Control Module Data
  - Collect EOCM Data
  - Collect FCM Data
- Test CDR Device Communications

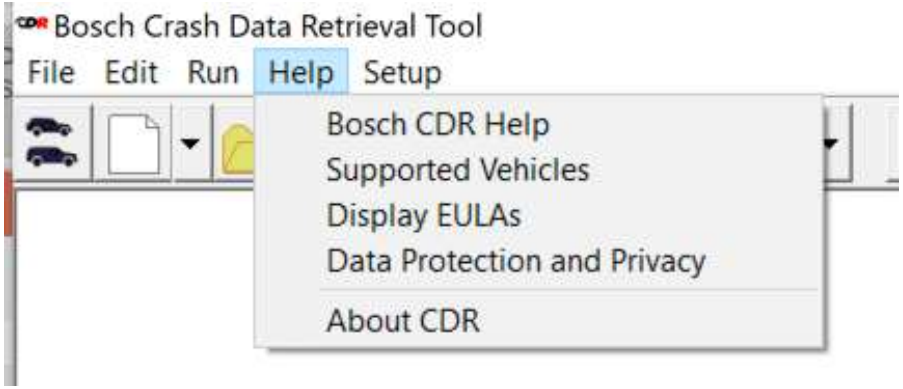
Bosch Crash Data Retrieval Tool

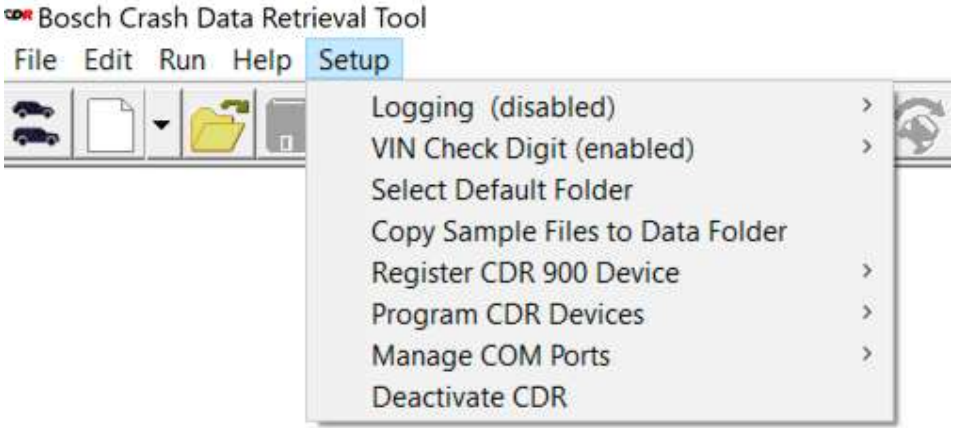
File Edit Run Help Setup

- Collect ACM Data
- Collect PCM Data
- Collect ROS Data
- Collect PPM Data
- Collect Active Safety Control Module Data
- Test CDR Device Communications
  - Connection Help
  - Test CANplus Interface
  - Test CDR 900
  - Test CDR 500 Adapter



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- Under Help:
  - Bosch CDR Help
  - Supported Vehicle List
  - Other Info
- Setup
  - Logging File on/off
  - Register CDR900
  - Program CDR Devices
  - Choose Comm. Port
  - Other Options

File Edit Run Help Setup
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What is a VIN Check Digit

A VIN check digit is the 9th character in a standard 17-digit **North American** Vehicle Identification Number (VIN) that is used to verify the VIN's validity. This digit is the result of a complex mathematical calculation involving the other 16 characters of the VIN, and its purpose is to detect errors or forgeries, ensuring the VIN is authentic. This is not always used in vehicles manufactured for outside of the US market.

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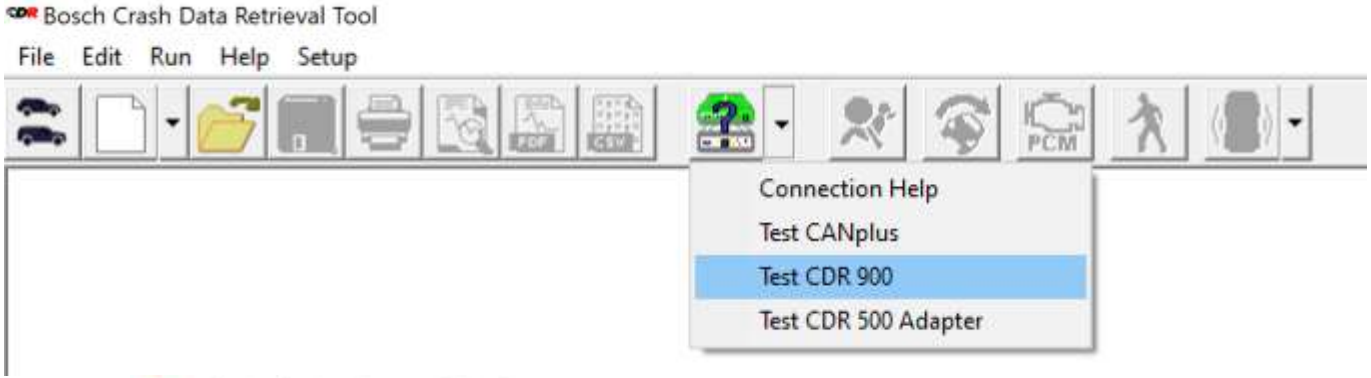




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Imaging Icons become “live” after a VIN is entered



Before entering a VIN, module icons are not “live”



Most supported OE’s will have ACM support



Some GM will have both ACM and Active Safety

GM only if ROS was available (but not necessarily installed)



may have RCM, PCM or both accessible



may have ACM, or ACM and PPM accessible

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File Edit Run Help Setup

Bosch Crash Data Retrieval Tool 2GNAXVEX6L6112470 Page 1 of 10 - [Report]

File Edit Run Help Setup

Report Navigation

Begin

First Page

Back One Page

Forward One Page

Last Page

Bosch Crash Data Retrieval Tool 2GNAXVEX6L6112470 Page 1 of 10 - [Report]

File Edit Run Help Setup

Report Navigation

CDR File Information

Hide Navigation

Bosch Crash Data Retrieval Tool 2GNAXVEX6L6112470 Page 1 of 10 - [Report]

File Edit Run Help Setup

CRASH DATA

Show Navigation

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Software version identification

- Support is extended to current software subscription holders for the current version of CDR

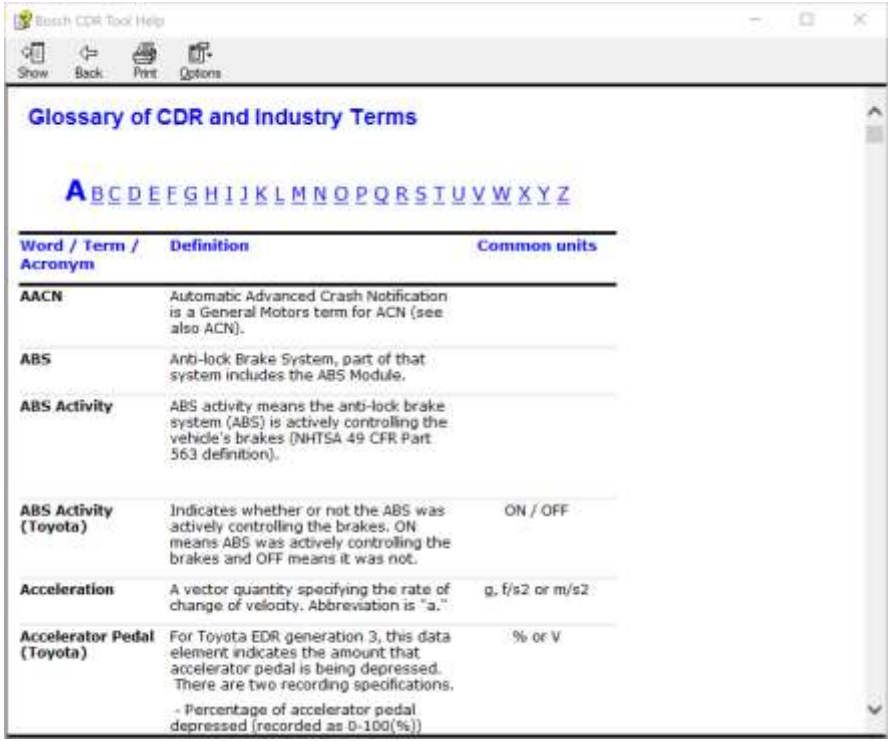


- The vehicles that can be accessed is directly related to the licensed version of the software you have or the users subscription

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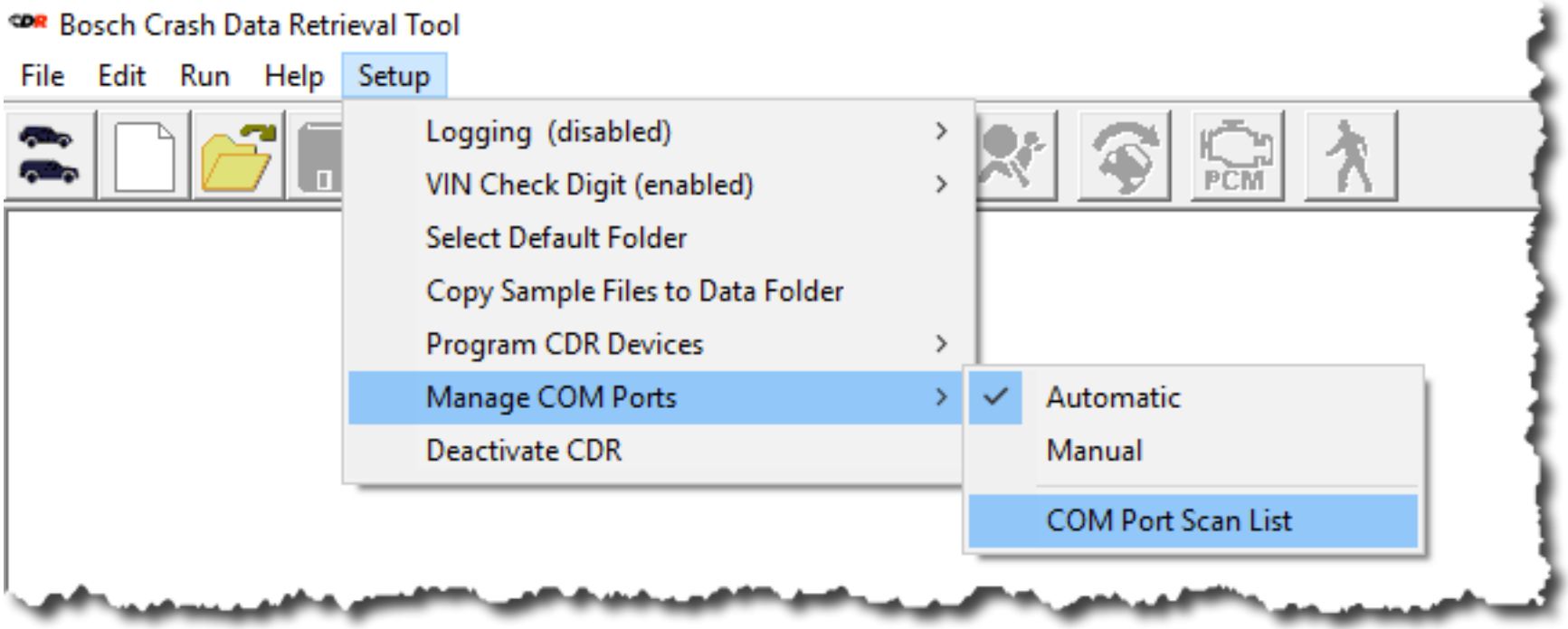
Glossary

- Located in the help File
- Contains terms and definitions that apply in general, and specific information that applies across all manufacturers



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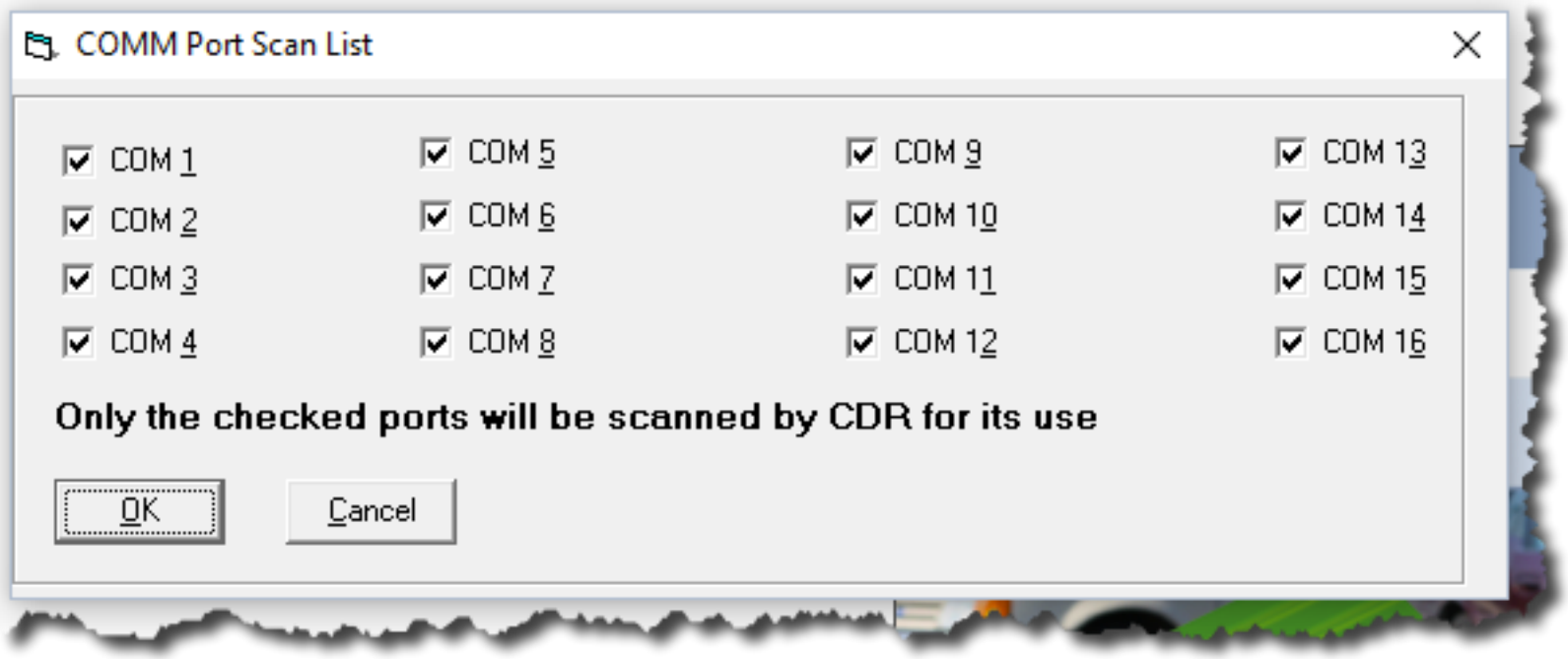
Comm. Port Management



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File Edit Run Help Setup

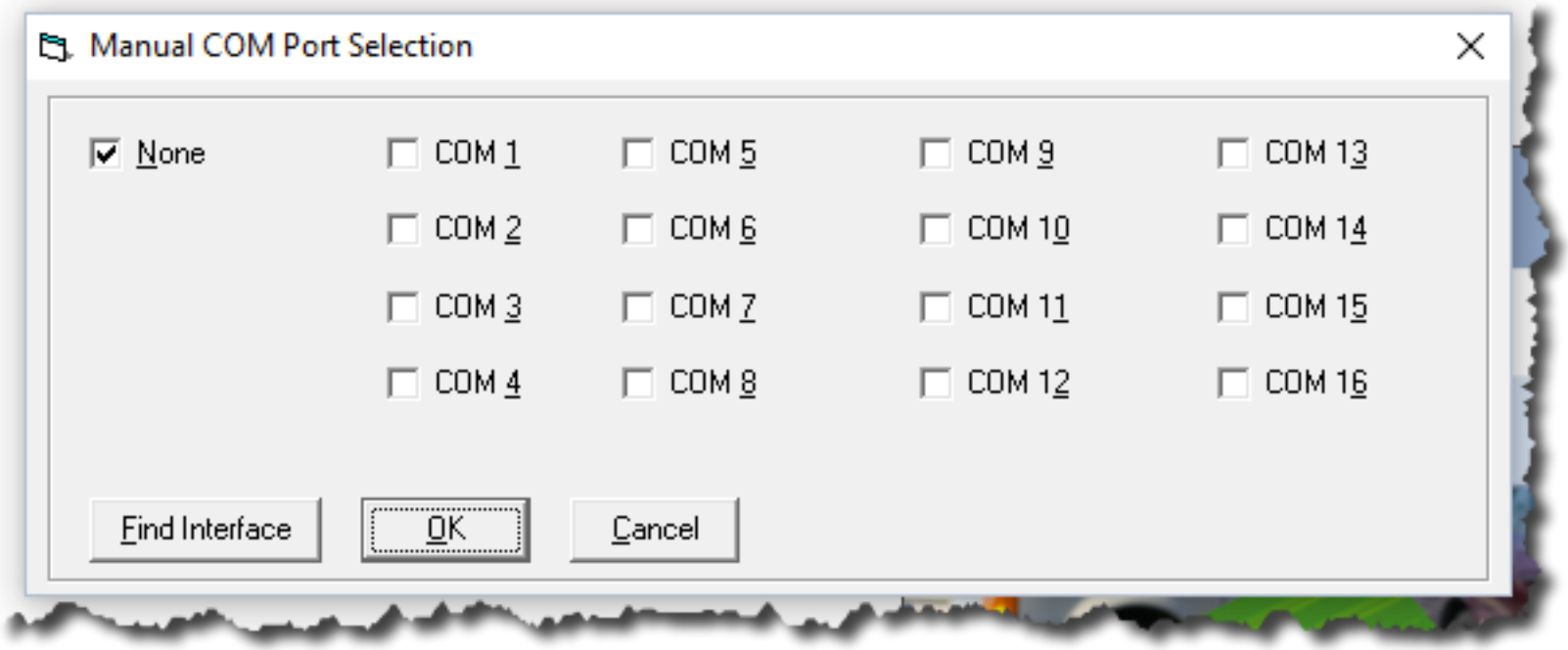
Choose Comm. Ports for Scan



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File Edit Run Help Setup

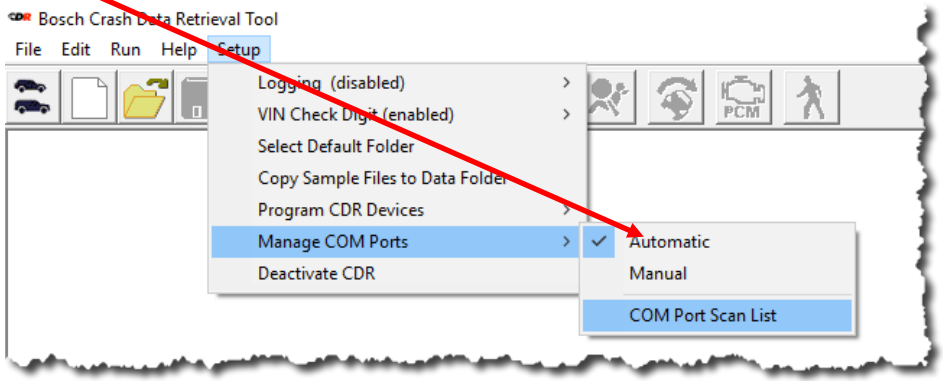
Manual Selection / Find Interface



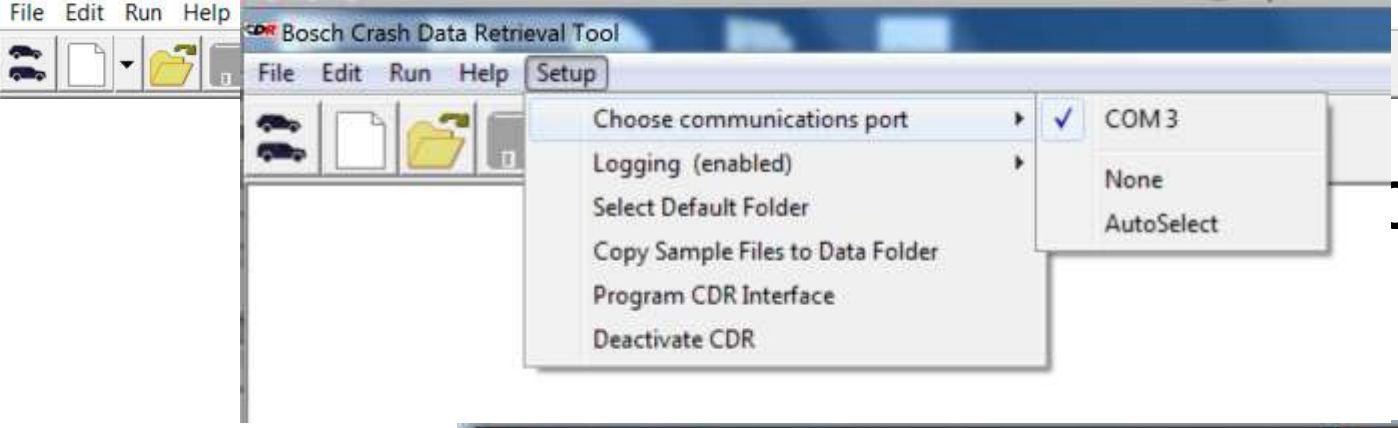
File Edit Run Help Setup © 2024 - Crash Data Specialists LLC

Auto-selecting a Com Port

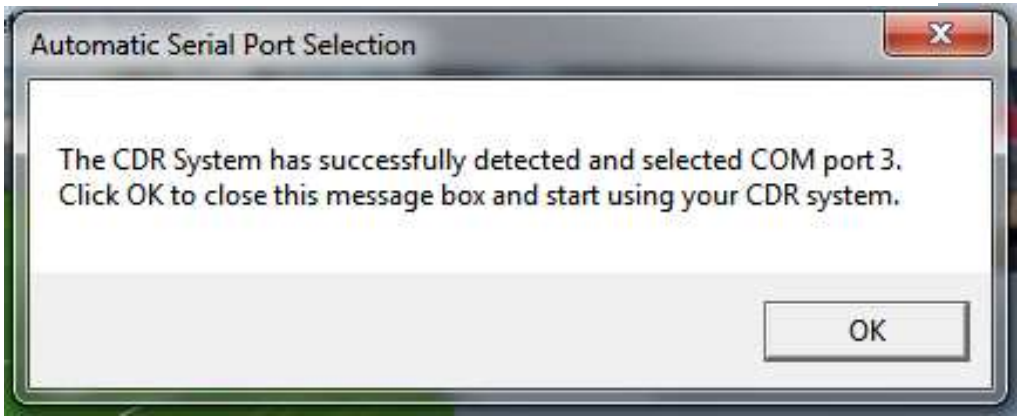
- Connect and power the CDR Interface and select the AutoSelect option under the Setup toolbar button



- The software detects the interface and selects the proper communications port
- May not work if your PC has Bluetooth “on”


File Edit Run Help
Bosch Crash Data Retrieval Tool
File Edit Run Help Setup
Choose communications port
Logging (enabled)
Select Default Folder
Copy Sample Files to Data Folder
Program CDR Interface
Deactivate CDR
COM 3
None
AutoSelect

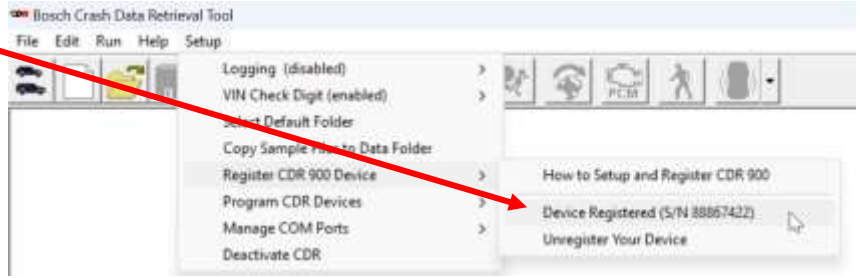
ort


Automatic Serial Port Selection
The CDR System has successfully detected and selected COM port 3.
Click OK to close this message box and start using your CDR system.
OK

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CDR900 – Register / Unregister

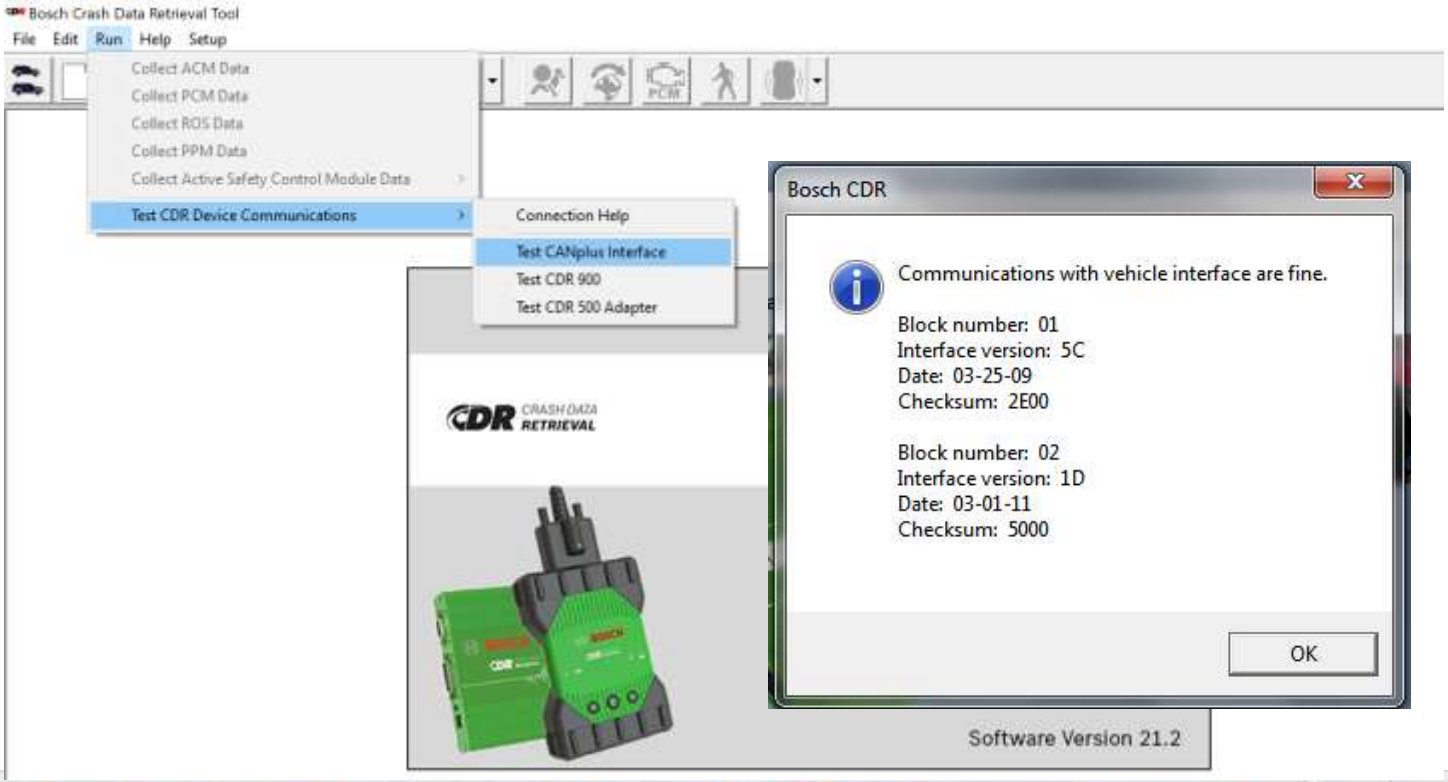
- CDR900 Must be “Registered” before first use
- CDR900 can be “unregistered” and a different CDR900 registered as needed
- When CDR accesses a CDR900 it checks that you are using the currently registered VCI





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Checking the interface



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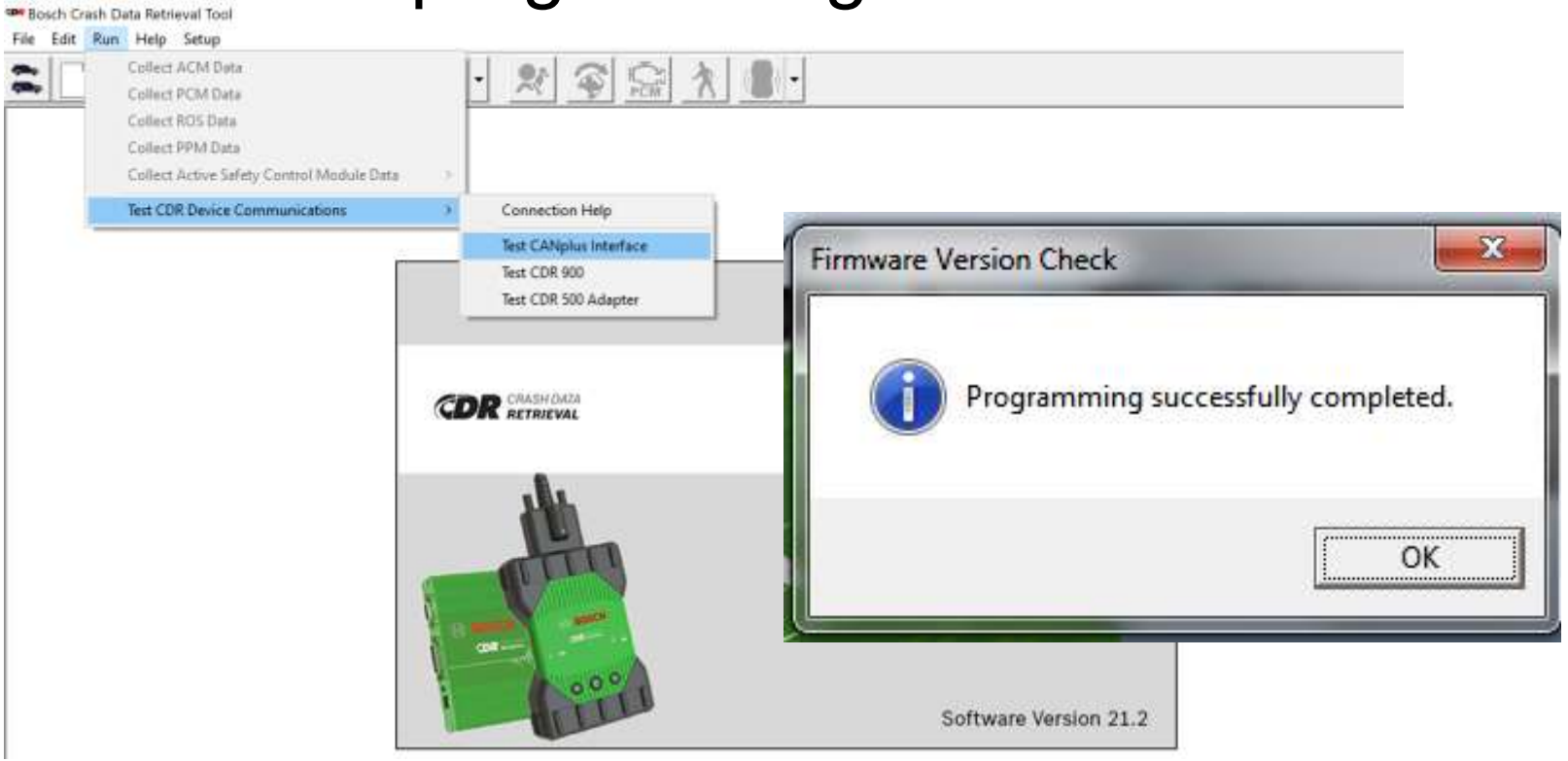


Reprogramming the Interface

- If the software asks if you want to reprogram the interface always select yes
  - This happens when the interface hasn't seen the software version currently on the computer or when it has been used on another computer with a dissimilar software version
  - This message displays when testing communication with the module or attempting to download

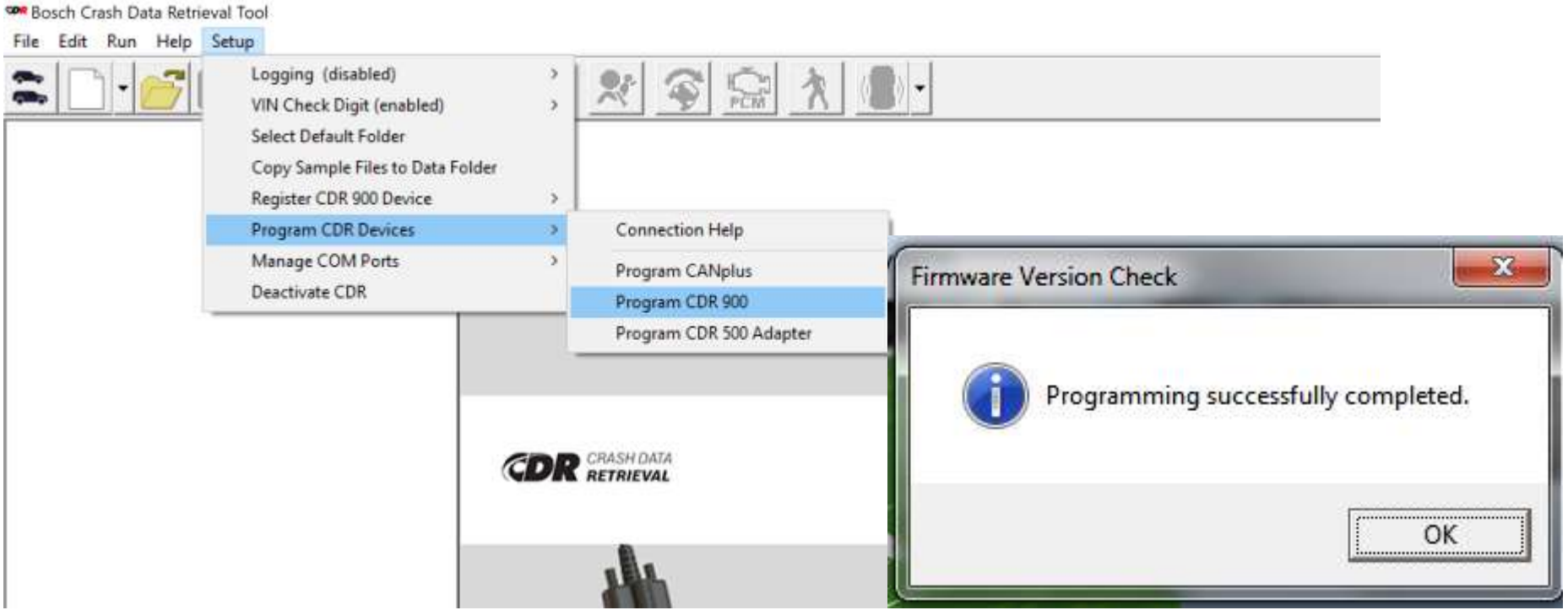
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Reprogramming the CDR Interface



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Reprogramming the CDR Interface



File Edit Run Help Setup © 2024 - Crash Data Specialists LLC

CDR File Information

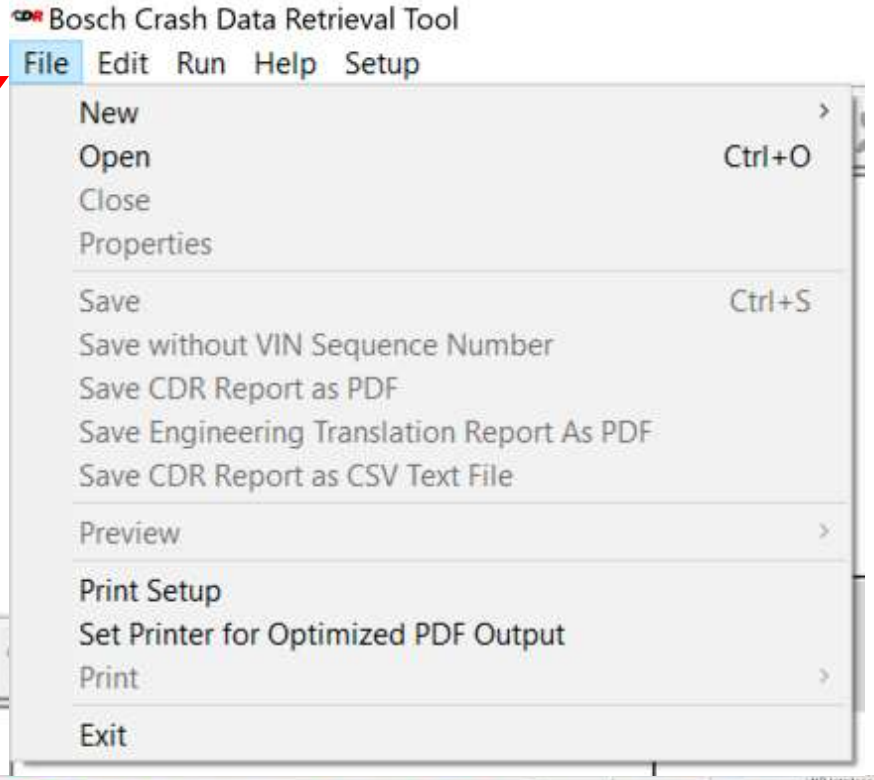
CDR File Information

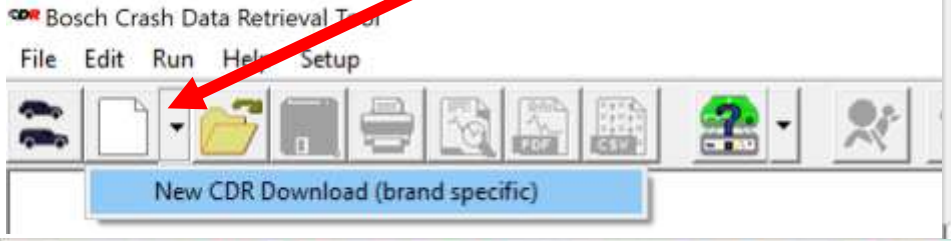
User Entered VIN	1GC4KYB67AF105385
User	B. Muir for M. Phillips
Case Number	NP16120079
EDR Data Imaging Date	12/07/2016
Crash Date	10/03/2016
Filename	2010 CHEVROLET SILVERADO -OPP - 1GC4KYB67AF105385_ACM.CDRX
Saved on	Wednesday, December 7 2016 at 16:16:40
Imaged with CDR version	Crash Data Retrieval Tool 17.1
Imaged with Software Licensed to (Company Name)	Crash Data Specialists
Reported with CDR version	Crash Data Retrieval Tool 17.1
Reported with Software Licensed to (Company Name)	Crash Data Specialists
EDR Device Type	Airbag Control Module
Event(s) recovered	Non-Deployment

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Download / Imaging Process

- Open the Bosch CDR Software and select New from the File dropdown menu
  - (Note arrow beside New – must click with mouse!)
- or select the New icon from the menu bar

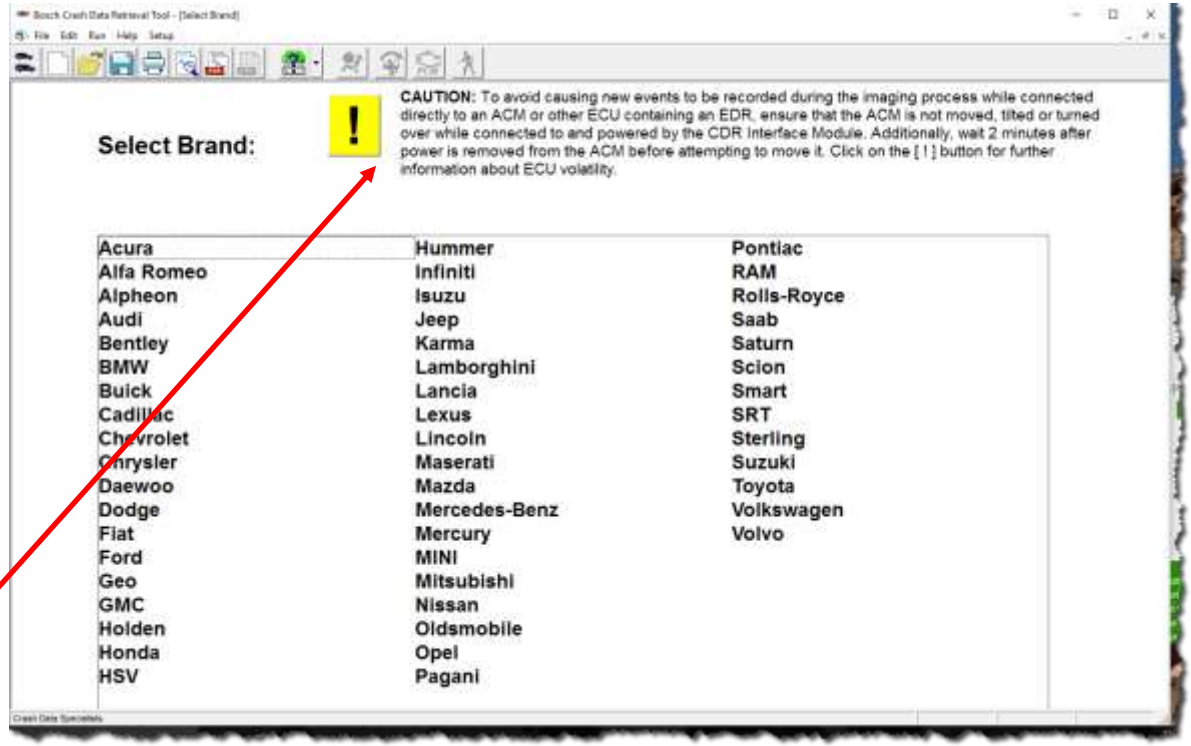




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Brand Selection

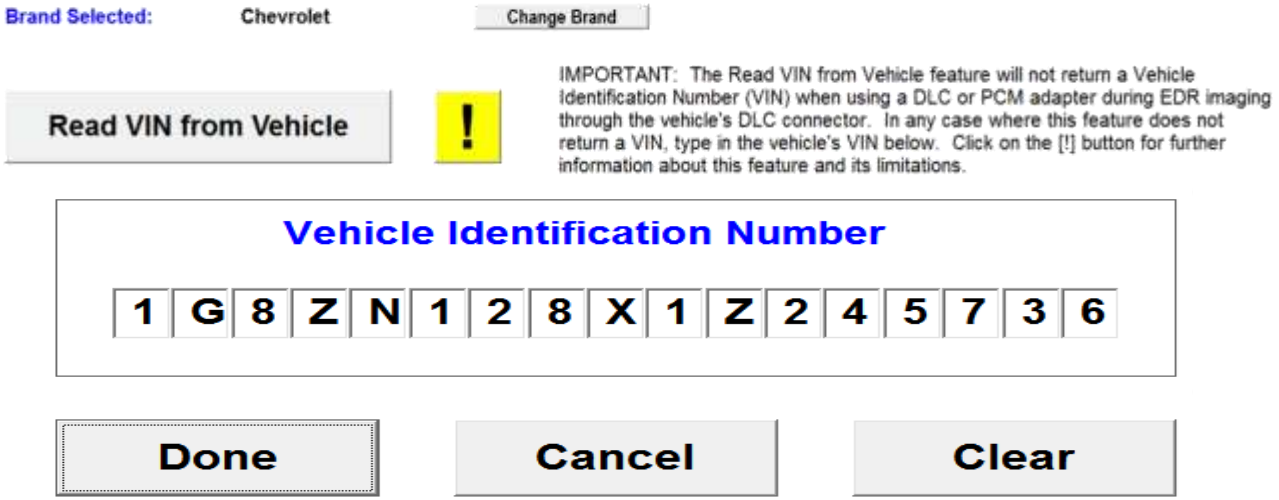
- Additional Step upon selecting new imaging sessions
- Necessary for International coverage due to different VIN standards / implementation
- *NEW* Warning Notice



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File Edit Run Help Setup

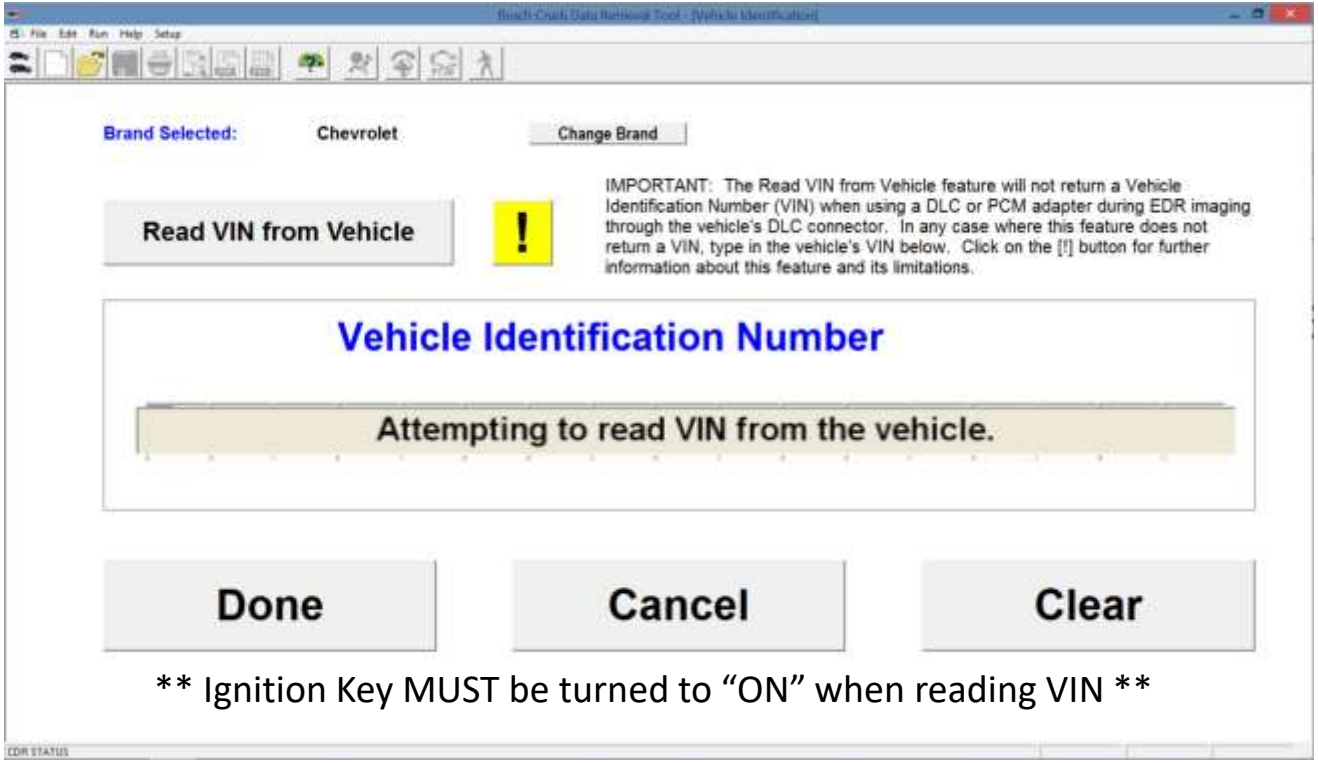
Enter the Vin – Manual Entry



- Enter the VIN from the vehicle
- If you make a mistake press clear or cancel
- Press Done when finished

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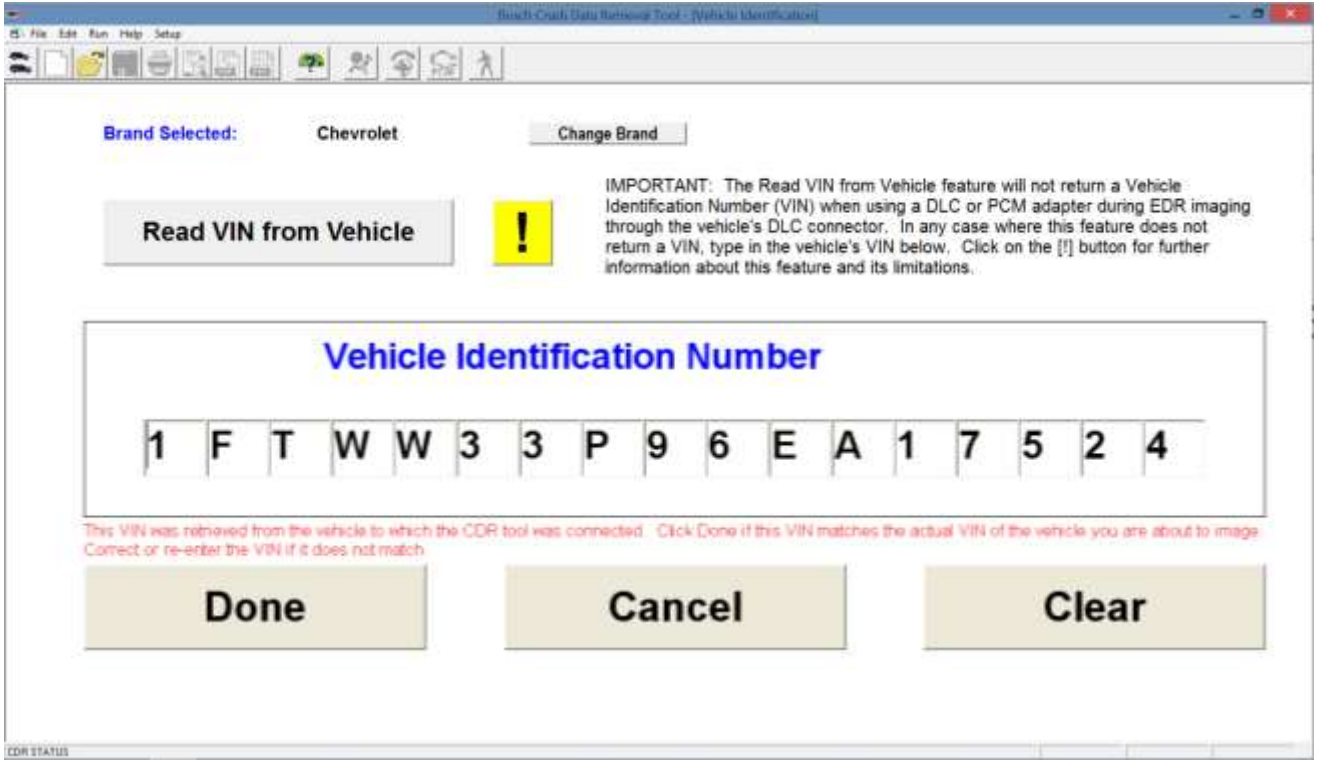
Read VIN from Vehicle



** Ignition Key MUST be turned to “ON” when reading VIN **

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Read VIN from Vehicle



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File Edit Run Help Setup

Case Information



- Enter the information from your case and press done when finished

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File Edit Run Help Setup

Notes



- Enter any notes you want and press done when finished

File Edit Run Help Setup
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Sample Notes

- Entries are printed on the last page of the CDR report and can be edited later

- If the download was DLC, Direct to Module or Benchtop
- Names of anyone present during download
- How the system was powered and what adapters or cables were used
- Location of download, address or business name
- Documentation of how many times ignition was cycled or power was applied to module or interface
- For DLC downloads a description of electrical switches and indication of power levels
- Manufacturer tire size sticker or recommendation along with actual sizes of all tires
- Powertrain reprogramming stickers or indication of indication of recalls completed
- Indication of mounting conditions or damage for direct to module downloads
- Position of ignition key for both ACM and PCM downloads
- DTC’s present at time of download if scan tool was used

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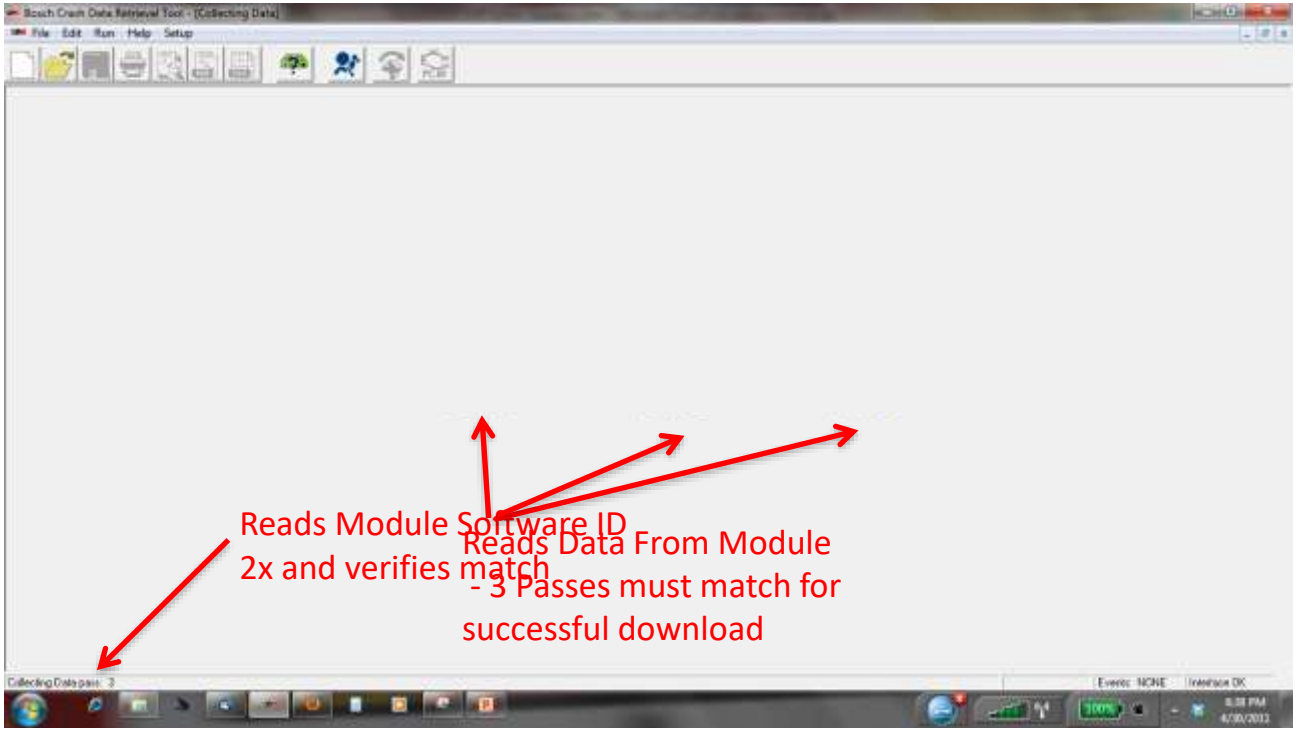
File Edit Run Help Setup

CDR Software operation
- Ford Only - “Privacy Question”



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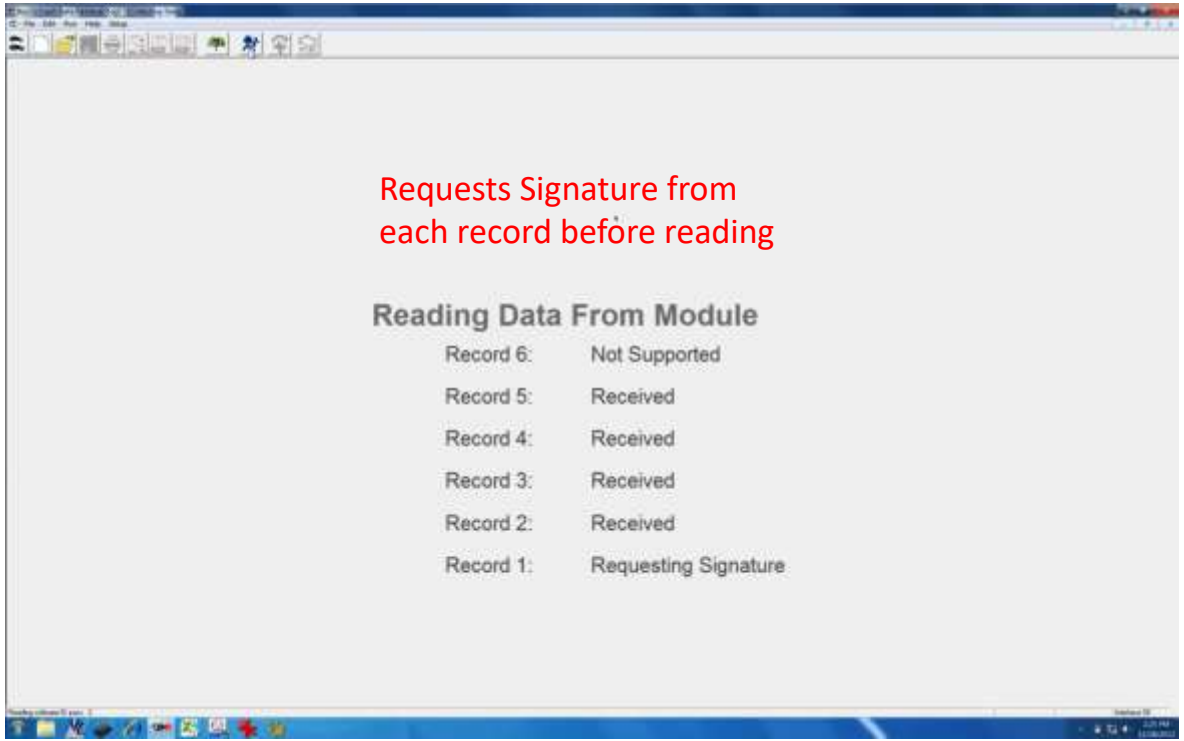
Reading Data From Module



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File Edit Run Help Setup

BMW / Mercedes Specific Read Process



Crash Data Retrieval 2 - v23.2.327

Home

BOSCH

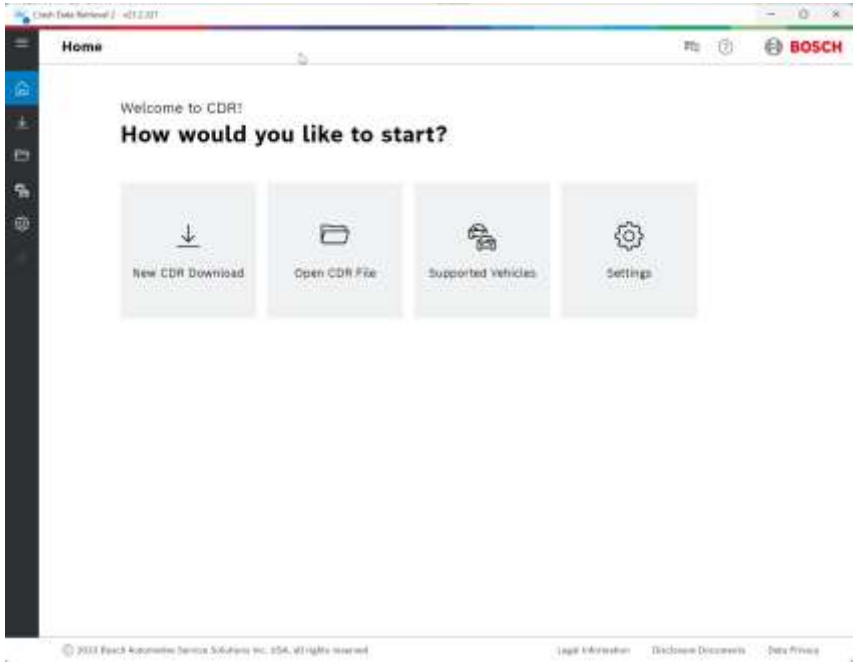


CDR 2 – What's New!!

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CDR-2

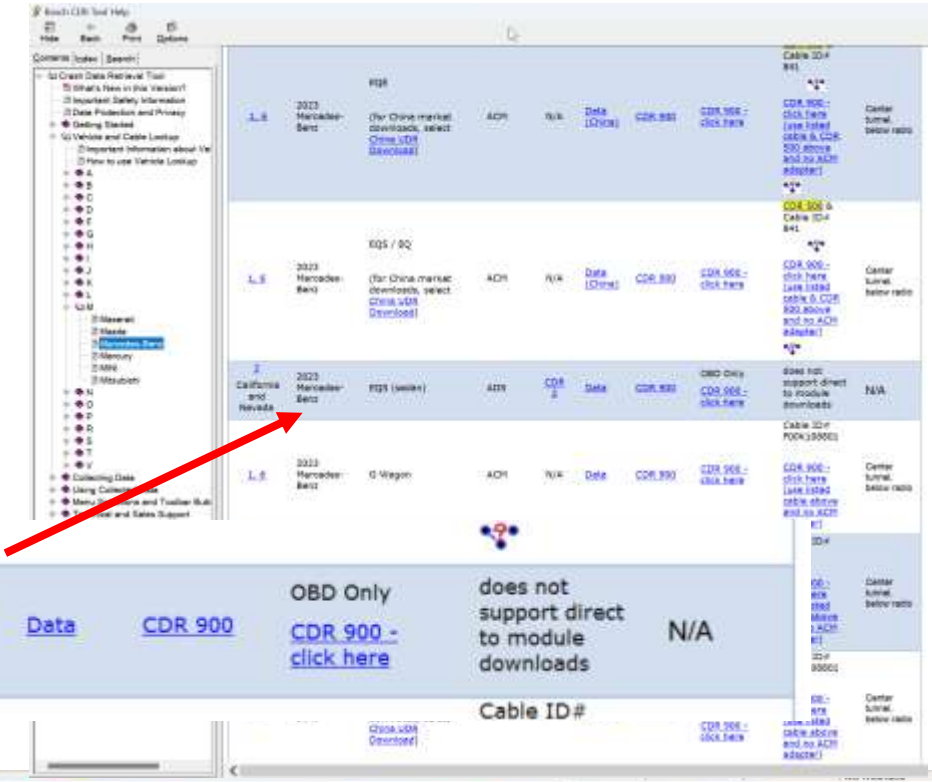
- Original Info was Version would switch based on VIN entered
- Changed – you must choose which version based on Help File listing
- Limited coverage in CDR2



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CDR-2

- Help File coverage indicats use CDR2
- Clicking the CDR2 blue Hyperlink will launch CDR2
- Currently limited to Mercedes and Mazda/Fiat in the EU Market



2 California and Nevada	2023 Mercedes-Benz	EQS (sedan)	ADS	CDR 2	Data	CDR 900	OBD Only CDR 900 - click here	does not support direct to module downloads	N/A

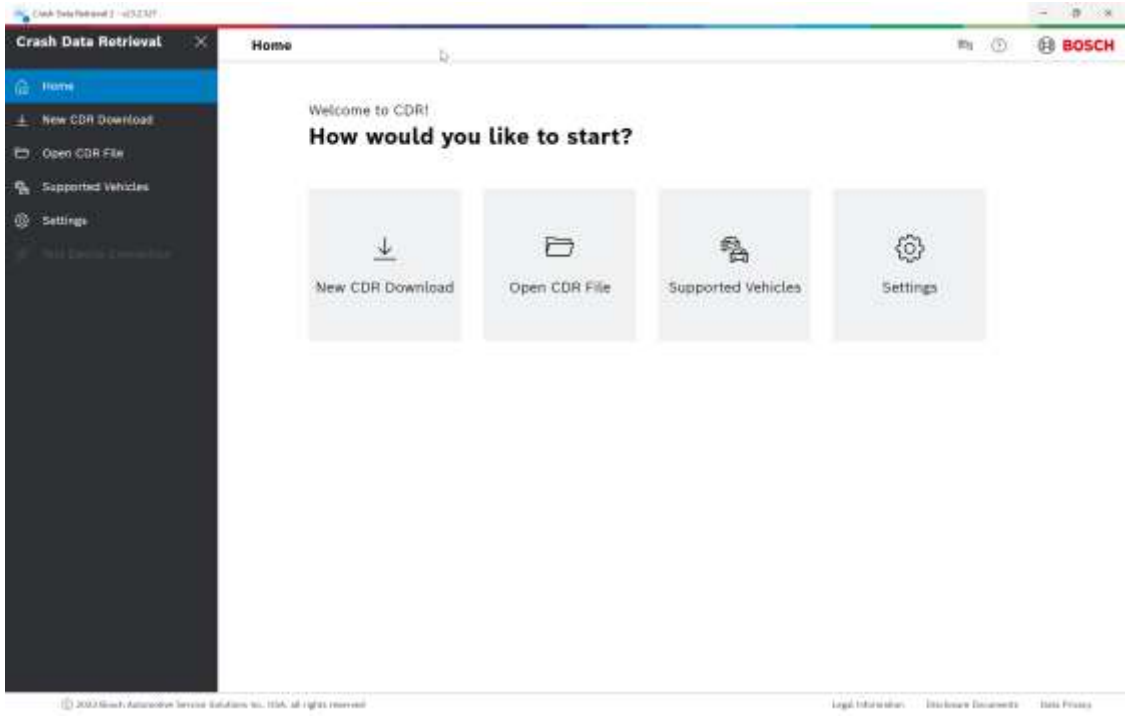
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BOSCH

CDR-2

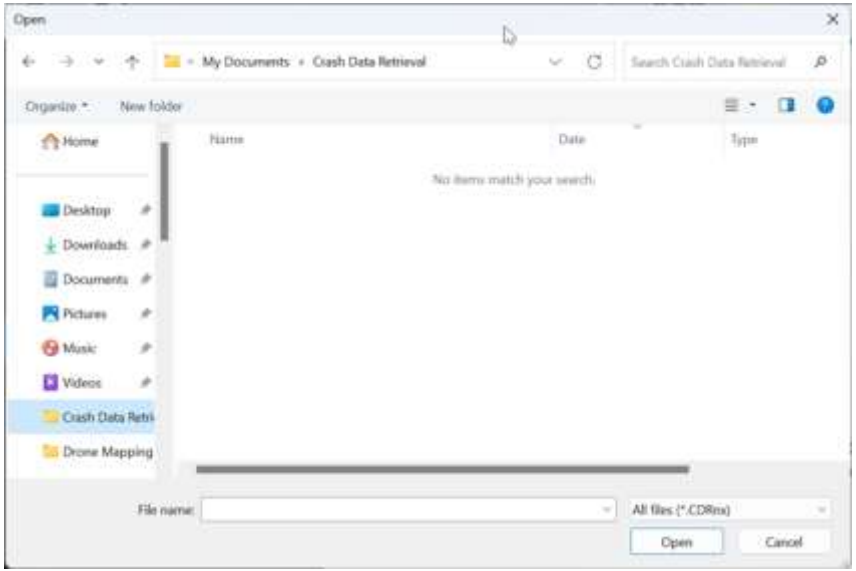


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CDR-2

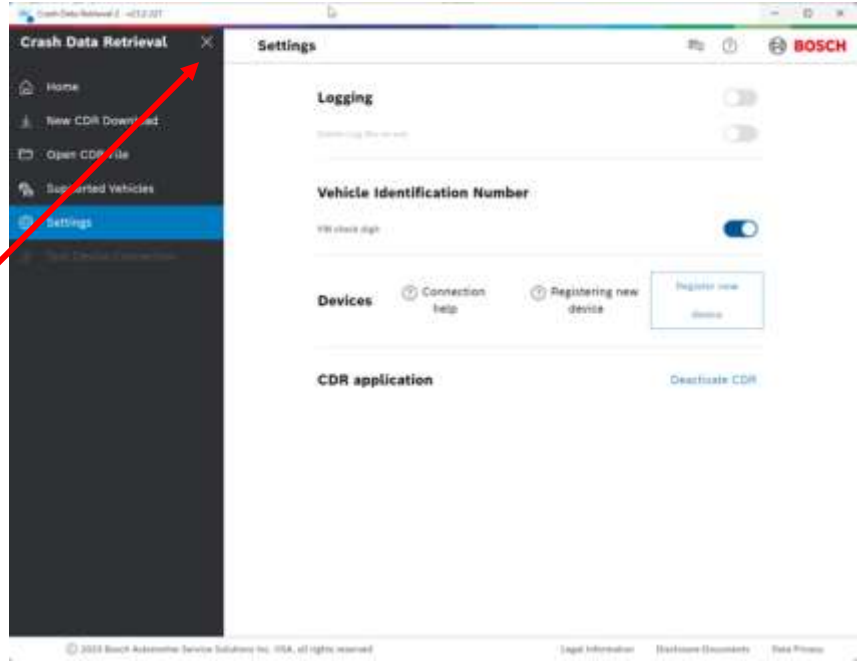
- Open File in CDR2 only opens CDR2 files
- New File Extension: “.CDRnx”
- Will NOT open .CDRx files
  - Use legacy version to view



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CDR-2

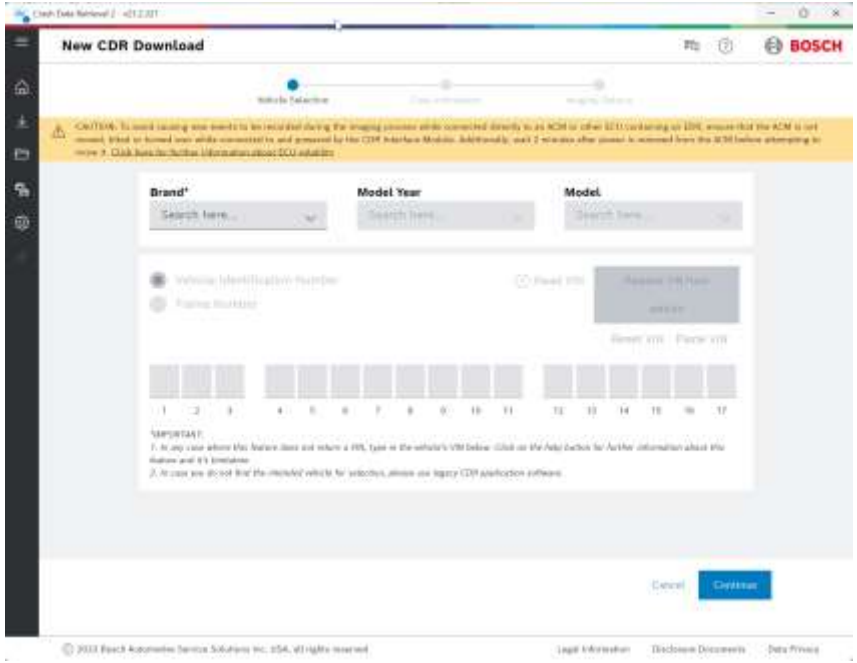
- Limited Icons on side bar
- Other Actions available by clicking “Home” to expand side bar
- Click the “X” to close the side bar



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CDR-2

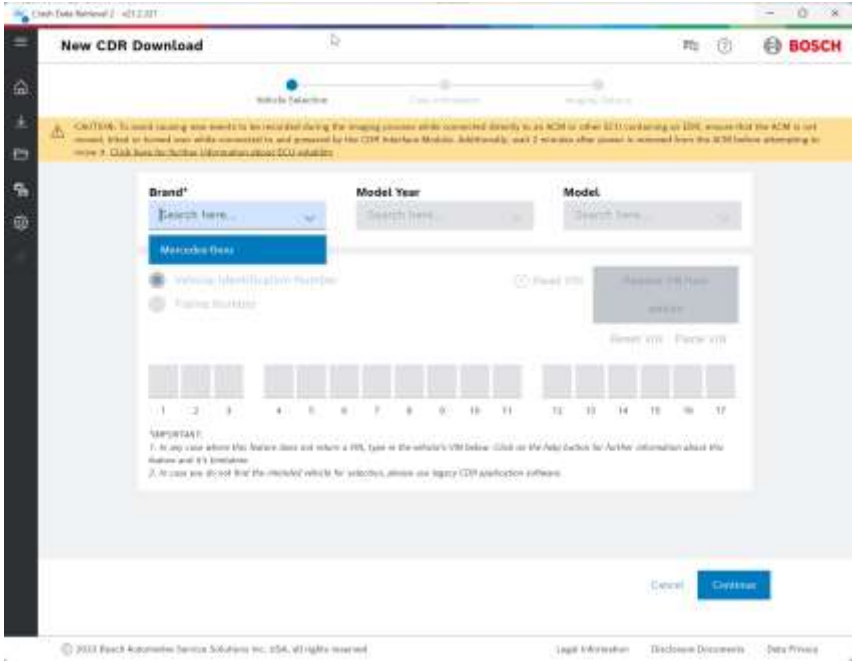
- New look and fell for selecting Brand / Model Year / Model
- “Paste VIN” feature added
- Note warning about powered ECU’s at the top!



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CDR-2

- Currently only Mercedes selectable in North America
- Mazda and Fiat (some) available to select but only in EU
- CDR operations may change based on licenced region



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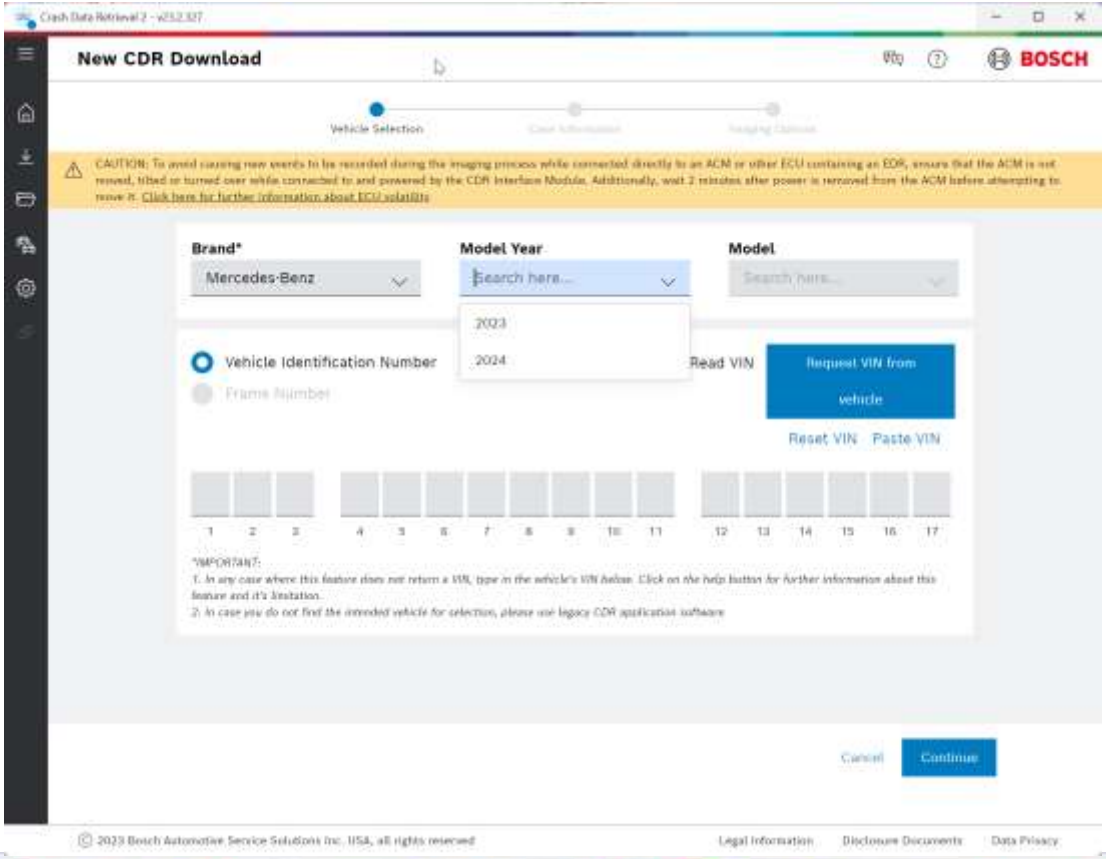
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BOSCH

CDR-2



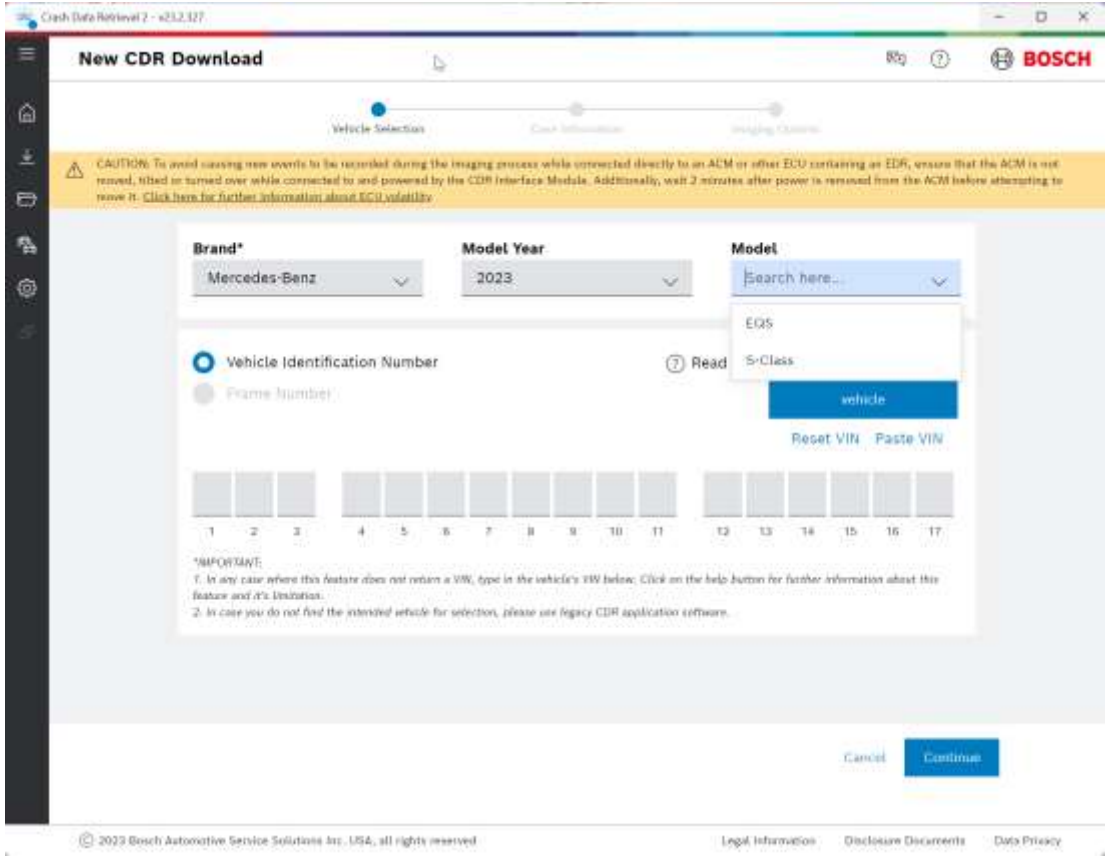
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CDR-2



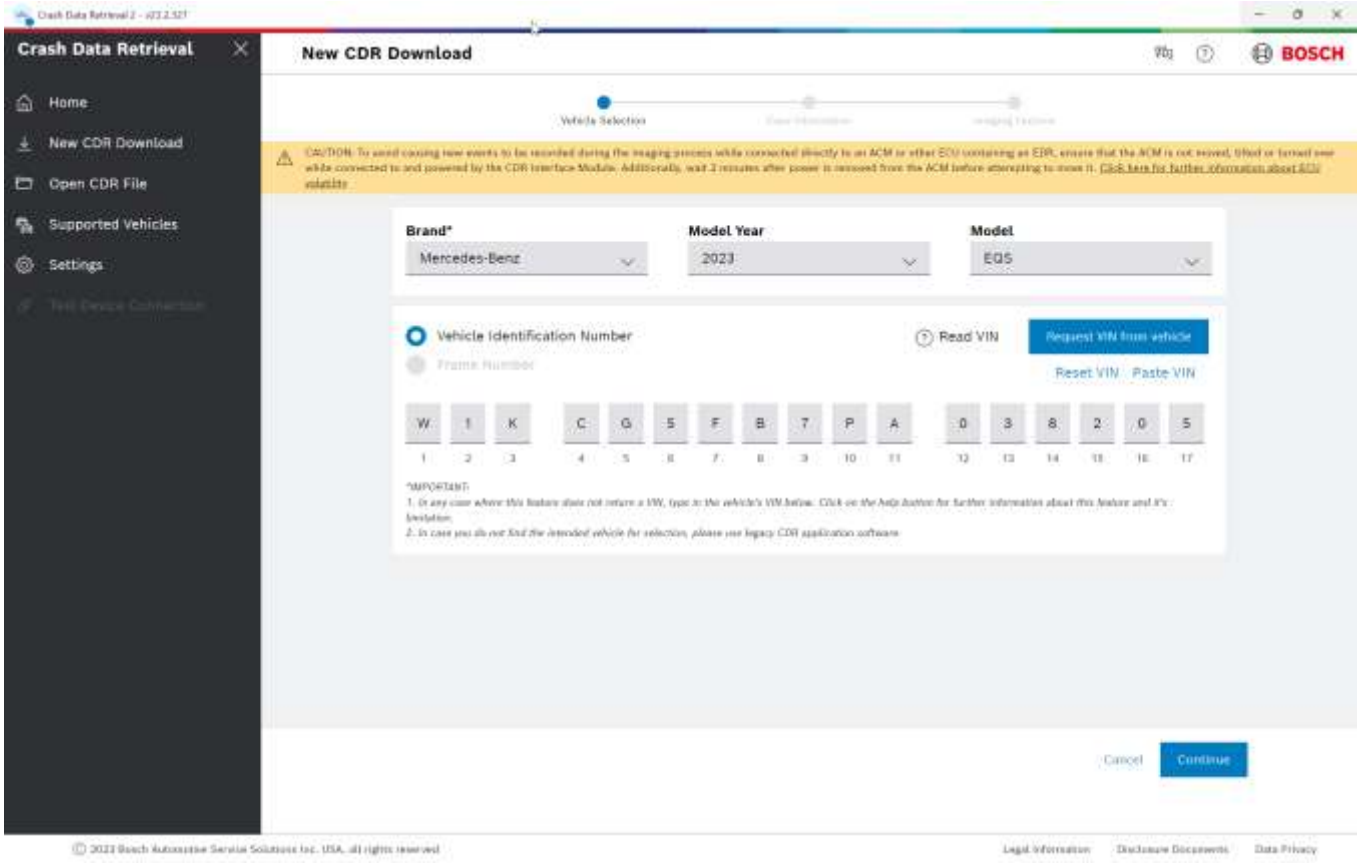
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CDR2



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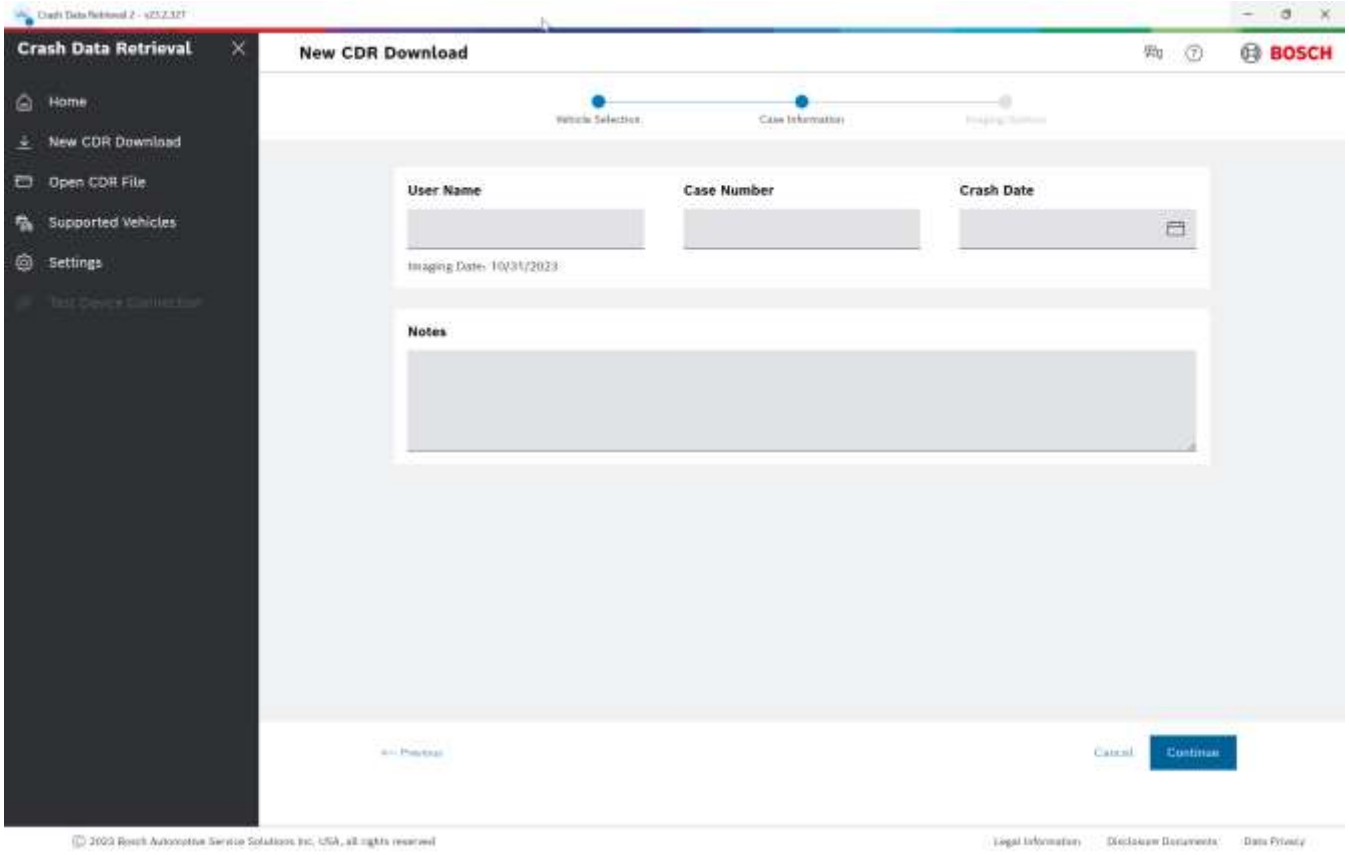
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CDR-2

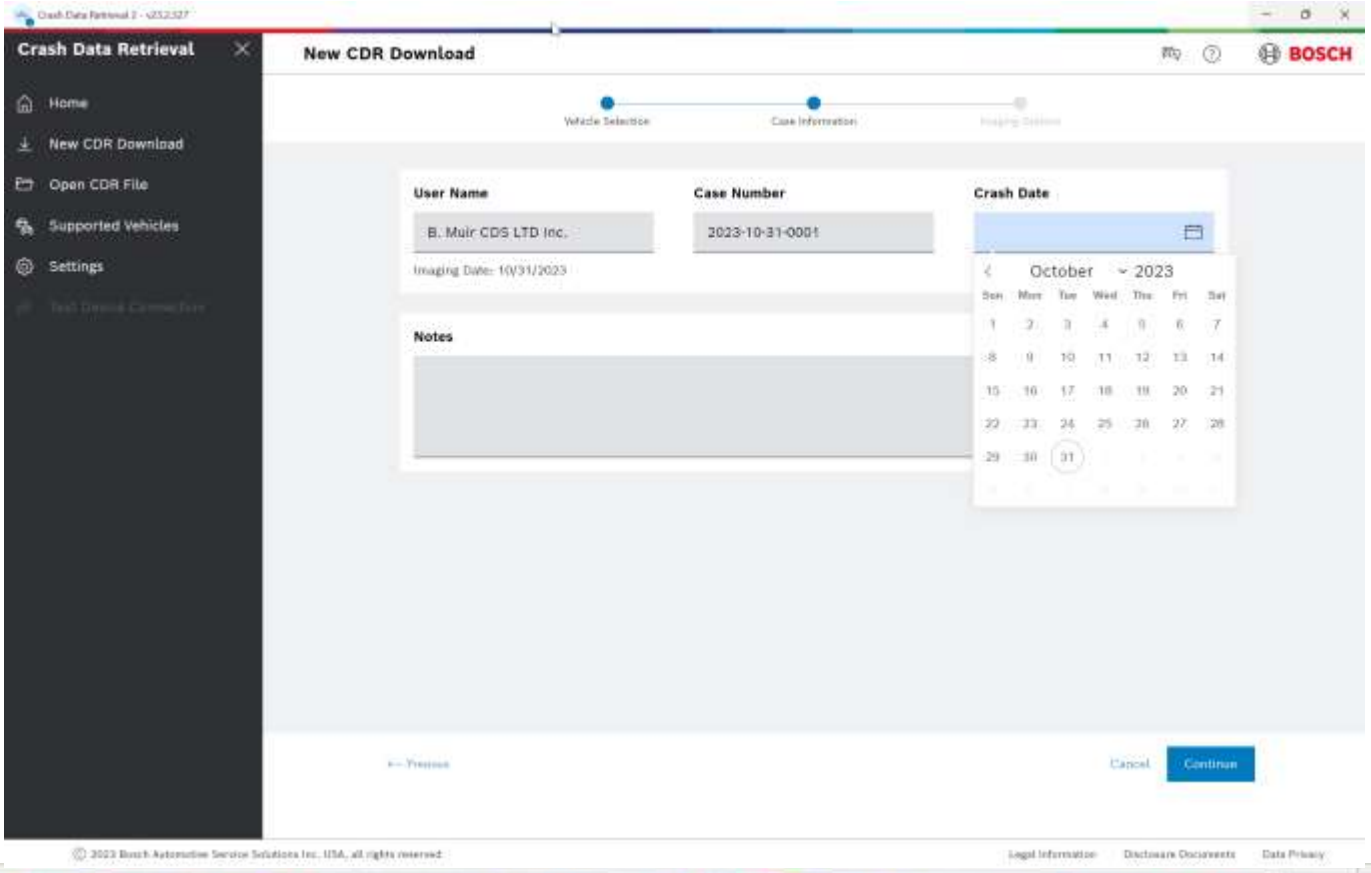
BOSCH



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CDR-2

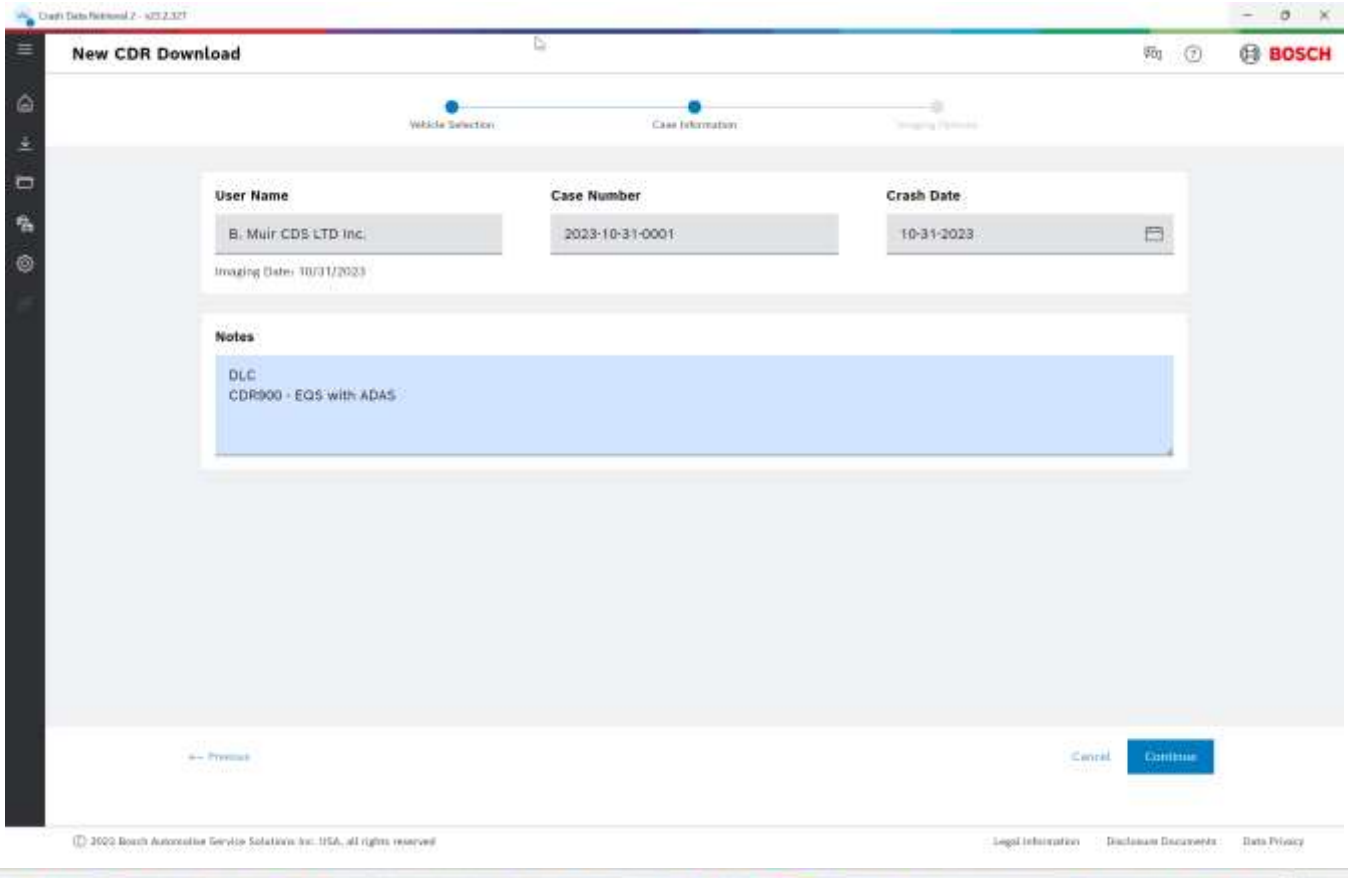


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CDR-2



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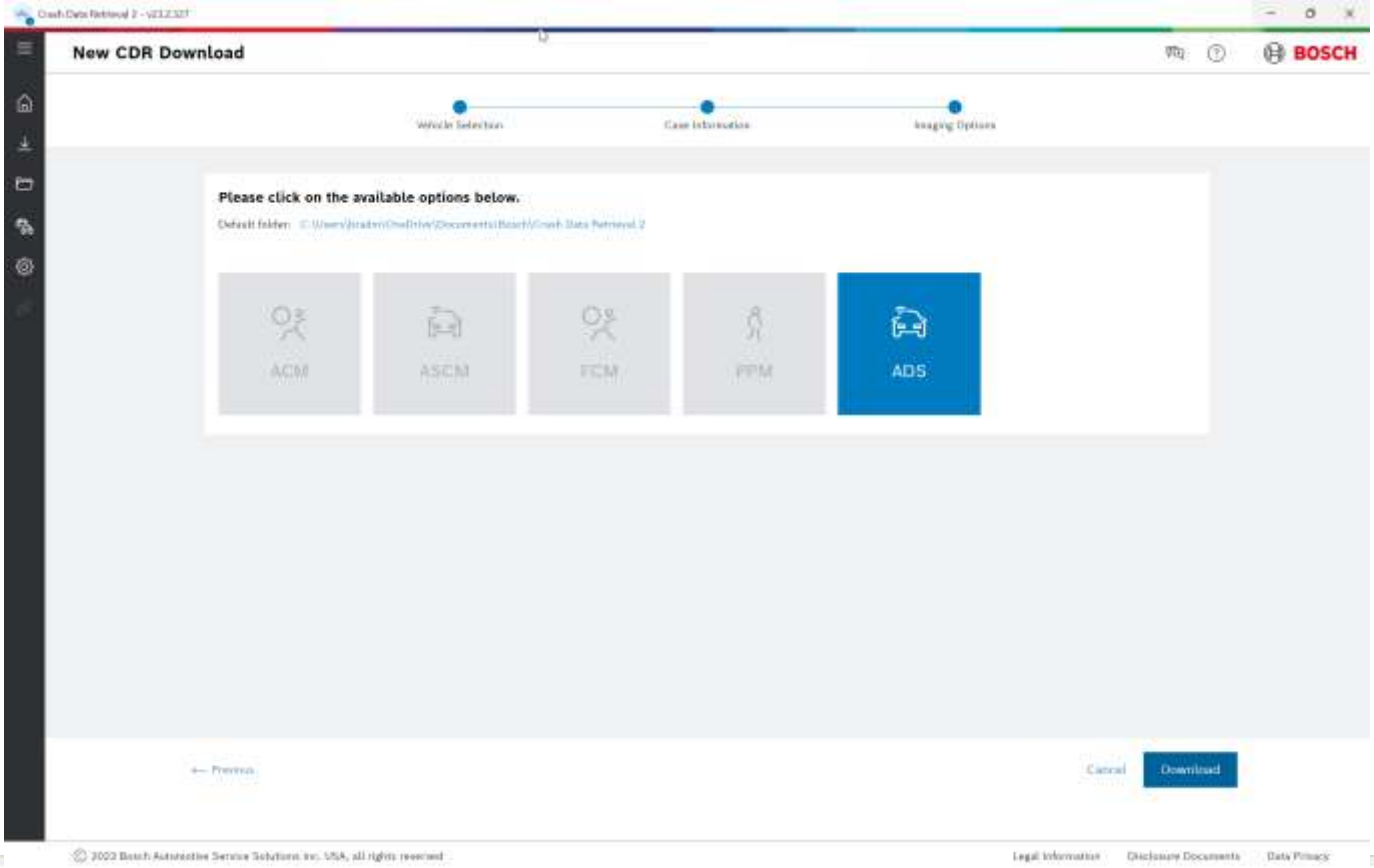
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BOSCH

CDR-2



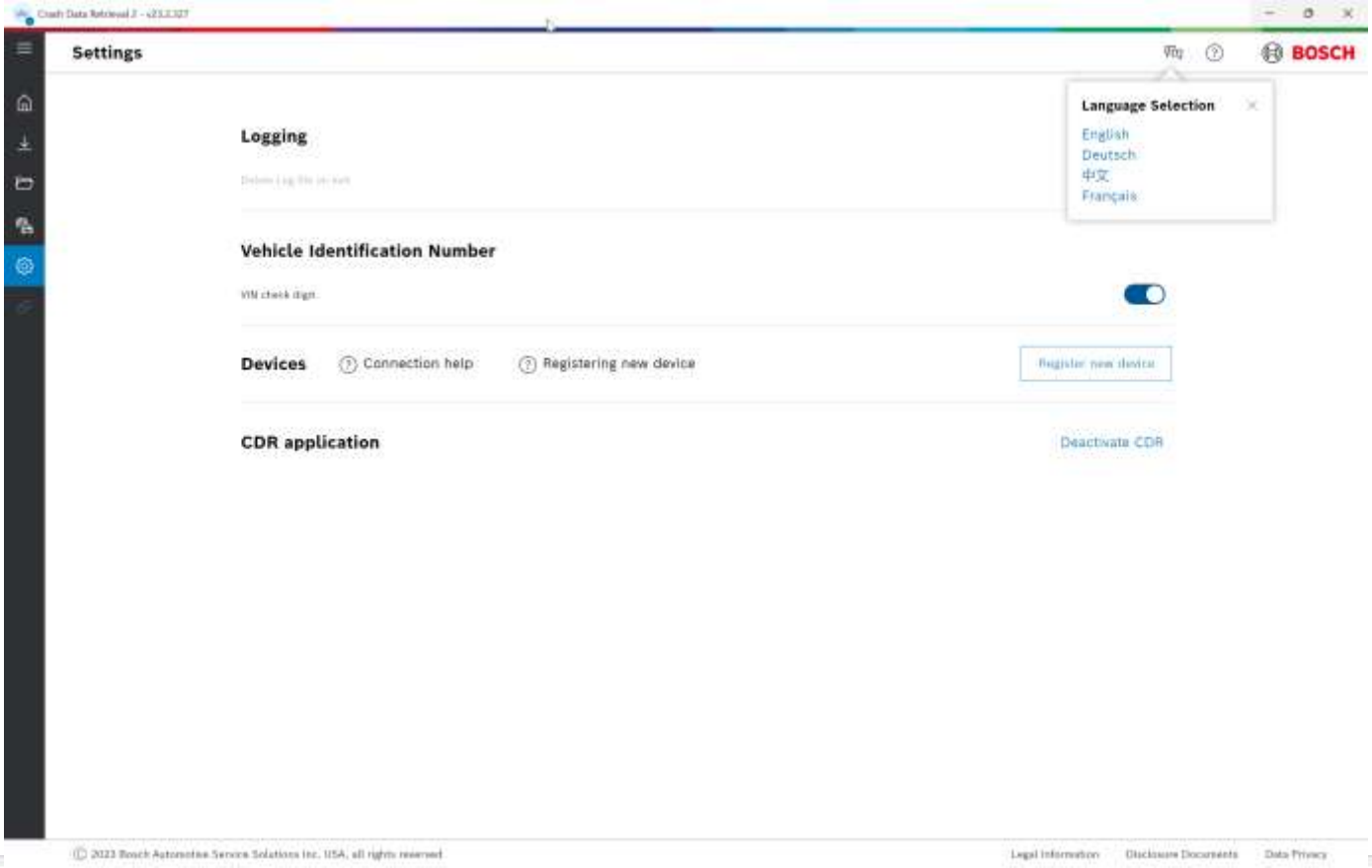
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CDR-2



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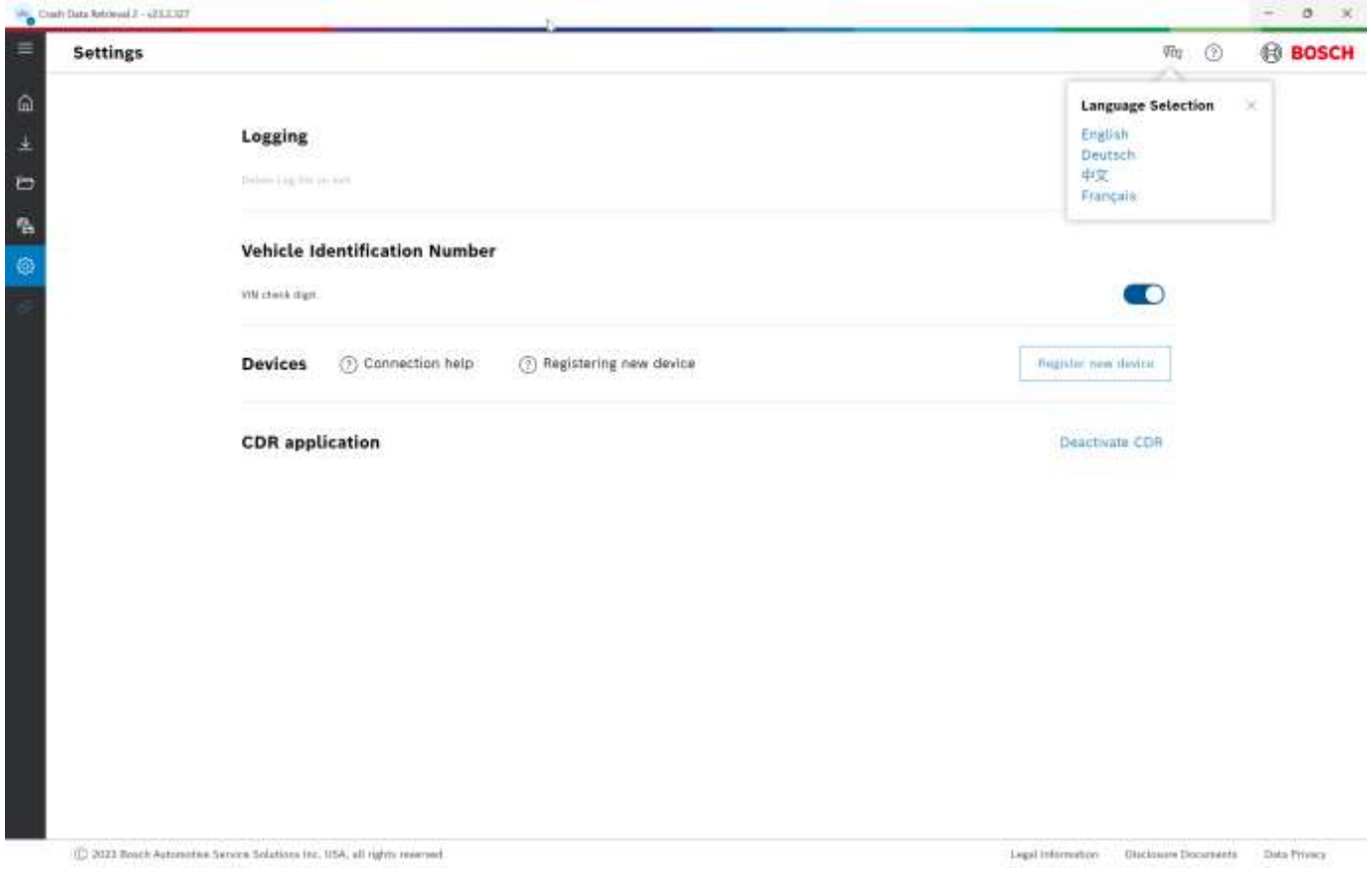
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Home

CDR-2

BOSCH



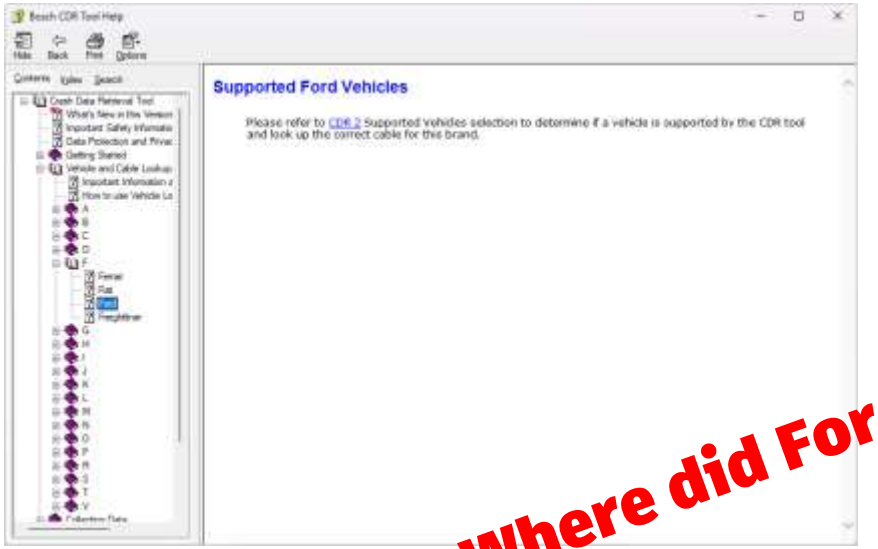
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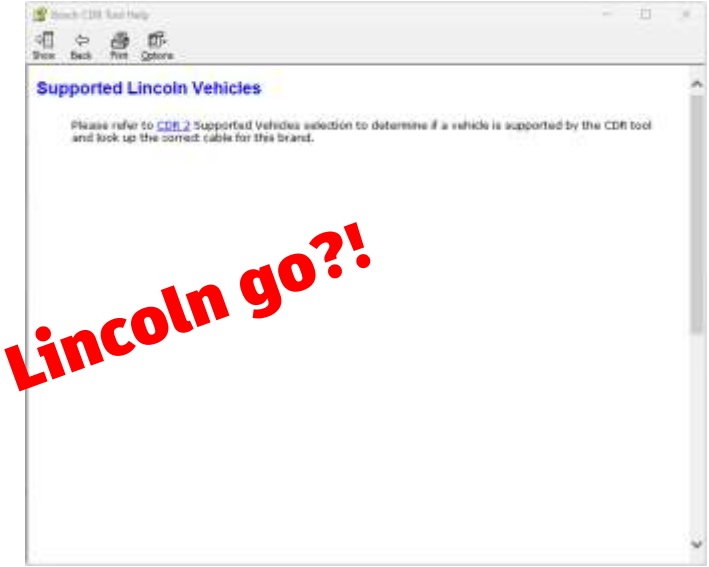
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File Edit Run Help Setup

But wait! There's more!

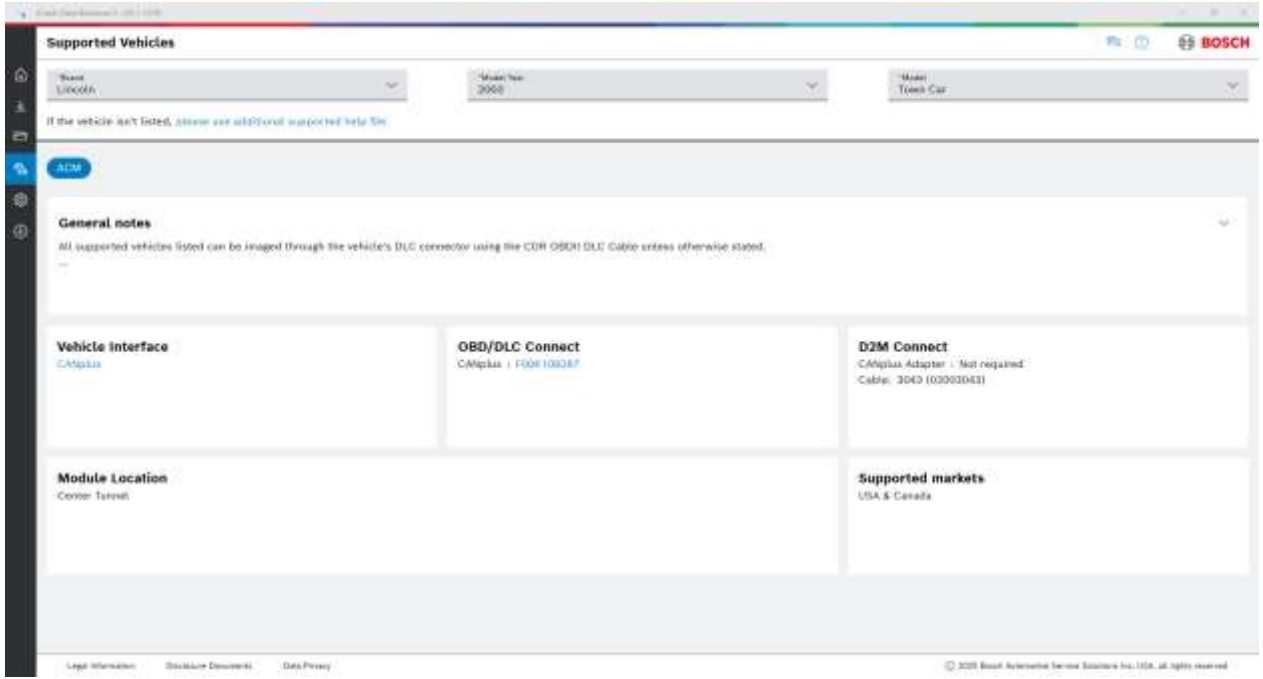




Where did Ford and Lincoln go?!

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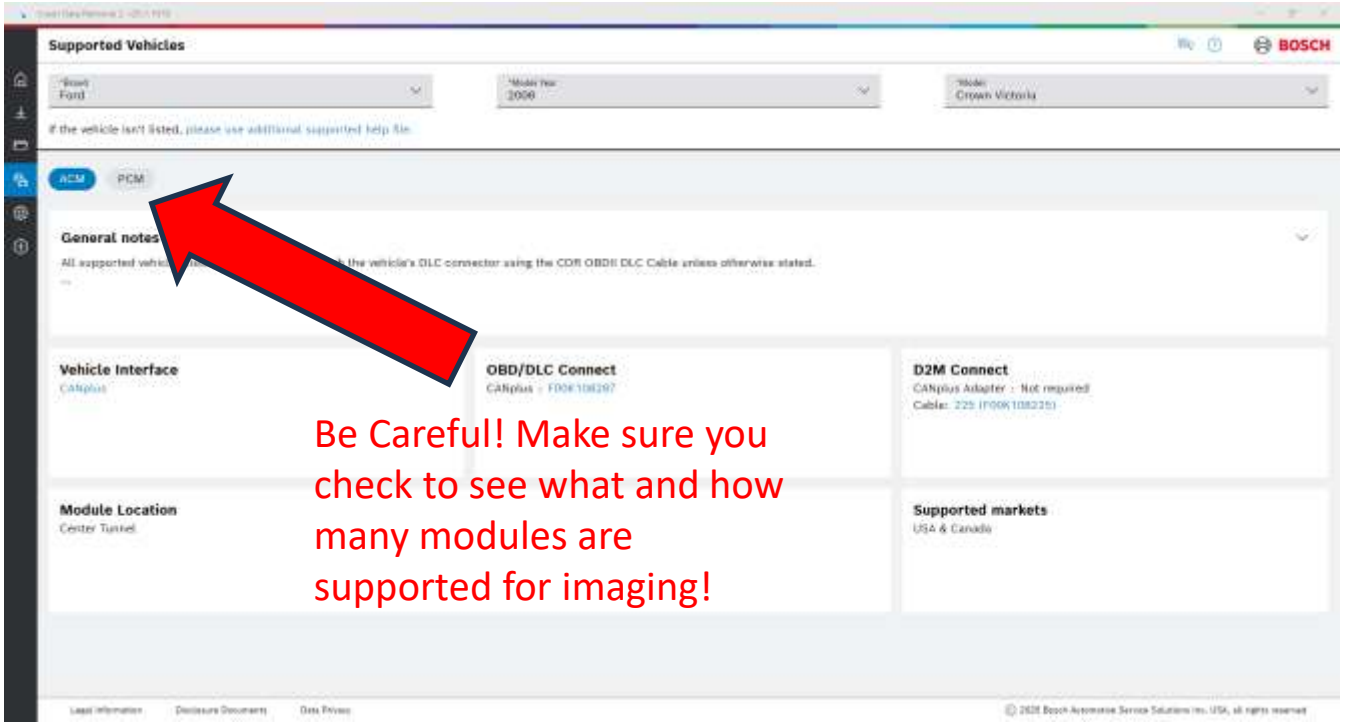
- They have been migrated over to CDR2
- We should expect to continue to see this with other manufacturers as new updates are released.



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File Edit Run Help Setup

CDR2 Supported Vehicle File



File Edit Run Help Setup

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CDR System Operators Course



Saving Files – After Data Collection

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Event Names

- Event names vary by OE and generation of module
- After successful collection of data and saving the .CDRX file, an indication of events downloaded can be found in the CDR Software Status bar at the lower right of the screen:



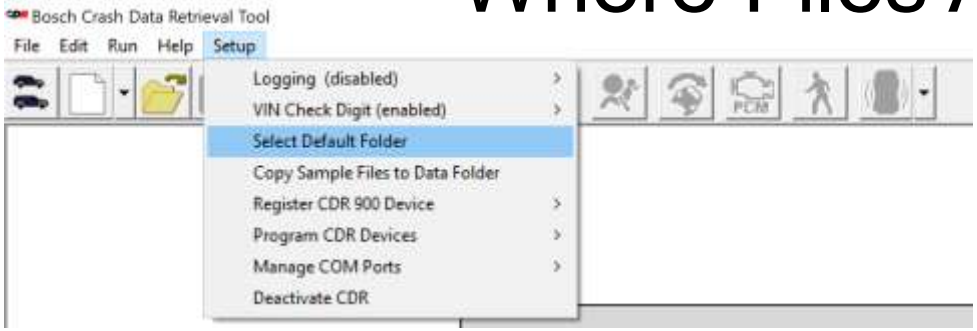
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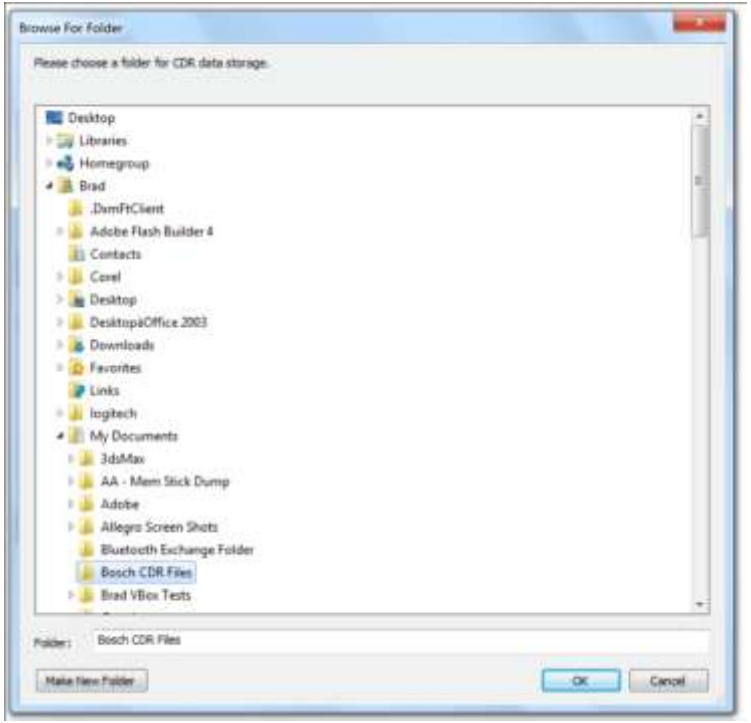
Save data

- Names MAY include other text characters in accordance with Windows file naming conventions but are easiest to search by Year_Make_Model_Case Info_VIN
- Default file name conventions are:
  - (VIN) _ACM.CDRX
  - (VIN)_PCM.CDRX
  - (VIN)_ROS.CDRX
  - (VIN)_ASCM.CDRX
  - (VIN)_FCM.CDRX
  - (VIN)_PPM.CDRX

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Where Files Are Saved

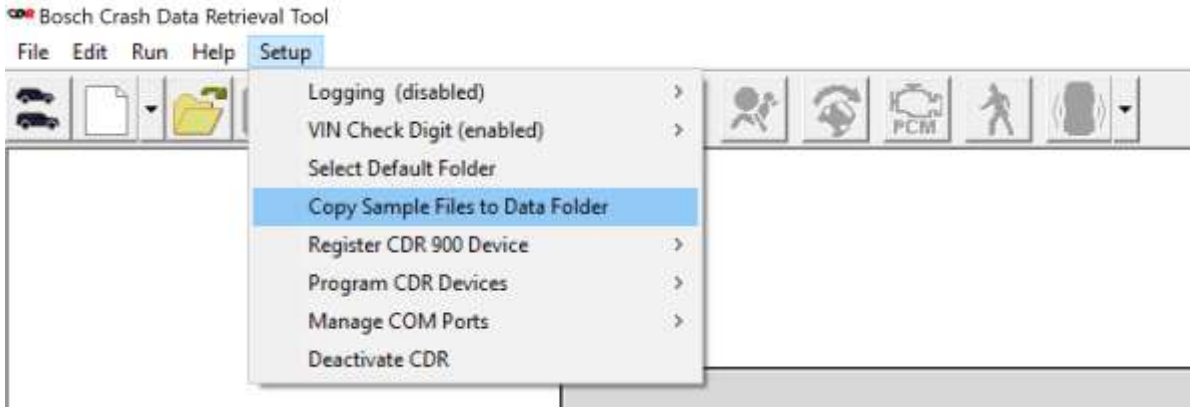




- The “Best Practice” is to save .CDRX files in “Documents\Crash Data Retrieval” by default
- For CDR2 CDRnX files are saved in Documents\Bosch/Crash Data Retrieval 2

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Copy .CDRX Sample Files



- Use this selection to Copy the .CDRX files that come with the CDR Application to your chosen storage directory

File Edit Run Help Setup
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Saving Files

- Good idea to make a backup ASAP
  - USB “thumb drive”
  - Email to desktop computer
- Saving only as PDF incorrect
  - Can not be opened in later version of the software
  - PDF only may result in data exclusion
- Saving only as CSV equally incorrect

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Can I accidentally alter the data?

- Properly downloading data does not alter crash data stored (if present) in a module
  - In the context of “downloading / imaging,” the Operator is not removing it from the module or moving from the module to your computer
  - Each download simply images what’s stored leaving the original data for the next Operator to gather the same crash information

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Can I accidentally alter the data?

- Once downloaded the file should only be opened in the CDR Software
- If opened in another program and accidently saved there the file will be unusable in the CDR Software.

File Edit Run Help Setup

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CDR System Operators Course

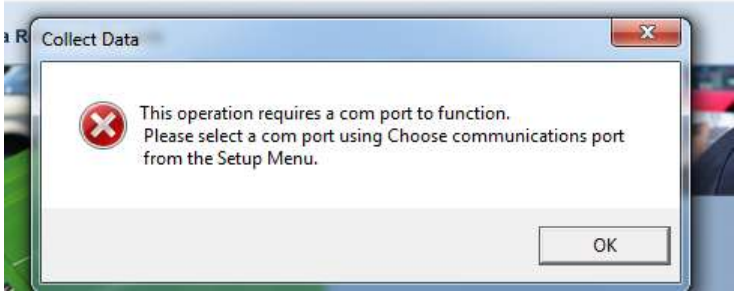


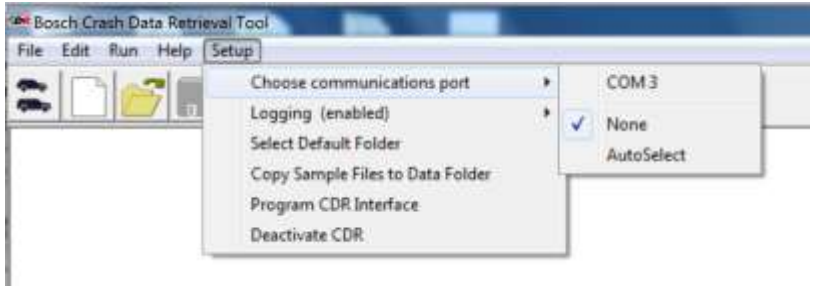
Troubleshooting

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No Communication Port Selected

- Upon trying to Collect Data when No COMM port has been selected
  - Normally occurs when USB to Serial Converter plugged in AFTER CDR program is started
  - Use /Setup/Choose Communications Port
- NEW Comm. Port Management function should eliminate these issues



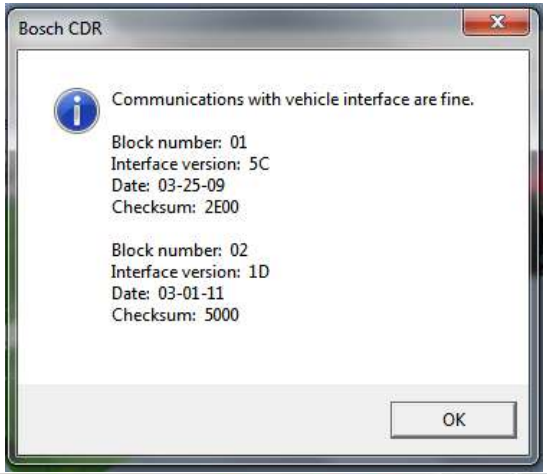


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No Communication with Interface

- Most common Causes:
  - No Power to the CDR interface
  - Serial cable not connected properly
  - Check connections and power level then re-test communications until communications are confirmed

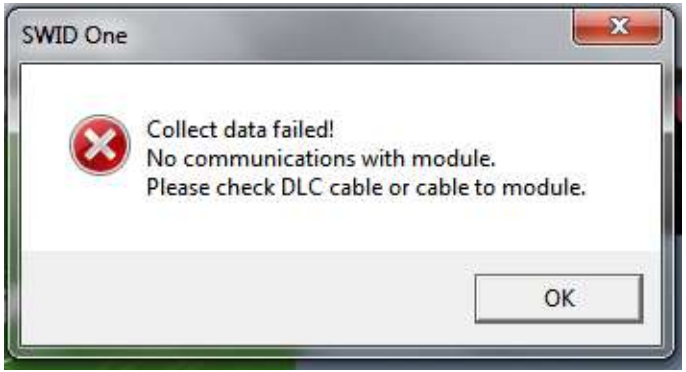


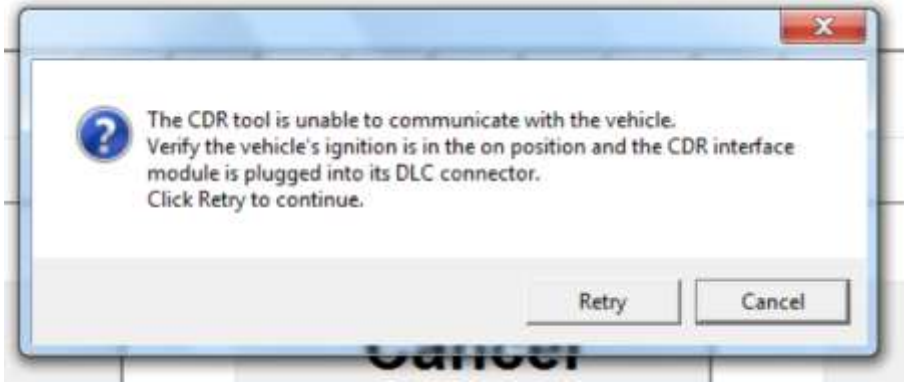


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No communication with the Module

- Collect Data Failed
- Common causes:
  - Key on when required?
  - umbilical or DLC cable not connected
- May indicate damage to the module or vehicle wiring
- Check cables and try again





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Errors Related to VIN input



- **VIN entered is invalid.**
  - The VIN did not pass the VIN check digit test.
  - Some unsupported vehicles may successfully pass this test.



- **This vehicle is not presently supported.**
  - means the VIN as entered belongs to a vehicle that is not currently supported for access in the current software version

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File Edit Run Help Setup

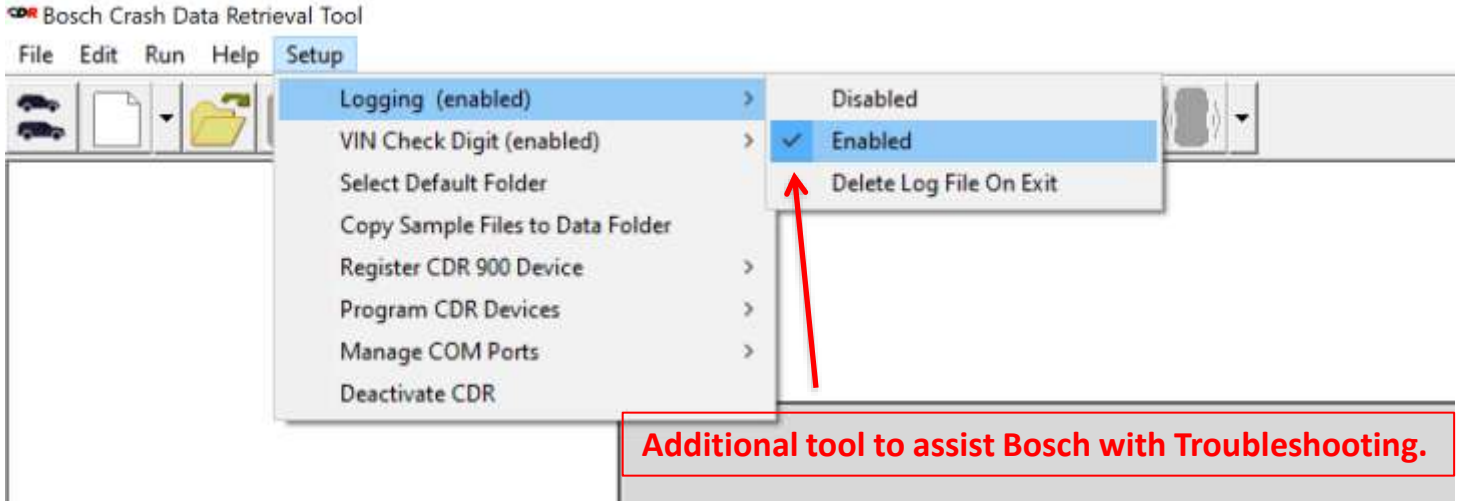
Module's Software ID not supported

- This error message may occur even when the vehicle is listed as a supported vehicle in the Help file
- Operator should record the software ID and report it to Bosch Technical Support
  - In this example the Software ID is "$A273"



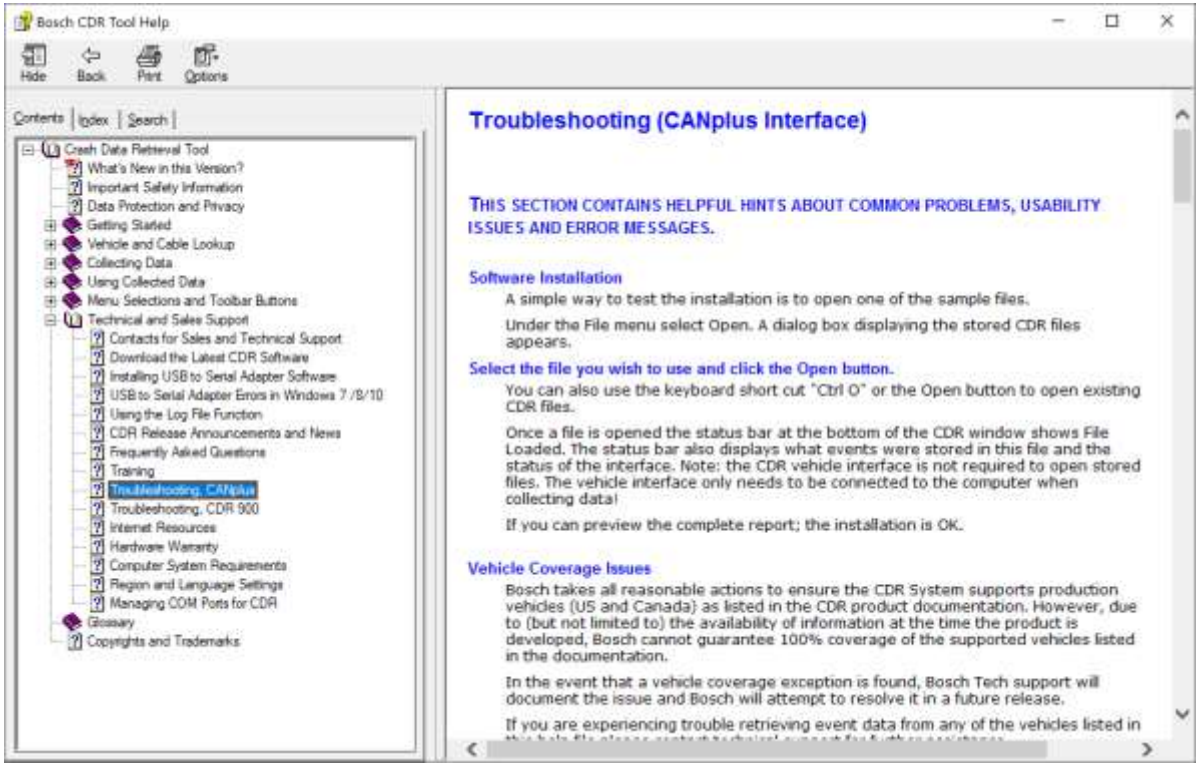
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Log File Enable/Disable



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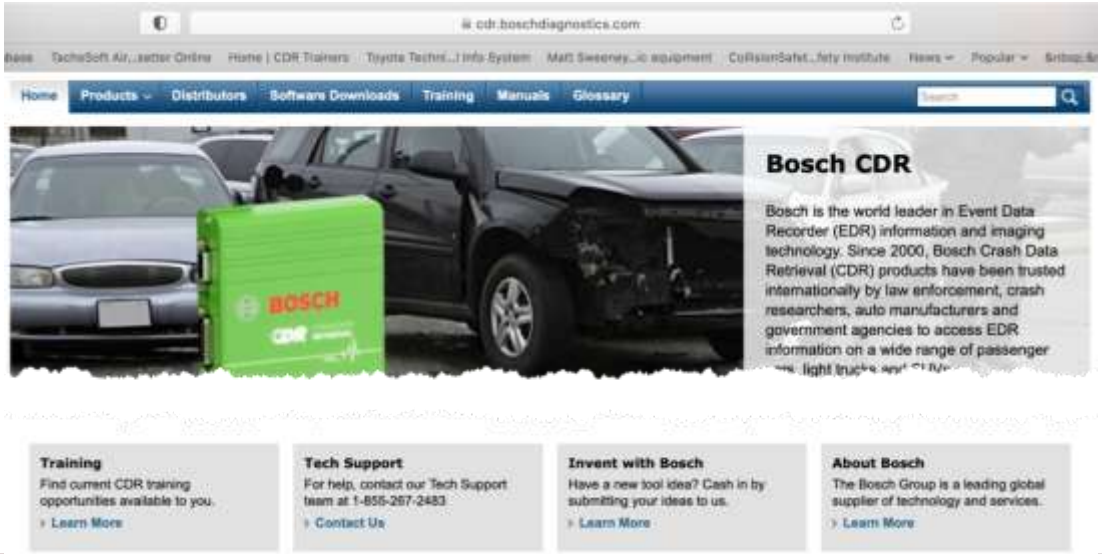
Bosch Help File Troubleshooting



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Contacting Bosch

- You can either call or email Bosch Tech Support using the information below



Contact Us

Submissions for this form are closed.

Customer Support (i.e. customer orders, invoicing, credits)
Tel: 1-800-321-4889, Ext 4 Option 2
Fax: 1-800-283-8665
dgwscs@us.bosch.com

Technical Support (i.e. product support, warranty)
Tel: 1-855-BoschTech (1-855-267-2483)
Fax: 1-800-955-8329
techsupport.diagnostics@us.bosch.com

Repair Information
Bosch Repair Center
Tel: 1-800-321-4889, Ext 7
Fax: 1-805-966-9540
servicecenter.vetronix@us.bosch.com

International Sales Inquiries
Tel: 1-724-489-5267
Fax: 1-507-455-7063
Intl.inquiries@service-solutions.com

– EDR (KIA) –
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Retrieve

Kia and Hyundai EDR Tool

EDR

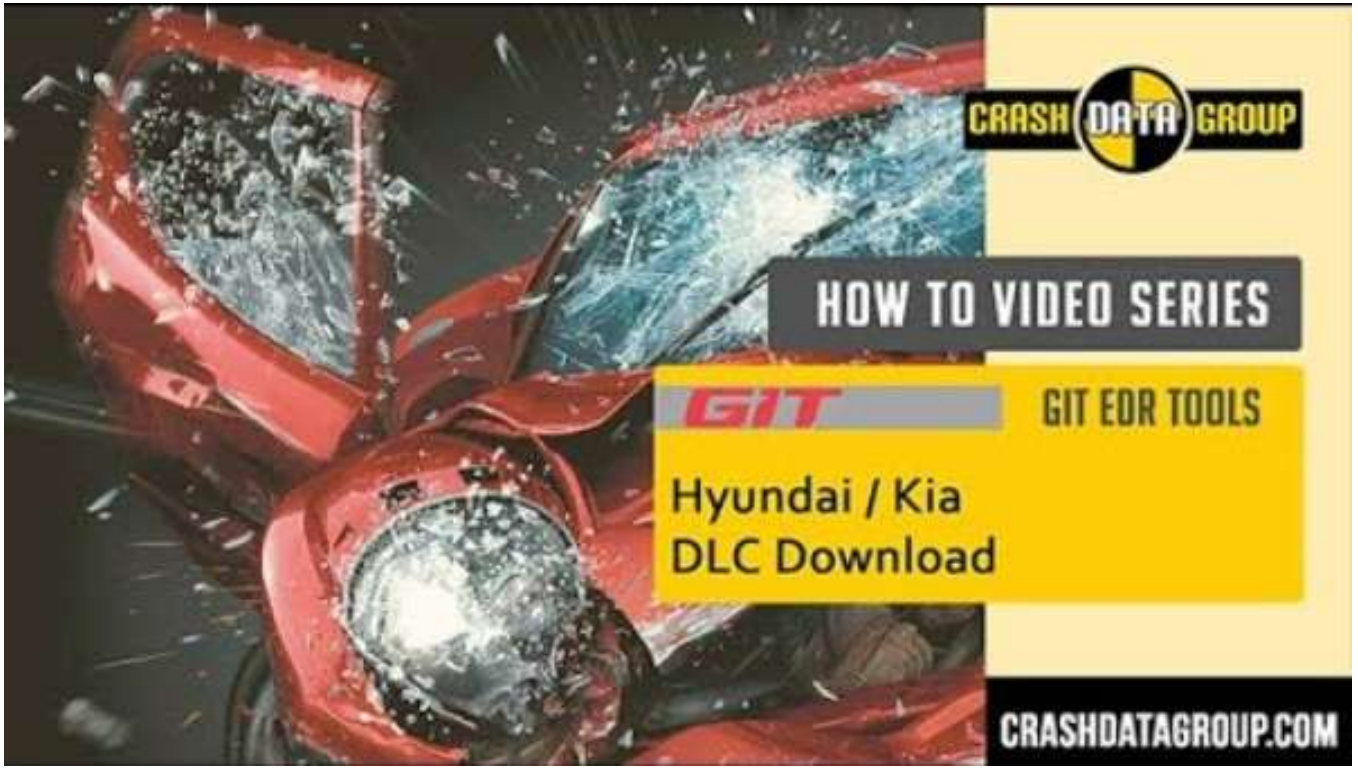


Kia and Hyundai Vehicles with accessible EDR Data

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Retrieve

Courtesy of Crash Data Group



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Hyundai/Kia Hardware

Old Style VCI Interfaces





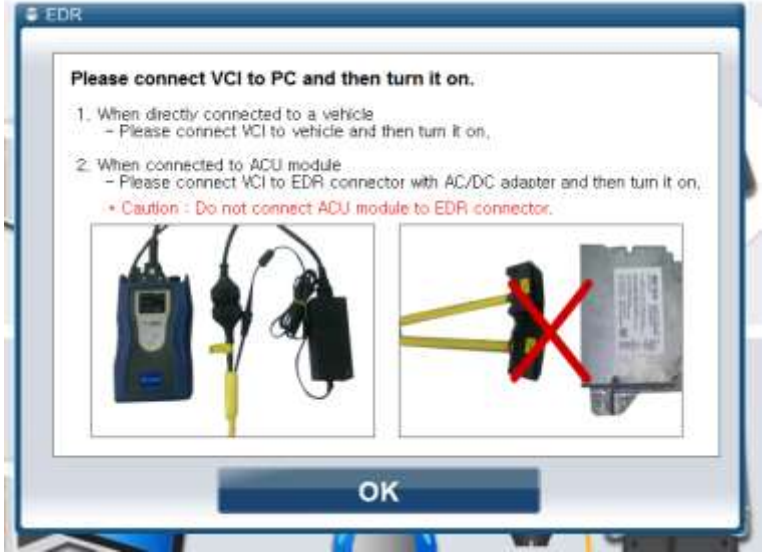
Current EDR 2.0



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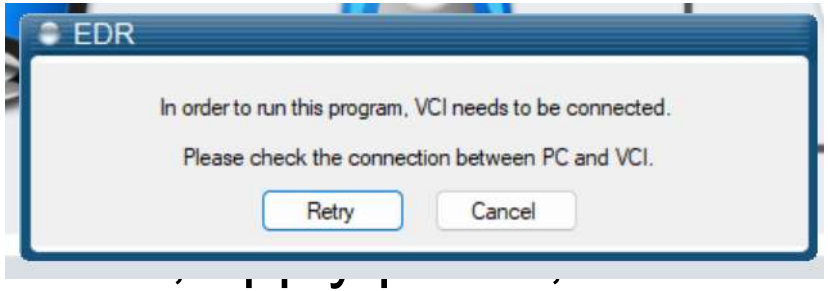
EDR Software

- Kia and Hyundai Software requires Interface to be detected before it loads



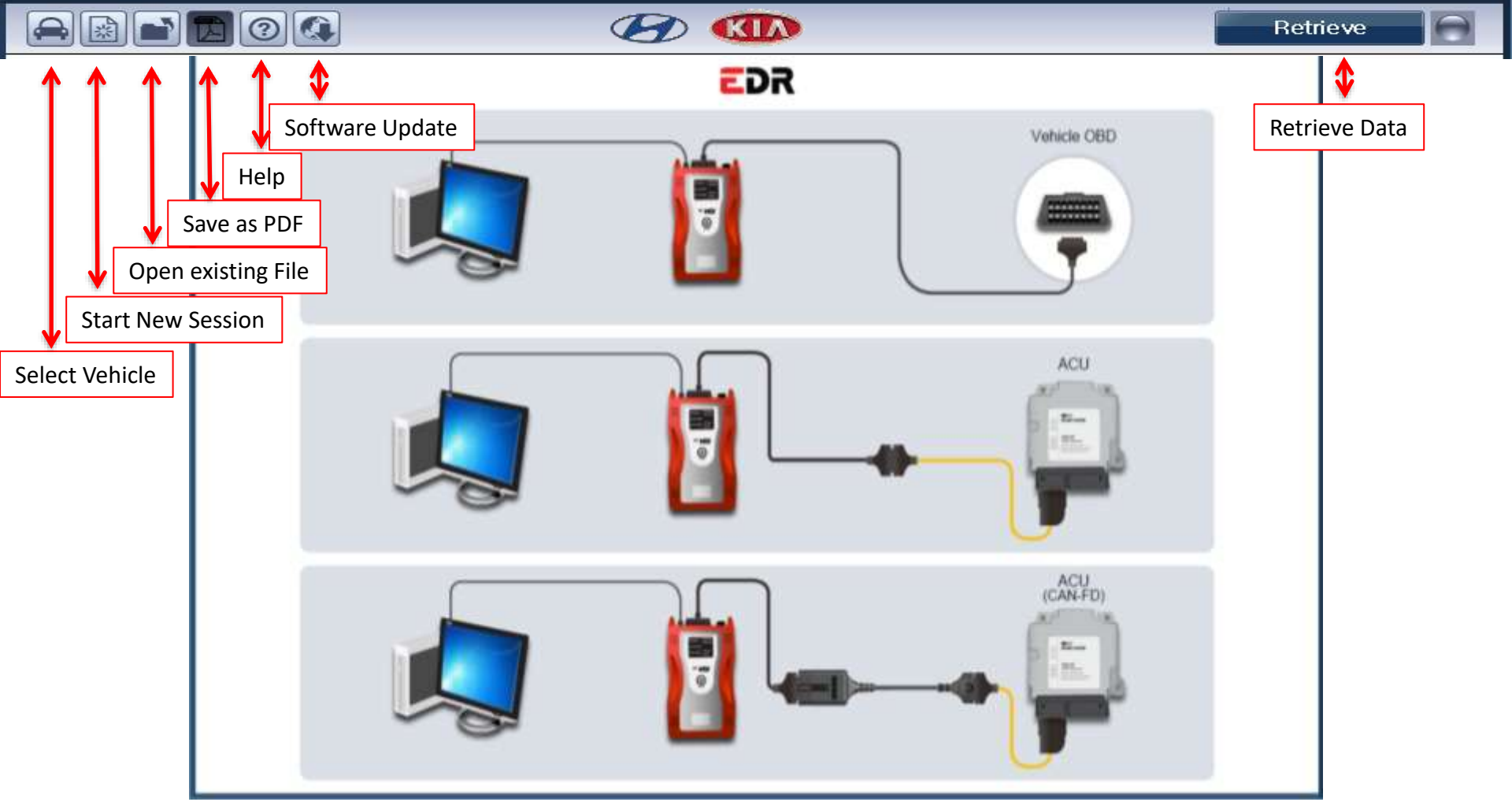
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- If the Hyundai or Kia VCI is not connected with power applied you will receive this error message:



- Connect any the VCI ON

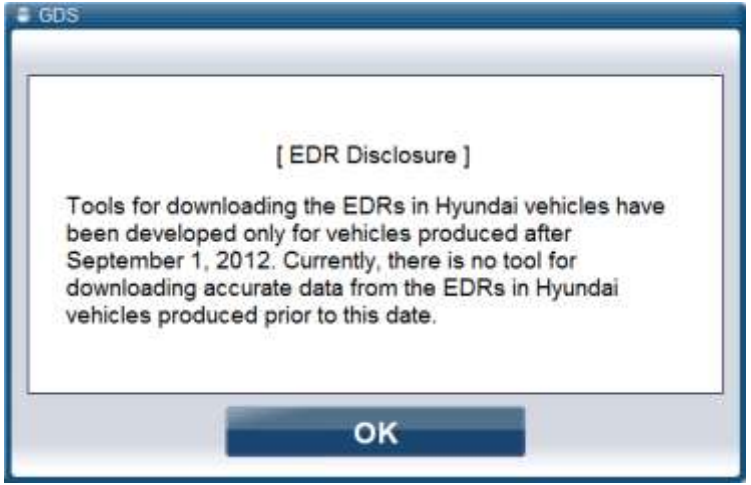
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EDR Software

- Warns that tool officially supports only vehicles made after Sept 1, 2012
- Other have data (back to 2010)
- More on that later!

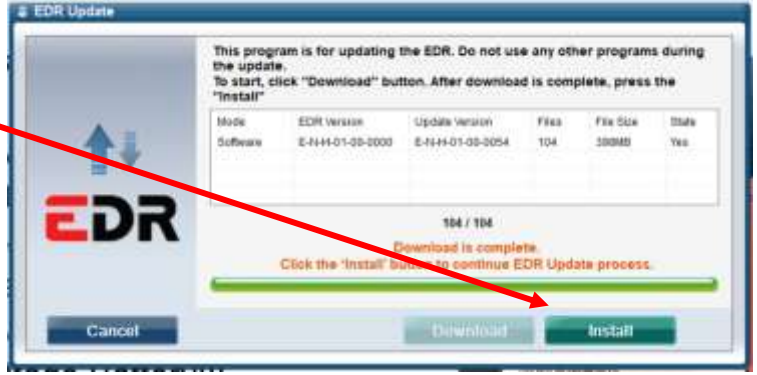


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Software Updates

- After Logging in the Updates will download automatically
- Click Install to apply the updates





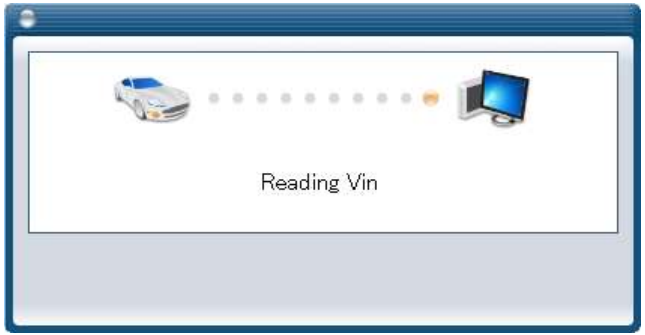
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Software Operation - DLC

- Click “New” and the software immediately tries to read a VIN from vehicle
- Ignition Key MUST be “ON”
- EDR Tool reads Vehicle VIN
- ** if downloading a 2010-2012 vehicle, do NOT turn key on. This forces VIN read to fail **



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Software Operation – DLC 2013+

- After successful VIN read from Vehicle, D2M Cable required is displayed
- Vehicle Model will be auto selected based on VIN read
- Click “OK” to Proceed



– EDR (KIA) –
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Software Operation – DLC 2010-2012

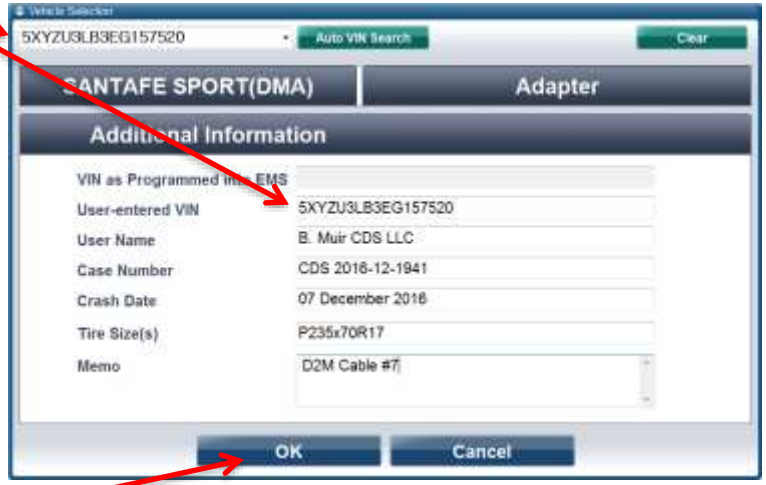
- After VIN read FAILS manually enter VIN
- Change 10th Digit to “D” for MY2013
- Click “Auto VIN Search”
- Select Vehicle Model from list (refer to next slide)
- Click “OK” to Proceed



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Software Operation – DLC 2010- 2012

- VIN as entered by user
- Enter User Info and comments
- Note prompt for tire size!
- In “Memo” input ACTUAL VIN and note that 10th digit changed to “D” to permit data retrieval
- Click OK



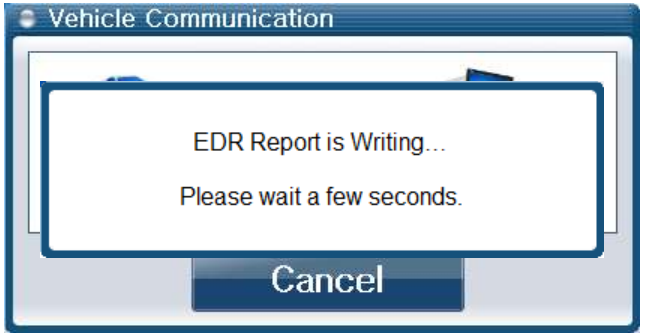
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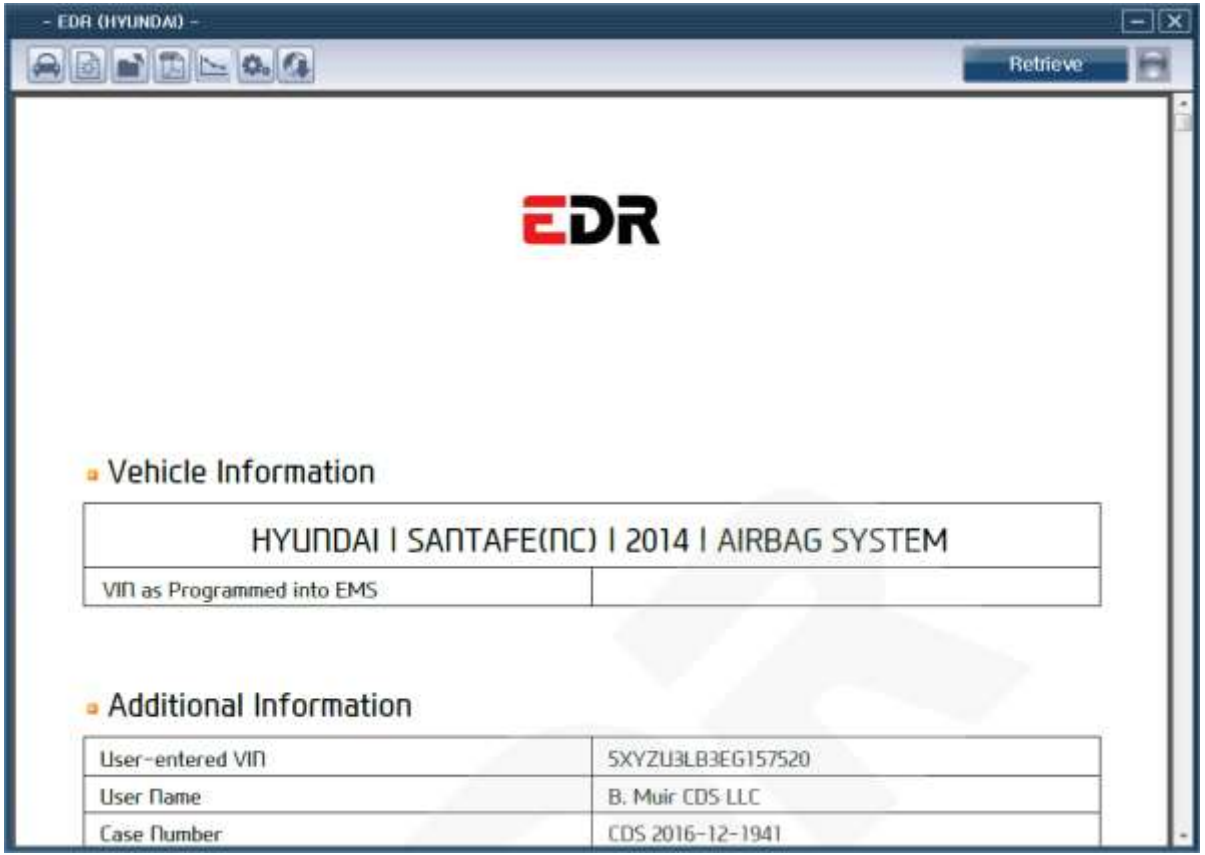




- Click Retrieve
- Lights on interface should blink
  - 1 rapidly for a few seconds (Green)



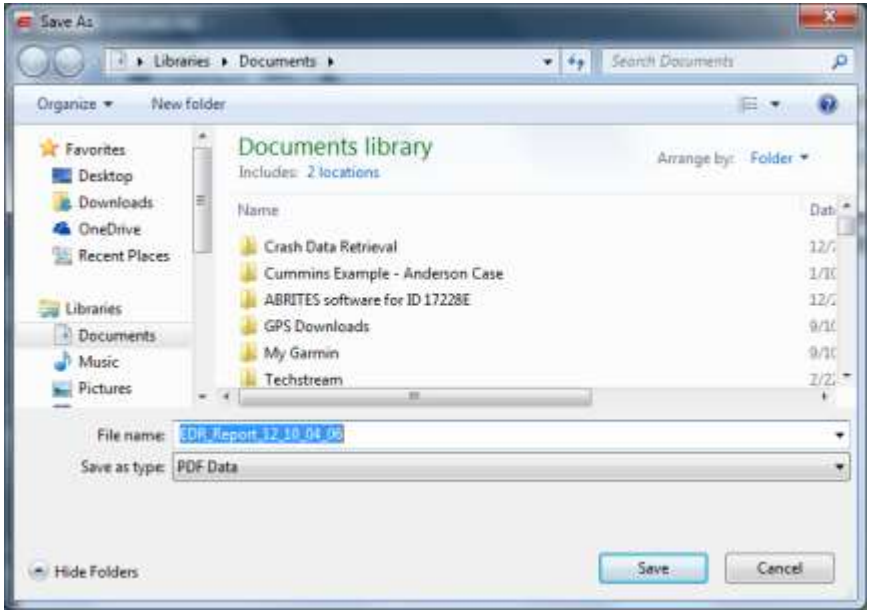
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Software Operation

- “Save as PDF”
- Use your naming convention
- Default is EDR_REPORT_MM_SS_HH_MM



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Software Operation – D2M

- Click “New” and the software immediately tries to read a VIN from vehicle
- on D2M download – wait for error



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Software Operation – D2M

- Must enter VIN
- Auto VIN Search" will populate Model (MY2013+)
- Required D2M Cable is indicated
- Click on Highlighted Model
- Click OK to Proceed



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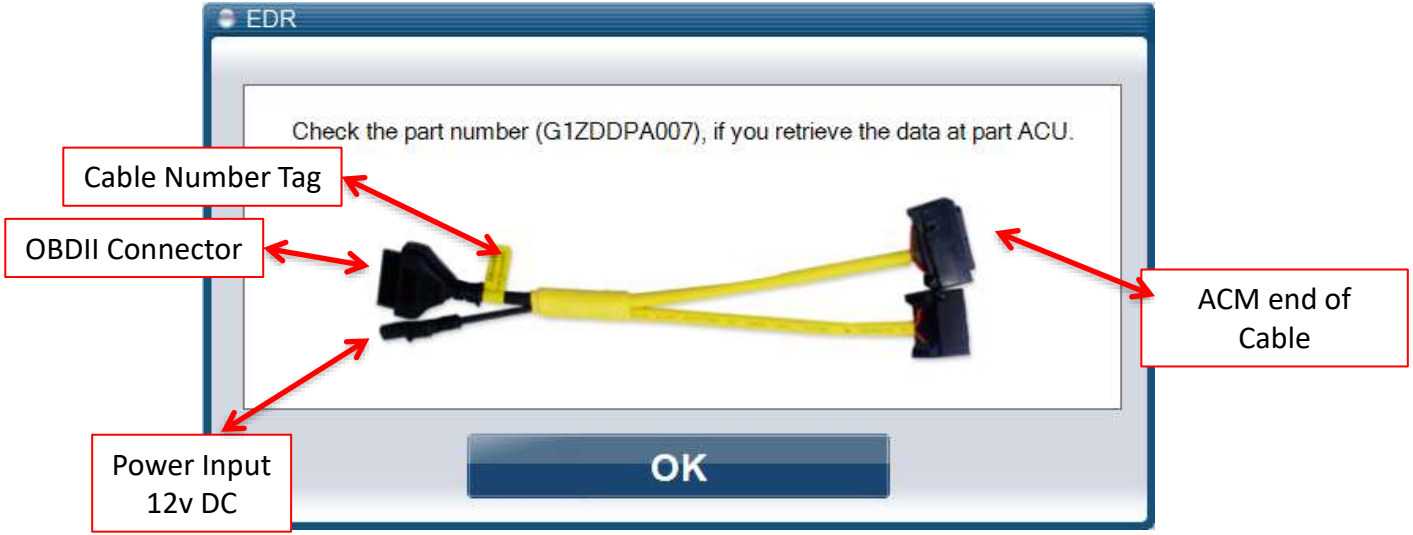
Software Operation – D2M 2010-2012

- After VIN read FAILS manually enter VIN
- Change 10th Digit to “D” for MY2013
- Click “Auto VIN Search”
- Select Vehicle Model from list (refer to next slide)
- Required D2M Cable is indicated
- Click “OK” to Proceed



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Software Operation – D2M



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CFCI Adapter



- Some newer Hyundai and Kia vehicles require the CFCI Adapter for D2M downloads
- CAN FD Support
- Installs in-line between interface and D2M Cable

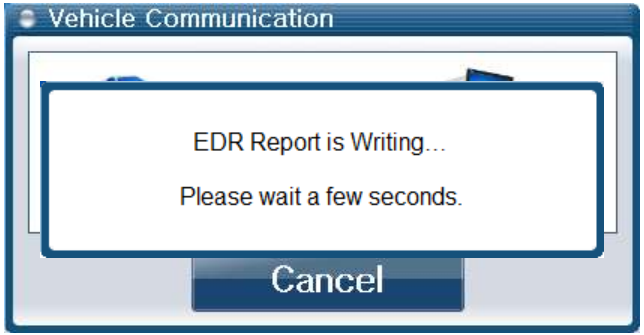
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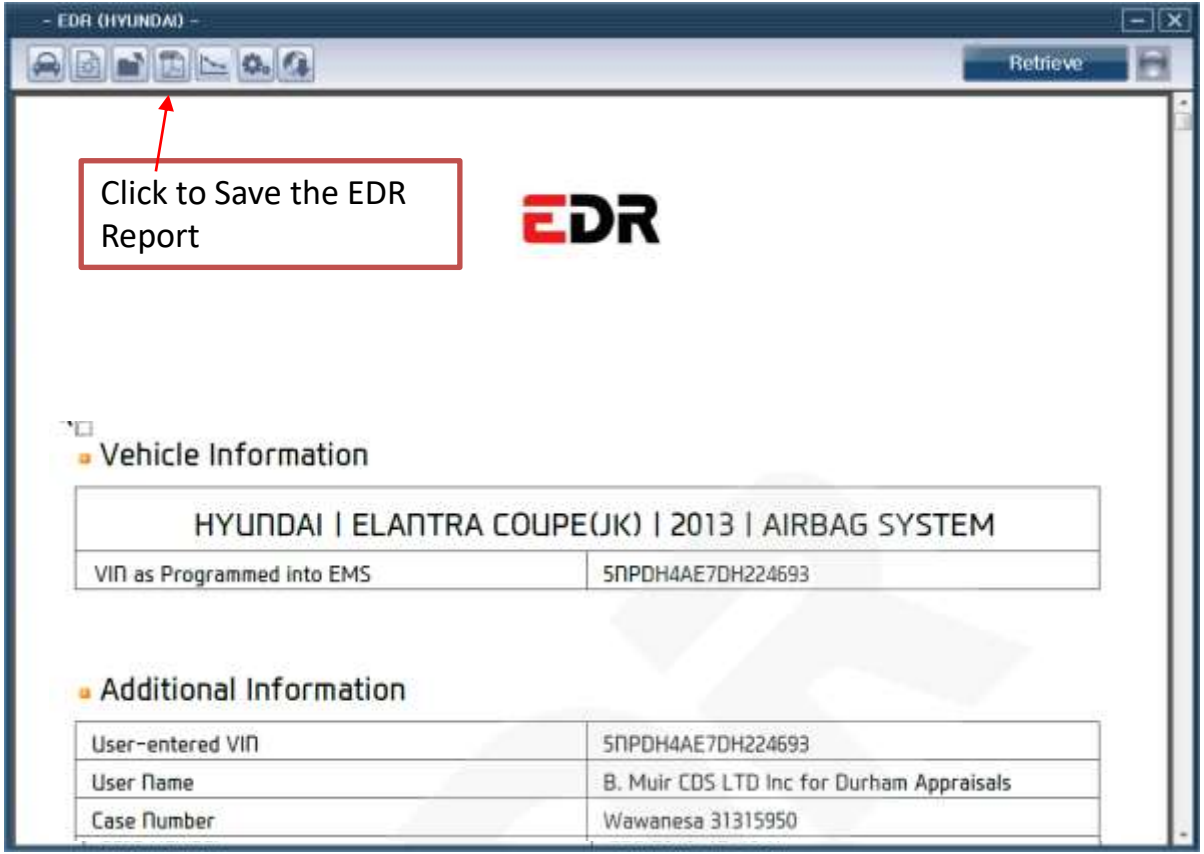




- Click Retrieve
- Lights on interface should blink
  - 1 rapidly for a few seconds (Green)



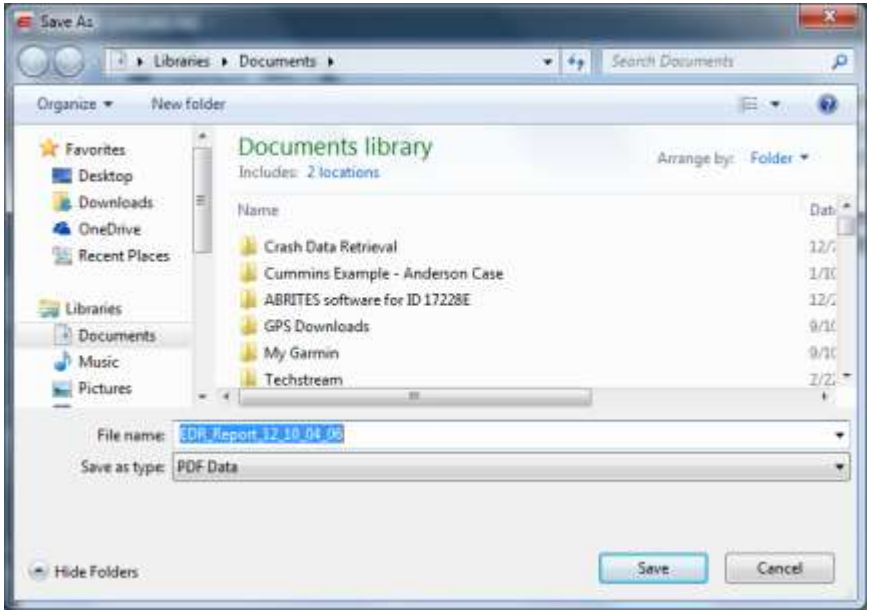
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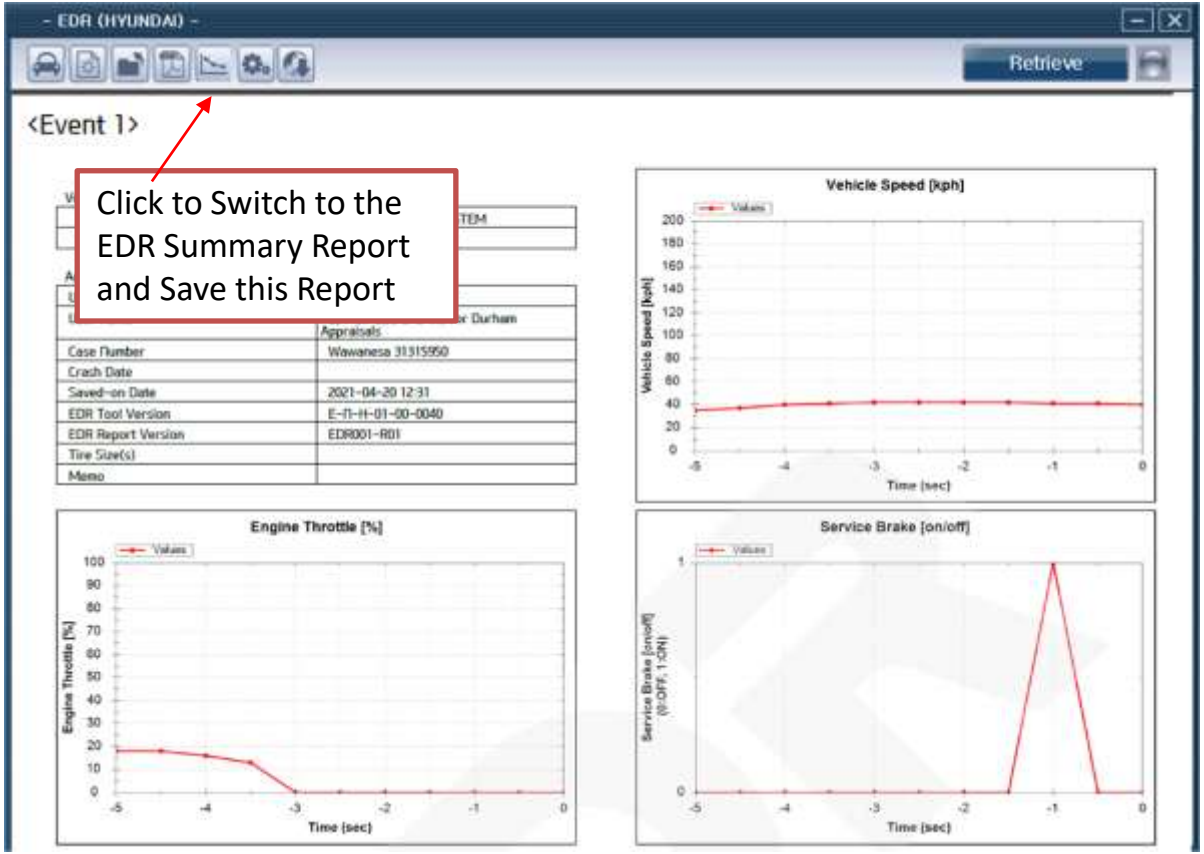
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Software Operation

- “Save as PDF”
- Use your naming convention
- Default is EDR_REPORT_MM_SS_HH_MM



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EDR Report

Vehicle Information

HYUNDAI | SANTAFE SPORT(DMA) | 2014 | AIRBAG SYSTEM	
VIN as Programmed into EMS	

General Information:

Tools for downloading and interpreting the EDRs in Hyundai vehicles have been developed for vehicles produced after September 1, 2012. Currently, there is no tool for downloading and accurate interpreting data from the EDRs in Hyundai vehicles produced prior to this date.

The EDR Report requires Adobe Reader Version 9.00 or higher to open.



– EDR (KIA) –
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Retrieve

Validating Hyundai / Kia Data

EDR



ACM Data from Models Not “Officially” Supported

– EDR (KIA) –
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Retrieve

Hyundai Sonata – Hyundai EDR Tool Data

- First documented Crash Test with Hyundai vehicle with EDR outside of NHTSA / IIHS
- MY2011 Hyundai not officially supported by Hyundai EDR Tool

Vehicle Information

HY | SONATA(YF) | 2013 | ALL | AIRBAG SYSTEM

Additional Information

User : B. Muir CDS LLC
CaseNumber : IATAI 2015 Test #3
CrashDate : 09/16/2015
Part No. : 95910 - 3Q010
Save on : Thursday, September 17, 2015, at 10:26:11
G-EDR Software Version : 1.00

– EDR (KIA) –
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Retrieve

Hyundai Sonata – Hyundai EDR Tool Data

EDR Data Limitations

Tools for downloading the EDRs in Hyundai vehicles have been developed only for vehicles produced after September 1, 2012. Currently, there is no tool for downloading accurate data from the EDRs in Hyundai vehicles produced prior to this.

The retrieval of his EDR data has authorized by the vehicle's owner, or other legal authority.

The airbag ECU can store up to two events. Deployment events cannot be overwritten or cleared from airbag ECU. Non-deployment events(which did not qualify as deployable events) can be overwritten by subsequent events.

The specifications for EDR are designed to be compatible with NHTSA 49 CFR Part 563 rule. The EDR data recording specifications of airbag ECU are divided into the following four categories.
- For the Event#1: Event#1-1 Event#1-2 Event#1-3 Event#1-4 Real-Time Data.
- For the Event#2: Event#2-1 Event#2-2 Event#2-3 Event#2-4 Real-Time Data.
The airbag ECU records data for all or some of the following crash(event)
But, depending on the installed airbag ECU, data for side crash and/or rollover crash(event) may not be recored.

Ignition cycle counter(download) will increment by 1 every time when the power mode cycles is changed from OFF/Accessory to IGN/RUN or EDR data is downloaded by using the retrieval tool.

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Retrieve

Standards Publications News Attend Learn Parti

Browse » Publications » Technical Papers » 2014-01-0502

2014-04-01

Accuracy of Translations Obtained by 2013 GIT Tool on 2010-2012 Kia and Hyundai EDR Speed and Delta V Data in NCAP Tests

2014-01-0502

Kia and Hyundai released publicly available tools in the spring of 2013 to read model year (MY) 2013 vehicle event data recorders (EDRs). By empirical testing, this study determined the tools also read data from some 2010-2012 models as EDRs were phased in by the manufacturer. Fifty-four (54) MY 2010-2012 airbag control module EDRs from the National Highway Traffic Safety Administration's (NHTSA) New Car Assessment Program (NCAP) crash tests were downloaded direct-to-module. The vehicles analyzed were exposed to frontal, side moving deformable barrier (MDB), and side pole tests. The EDR data was compared to the reference instrumentation for speed and Delta V data. Other data elements were also tabulated but are not evaluated for accuracy because they were not fully exercised during the crash tests, the reference instrumentation was not available, or they were outside the scope of this paper.

Tesla EDR



Tesla Harware Kit



Interface (PCAN)

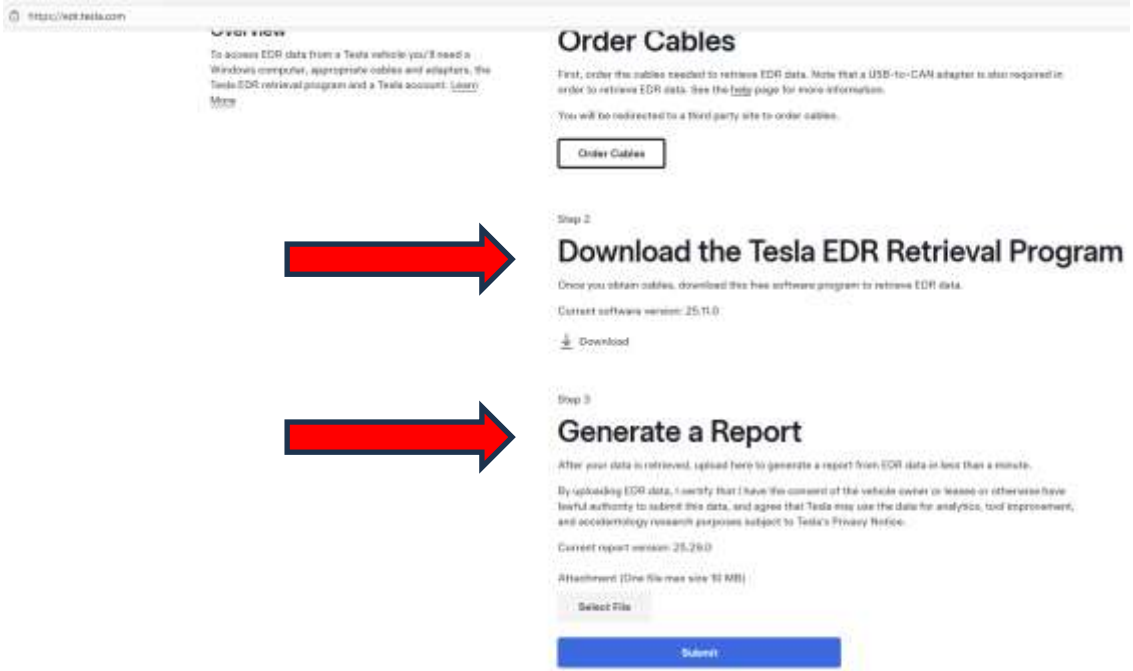
- Required for all Tesla Downloads.
- Similar to Bosch CanPlus and CDR900



Courtesy of Crash Data Group



edr.tesla.com



edr.tesla.com/help

- Model Specific Download Guides

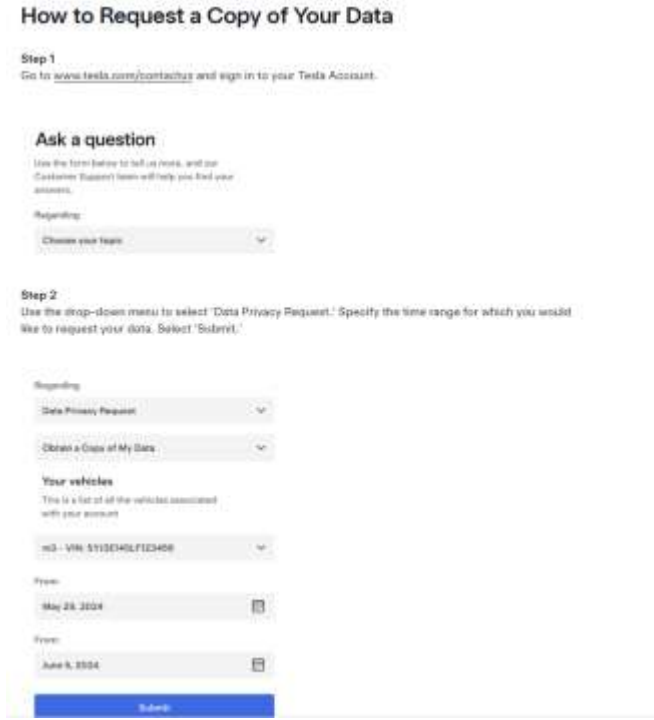
Data Retrieval

EDR Data can be retrieved from all Tesla Model S, Model X, Model 3 and Model Y vehicles. Tesla Roadster, which was produced from 2008 to 2012, is equipped with an RCM but does not have an EDR.

Model	Manufacture Date	Retrieval Guide
Model S	Before 2/26/19	Download
Model S	2/26/19 - 6/9/21	Download
Model S	On or after 6/10/21	Download
Model X	Before 2021	Download
Model X	On or after 2021	Download
Model 3	Before 10/5/20	Download
Model 3	On or after 10/5/20	Download
Model Y	All MFG dates	Download
Cybertruck	All MFG dates	Download

tesla.com/support/privacy

- Owners can request a copy of their vehicle data.



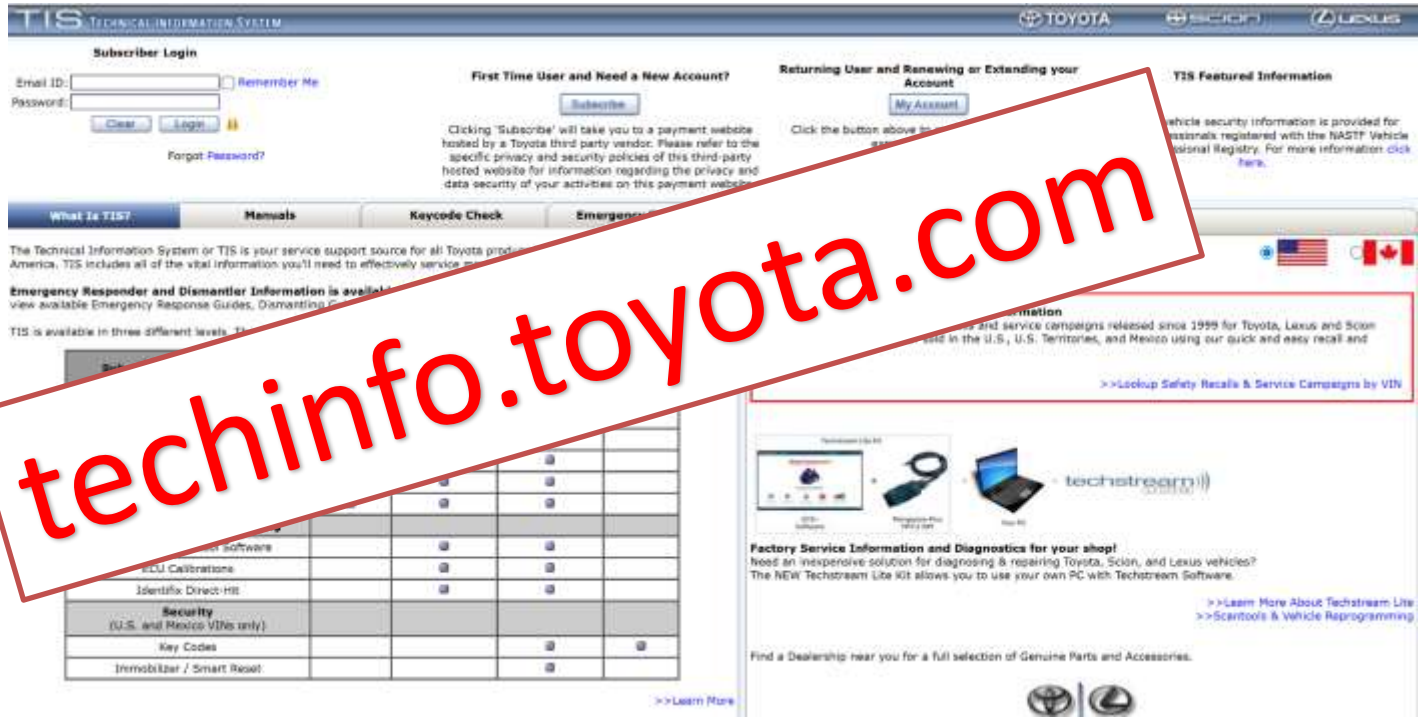
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Toyota Global Techstream Plus



File Edit Run Help Setup
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Toyota Global Techstream Plus (GTS+)



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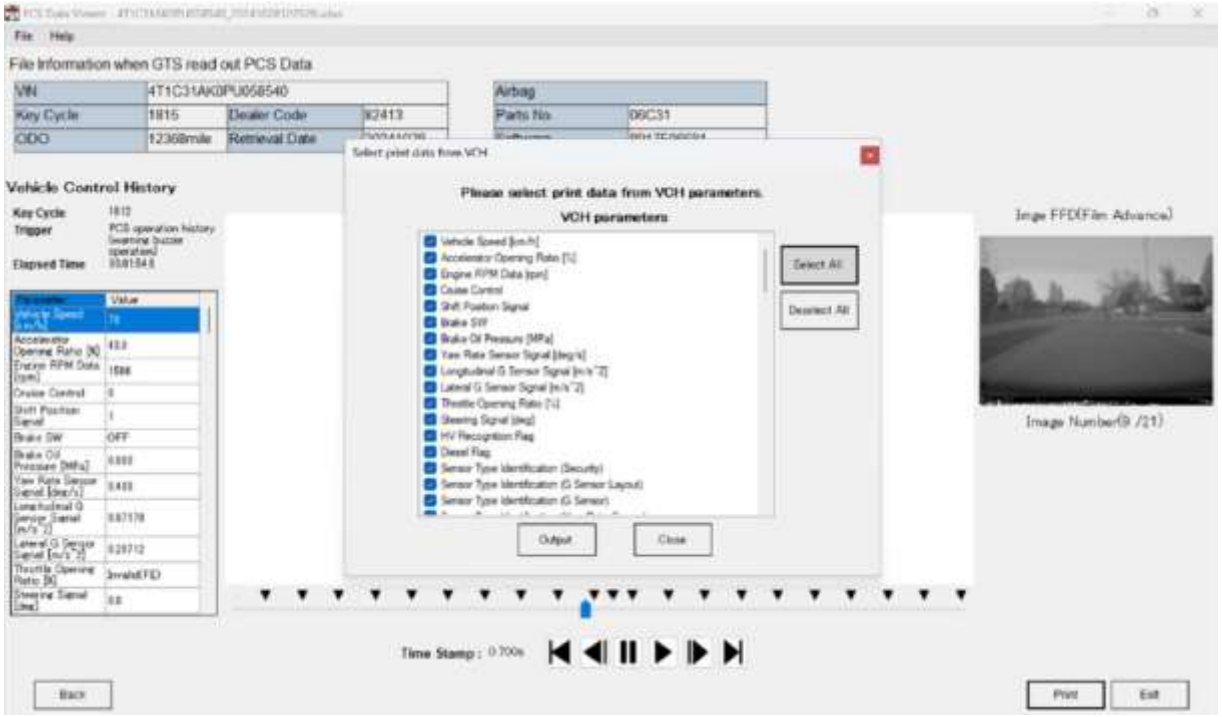
File Edit Run Help Setup

Toyota Global Techstream Plus



File Edit Run Help Setup
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Toyota Global Techstream Plus



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CDR System Operators Course



Introduction to CDR Reports

File Edit Run Help Setup
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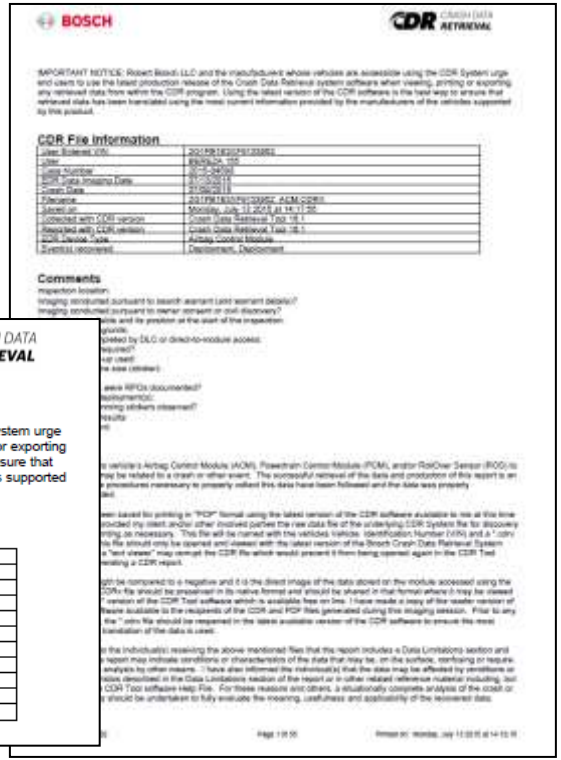
Sample Report

- This is a “Walk in the Park” Only
  - Simply to familiarize you with what a typical file should look like when you complete a download
- **PRIOR** to analyzing any CDR File, an investigator must have taken a CDR Analysis and Application course
- Such Analysis training addresses
  - Data concerns
  - Data limitations
  - Vehicle systems and behavior
  - Proper application to a traffic crash reconstruction
  - Pitfalls and problems…the list goes on and on

File Edit Run Help Setup
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Sample Report

- Cover Page
  - Warning to only use the most current version of CDR software when viewing!



BOSCH

CDR CRASH DATA RETRIEVAL

IMPORTANT NOTICE: Robert Bosch LLC and the manufacturers whose vehicles are accessible using the CDR System urge end users to use the latest production release of the Crash Data Retrieval system software when viewing, printing or exporting any retrieved data from within the CDR program. Using the latest version of the CDR software is the best way to ensure that retrieved data has been translated using the most current information provided by the manufacturers of the vehicles supported by this product.

CDR File Information

User Entered VIN	2G1FB1E3XF9133952
User	BEREZA 155
Case Number	2015-04598
EDR Data Imaging Date	07/13/2015
Crash Date	07/09/2015
Filename	2G1FB1E3XF9133952_ACM.CDRX
Saved on	Monday, July 13 2015 at 14:17:55
Collected with CDR version	Crash Data Retrieval Tool 16.1
Reported with CDR version	Crash Data Retrieval Tool 16.1
EDR Device Type	Airbag Control Module
Event(s) recovered	Deployment, Deployment

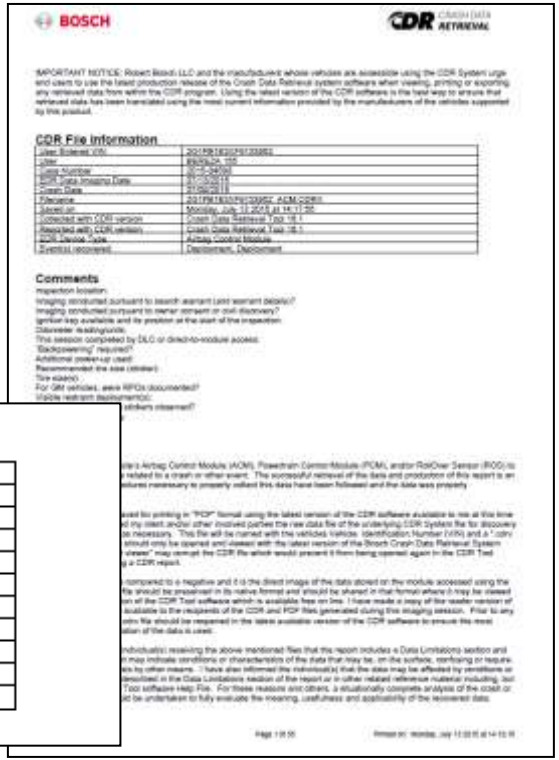
File Edit Run Help Setup © 2024 - Crash Data Specialists LLC

Sample Report

- Cover Page
  - Case File Information
    - Data entered by the user
    - Computer and Software data

CDR File Information

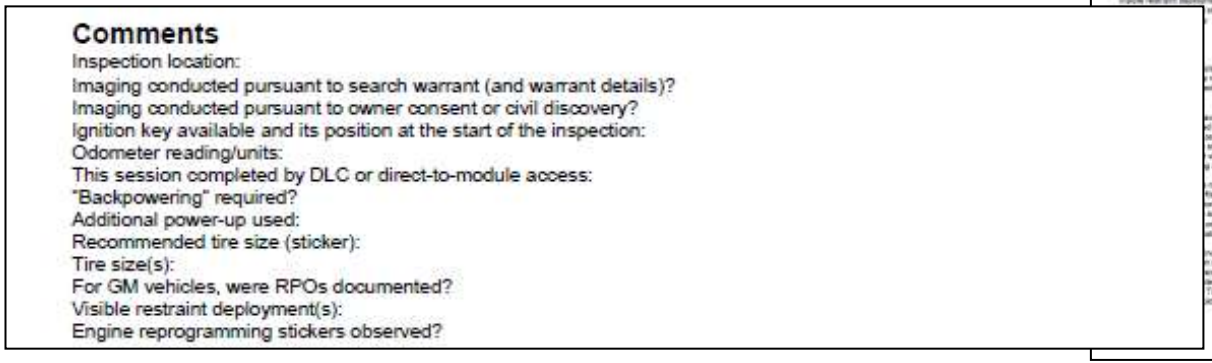
User Entered VIN	2G1FB1E3XF9133952
User	BEREZA 155
Case Number	2015-04598
EDR Data Imaging Date	07/13/2015
Crash Date	07/09/2015
Filename	2G1FB1E3XF9133952_ACM.CDRX
Saved on	Monday, July 13 2015 at 14:17:55
Collected with CDR version	Crash Data Retrieval Tool 16.1
Reported with CDR version	Crash Data Retrieval Tool 16.1
EDR Device Type	Airbag Control Module
Event(s) recovered	Deployment, Deployment



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Sample Report

- Cover Page
  - Comments
    - Information the investigator enters about the downloading process



Comments

Inspection location:
Imaging conducted pursuant to search warrant (and warrant details)?
Imaging conducted pursuant to owner consent or civil discovery?
Ignition key available and its position at the start of the inspection:
Odometer reading/units:
This session completed by DLC or direct-to-module access:
"Backpowering" required?
Additional power-up used:
Recommended tire size (sticker):
Tire size(s):
For GM vehicles, were RPOs documented?
Visible restraint deployment(s):
Engine reprogramming stickers observed?



BOSCH

CDR CRASH DATA RETRIEVAL

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CDR File Information

User Entered VIN	
User	
Case Number	
CDR Data Imaging Date	
Crash Date	
Filename	
Saved on	Monday, July 12 2021 at 14:17:55
Collected with CDR version	Crash Data Retrieval Tool 18.1
Reported with CDR version	Crash Data Retrieval Tool 18.1
SDM Device Type	Airbag Control Module
Event(s) recovered	Deployment, Deployment

Comments

Inspection location:
Imaging conducted pursuant to search warrant (and warrant details)?
Imaging conducted pursuant to owner consent or civil discovery?
Ignition key available and its position at the start of the inspection:
Odometer reading/units:
This session completed by DLC or direct-to-module access:
"Backpowering" required?
Additional power-up used:
Recommended tire size (sticker):
Tire size(s):
For GM vehicles, were RPOs documented?
Visible restraint deployment(s):
Engine reprogramming stickers observed?

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Sample Report

- Cover Page
  - Data Limitations
    - Module specific details
    - Dynamic text document
      - “Living Document” – IT MAY UPDATE with new releases
    - End user cautions
    - Extremely important section for any data analysis to be performed



BOSCH

CDR CRASH DATA RETRIEVAL

Data Limitations

Recorded Crash Events

There are two types of recorded crash events for Front, Side, and Rear (FSR) Events. The first is the Non-Deployment Event. A Non-Deployment Event records data but does not deploy the air bag(s). The minimum SDM Recorded Vehicle Velocity Change that is needed to record a Non-Deployment Event, is five MPH (8 km/h). A Non-Deployment Event contains Pre-Crash and Crash data. The deployment Non-Deployment event can be overwritten by a Deployment Event. If all three records are full and this Non-Deployment Event is not locked. Non-Deployment Events can be overwritten after approximately 250 ignition cycles. Also, a Non-Deployment event can be recorded if one of the following occurs without the Deployment of any of the frontal air bags, side air bags, or roll bars.

-Pretensioner(s) only Deployment
-Hood/ Knee Deployment
-Battery Dump Deployment

The second type of SDM recorded crash event for FSR Events is the Deployment Event. It also contains Pre-Crash and Crash data. Deployment Events cannot be overwritten or deleted by the SDM.

There are also two types of recorded crash events for Rollover Events. The first is the Non-Deployment (Non-rollover) Event. A Non-Deployment Event records data but does not deploy the air bag(s). A Non-Deployment Event contains Pre-Crash and Crash data. Non-Deployment Rollover event follow the same rules as FSR Non-Deployment events.

The SDM can store up to three Events.

Data:

For FSR Events, SDM Recorded Vehicle Velocity Change reflects the change in velocity that the sensing system experienced during the recorded portion of the event. SDM Recorded Vehicle Velocity Change, is the change in velocity during the recording time and is not the speed the vehicle was traveling before the event, and is also not the Barrier Equivalent Velocity. For Deployment and Non-Deployment Events, the SDM will record 300 milliseconds of data after time zero. The SDM will also record 300 milliseconds of Vehicle Acceleration data after time zero.

For Rollover Events, the SDM may record Lateral Acceleration, Vertical Acceleration, and Roll Rate data, if the SDM is rollover capable. This data reflects what the sensing system experienced during the recorded portion of the event. For Rollover Deployment Events, the SDM will record up to 750 milliseconds of data before the Deployment criteria is met and 300 milliseconds after the Deployment criteria is met.

-Deployment logic may be disabled as being deployed in a Non-Deployment event record. If a Deployment event is qualified during the Non-Deployment event. That is, if two or more events are occurring at the same time and one is a Non-Deployment event and one of the others is a Deployment event, and this Deployment event is qualified while the Non-Deployment is still active, the deployed loops may be recorded in the Non-Deployment event record.

-Deployment loops can only be deployed once per ignition cycle.

-The Indexed Events is recorded in 10 msec intervals and is displayed in seconds for a maximum time of 655.33 seconds. The counter measures the time from the start of one event to the start of the next event if both events occur within the same ignition cycle.

-The Maximum SDM Recorded Vehicle Velocity Change may occur between the recorded 10 milliseconds sample points of the SDM Recorded Vehicle Velocity Change.

-Event Recording Completed will indicate if data from the recorded event has been fully written to the SDM memory or if it has been interrupted and not fully written.

-SDM Recorded Vehicle Speed accuracy can be affected by various factors, including but not limited to the following:

-Significant changes in the tire's rolling radius
-Final drive axle ratio changes
-Wheel lockup and wheel slip

Brake Switch Circuit Status indicates the open/closed state of the brake switch circuit.

-Pre-Crash data is recorded asynchronously. The 0.5 second Pre-crash data value, most recent recorded data point, is the data point last sampled before Time Zero. That is to say, the last data point may have been captured just before Time Zero but no more than 0.5 second before Time Zero. All subsequent Pre-crash data values are referenced from this data point.

Pre-Crash Electronic Data Validity Check Status indicates "Data Invalid if '1'

-The SDM receives a message with an "invalid" flag from the module sending the pre-crash data

Pre-Crash Electronic Data Validity Check Status indicates "Data Not Available '1'

-No data is received from the module sending the pre-crash data.

Start Switch Circuit Status indicates the status of the input start switch circuit.

-The ignition cycle counter will increment when the power mode cycles from OFF/Accessory to Run. Applying and removing of battery power to the module will not increment the ignition cycle counter.

-Ignition Cycles Since DTCs Were Last Cleared can record a maximum value of 255 cycles and can only be reset by a scan tool.

-Deployment Event Counter tracks the number of Deployment events that have occurred during the SDM's lifetime.

-Event Counter tracks the number of qualified events (either Deployments, Non-deploy, or Rollover events) that have occurred during the SDM's lifetime.

-The Time Zero in Deployment Command Criteria Met times for the following will be indicated for rollover/over turns first:

-Driver Time to of Driver Curtain
-Passenger Thorax or Passenger Curtain
-Driver Pretensioner Loop #1 or Driver Pretensioner Loop #2
-Passenger Pretensioner Loop #1 or Passenger Pretensioner Loop #2

-For Deployment Events, DTC B0051 (Deployment commanded), shall be recorded with the remainder of the data for this event even though it occurred after Event Ended.

-Once a firing loop has been commanded to be deployed, it will not be commanded to be deployed again during the same ignition cycle. Firing loop times for subsequent deployment type events, during the same ignition cycle, will record the deployment times as N/A.

-The SIM parameter name is displayed in parentheses after the HHTOS. Part 563 parameter name.

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Sample Report

- Cover Page
  - Data Limitations
    - Analysis should be used in conjunction with a reconstruction!
    - Vehicle and Scene Evidence Analysis


BOSCH

CDR CRASH DATA RETRIEVAL

Data Limitations

Recorded Crash Events

There are two types of recorded crash events for Front, Side, and Rear (FSR) Events. The first is the Non-Deployment Event. A Non-Deployment Event records data but does not deploy the air bag(s). The minimum SDM Recorded Vehicle Velocity Change that is needed to record a Non-Deployment Event, is five MPH (8 km/h). A Non-Deployment Event contains Pre-Crash and Crash data. The second Non-Deployment event can be overwritten by a Deployment Event. If all three records are full and this Non-Deployment Event is not locked. Non-Deployment Events can be overwritten after approximately 230 ignition cycles. Also, a Non-Deployment event can be recorded if one of the following occurs without the Deployment of any of the frontal air bags, side air bags, or roll bars.

- Pretensioner(s) only Deployment
- Heated Seat Deployment
- Battery Dump Deployment

The second type of SDM recorded crash event for FSR Events is the Deployment Event. It also contains Pre-Crash and Crash data. Deployment Events cannot be overwritten or deleted by the SDM.

There are also two types of recorded crash events for Rollover Events. The first is the Non-Deployment (Non-eligible) Event. A Non-Deployment Event records data but does not deploy the air bag(s). A Non-Deployment Event contains Pre-Crash and Crash data. Non-Deployment Rollover event follow the same rules as FSR Non-Deployment events.

The SDM can store up to three Events.

Data:

For FSR Events, SDM Recorded Vehicle Velocity Change reflects the change in velocity that the sensing system experienced during the recorded portion of the event. SDM Recorded Vehicle Velocity Change, is the change in velocity during the recording time and is not the speed the vehicle was traveling before the event, and is also not the Barrier Equivalent Velocity. For Deployment and Non-Deployment Events, the SDM will record 300 milliseconds of data after time zero. The SDM will also record 300 milliseconds of Vehicle Acceleration data after time zero.

For Rollover Events, the SDM may record Lateral Acceleration, Vertical Acceleration, and Roll Rate data, if the SDM is rollover capable. This data reflects what the sensing system experienced during the recorded portion of the event. For Rollover Deployment Events, the SDM will record up to 750 milliseconds of data before the Deployment criteria is met and 300 milliseconds after the Deployment criteria is met.

Deployment logic may be displayed as being deployed in a Non-Deployment event record. If a Deployment event is qualified during the Non-Deployment event. That is, if two or more events are occurring at the same time and one is a Non-Deployment event and one of the others is a Deployment event, and this Deployment event is qualified while the Non-Deployment is still active, the displayed loops may be recorded in the Non-Deployment event record.

Deployment loops can only be displayed once per module cycle/ cycle.

Time between events is recorded in 10 msec intervals and is displayed in seconds for a maximum time of 655.33 seconds. The counter measures the time from the start of one event to the start of the next event if both events occur within the same ignition cycle.

The Maximum SDM Recorded Vehicle Velocity Change may occur between the recorded 10 milliseconds sample points of the SDM Recorded Vehicle Velocity Change.

If Event Recording Complete self indicates it data from the recorded event has been fully written to the SDM memory or if it has been interrupted and not fully written.

SDM Recorded Vehicle Speed accuracy can be affected by various factors, including but not limited to the following:

- Tire's rolling radius

-The reported range of the longitudinal and lateral acceleration values is approximately ± 50 g.
-All data should be examined in conjunction with other available physical evidence from the vehicle and scene.

Data Source:
All SDM recorded data is measured, calculated, and stored internally, except for the following:
-Vehicle Status Data (Pre-Crash) is transmitted by the Body Control Module, via the vehicle's communication network.
-The Belt Switch Circuit is wired directly to the SDM.

Data Element Sign Convention:
The following table provides an explanation of the sign notation for data elements that may be included in this CDR report. Directional references to sign notation are all from the perspective of the driver when seated in the vehicle facing the direction of forward vehicle travel.

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Sample Report

- Data Within
  - System Data
  - Event Counters and overall system information
  - Deployment Event

System Status At Deployment

SIR Warning Lamp Status	OFF
Driver's Belt Switch Circuit Status	BUCKLED
Passenger's Belt Switch Circuit Status	BUCKLED
Passenger Seat Position Switch Circuit Status	Rearward
Ignition Cycles At Deployment	14313
Ignition Cycles At Investigation	14315
Maximum SDM Recorded Velocity Change (MPH)	-45.92
Algorithm Enable to Maximum SDM Recorded Velocity Change (msec)	147.5
Driver 1st Stage Time From Algorithm Enable to Deployment Command Criteria Met (msec)	7.5
Driver 2nd Stage Time From Algorithm Enable to Deployment Command Criteria Met (msec)	10
Passenger 1st Stage Time From Algorithm Enable to Deployment Command Criteria Met (msec)	7.5
Passenger 2nd Stage Time From Algorithm Enable to Deployment Command Criteria Met (msec)	10
Time Between Non-Deployment And Deployment Events (sec)	N/A
Frontal Deployment Level Event Counter	1
Event Recording Complete	Yes
Multiple Events	No
Multiple Events Not Recorded	No

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Sample Report

- Vehicle Data
  - Full analysis BEFORE using such data!
  - It is not simply black and white text and definitive!

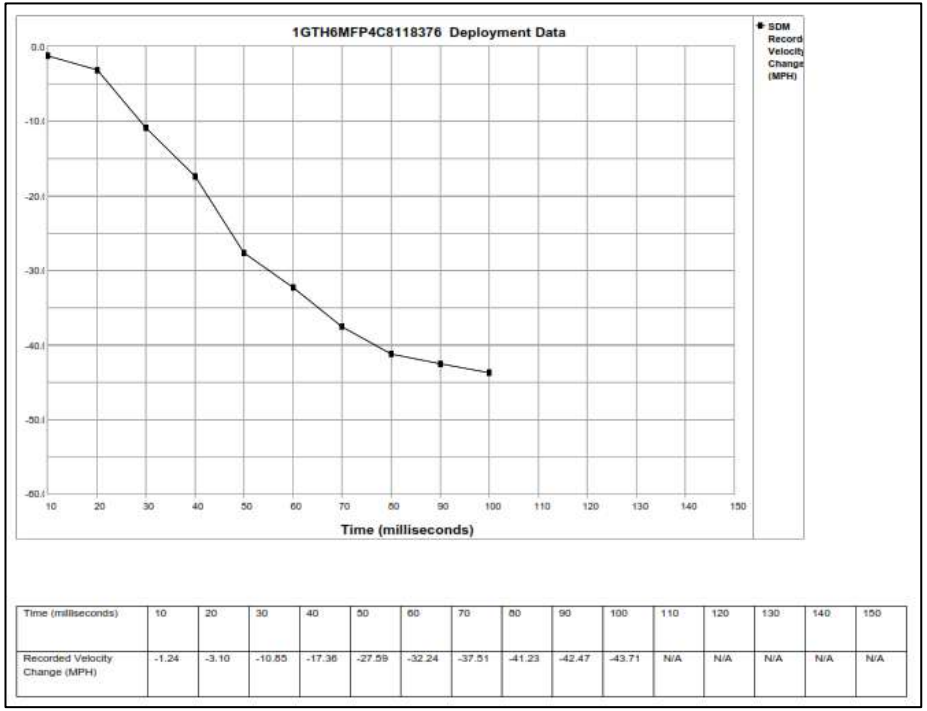
Seconds Before AE	Vehicle Speed (MPH)	Engine Speed (RPM)	Percent Throttle
-5	58	1792	0
-4	58	1792	0
-3	58	1792	0
-2	58	1792	0
-1	57	1728	0

Seconds Before AE	Brake Switch Circuit State
-8	OFF
-7	OFF
-6	OFF
-5	OFF
-4	OFF
-3	OFF
-2	ON
-1	ON

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Sample Report

- Crash Pulse Data
  - Changes in velocity
  - Acceleration
  - Collision time



Time (milliseconds)	10	20	30	40	50	60	70	80	90	100	110	120	130	140	150
Recorded Velocity Change (MPH)	-1.24	-3.10	-10.85	-17.36	-27.59	-32.24	-37.51	-41.23	-42.47	-43.71	N/A	N/A	N/A	N/A	N/A

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Sample Report

- Non-Deployment Event
  - Lots more to learn!

System Status At Non-Deployment

SIR Warning Lamp Status	ON
Driver's Belt Switch Circuit Status	BUCKLED
Passenger's Belt Switch Circuit Status	BUCKLED
Passenger Seat Position Switch Circuit Status	Rearward
Ignition Cycles At Non-Deployment	14313
Ignition Cycles At Investigation	14315
Maximum SDM Recorded Velocity Change (MPH)	-1.11
Algorithm Enable to Maximum SDM Recorded Velocity Change (msec)	52.5
Crash Record Locked	No
Event Recording Complete	Yes
Multiple Events	No
Multiple Events Not Recorded	No

Seconds Before AE	Vehicle Speed (MPH)	Engine Speed (RPM)	Percent Throttle
-5	58	1792	0
-4	58	1792	0
-3	58	1792	0
-2	57	1728	0
-1	50	0	0

Seconds Before AE	Brake Switch Circuit State
-8	OFF
-7	OFF
-6	OFF
-5	OFF
-4	OFF
-3	ON
-2	ON
-1	OFF

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Sample Report

- The information continues to evolve as vehicles develop

Pre-Crash Data -5.0 to 0 sec (100 msec) (Most Recent Event (Non-Deployment)) - Table 1 of 8

Time (sec)	Pre-Crash Recorder Status	Speed, Vehicle Indicated (MPH [km/h])	Accelerator Pedal, % Full (%)	Raw Pedal #1 (V)	Raw Pedal #2 (V)	Engine Throttle, % Full (%)	Raw Throttle #1 (V)	Raw Throttle #2 (V)
-5.0	Complete	17 [28]	8	0.859	0.429	7	0.781	4.199
-4.9	Complete	17 [28]	8	0.859	0.410	7	0.781	4.199
-4.8	Complete	17 [28]	8	0.859	0.410	6	0.761	4.199
-4.7	Complete	17 [28]	8	0.839	0.410	6	0.761	4.218
-4.6	Complete	17 [28]	8	0.859	0.410	6	0.742	4.238
-4.5	Complete	17 [28]	8	0.859	0.410	6	0.742	4.238
-4.4	Complete	17 [28]	8	0.859	0.410	6	0.742	4.238
-4.3	Complete	17 [28]	8	0.859	0.410	6	0.742	4.238
-4.2	Complete	17 [28]	10	0.878	0.429	5	0.722	4.257
-4.1	Complete	17 [28]	12	0.976	0.488	6	0.742	4.218
-4.0	Complete	17 [28]	12	0.976	0.488	7	0.800	4.179
-3.9	Complete	17 [28]	12	0.976	0.488	7	0.800	4.179
-3.8	Complete	17 [28]	12	0.957	0.468	7	0.800	4.179

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Sample Report

- Vast amounts of information for analysis

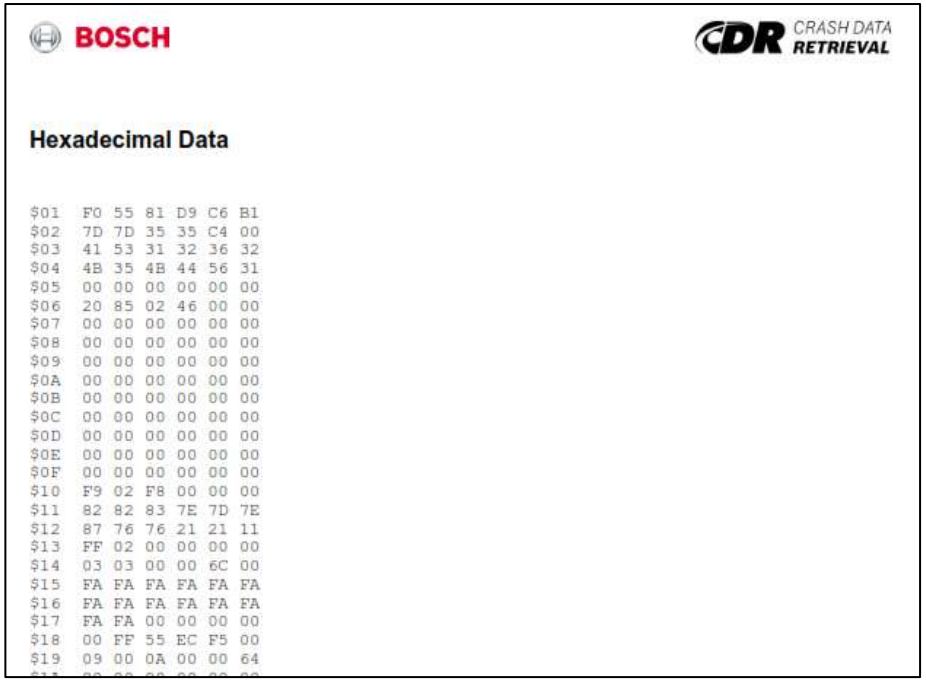
Pre-Crash Data -5.0 to 0 sec (100 msec) (First Prior Event (Non-Deployment)) - Table 4 of 8

Time (sec)	Wheel Speed LR (RPM)	Wheel Speed RR (RPM)	Steering Angle (deg)	Steering wheel angle sensor status	EPB Warning Lamp	Parking Brake Status	EPB Actual Force (daN)	ESS Engine State
-5.0	290	290	-7	Sensor OK	OFF	REL	0	Run
-4.9	292	291	-10	Sensor OK	OFF	REL	0	Run
-4.8	293	293	-14	Sensor OK	OFF	REL	0	Run
-4.7	296	295	-15	Sensor OK	OFF	REL	0	Run
-4.6	299	298	-19	Sensor OK	OFF	REL	0	Run
-4.5	303	300	-20	Sensor OK	OFF	REL	0	Run
-4.4	304	302	-20	Sensor OK	OFF	REL	0	Run
-4.3	307	305	-22	Sensor OK	OFF	REL	0	Run
-4.2	310	309	-25	Sensor OK	OFF	REL	0	Run
-4.1	314	311	-29	Sensor OK	OFF	REL	0	Run

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Sample Report

- Hexadecimal Data
  - Raw, untranslated data which is translated by the CDR Software
  - Proprietary between Bosch and Manufacturer
  - Translation is not taught



BOSCH

CDR CRASH DATA RETRIEVAL

Hexadecimal Data

```
$01   F0 55 81 D9 C6 B1
$02   7D 7D 35 35 C4 00
$03   41 53 31 32 36 32
$04   4B 35 4B 44 56 31
$05   00 00 00 00 00 00
$06   20 85 02 46 00 00
$07   00 00 00 00 00 00
$08   00 00 00 00 00 00
$09   00 00 00 00 00 00
$0A   00 00 00 00 00 00
$0B   00 00 00 00 00 00
$0C   00 00 00 00 00 00
$0D   00 00 00 00 00 00
$0E   00 00 00 00 00 00
$0F   00 00 00 00 00 00
$10   F9 02 F8 00 00 00
$11   82 82 83 7E 7D 7E
$12   87 76 76 21 21 11
$13   FF 02 00 00 00 00
$14   03 03 00 00 6C 00
$15   FA FA FA FA FA FA
$16   FA FA FA FA FA FA
$17   FA FA 00 00 00 00
$18   00 FF 55 EC F5 00
$19   09 00 0A 00 00 64
```

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Sample Report

- Attend the CDR Analysis and Application course next
- Do not apply any data without training
- Apply the data to a full crash investigation and reconstruction
- Be cautious of anomalies
- Always think about the data and the overall situation – does it make sense?

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Types of Events

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Events

- An “Event” is:
  - The occurrence of some level of acceleration which causes an ACM to evaluate available data and decide whether or not to deploy restraint device(s). A crash or other physical occurrence that causes a module's trigger threshold to be met or exceeded (NHTSA 49 CFR Part 563 definition).

- CDR related Events include:
  - Deployment Event
  - Non-Deployment Event
  - Deployment Level Event
  - Near-Deployment Event (Do not use – out of date!)

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Events

- Deployment Event
  - Acceleration observed along one of the car's axes sufficient to cause the control module's crash sensing algorithm to "enable" or "wake up" and which is sufficient to warrant a commanded deployment.
  - Airbag Deployment
    - Frontal, side, torso, knee bolster, etc
  - Seatbelt Pretensioner
  - Headrest
  - Others?





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Events

- Non-Deployment Event
  - Acceleration observed along one of the car's axes sufficient to cause the module's crash sensing algorithm to "enable" or "wake up" but which does not warrant a commanded deployment
  - The system wakes up, but the event is not severe enough to reach the deployment level threshold
    - Set by manufacturer, difficult or nearly impossible to obtain level data

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Events

- Deployment Level Event
  - Acceleration is observed along one of the car's axes which is sufficient to cause the GM SDM's crash sensing algorithm to "enable" and anticipate a collision severity which otherwise warrants a deployment for that vehicle, but a deployment had been previously commanded
  - Nothing left to give

- Near Deployment – Don't Use
  - An out-of-date and inappropriate term replaced by the more descriptive and more accurate term "non-deployment."

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Do I need a Warrant?

What about Consent?

Operator specific considerations

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Operator specific considerations

- Training
  - Have you been trained to safely and properly image the data and preserve it for later use?
- Spoliation Issues
  - Does your imaging method / access expose the data to a risk of spoliation?
  - Consider:
    - Did you move the seat or tilt steering?
    - Removal or movement of other vehicle components?

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Operator specific considerations

- What authority or permission to access and download the data?
  - Search Warrant?
  - Owner/Driver Consent?
  - As part of Civil discovery process?
  - What constitutes “informed consent”?



- How will you preserve the data?
  - Electronic .CDRx file and related printouts?



- Will you remove and store the affected module?

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Operator specific considerations

- Will you remove and store the module?
  - Or the vehicle?
  - Newly supported OE considerations
- How long will you keep it?
  - Disposal when case complete?
- Have you considered potential spoliation issues like setting new DTCs?

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Legal Details

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Driver Privacy

- Driver Privacy Act of 2015
  - Contained within the Fixing America's Surface Transportation Act
    - "FAST Act"
  - Section 24301 – Driver Privacy Act of 2015
    - LIMITATIONS ON DATA RETRIEVAL FROM VEHICLE EVENT DATA RECORDERS
      - (a) OWNERSHIP OF DATA.—**Any data** retained by an event data recorder (as defined in section 563.5 of title 49, Code of Federal Regulations), regardless of when the motor vehicle in which it is installed was manufactured, **is the property of the owner**, or, in the case of a leased vehicle, **the lessee** of the motor vehicle in which the event data recorder is installed.

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Driver Privacy

- Driver Privacy Act of 2015
  - (b) PRIVACY—Data recorded or transmitted by an event data recorder described in subsection
    - (a) may not be accessed by a person other than an owner or a lessee of the motor vehicle in which the event data recorder is installed unless—
      - (1) **a court or other judicial** or administrative authority having jurisdiction— (A) authorizes the retrieval of the data; and (B) to the extent that there is retrieved data, the data is subject to the standards for admission into evidence required by that court or other administrative authority;
      - (2) **an owner or a lessee** of the motor vehicle **provides written, electronic, or recorded audio consent** to the retrieval of the data for any purpose, including the purpose of diagnosing, servicing, or repairing the motor vehicle, or by agreeing to a subscription that describes how data will be retrieved and used;

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Driver Privacy

- Owners Manuals have information contained within about the information the vehicle may record!
- GM Example:

Vehicle Data Collection and Event Data Recorders

Your vehicle, like other modern motor vehicles, has a number of sophisticated computer systems that monitor and control several aspects of the vehicle's performance. Your vehicle uses on-board vehicle computers to monitor emission control components to optimize fuel economy, to monitor conditions for airbag deployment and, if so equipped, to provide anti-lock braking and to help the driver control the vehicle in difficult driving situations. Some information may be stored during regular operations to facilitate repair of detected malfunctions; other information is stored only in a crash event by computer systems, such as those commonly called event data recorders (EDR).

In a crash event, computer systems, such as the Airbag Sensing and Diagnostic Module (SDM) in your vehicle may record information about the condition of the vehicle and how it was operated, such as data related to engine speed, brake application, throttle position, vehicle speed, safety belt usage, airbag readiness, airbag performance, and the severity of a collision. This information has been used to improve vehicle crash performance and may be used to improve crash performance of future vehicles and driving safety. Unlike the data recorders on many airplanes, these on-board systems do not record sounds, such as conversation of vehicle occupants.

To read this information, special equipment is needed and access to the vehicle or the device that stores the data is required. GM will not access information about a crash event or share it with others other than:

- with the consent of the vehicle owner or, if the vehicle is leased, with the consent of the lessee,
- in response to an official request of police or similar government office,
- as part of GM's defense of litigation through the discovery process, or
- as required by law.

In addition, once GM collects or receives data, GM may:

- use the data for GM research needs,
- make it available for research where appropriate confidentiality is to be maintained and need is shown, or
- share summary data which is not tied to a specific vehicle with non-GM organizations for research purposes.

Others, such as law enforcement, may have access to the special equipment that can read the information if they have access to the vehicle or the device that stores the data.

If your vehicle is equipped with OnStar®, please check the OnStar® subscription service agreement or manual for information on its operations and data collection.

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Driver Privacy

- Owners Manuals have information contained within about the information the vehicle may record!
- Another Example

Event Data Recording

Other modules in your vehicle — event data recorders — are capable of collecting and storing data during a crash or near crash event. The recorded information may assist in the investigation of such an event. The modules may record information about both the vehicle and the occupants, potentially including information such as:

- how various systems in your vehicle were operating;
- whether or not the driver and passenger seatbelts were buckled;
- how far (if at all) the driver was depressing the accelerator and/or the brake pedal;
- how fast the vehicle was traveling; and
- where the driver was positioning the steering wheel.

To access this information, special equipment must be directly connected to the recording modules. Ford Motor Company and Ford of Canada do not access event data recorder information without obtaining consent, unless pursuant to court order or where required by law enforcement, other government authorities or other third parties acting with lawful authority. Other parties may seek to access the information independently of Ford Motor Company and Ford of Canada.

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Driver Privacy

- Best Practice
  - Obtain legal authority (Search Warrant)
  - Obtain written consent from the vehicle owner/lessee
  - Obtain audio recorded permission from the owner
    - Body Worn Camera
  - Have a plan in place BEFORE you need one!
    - Discuss with your legal counsel, State's Attorney, District Attorney, etc.

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Part 563

- 49 CFR Part 563 – Event Data Recorders
  - NHTS final rule for EDR implementation, September 2012
    - Some manufacturers prior, many after
  - First petition for a rule began in 1998 and continued until 2001
  - Primary goal is to develop future automated crash notifications, enhance safety and gain a better understanding of crash elements and causation
  - Current police investigations of serious crashes
    - Many agencies report downloads available in over 90% of crash investigations in the United States

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Part 563

- 49 CFR Part 563 – Event Data Recorders
  - Many definitions are contained within Part 563, a great place to start
  - Event Data Recorder
    - A device or function in a vehicle that records the vehicle's dynamic time-series data during the time period just prior to a crash event (e.g., vehicle speed vs. time) or during a crash event (e.g., delta-V vs. time), intended for retrieval after the crash event.
    - For purposes of this definition, the event data do not include audio and visual data.
    - The definition was changed to specifically exclude freeze frame system data

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Part 563

- 49 CFR Part 563 – Event Data Recorders
  - Event
    - An impact or other physical occurrence that causes the trigger threshold to be met or exceeded
  - Trigger Threshold
    - Change in velocity, longitudinally or laterally, that equals or exceeds 5 mph within a 150-millisecond period (0.150 sec)
    - A recordable event that is recognized by the EDR as suitable for further analysis
    - NOT tied to deployments

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Part 563

- 49 CFR Part 563 – Event Data Recorders
  - Data from the CDR Reports often vary (most are not the same)
  - Required Data Requirements – if EDR equipped in the vehicle
    - Change in velocity
    - Speed
    - Service Brake
    - Ignition Cycles
    - Seatbelt Status
    - Throttle, event timing, file recording information

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Part 563

- 49 CFR Part 563 – Event Data Recorders
  - If equipped, the recording of the below takes place
    - Acceleration – longitudinal and lateral
    - RPM, Roll Angle, ABS Activity, Stability Control
    - Steering
    - Timing data
    - Pretensioner data
    - Seat track position, occupant size, classifications
  - Lastly, the recording requirements
    - Specifics on how data is recorded and when

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Part 563 New Requirements!

For large manufacturers, vehicles manufactured after September 1, 2027 will now be required to provide 20 seconds of data every .1 second where it had been 5 seconds of data every .5 second on certain pre-crash data!

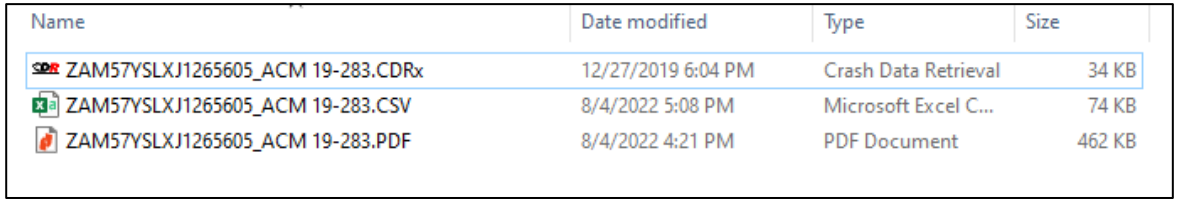
Speed, vehicle indicated		−20.0 to 0 sec [4]	[4] 10
Engine throttle, % full (or accelerator pedal, % full)		−20.0 to 0 sec [4]	[4] 10
Service brake, on/off		−20.0 to 0 sec [4]	[4] 10
Engine rpm	If recorded	−20.0 to 0 sec [5]	[5] 10
ABS activity (engaged, non-engaged)	If recorded	−20.0 to 0 sec [5]	[5] 10
Stability control (on, off, or engaged)	If recorded	−20.0 to 0 sec [5]	[5] 10
Steering input	If recorded	−20.0 to 0 sec [5]	[5] 10

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Evidence Files

- Preserve all information downloaded as evidence
  - Use the most current version of CDR Tool Software for downloading

- Each download should have two minimum files, many times three or more
  - 1. Original File - .CDRx
  - 2. PDF of Report
  - 3. Additional .CSV file

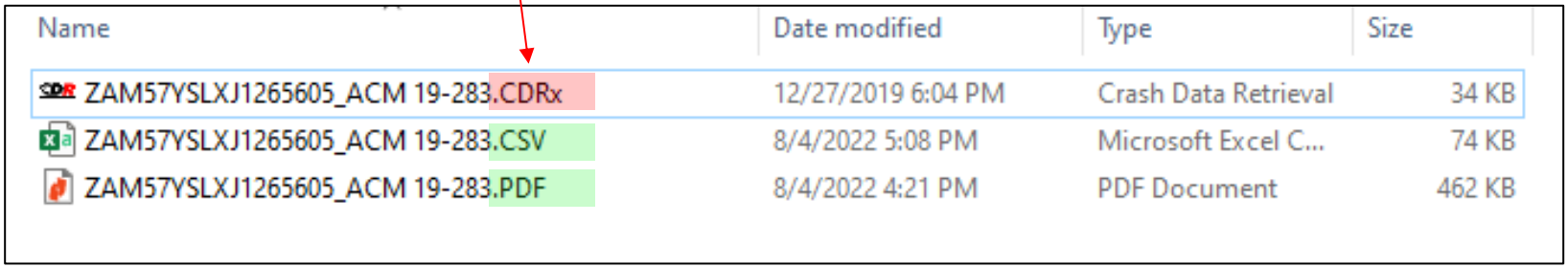


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Evidence Files

- Most Important - .CDRx File



- Others can be recreated using the original file

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Advanced Methods



Introduction to Backpowering for DLC Downloads

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Why Teach These Methods?

- Feedback from Manufacturers wanting greater emphasis on DLC Imaging
- Lessen potential of adding/overwriting DTC
- Allows for Imaging without an available umbilical cable
- Negates need for removing ACM for bench top imaging

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Advanced Methods

- When would to use these techniques?
  - No Keys (when key is required) or broken ignition switch
  - Problems with the vehicles electrical system
    - Crash Damage
    - Potential for fire
  - When setting further DTC's is a concern
- *Not recommended for PCM access*

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Assess….

- Is the DLC available?
- Are the DLC wires intact?
- Key Available? – Do I need the key?
- Can I access the Ignition Switch?
- Is the Fuse Panel / Power Distribution Center intact?
- Is the wiring to the ACM intact?

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12v Accessory Plug to Spade Terminal

- 12 volt accessory plug
- Clip for Ground
- Fused 12volt lead to spade terminal
  - Use with fuse block adapter

Extra female adapter to power Interface





To interface

Repowering rigs can be built or purchased through online vendors

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12 volt Y-adapter

- Provides more than one 12v outlet from single outlet source
- Useful when your booster only has one outlet
- Power can be supplied to interface and module





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DC Power Supply

- A “Cigarette Lighter” style socket adapter.
- Supplies 12v from a jump box (Dead vehicle) or a live DC Port in a vehicle. (Power in vehicle but no Key)





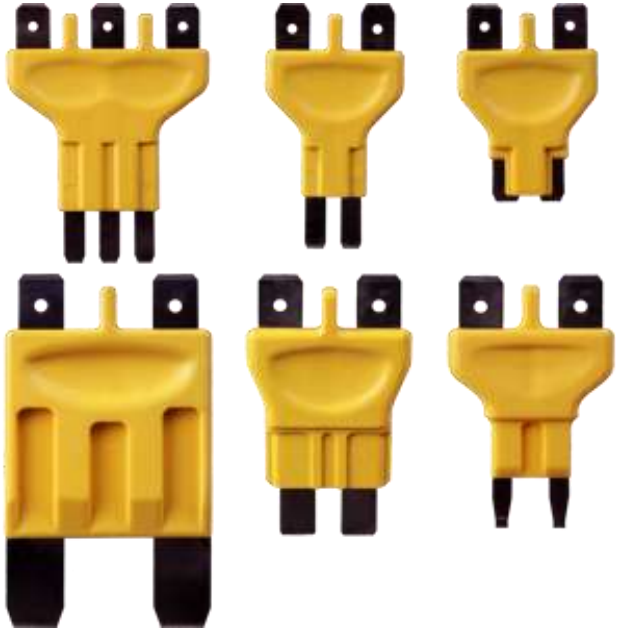


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Fuse Panel Connectors

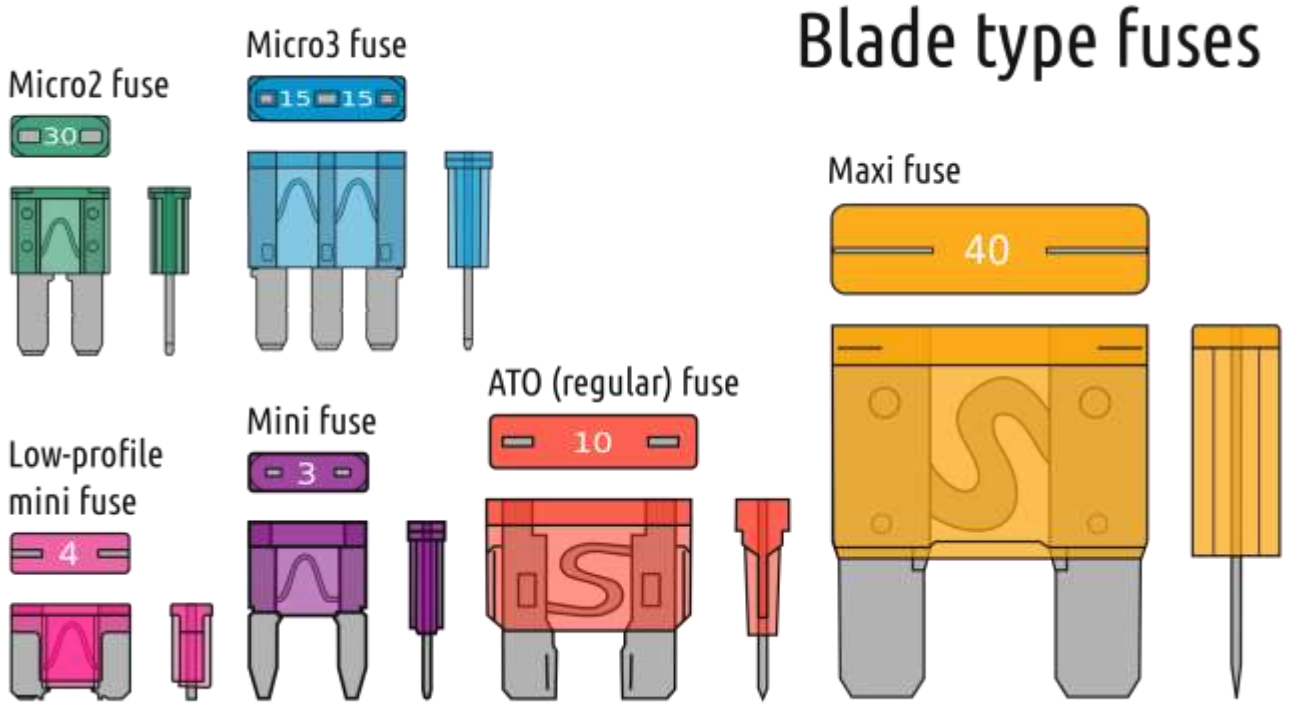
GTC063 Fuse Socket Adapters

- Now Available in ALL Fuse Types
- Amazon = best source ~ $20.00 / set
- Made by General Technologies
- Recommend 2 x sets
  - Allows for more comlex back-powering requiring 2 x fuses



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Fuse Types



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Using 12volt Accessory to Spade Terminal Cable

- Protected by Fuse
- Ground Clip
- Can make with multiple ends
- May be double fused
  - Inline Fuse
  - Fused Accessory Plug



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Required Information

- Fuse Panel Locations and Diagrams
  - Fuse Panel / PDC diagram
  - Owners Manual
  - Online Service Information
    - Some have online owners manuals
- Some have more than one fuse for ACM
  - constant power to ACM
  - power present when Ignition switch = On

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Vehicle Battery

- Currently supported vehicles are 12 volt systems
  - Hybrids use 36v or other, but for propulsion only
- Location of battery may vary widely
  - Normally in engine compartment
  - Location within varies widely
- Other locations
  - Passenger compartment
  - Trunk

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Fuse Panel / Box

- More than one may be in a given vehicle
  - Distributes power to the component level
  - inputs / outputs may be switched by the ignition key
- Fuse Panels may be in multiple locations
  - Under hood, under dash, kick panel or trunk

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Finding the Fuse Box

- May not be located in a “standard” location
  - Under hood in engine compartment
  - In occupant compartment
    - At base of “A” Pillar
    - In Centre Console
    - Under Dash
    - Under Back Seat
- May be more than one location for ACM fuse or multiple ACM fuses in one location

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Locating Owners Manuals

- BMW:
  - www.bmwusa.com
  - NO Fuse Maps
  - Pre-2016/17 BMW Can NOT be back-powered
- Fiat Chrysler Automobiles:
  - Chrysler / Dodge / Ram / Jeep / Fiat
  - www.moparownercenter.com
- Ford / Lincoln / Mercury
  - Fleet.ford.com
- my.cadillac.com
- my.chevrolet.com
- my.GMC.com

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Owners Manual Example

2012 Chevrolet Cruze Owner Manual

Driving and Operating 9-1
Driving Information 9-2
Starting and Operating 9-15
Engine Exhaust 9-27
Automatic Transmission 9-29
Manual Transmission 9-33
Brakes 9-35
Ride Control Systems 9-37
Cruise Control 9-40
Object Detection Systems 9-42
Fuel 9-44
Towing 9-48
Conversions and Add-Ons 9-55

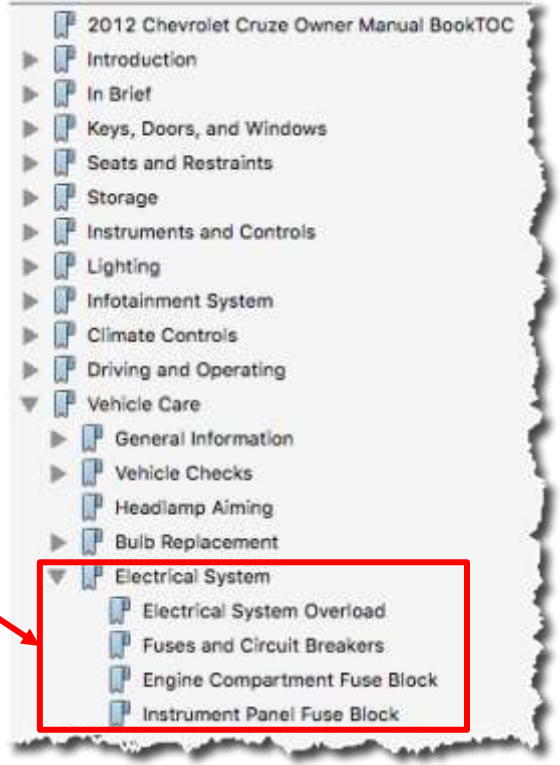
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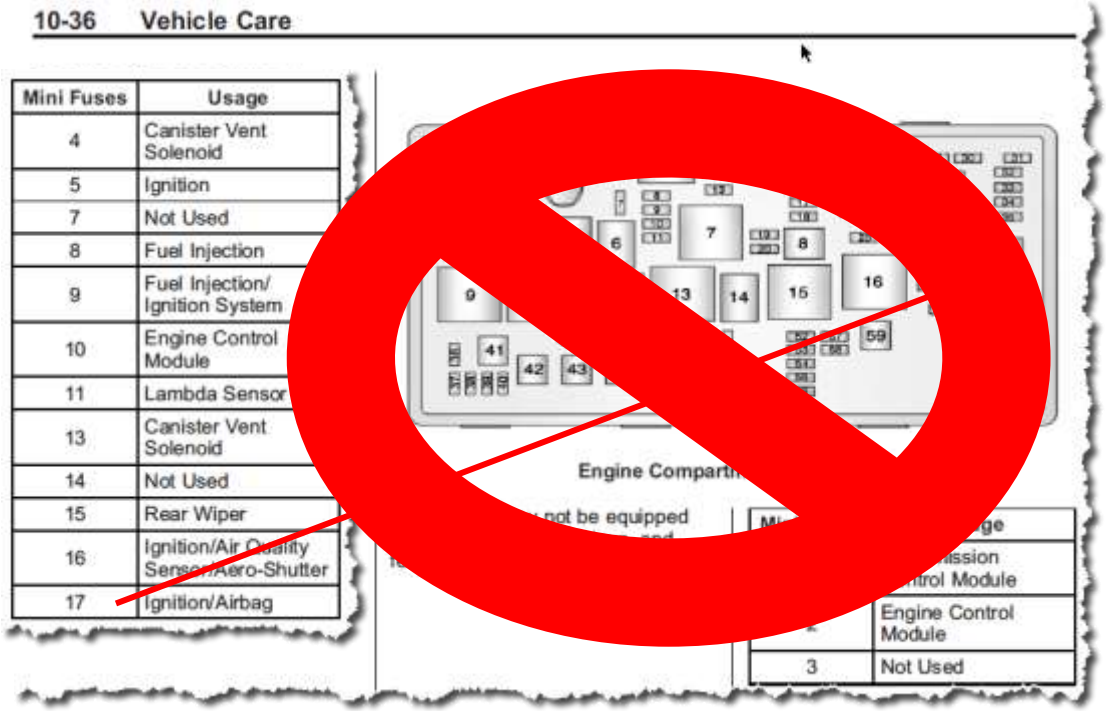
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Owners Manual Example



10-36 Vehicle Care

Mini Fuses	Usage
4	Canister Vent Solenoid
5	Ignition
7	Not Used
8	Fuel Injection
9	Fuel Injection/ Ignition System
10	Engine Control Module
11	Lambda Sensor
13	Canister Vent Solenoid
14	Not Used
15	Rear Wiper
16	Ignition/Air Quality Sensor/Aero-Shutter
17	Ignition/Airbag

- For GM Vehicles with TWO fuses:
  - Fuse labeled Airbag/Ignition is NOT the correct fuse
  - Correct Fuse ***normally in occupant compartment***

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Owners Manual Example

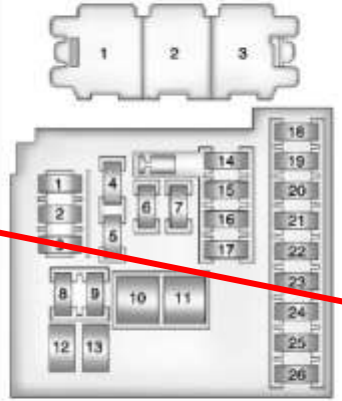
Instrument Panel Fuse Block



The instrument panel fuse block is in the driver side of the instrument panel. To access the fuses:

1. Open the fuse block cover by pulling out at the top.
2. Remove the lower edge of the cover.
3. Remove the cover.

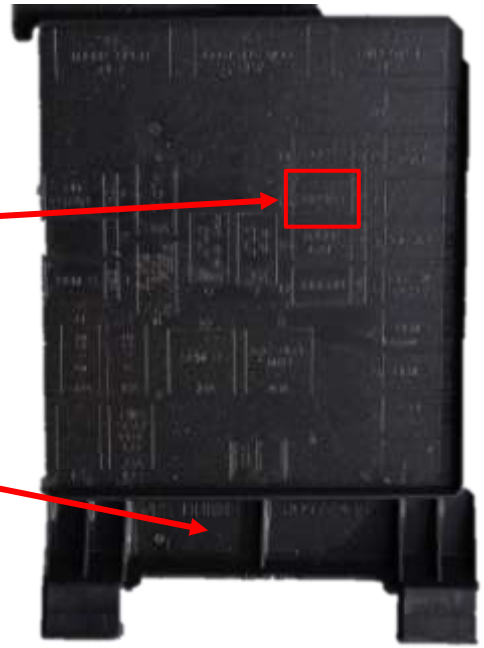
To reinstall the cover, reverse the steps above.



Instrument Panel Fuse Block

The vehicle may not be equipped with all of the fuses, relays, and features shown.

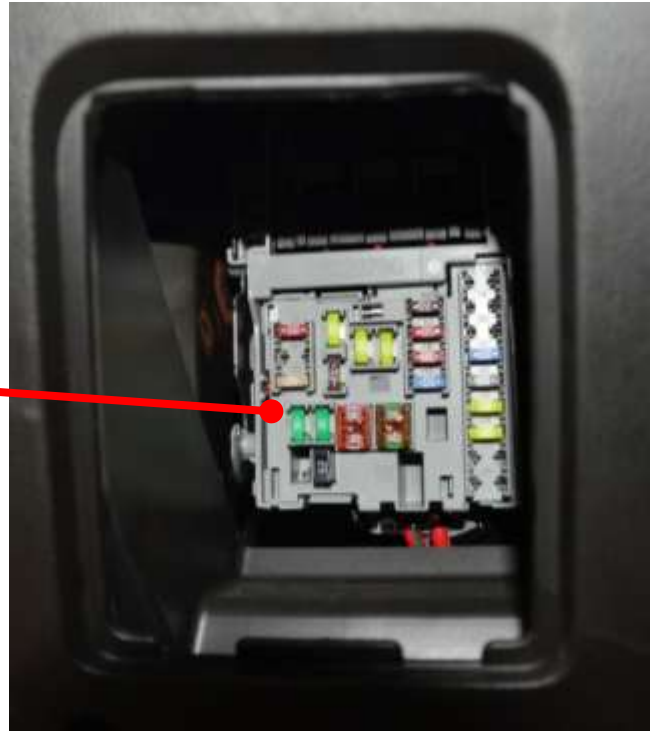
Fuses	Usage
11	Interior Fan
12	Not Used
13	Power Seat
14	Diagnostic Connector
15	Airbag
16	Central Locking System/Tailgate
17	Air Conditioning System
18	Not Used
19	Not Used



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Owners Manual Example





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Power Source

- *<u>WHEREVER POSSIBLE</u>* Use the same power source for:
  - CDR Interface Module
  - Power to ACM / Fuse Panel
- Try to avoid two different Ground paths
  - Vehicle communications are sensitive and ***<u>may</u>*** be affected

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Chassis Ground



- Ground completes the circuit
  - Every circuit is grounded back to Battery – side
  - Engine is grounded to chassis “body” of vehicle
  - ACM is grounded through its metal case to the chassis
    - Even “plastic” ACM is grounded through mounting holes



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Disconnecting Vehicle Battery



- If at any stage you must disconnect the battery:
  - Treat all batteries as “live”
  - Use Battery Wrench
  - Disconnect the negative first
    - Reduces chance of spark
    - Shorting Wrench to Ground poses no danger
  - Disconnect Positive last
    - If done first there is a risk of spark and shorting wrench to ground

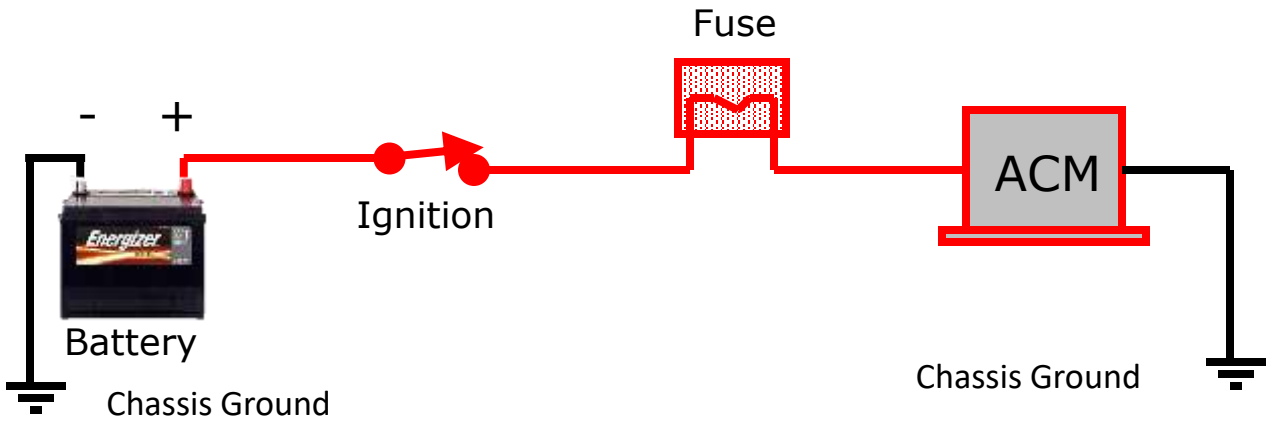




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Powering thru the fuse socket

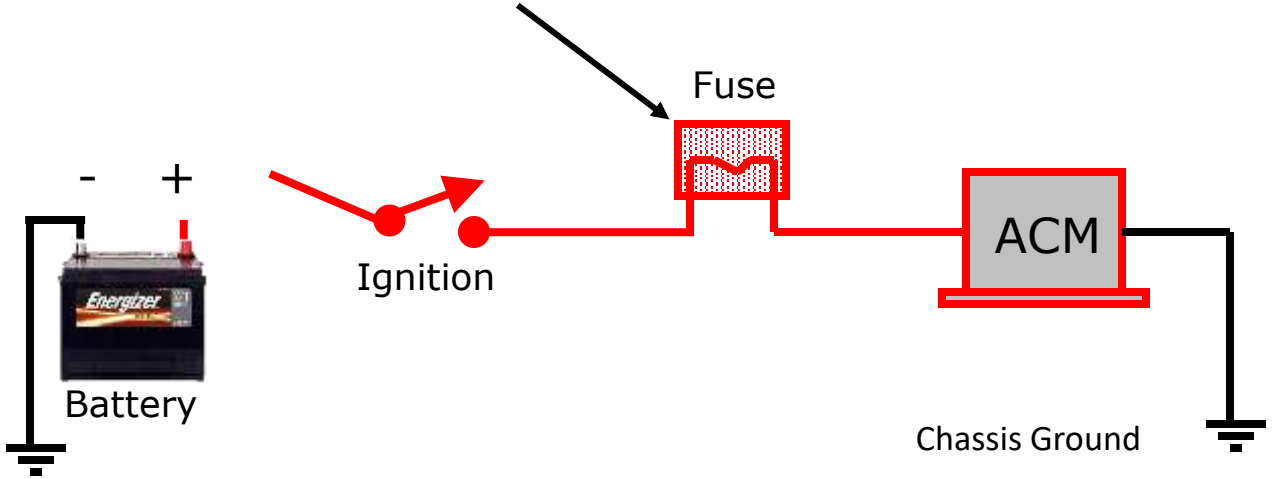
Typical ACM Wiring Schematic



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Powering thru the fuse socket

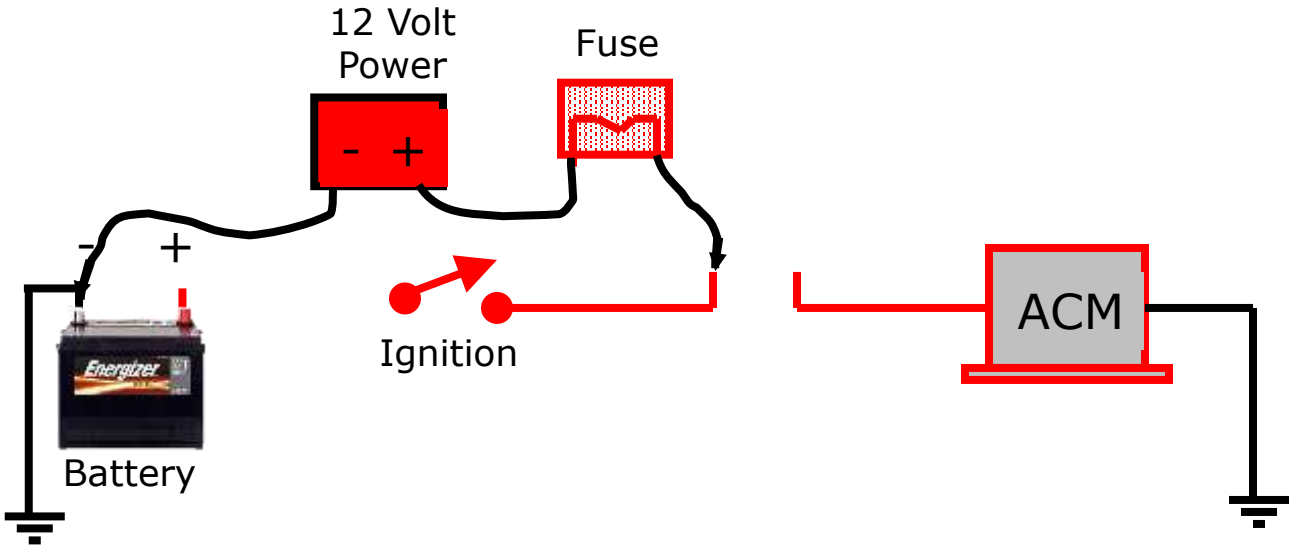
- Remove Battery from circuit
- Remove ACM Power Feed Circuit Fuse(s)



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Powering Thru the Fuse socket – No Meter Check

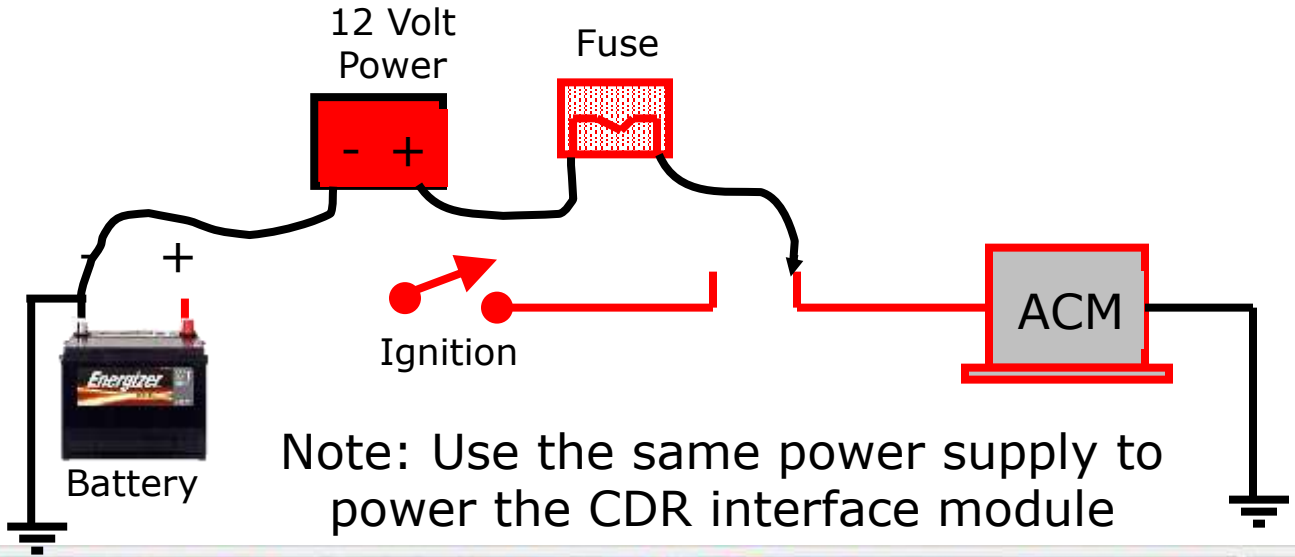
Try connecting a 12 volt fused power source to one side of the fuse socket. Can you read data?



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Powering Thru the Fuse socket – No Meter Check

Connect a 12 volt fused power source to the other side of the fuse socket. Can you read data now?



Note: Use the same power supply to power the CDR interface module

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Backpowering Video



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Backpowering Exceptions / Not Tested

- Most BMW Vehicles Prior to MY2017 Can NOT be Back-Powered
  - See next slide
- FCA – most 2011+ Vehicles can be back-powered
  - Look for single “ORC” fuse
  - Double-pack fuse = may not be successful
- Toyota
  - May not list ACM fuse only
  - May be on IGN fuse
- Not Tested:
  - Rolls Royce
  - Mini
  - Smart
  - Pagani
- Difficult to Back-Power:
  - Mercedes
  - Lamborghini

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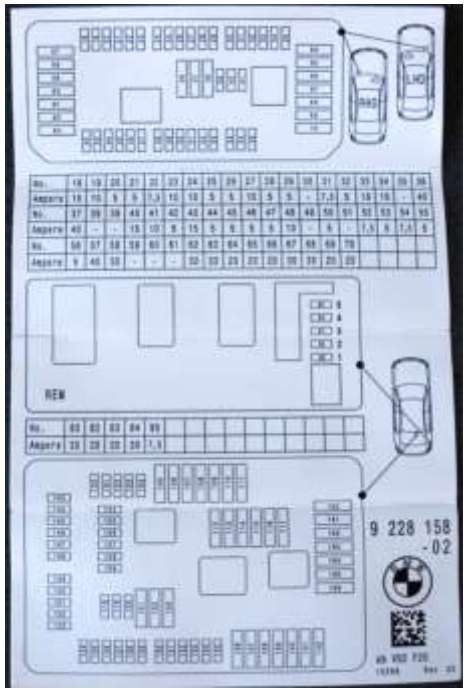
Backpowering Reference Section



Vehicle Specific Examples

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BMW Fuse Guides



- Fused items indicated as Symbols
- Multiple fuses possible for each system
- Reference best made to circuit diagrams



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2017 + BMW Backpowering

- Battery (some) back under the hood
- *MAY* have more than one battery
- Under cowl at rear of engine bay
- Note RED boosting post



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2017 + BMW Backpowering

- “Negative” battery terminal equipment with battery monitoring device
- Cuts off power to preserve “start” power
- *MAY* be damaged if “-” terminal removed first
- MUST remove battery from circuit for “Key By-Pass”



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Chrysler Repowering Considerations

- Starting in 2012 Chrysler changed their power management strategy
- Single Fuse for ORC (ACM)
- Back-powering now possible

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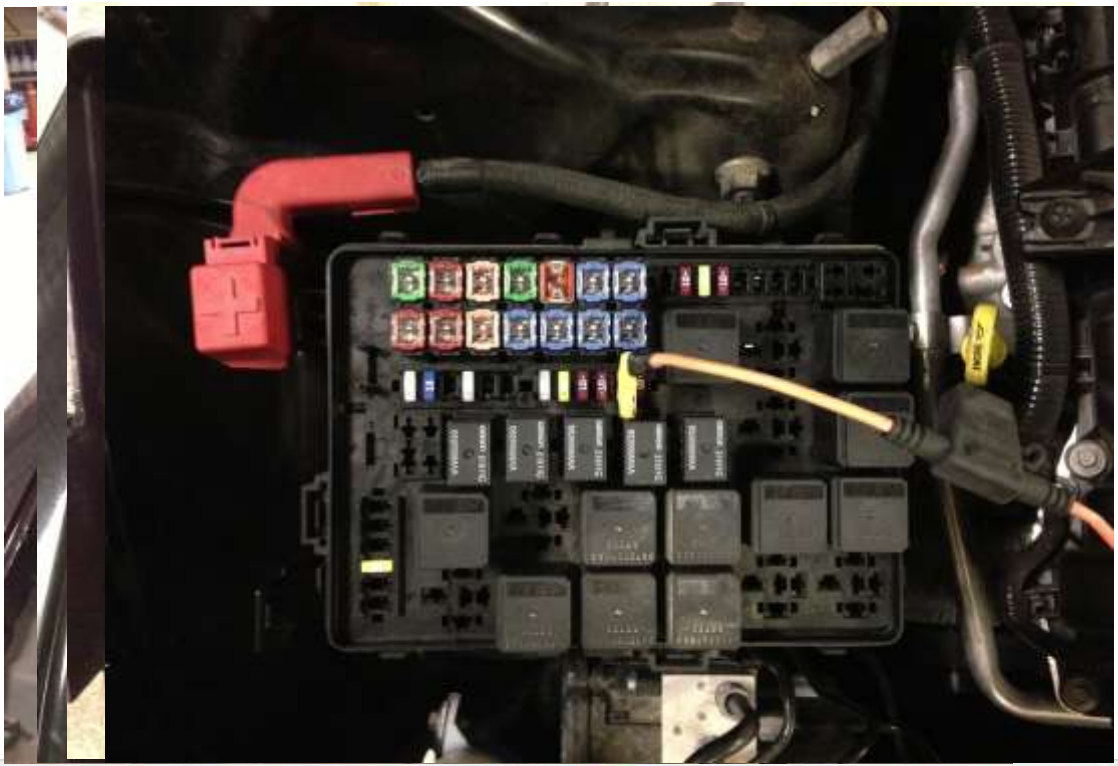
Fiat 500 Example



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2012 Charger Example



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FCA Cybersecurity Module

- 2018+ FCA vehicles
- Communication between vehicle ECU's is encrypted
- Access via "Gateway" ECU
- Invisible to CDR when communicating with vehicle on DLC or D2M download
- MUST be powered for D2M via back-power



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2019 RAM 1500 – Backpowering





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FCA Owners Manual Listing

Cavity	Cartridge Fuse	Micro Fuse	Description
F49	–	15 Amp Blue	Mod CVIM / SNSR Blind Spot / HDLP Adaptive Front Lighting Sensor (AFLS)
F50A	–	10 Amp Red	Battery PACK Control Mod
F51 A&B	–	–	Spare
F52	20 Amp Blue	–	Direct Battery Feed
F53	–	15 Amp Blue	Mod ICS Switch / Mod HVAC CTRL / Sw Bank Upper / Sw EPB / Mod Control Steering
F54B	–	20 Amp Yellow	Power Outlet Center Seat
F55	25 Amp White	–	Upfitter
F56	30 Amp Pink	–	Mod Network Interface
F57	20 Amp Blue	–	Direct Battery Feed
F58	20 Amp Blue	–	Direct Battery Feed
F60	50 Amp Red	–	Mod Inverter
F61	–	–	Spare
F62 A&B	–	10 Amp Red	ITBM / Mod Occupant Class / Mod IAIR Suspension / Mod HVAC Snsr Incar Temp / Rear Coolant Temp / PTS / Mod IRCM
F63	–	–	Spare
F64	–	–	Spare
F65	–	10 Amp Red	Mod ORC
F66		10 Amp Red	Run - Accessory Feed

Cavity	Cartridge Fuse	Micro Fuse	Description
F17	–	20 Amp Yellow	Spare
F18	30 Amp Pink	–	Motor Sunshade Sunroof
F19	–	–	Spare
F20	–	20 Amp Yellow	Comfort Rear Seat Module (CRSM) (Heat Rear RT)
F21	–	–	Spare
F22	–	–	Spare
F23	–	–	Spare
F24	–	15 Amp Blue	Mod RF Hub / Mod Ignition / Mod Cluster
F25	40 Amp Green	–	Mod Integrated Trailer Brake
F26	–	15 Amp Blue	Mod Cluster CCN / Mod Cyber Security
F27	–	5 Amp Tan	Mod Cluster CCN / Mod SGW
F28	–	10 Amp Red	Mod ORC
F29	–	20 Amp Yellow	Mod CRSM (Heat Rear LT)
F30	30 Amp Pink	–	Mod DTCM / Mod Tailgate
F31	30 Amp Pink	–	Mod CBC I Interior Light
F32	–	–	Spare / Sunshade
F33	–	10 Amp Red	Assy Overhead Console / Switch 911 / Switch Assist / Sunshade
F34	–	15 Amp Blue	Frt & RR Ventilated Seat Motor

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2019 RAM 1500 – Backpowering





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Newer GM Vehicles With Gateway

- Newer GM vehicles use a Gateway for various ECU's to communicate through the DLC
- May require power for ACM / FCM and ASCM
- Consult owners manual!!
- If a gateway is listed, it likely needs to be powered



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Gateway Module

- Newer GM vehicles
  - Starting with #845 cable
- No longer Note 6
  - Which means what?

2019 Chevrolet	Corvette (note 6)

- Have gateway modules
  - CGM
  - Between DLC & ACM
- CGM only has power when ignition is on
  - Unlike Ford gateways

Mkt	Year/Make	Model	Module		Vehicle Interface	OBD/DLC Connect Adapter/Cable	D2M Connect Adapter/Cable	Module Location
4	2023 Chevrolet	Corvette	ACM	Data	CDR 900	CDR 900 - click here	Cable ID# 845	Center tunnel, forward

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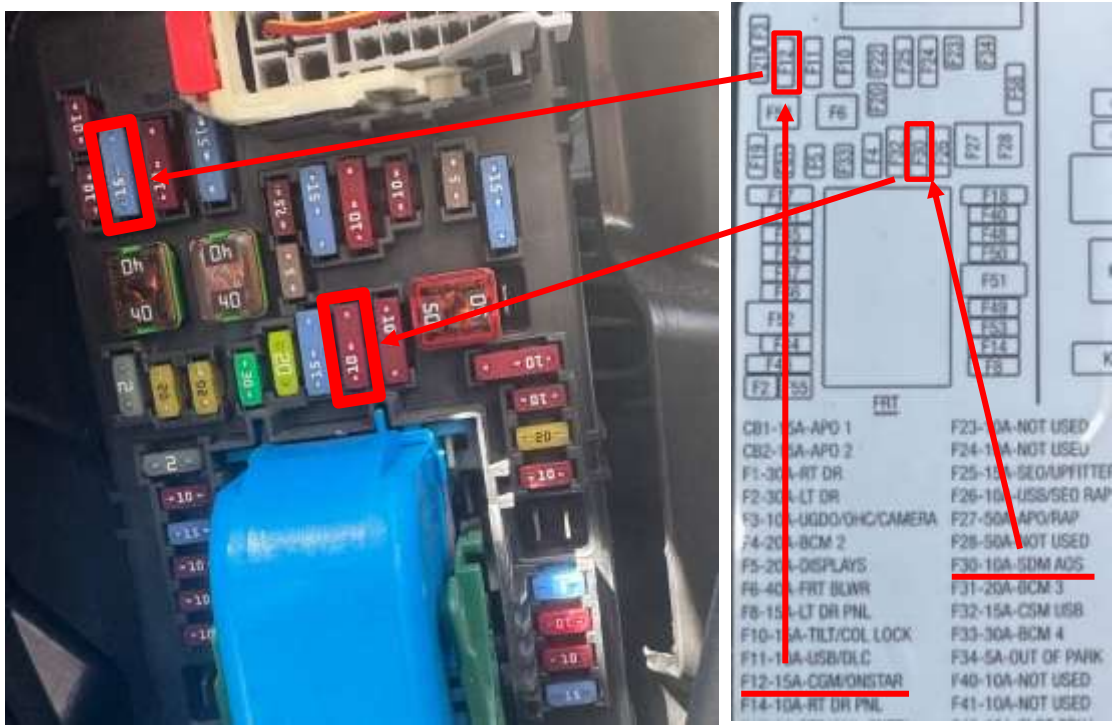
2021 Chevrolet Tahoe





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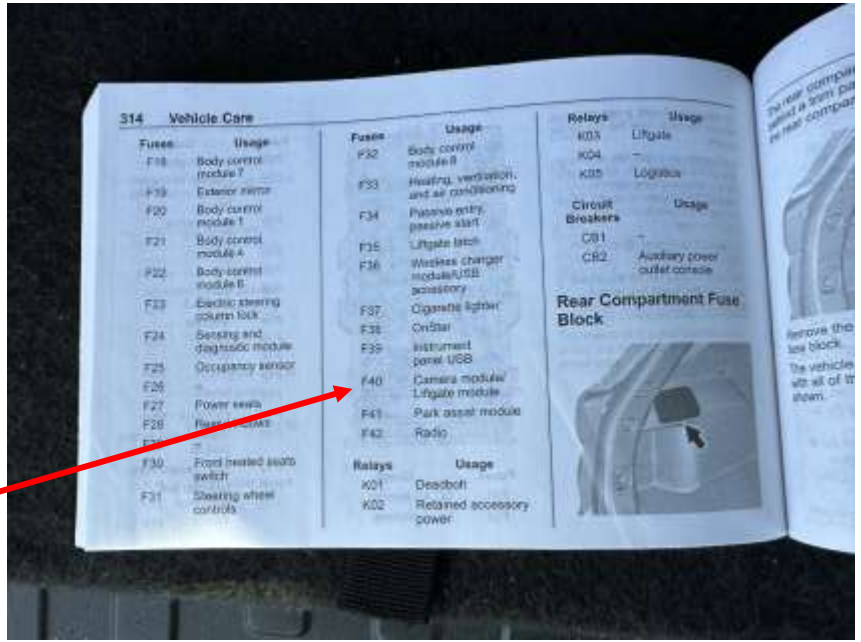
2021 Chevrolet Tahoe



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Backpower FCM?

- Why?
  - Readily available on windshield for D2M on-vehicle
  - Cable availability?
  - DLC Kit only users
  - Cases where removal or direct access not an option
- Listed in Owners Manual
  - Front Left Occupant Compartment
    - Fuse 40

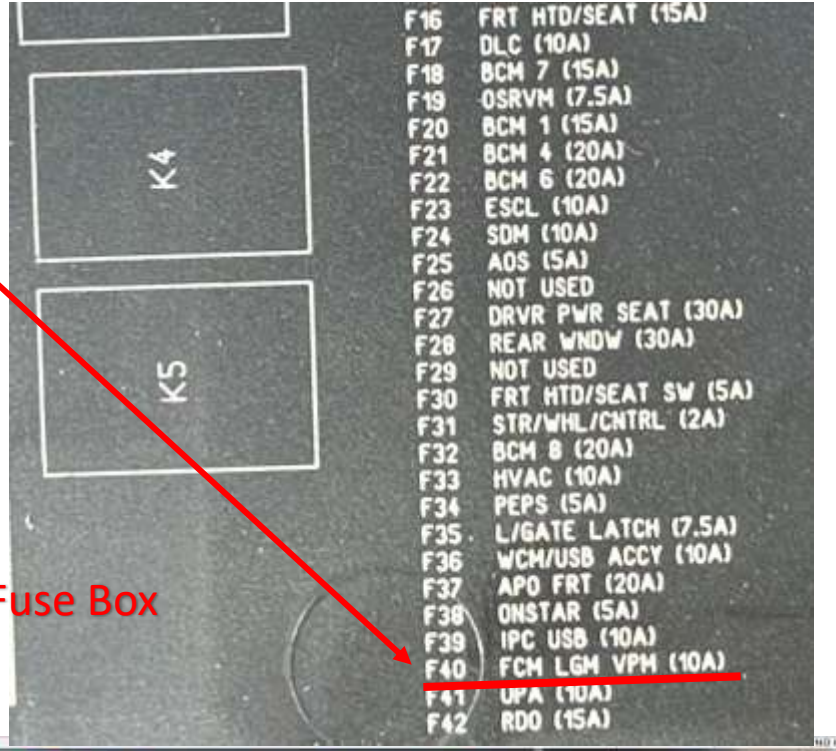


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2020 Equinox - Backpower FCM

- Plastic Fuse Box Cover
  - Lists as FCM / LGM / VPM
  - FCM = Front Camera Module
  - LGM = Lift Gate Module
  - VPM = Video Processing Module
    - Additional Fuse in rear for this ECU

Instrument Panel/Passenger Compartment Fuse Box



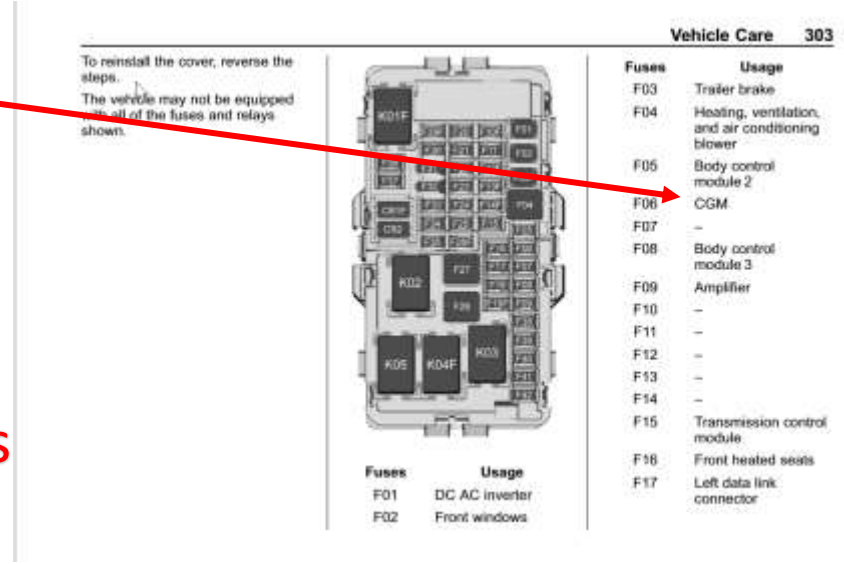
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2020 Equinox - CGM

- CGM Must also be powered

Rear Compartment Fuse Box

2 Fuses in two different Locations



To reinstall the cover, reverse the steps.

The vehicle may not be equipped with all of the fuses and relays shown.

Fuses	Usage
F01	DC AC inverter
F02	Front windows

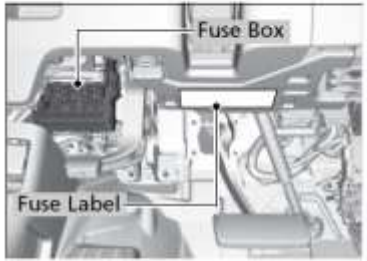
Fuses	Usage
F03	Trailer brake
F04	Heating, ventilation, and air conditioning blower
F05	Body control module 2
F06	CGM
F07	–
F08	Body control module 3
F09	Amplifier
F10	–
F11	–
F12	–
F13	–
F14	–
F15	Transmission control module
F16	Front heated seats
F17	Left data link connector

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HONDA

Interior Fuse Box

Located under the dashboard.



Fuse locations are shown on the label on the cover.

Locate the fuse in question by the fuse number and label number.

Circuit protected and fuse rating

	Circuit Protected	Amps
1	–	–
2	ACG	10 A
3	SRS	10 A
4	Fuel Pump	15 A
5	Meter	10 A
6	Power Window	7.5 A
7	VB SOL	7.5 A
8	Door Lock Motor 2 (Unlock)	15 A
9	Door Lock Motor 1 (Unlock)	10 A
10	Trailer*	(7.5 A)
11	Moonroof*	(20 A)
12	Accessory Power Socket (Center Console)*	(15 A)
13	–	–
14	Seat Heaters*	(20 A)
15	Driver's Door Lock Motor (Unlock)	10 A
16	Driver's Power Seat Sliding*	(20 A)
17	Driver's Power Seat Reclining*	(20 A)
18	–	–
19	ACC	7.5 A
20	ACC Key Lock	7.5 A
21	Daytime Running Lights	7.5 A
22	A/C	7.5 A
23	Rear Wiper	10 A
24	ABS/VSA	7.5 A
25	Audio	10 A
26	–	–
27	Accessory Power Socket (Front)	15 A
28	Washer	15 A
29	ODS	7.5 A
30	Driver's Door Lock Motor (Lock)*	10 A
31	–	–
32	Door Lock Motor 2 (Lock)	15 A
33	Door Lock Motor 1 (Lock)	15 A
34	Small Lights	10 A
35	Illumination	7.5 A
36	–	–
37	–	–
38	Left Headlight High Beam	10 A
39	Right Headlight High Beam	10 A
40	TPMS	7.5 A
41	Door Lock	20 A
42	Driver's Power Window	20 A
43	Rear Passenger's Side Power Window	20 A
44	Front Passenger's Side Power Window	20 A
45	Rear Driver's Side Power Window	20 A
46	–	–

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Honda Fuse Panel - Civic



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Honda Fuse Panel - Odessy



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2018 Honda CRV





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Newer Honda Repowering Considerations

- “Battery Cut-off” contained in under-hood Power Distribution Center
- Power must be jumped across fusible link



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2018 Honda CR-V

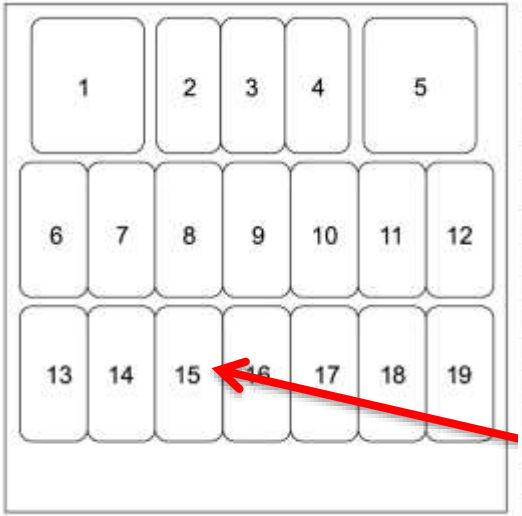
- “Battery Cut-off” contained in under-hood Power Distribution Center
- Power must be jumped across fusible link



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Mazda

Fuse block (Passenger's side)



	DESCRIPTION	FUSE RATING	PROTECTED COMPONENT
1	P/W	30 A	Power window
2	M.DEF	—	—
3	STARTER	10 A	Engine control system
4	ENG3	20 A	Engine control system
5	P/W	—	—
6	P.OUTLET	15 A	Accessory sockets (Cargo compartment)
7	SHIFT/L	5 A	—
8	CIGAR	15 A	Accessory sockets (Dashboard)
9	MIRROR	7.5 A	Power control mirror
10	A/C	10 A	Air conditioner
11	F.WIP	25 A	Windshield wipers and washer
12	R.WIP	15 A	Rear window wiper
13	ENG	—	—
14	METER	10 A	Instrument cluster
15	SAS	10 A	Air bag
16	S.WARM	15 A	Seat warmer*
17	ABS/DSC	—	—
18	EHPAS	5 A	Power assist steering
19	ENG2	15 A	Engine control system

*Some models.

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Mazda Fuse Panels



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Mercedes Example

- Paper Fuse Guide in vehicle
- Similar to BMW
- Many fuses listed for “Airbag”
- For THIS vehicle, Fuses 94/95 were the only ones ignition switched

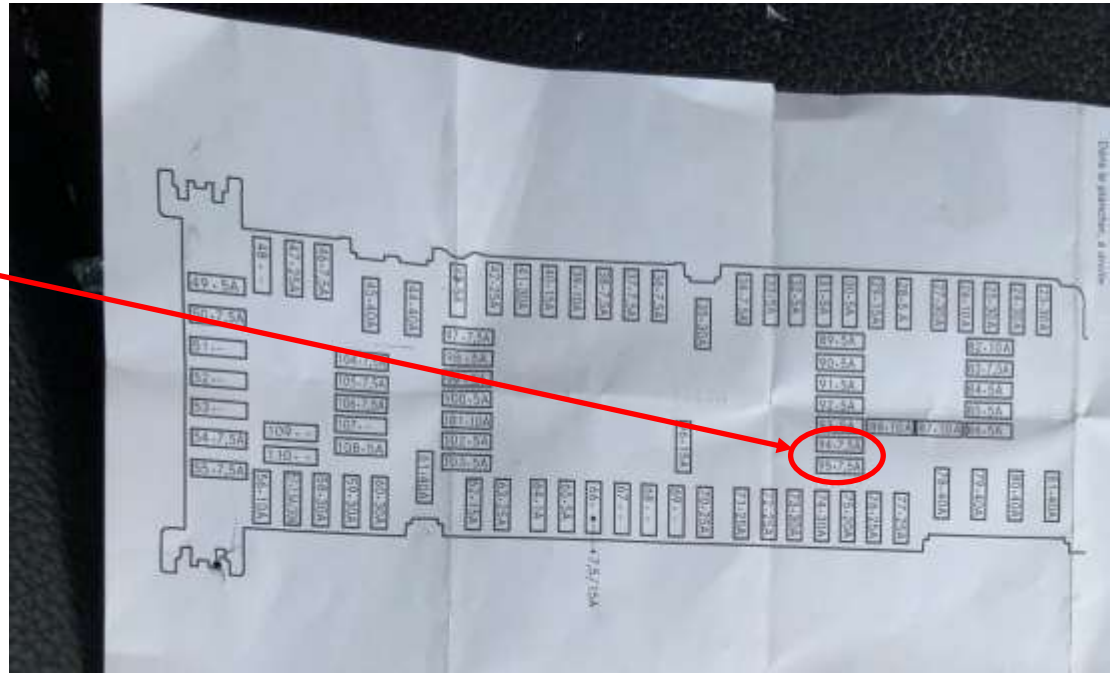




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Mercedes Example

Function	Fuse No.
4-Wheel drive	31, 101
Automatic transmission	27, 100, 217, 220
A/C system/ heater	34, 81, 102, 202
Airbag Systems	38, 44, 45, 94, 95
Anti-theft alarm system	85, 201
CBC/SAM	25, 79, 80, 98, 236
Central locking	55, 86
Central consol socket	70
Cigar lighter / socket front	72
Damper adjustment	47
Diagnosis	54, 87, 221, 225



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Mercedes Example

- 2015 Mercedes CLA
- Fuse Panel under right front dash
- Under carpet
- No Fuse map on plastic cover



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Mercedes Example

- Both Fuses together were powered
- Likely redundant fuses
- Need back-power “Y” cable and two fuse adapters




40
40
40
40
40
40

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Mercedes Example

- Small battery for “Eco-Start” system located near fuse panel
- May allow for additional powering rig that clips to battery!



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Nissan Fuse Panels







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Nissan Airbag Fuse

- 10 amp fuse labeled “Airbag” in Occupant Compartment Fuse Panel
- Remember Owners Manuals do not have fuse maps!!
- See Nissan Fuse Guide in DropBox



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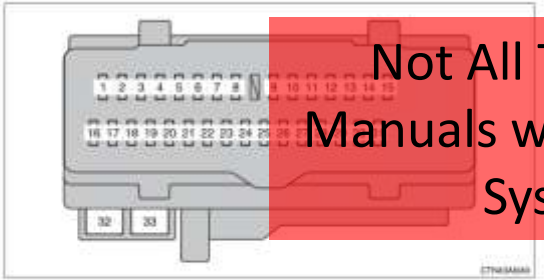
Nissan Fuse Panel Location Chart

Vehicle	Fuse Panel Location	Fuse Label	Fuse Size	Fuse Location
370Z	Drivers side kick panel	Airbag	10A	Left side, top fuse
Altima	Left Side of Dashboard above Knee Bolster	Airbag	10A	Bottom Row, 1st Fuse from right
Armada	not available at this time			
Cube	Left side of Dashboard	Airbag	10A	Right Side, top fuse
Frontier	Inside Glove Box	Airbag	10A	2nd from bottom, right side
Juke	Left side of Dashboard	Airbag	10A	Ride Side, bottom fuse
Maxima	Left Side of Dashboard above Knee Bolster	Airbag	10A	Left Side, Second fuse from top
Murano	Left Side of Dashboard above Knee Bolster	Airbag	10A	Left Side, Second fuse from top
NV Van/Bus	Left Side of Dashboard above Knee Bolster	Airbag	10A	Left Side, Second fuse from top
Pathfinder	not available at this time			
Quest	Left Side of Dashboard above Knee Bolster	Airbag	10A	Left Side, Second fuse from top
Rogue	Left Side of Dashboard above Knee Bolster	Airbag	10A	Bottom Row, second fuse from right
Titan	Right side of Dash beside Glove Box	Airbag	10A	Right Side, second fuse from bottom
Versa	Left Side of Dashboard above Knee Bolster	Airbag	10A	Top Row, 1st Fuse from left
Xterra	Right side of Dash beside Glove Box	Airbag	10A	Right Side, second fuse from bottom

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Toyota Considerations

■ **Under the instrument panel**



	Fuse	Ampere	Circuit
1	RR DOOR RH	25 A	Rear right power window
2	RR DOOR LH	25 A	Rear left power window
3	FUEL OPN	7.5 A	No circuit
4	FR FOG	15 A	Front fog lights
5	OBD	7.5 A	On-board diagnosis system
6	ECU-B NO.2	7.5 A	Power windows
			High mounted stoplight, shift lock control system, multiport fuel injection system/sequential multiport

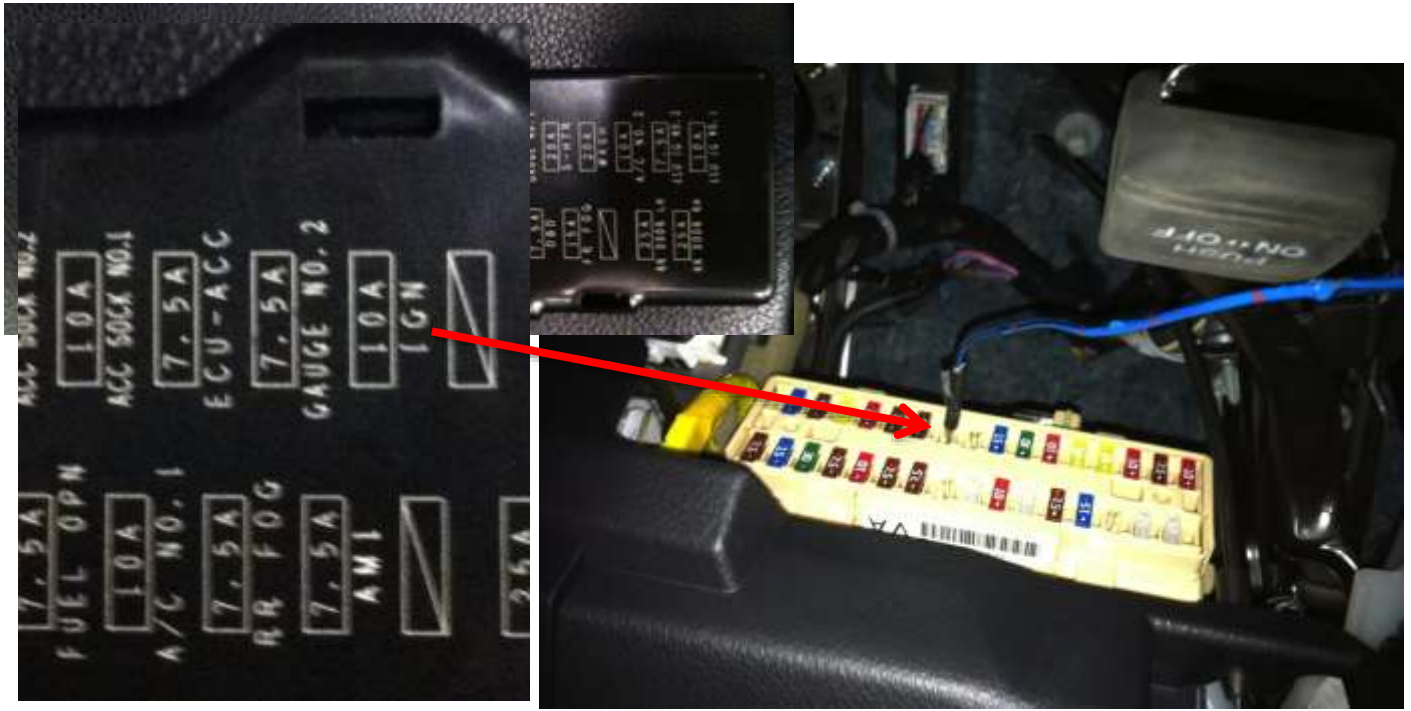
4-3. Do-it-yourself maintenance

	Fuse	Ampere	Circuit
22	WIP	25 A	Windshield wipers and washer
23	H-LP LVL	7.5 A	No circuit
24	IND	15 A	No circuit
25	IGN	10 A	Multiport fuel injection system/ sequential multiport fuel injection system, SRS airbag system, steering lock system, front passenger occupant classification system, smart key system, electronic throttle control system
26	GAUGE NO.2	7.5 A	Gauges and meters, multi-information display, clock
27	ECU-ACC	7.5 A	Clock, main body ECU, shift lock control system, outside rear view mirror, smart key system
28	CIG	20 A	No circuit
29	PWR OUTLET	20 A	Power outlet
30	RADIO NO.2	7.5 A	Audio system, navigation system
31	MIR HTR	10 A	Outside rear view mirror defoggers
32	POWER	30 A	Power windows
33	P/SEAT	30 A	Power seats

Not All Toyota Owners Manuals will indicate Airbag System Fuse

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Toyota Considerations



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Volvo Fuse Box Locations

Location of the fuse boxes

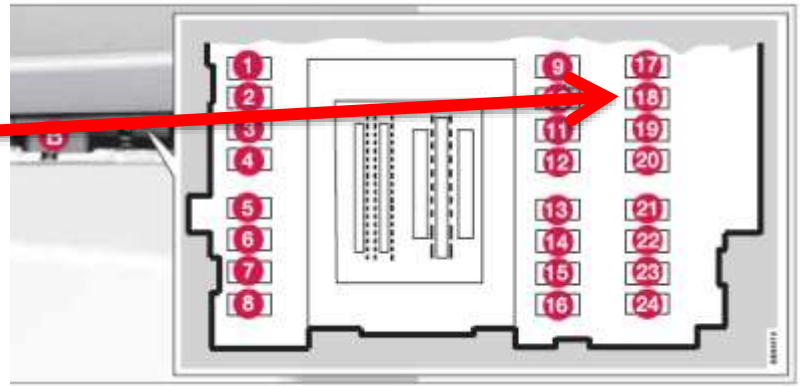


1. Engine compartment
2. Under the glove compartment
3. Under the glove compartment
4. Trunk
5. Not in use

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Airbag Fuses

Pos	Function	A
17	-	
18	Airbag system, occupant weight sensor	10
19	Collision warning system*	5
20	Throttle pedal, auto-dim mirror function, heated rear seats*	7.5
21	-	
22	Brake lights	5
23	Power moonroof*	20
24	Immobilizer	5



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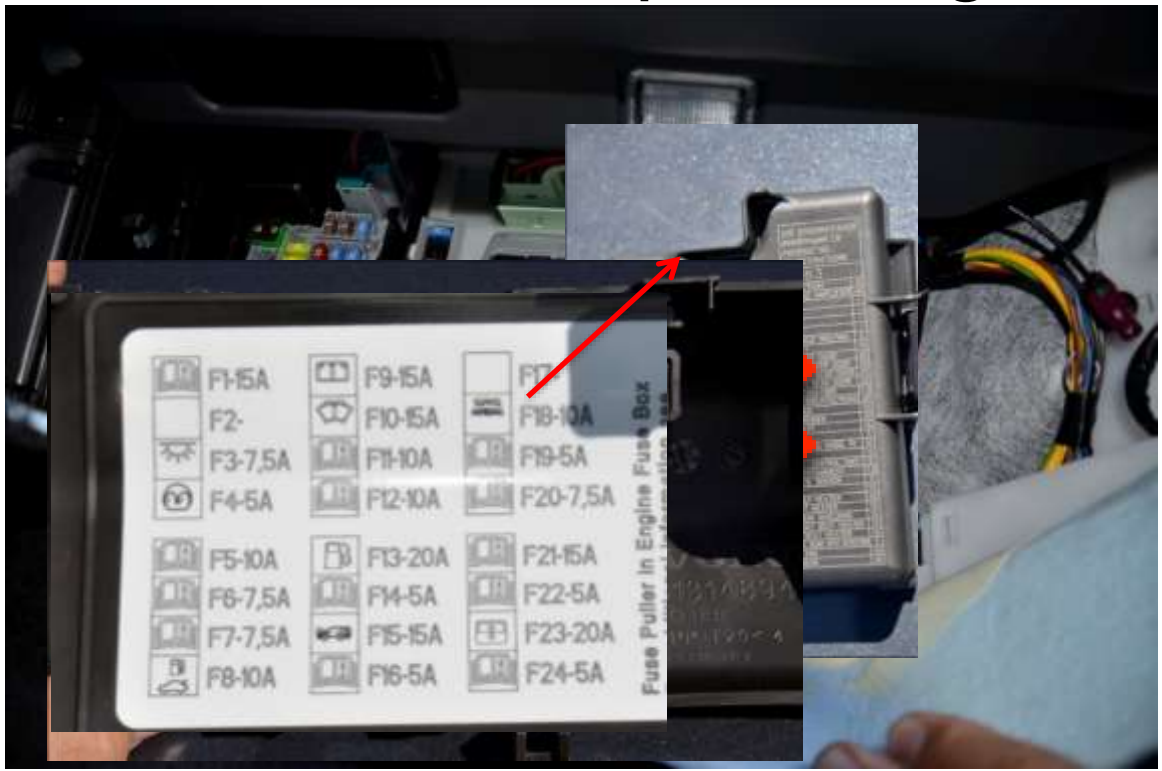
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Volvo Backpowering



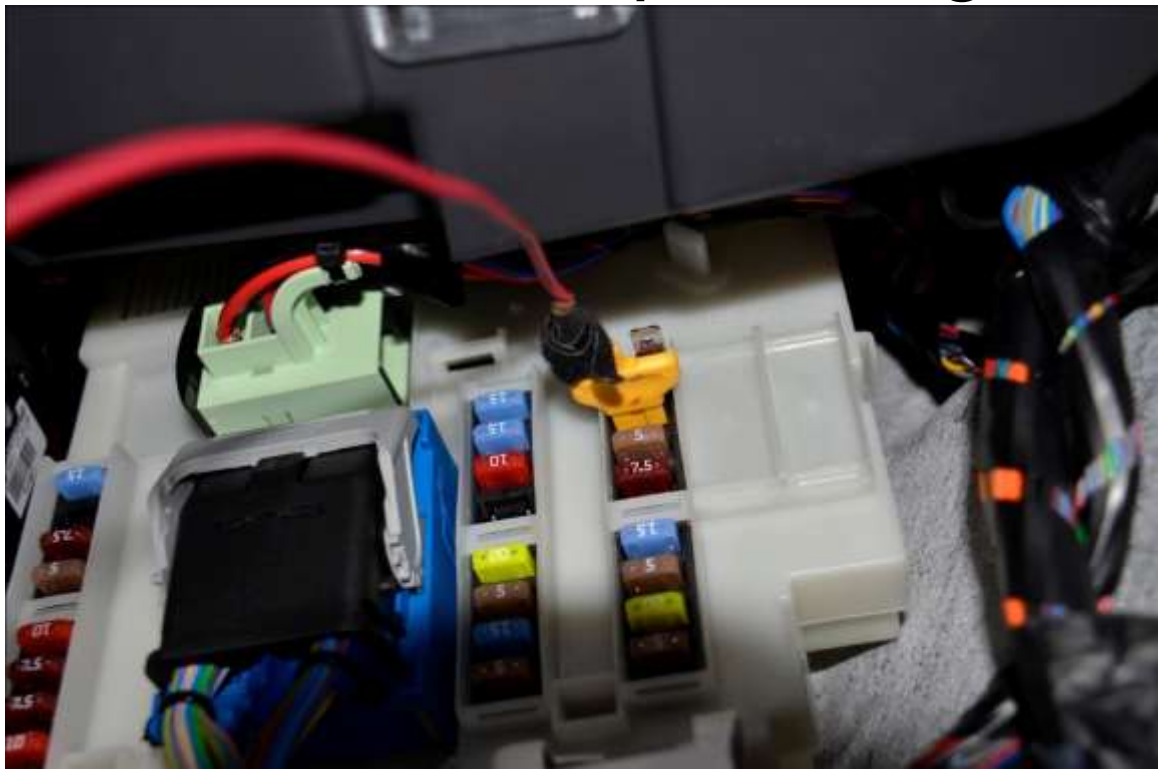
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Volvo Backpowering



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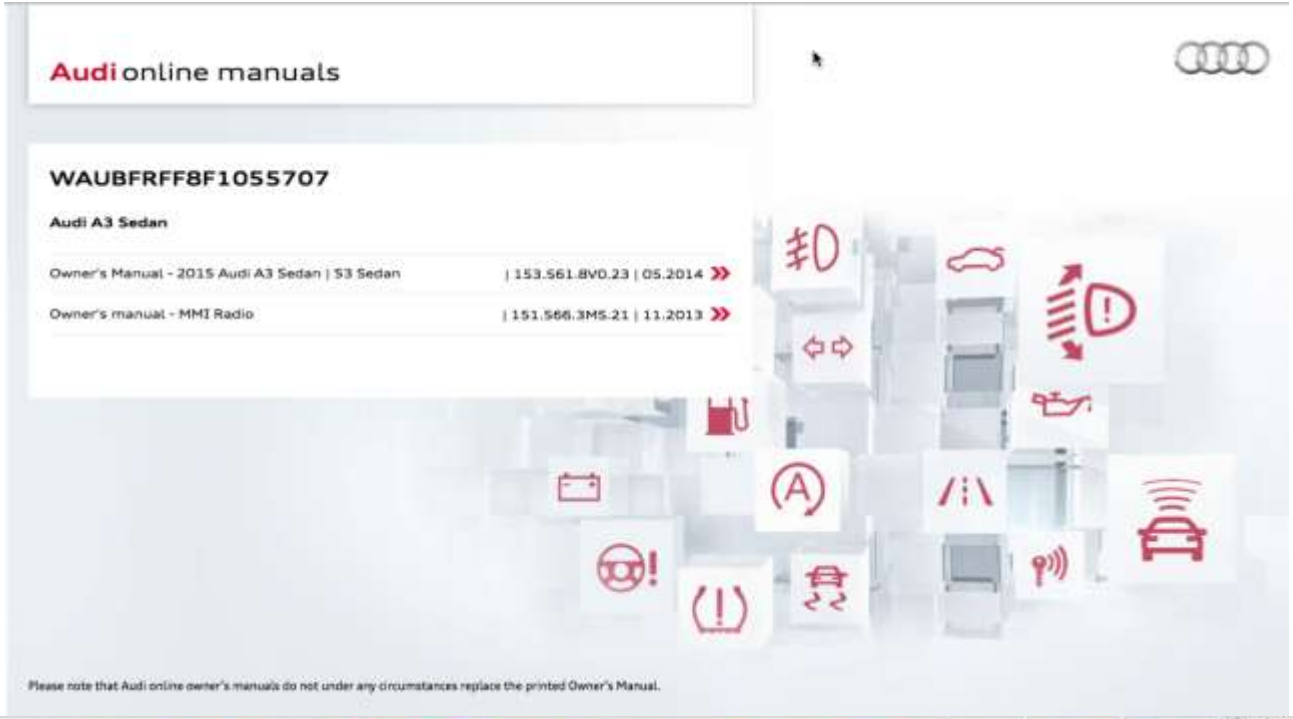
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Audi Backpowering

- Adobe Flash
- Can NOT be downloaded or saved locally!



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Audi Backpowering



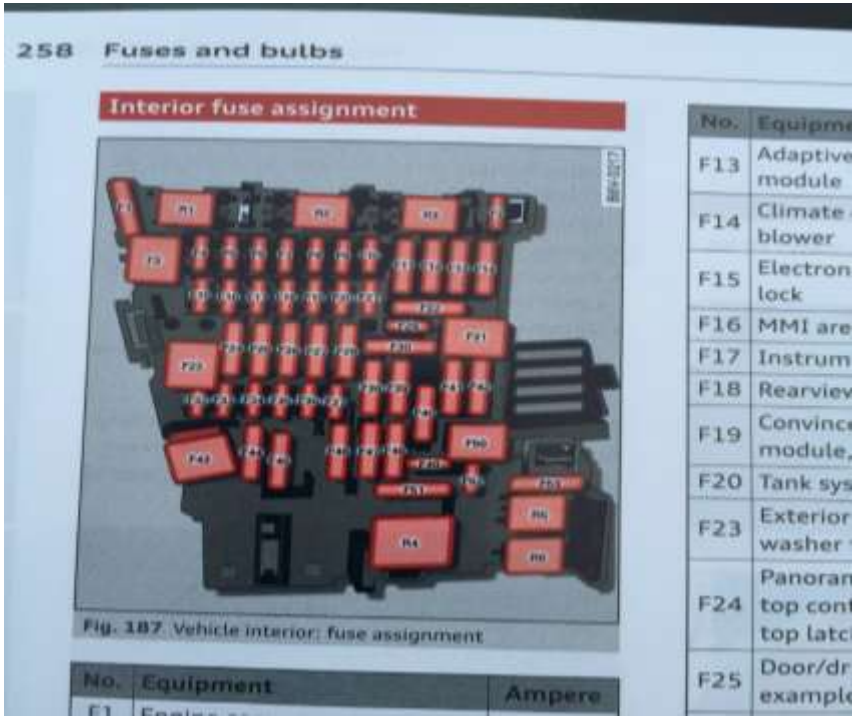


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Audi Backpowering



No.	Equipment	Ampere
F1	Engine components	30
F2	Seat adjustment	10
F3	Hydraulic pump cover (Cabriolet)	40
F4	MMI controls, MMI components	7.5
F5	Gateway	5
F6	Anti-theft alarm system	5
F7	Climate/heating control, selector lever (automatic transmission), parking heater, rear window heater relay coil	10
F8	Diagnosis, electrical parking brake switch, light switch, rain/light sensor, interior lighting	10
F9	Steering column switch module	1
F10	Display	5
F11	Reversible driver's side safety belt tensioners	25
F12	MMI area	
F13	module	30
F14	Climate control system blower	30
F15	Electronic steering column lock	10
F16	MMI area	7.5
F17	Instrument cluster	5
F18	Rearview camera	7.5
F19	Convince key system control module, tank system	7.5
F20	Tank system	7.5
F23	Exterior lighting, heated washer fluid nozzles	40
F24	Panorama sunroof / power top control module, power top latch (Cabriolet)	30/30
F25	Door/driver's side doors (for example power windows)	30
F26	Seat heating	30
F27	Sound amplifier	30
F28	Power top control module, electronics (Cabriolet)	5
F29	Interior lights	7.5
F31	Exterior lighting	40
F32	Driver assistance systems	7.5
F33	Airbag	5
F34	Button illumination, coils for upper cabin heating relay (Cabriolet) and socket relay, interior sound, back-up light switch, temperature sensor, oil level sensor	7.5
F35	Diagnosis, headlight range control system, air quality sensor, automatic dimming rearview mirror	10
F36	Right cornering light / right LED-headlight	15
F37	Left cornering light / left LED-headlight	15



258 Fuses and bulbs

Interior fuse assignment

Fig. 187 Vehicle interior: fuse assignment

No.	Equipment	Ampere
F1		

No.	Equipm
F13	Adaptive module
F14	Climate blower
F15	Electron lock
F16	MMI are
F17	Instrum
F18	Rearviev
F19	Convinc module,
F20	Tank sys
F23	Exterior washer
F24	Panoran top cont top latc
F25	Door/dr example

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Audi Backpowering





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Chip Off



- My vehicle looks like this…I guess we aren't getting any EDR data.
- If you have an important case don't be so quick to jump to conclusions

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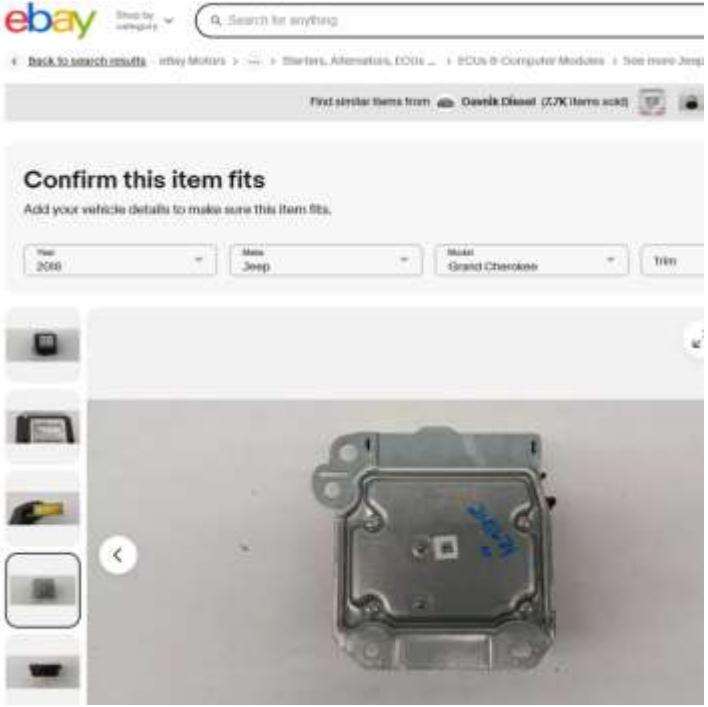
Chip Off





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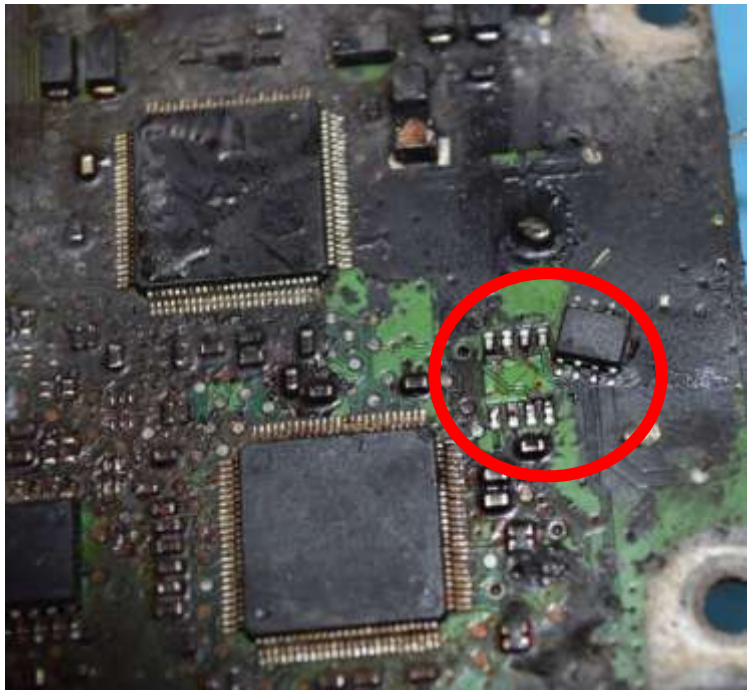
Chip Off – What to look for





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EEPROM from Damaged ACM transplanted to Surrogate ACM





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The Data

Pre-Crash Data (1st Prior Event - table 1 of 4)
(the most recent sampled values are recorded prior to the event)

Time Stamp (sec)	Pre-Crash Recorder Status	Speed, Vehicle Indicated (MPH [km/h])	Accelerator Pedal, % Full	Engine Throttle, % Full	Service Brake	Engine RPM	ABS Activity	Stability Control
-5.0	Complete	117 [188]	100	52	Off	5,225	No	On
-4.9	Complete	117 [188]	100	52	Off	5,232	No	On
-4.8	Complete	117 [188]	100	52	Off	5,230	No	On
-4.7	Complete	117 [188]	100	52	Off	5,232	No	On
-4.6	Complete	117 [188]	100	52	Off	5,236	No	On
-4.5	Complete	117 [188]	100	52	Off	5,236	No	On
-4.4	Complete	117 [188]	100	51	Off	5,244	No	On
-4.3	Complete	117 [188]	99	52	Off	5,246	No	On
-4.2	Complete	117 [188]	94	52	Off	5,242	No	On
-4.1	Complete	117 [189]	76	51	Off	5,248	No	On
-4.0	Complete	117 [189]	9	51	Off	5,244	No	On
-3.9	Complete	117 [188]	0	30	Off	5,203	No	On
-3.8	Complete	117 [188]	32	26	Off	5,201	No	On
-3.7	Complete	117 [188]	56	36	Off	5,259	No	On
-3.6	Complete	117 [188]	64	44	Off	5,265	No	On
-3.5	Complete	117 [188]	74	50	Off	5,249	No	On
-3.4	Complete	117 [188]	83	50	Off	5,242	No	On
-3.3	Complete	117 [188]	87	50	Off	5,234	No	On
-3.2	Complete	117 [188]	88	50	Off	5,234	No	On
-3.1	Complete	117 [188]	88	50	Off	5,244	No	On
-3.0	Complete	117 [189]	88	50	Off	5,248	No	On
-2.9	Complete	117 [189]	88	50	Off	5,249	No	On
-2.8	Complete	117 [189]	87	50	Off	5,254	No	On
-2.7	Complete	117 [189]	86	50	Off	5,258	No	On
-2.6	Complete	117 [189]	82	50	Off	5,256	No	On
-2.5	Complete	117 [189]	77	50	Off	5,261	No	On
-2.4	Complete	117 [189]	75	50	Off	5,263	No	On
-2.3	Complete	118 [189]	74	50	Off	5,259	No	On

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Other Event Data Recorder Overview

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Heavy Vehicles

- Other tools are available for heavy vehicle downloading
- Data preservation is similar to other vehicles, but the downloading collection methods are completely different
- The key is to recognize and preserve possible data!





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Heavy Vehicles

- Heavy vehicles have EDR capabilities contained within the Electronic Control Module (ECM) and potentially other locations
- The ECM is similar to the Powertrain Control Module (PCM) on a Ford product, and it acts as the brain of the engine, and it controls all aspects of the engine's operation
- Unlike the ACM, the purpose of the ECM is to control the electrical system
  - It also helps fleet managers with performance, mechanics and monitoring

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Heavy Vehicles

- Light vehicles are covered by their manufacturer
- Heavy vehicle coverage is by ENGINE manufacturer, not truck make or model
- Data recording depends on a lot
  - Brand of engine
  - Type of ECM
  - End-user settings
  - Type of incident

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Heavy Vehicles

- Coverage by Engine Brand
  - Detroit Diesel, 1997 to present
  - Mercedes-Benz, 2000 to present
  - Caterpillar, 1997 to present
  - Cummins, 1999 to present
  - International, 2010 to present
  - Mack, 1998 to present
  - PACCAR, 2010 to present
  - Volvo, 2006 to present

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Heavy Vehicles

- Determining Coverage
  - Examine the engine for make, model, etc.
  - Labels on engine and door
  - Check the VIN using software (APP: VINTRUCKS by Vear)

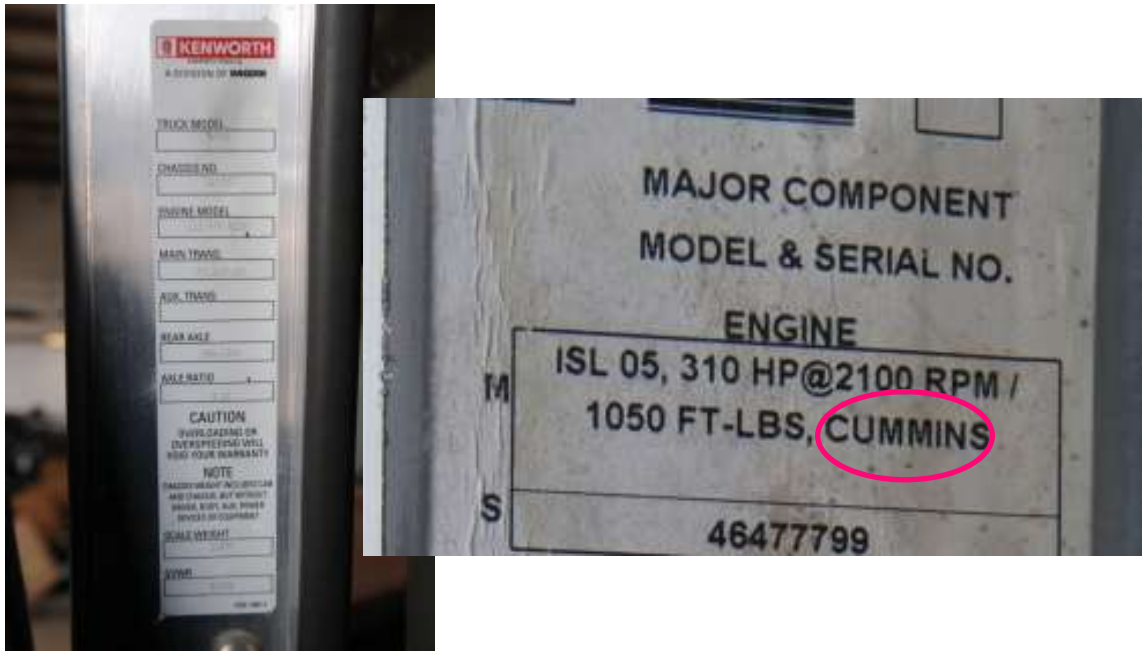




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Heavy Vehicles

- Determining Coverage



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Heavy Vehicles

- Information Possible from ECM Downloads
  - “Events” may be created similar to light vehicles based upon certain circumstances
  - Information may include:
    - Vehicle drive time, idle time
    - Fuel consumption, speeds
    - Odometer, clock time
    - Hard brake incidents, brake switch information, RPM

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Heavy Vehicles

- Preserving the Data
  - Turn off the truck and take the key
  - Secure the key and do not permit movement
  - Tow from the scene
    - Major parts may need disconnecting
    - Brakes may need caging
  - Apply power again for download, then do other inspection

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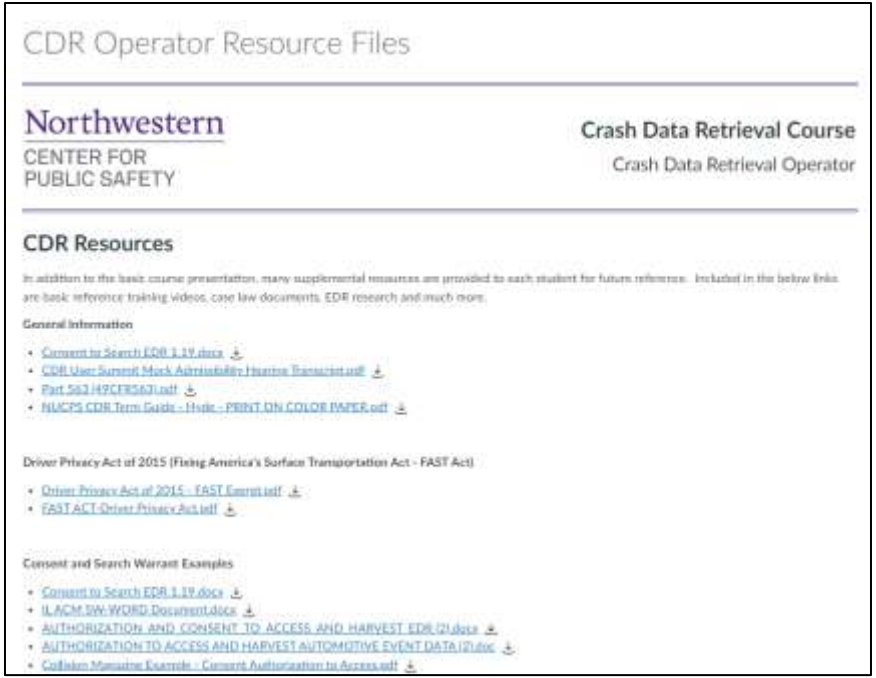


Resources Provided

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Resources

- Canvas Data and Documents
  - CDR Resources Page
  - Hyperlinks to various resources and documents
  - General data
  - Case law
  - Search documents
  - Vast amounts of resources for all operators to utilize



CDR Operator Resource Files

Northwestern
CENTER FOR
PUBLIC SAFETY

Crash Data Retrieval Course
Crash Data Retrieval Operator

CDR Resources

In addition to the basic course presentation, many supplemental resources are provided to each student for future reference. Included in the below links are basic reference training videos, case law documents, EDR research and much more.

General Information

- Consent to Search EDR 1.19.docx
- CDR User Summit Mock Admissibility Hearing Transcript.pdf
- Part 563 (49CFR563).pdf
- NUCPS CDR Term Guide - Hyde - PRINT ON COLOR PAPER.pdf

Driver Privacy Act of 2015 (Fixing America's Surface Transportation Act - FAST Act)

- Driver Privacy Act of 2015 - FAST Exempt.pdf
- FAST ACT-Driver Privacy Act.pdf

Consent and Search Warrant Examples

- Consent to Search EDR 1.19.docx
- ILACM SW WORD Document.docx
- AUTHORIZATION AND CONSENT TO ACCESS AND HARVEST EDR (2).docx
- AUTHORIZATION TO ACCESS AND HARVEST AUTOMOTIVE EVENT DATA (2).doc
- Collision Magazine Example - Consent Authorization to Access.pdf

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Resources

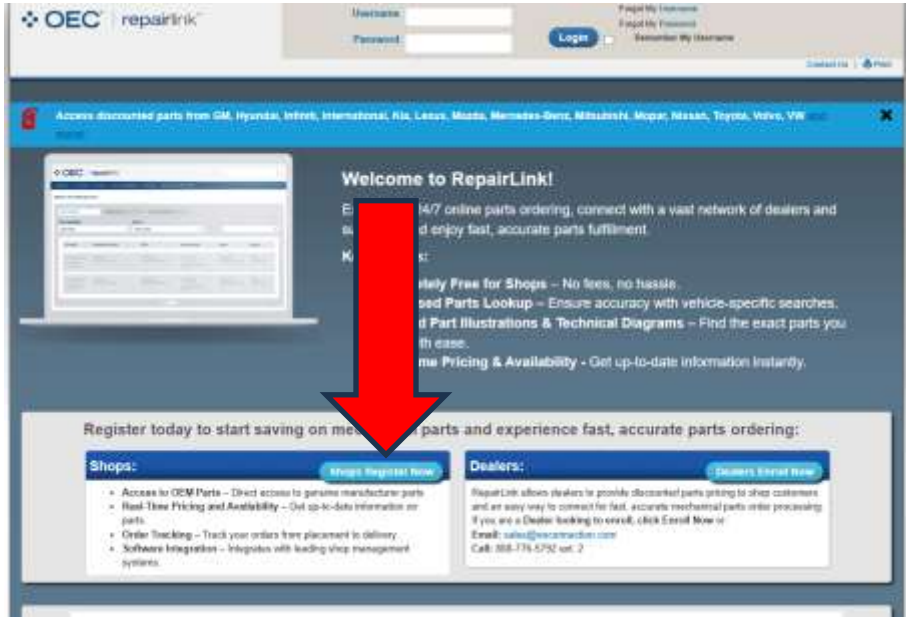
- CDR Tool
  - Groups IO
  - https://groups.io/g/cdrtool

  - Group accessible after completing this course
  - Posts with images, attachments and information from peers
  - Great problem-solving platform

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Resources

- RPO Codes – repairlinkshop.com



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Let's Recap!

- What is the hierarchy of Download Methods?
  1. DLC – DLC cable to vehicle OBD Port
  2. D2M – Direct to Module, module remains on vehicle
  3. D2M – Benchtop – module removed from vehicle and clamped prior to supplying power

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- What is the name of the component that stores crash data within an Airbag Control Module?

  Answer = EEPROM

- What does EEPROM stand for?

  Answer = Electrically Erasable Programmable Read-Only Memory

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- What position must a vehicle ignition be in for a DLC download in a non-note 6 vehicle?

  Answer = Run / On with engine off

- What is Part 563?

  Answer = The Federal regulations that govern the collection, recording, and retrievability of crash data.

- Who can give consent to download EDR data?

  Answer = Vehicle owner or lessee

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- Where is CDR located inside of a vehicle?

  Answer = CDR is the tool used for downloading vehicles. It is not a component inside of a vehicle.

- What does PCM stand for? What vehicles have data stored in a PCM?

  1. Powertrain Control Module
  2. Some Ford/Lincoln/Mercury vehicles between 2003 and 2011

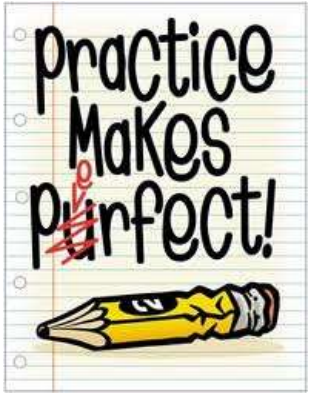
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Hands-On Exercises

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Student Exercises



- Each student must successfully complete
  - 2 downloads in a Direct to Module method
  - 1 downloads in a DLC method

- All downloads should be uploaded as a .CDRx or .CDRnx only to Canvas, per instructor instructions
  - Name the files with your last name and download number

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End of In-Class

- Selection of Available Modules in class for download
- “Vehicle Simulators” for back-power practice
- Parking lot for on-vehicle practice

